HUMAN ANATOMY AND PHYSIOLOGY

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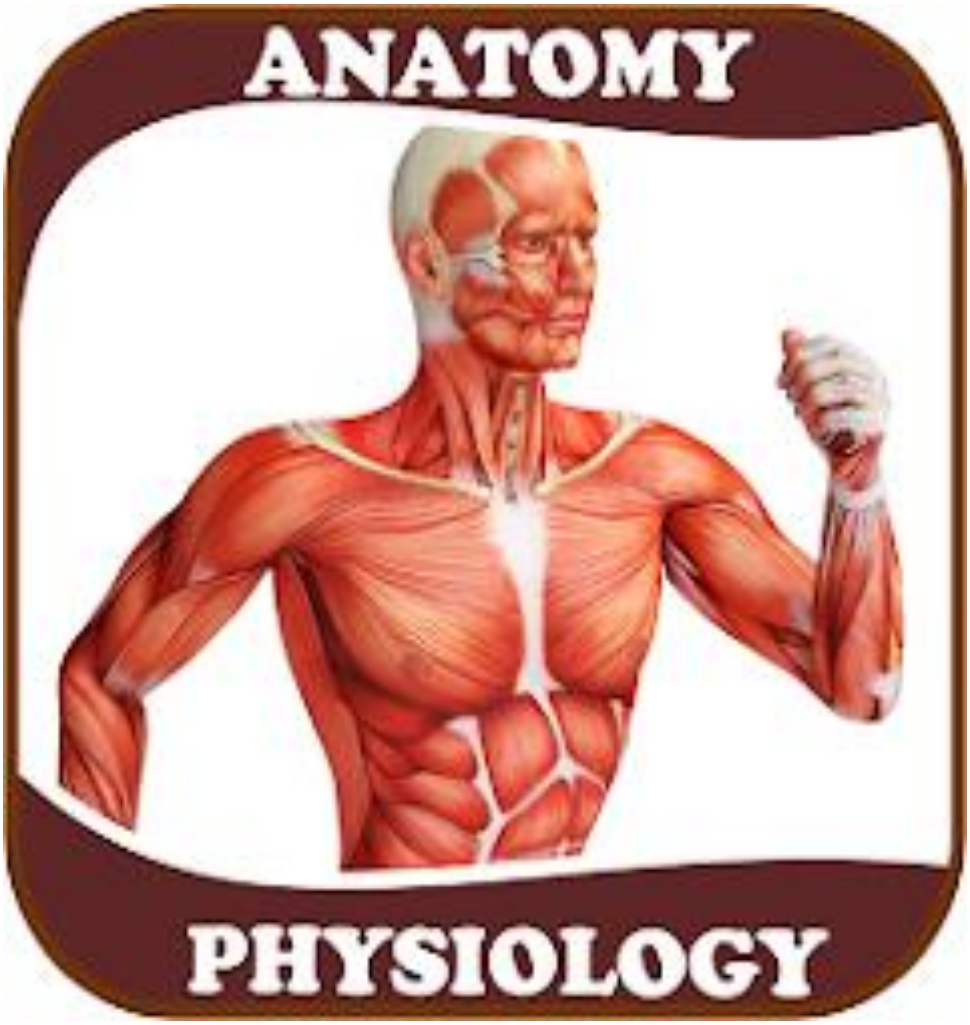

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Dedication

This work is lovingly dedicated to <u>My Family</u>

For their abundant support, for their patience and understanding, and for their love.

I would also like to dedicate this work to the immaculate spirit of <u>my daughter Lara</u>

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CHAPTER ONE

INTRODUCTION TO HUMAN ANATOMY AND PHYSIOLOGY

- **Anatomy** is the science of body structures and the relationships among them (structures of the body).

- **Physiology** is the science of body function (how the body parts work).

❖ **The study of anatomy includes many sub specialties:**

- **Gross anatomy** studies body structure without microscope.

- **Microscopic anatomy** (Histology) requires the use of microscope to study tissues that form the various organs of the body.

❖ **Level of structural organization of the body**

The human body has different structural levels of organization, starting with atoms molecules and compounds and increasing in size and complexity to cells, tissues, organs and the systems that make up the complete organism.

- **Atoms molecules and compounds**: At its simplest level, the body is composed of atoms. The most common elements in living organism are carbon, hydrogen, oxygen, nitrogen phosphorus and sulfur. (Atoms → Molecule → Compounds)

- **Cell:** The smallest independent units of life. All life depends on the many chemical activities of cells. Some of the basic functions of cell are: growth, metabolism, irritability and reproduction.

- **Tissue:** tissue is made up of many similar cells that perform a specific function. The various tissues of the body are divided in to four groups. These are epithelial, connective, nervous and muscle tissue.

1. Epithelial tissue: Found in the outer layer of skin, lining of organs, blood and lymph vessels and body cavities.

2. Connective tissue: Connects and supports most part of the body. They constitute most part of skin, bone and tendons.

3. Muscle tissue: Produces movement through its ability to contract. This constitutes skeletal, smooth and cardiac muscles.

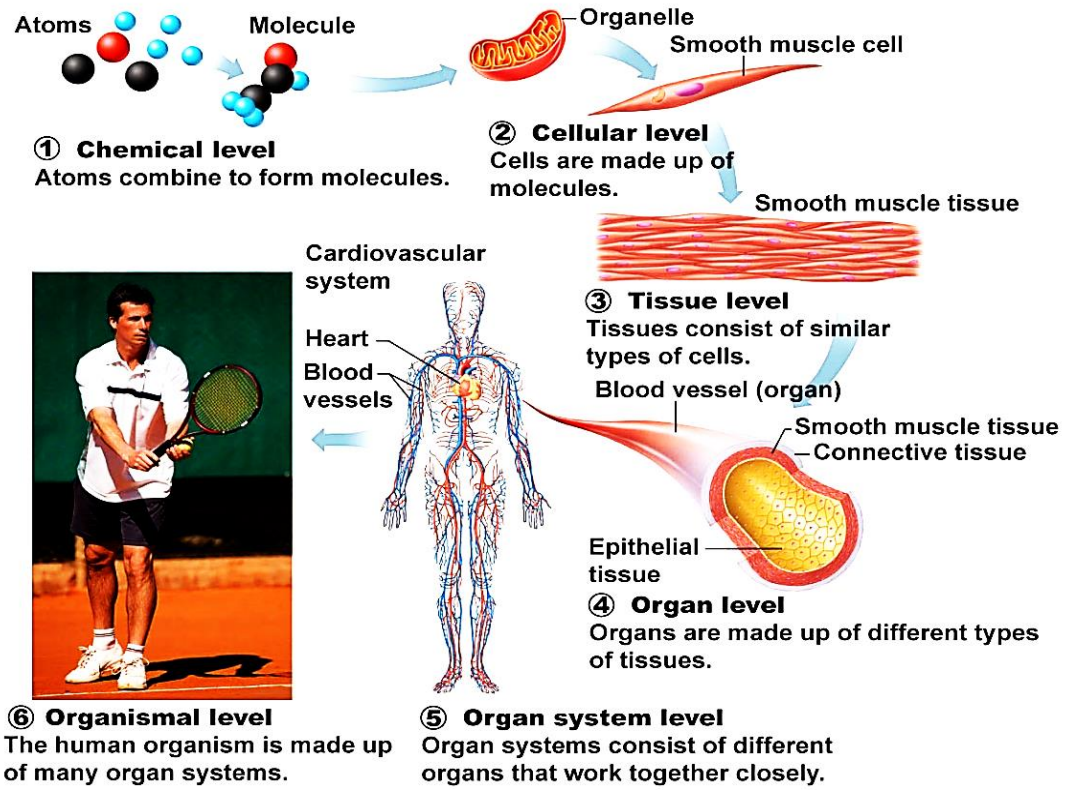
4. Nerve tissue: Found in the brain, spinal cord and nerves. It responds to various types of stimuli and transmits nerve impulses.

- **Organ:** Is an integrated collection of two or more kinds of tissue that works together to perform specific function. For example: Stomach is made of all type of tissues.

- **System:** Is a group of organs that work together to perform major function.

- For example: Respiratory system contains several organs.

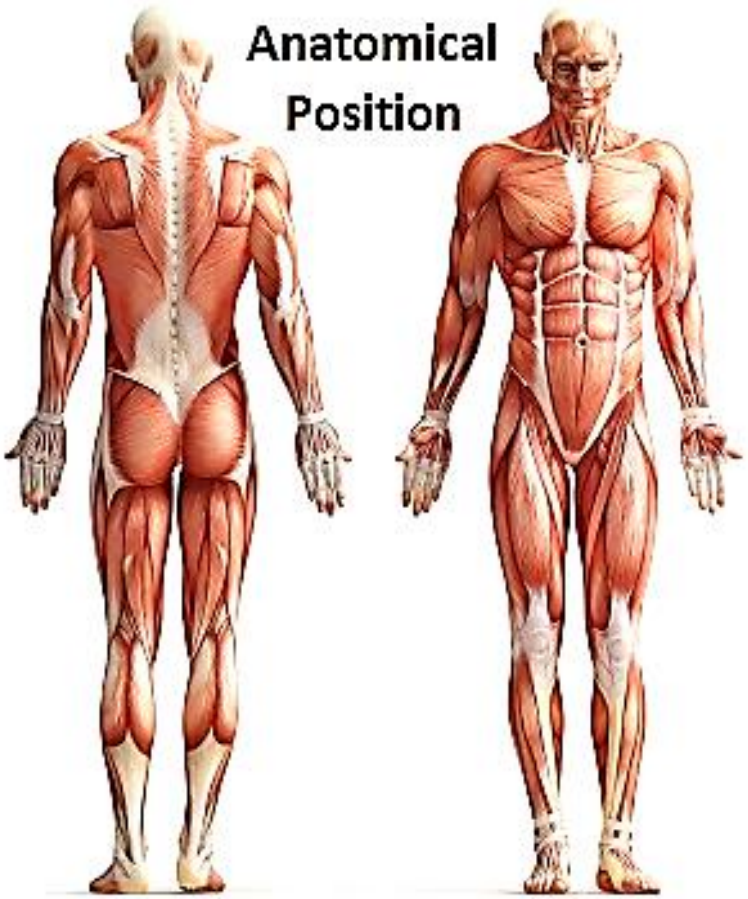
- **Organism level:** The various organs of the body form the entire organism.



Anatomical Position

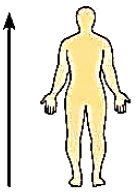
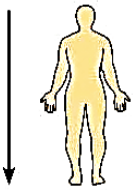
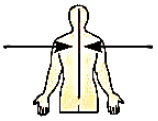
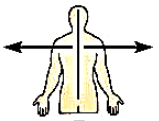
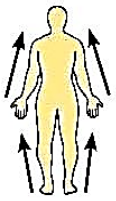
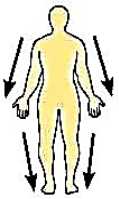
Anatomical positions are universally accepted as the starting points for positional references to the body.

In anatomical position the subject is standing erect and facing the observer, the feet are together, and the arms are hanging at the sides with the palms facing forward.



Anatomical directional terms

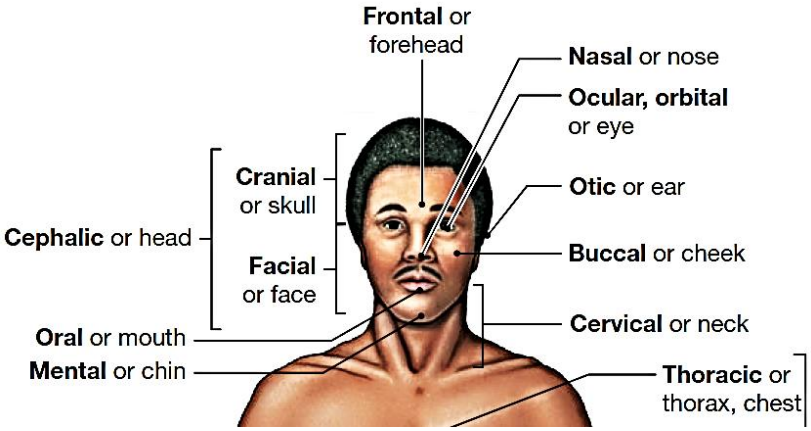
Directional terms, words that describe the position of one body part relative to another. Several directional terms are grouped in pairs that have opposite meanings, such as anterior (front) and posterior (back).

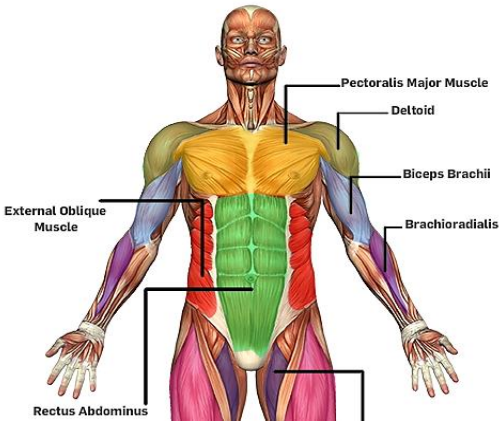
TERM	DEFINITION	EXAMPLE	
Superior (cranial)	Toward the head end or upper part of a structure or the body; above		The head is superior to the abdomen.
Inferior (caudal)	Away from the head end or toward the lower part of a structure or the body; below		The navel is inferior to the chin.
Ventral (anterior)*	Toward or at the front of the body; in front of		The breastbone is anterior to the spine.
Dorsal (posterior)*	Toward or at the back of the body; behind		The heart is posterior to the breast-bone.
Medial	Toward or at the midline of the body; on the inner side of		The heart is medial to the arm.
Lateral	Away from the midline of the body; on the outer side of		The arms are lateral to the chest.
Intermediate	Between a more medial and a more lateral structure		The collarbone is intermediate between the breastbone and shoulder.
Proximal	Closer to the origin of the body part or the point of attachment of a limb to the body trunk		The elbow is proximal to the wrist.
Distal	Farther from the origin of a body part or the point of attachment of a limb to the body trunk		The knee is distal to the thigh.
Superficial (external)	Toward or at the body surface		The skin is superficial to the skeletal muscles.
Deep (internal)	Away from the body surface; more internal		The lungs are deep to the skin.

Body Regions

The major body regions are: the head, neck, trunk, upper extremity, and lower extremity .

- **Head**: is divided into a facial region, which includes: the eyes, nose, and mouth, and a cranial region or cranium which covers and supports the brain .
- **Neck:** referred to as the cervical region, supports the head and permits it to move .

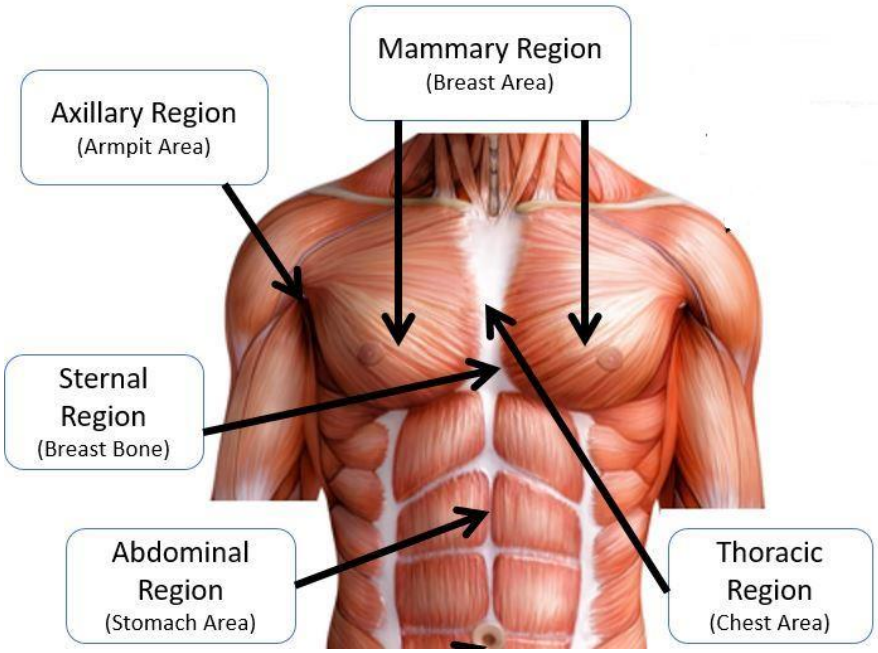




- **The trunk** (Torso) is the portion of the body to which the neck and upper and lower extremities attach. It divided into the thorax, abdomen, and pelvis.

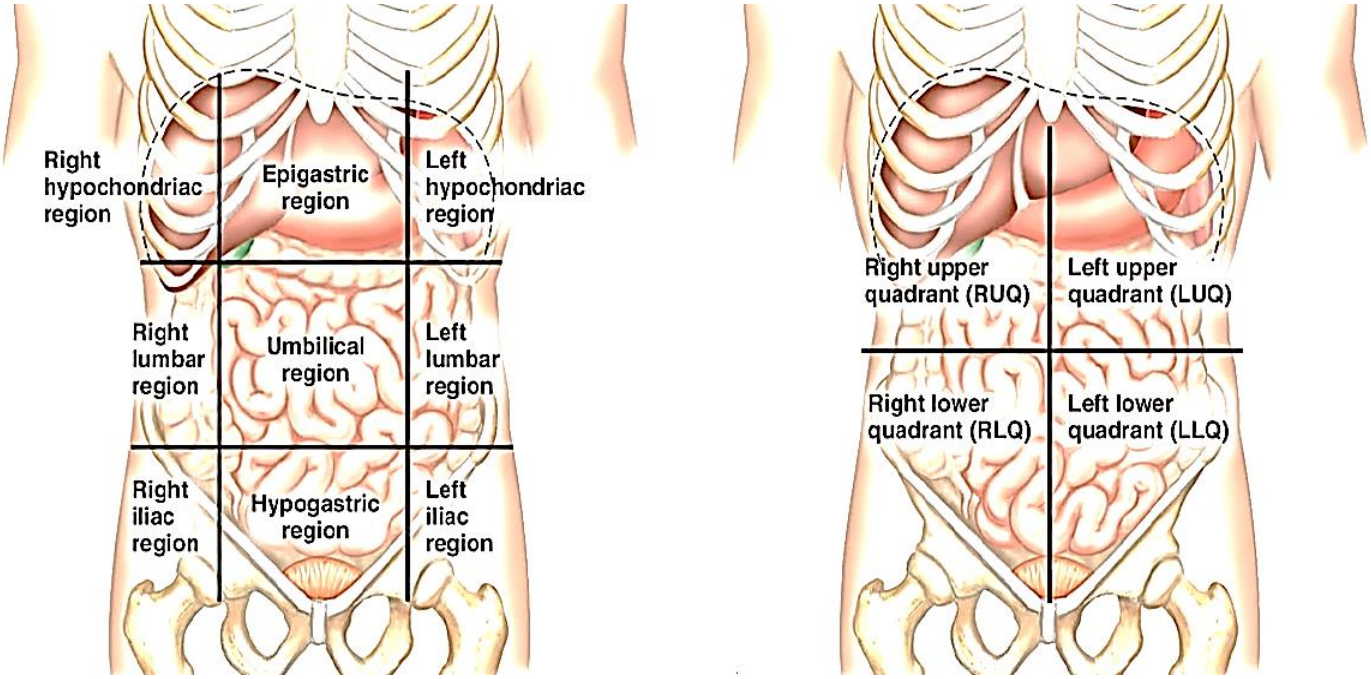
The trunk includes the following regions:

- ✓ **Thorax** or thoracic region, is commonly referred to as the chest.
- ✓ **The mammary region** of the thorax surrounds the nipple and in sexually mature females is enlarged as the breast.
- ✓ **Sternal region** located between the mammary regions.
- ✓ **The axillary region** (arm pit) is called the axillary fossa, or simply the axilla, and the surrounding area.
- ✓ **The vertebral region** extends the length of the back, following the vertebral column.



- ✓ **Abdomen** is located below the thorax. The abdomen has been divided into nine regions or four quadrants to describe the location of internal organs .
- ➢ **Appendicular body part:** It is the part of the body out of the axis line.

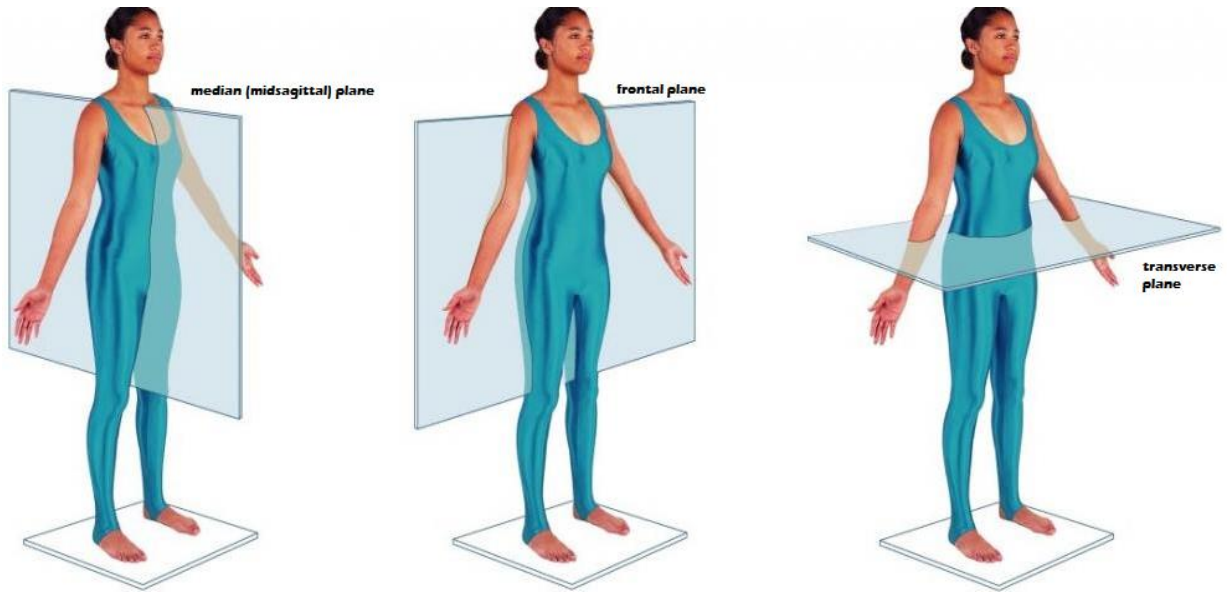
  This includes the upper and lower extremities.



Body planes and sections

Body planes are imaginary surfaces or planes lines that divide the body in to sections.

- ➢ **Sagittal plane**: divides the body into right and left half.
- ➢ **Frontal (Coronal) plane:** divides the body into anterior and posterior sections.
- ➢ **Transverse plane:** divides the body into upper and lower body section.
- ➢ **Oblique plane:** divides the body obliquely into upper and lower section.



Body Cavities

The cavities of the body house the internal organs, which commonly referred to as the viscera. The two main body cavities are:

- The larger ventral (anterior) cavity
- The smaller, dorsal (posterior) body cavity.

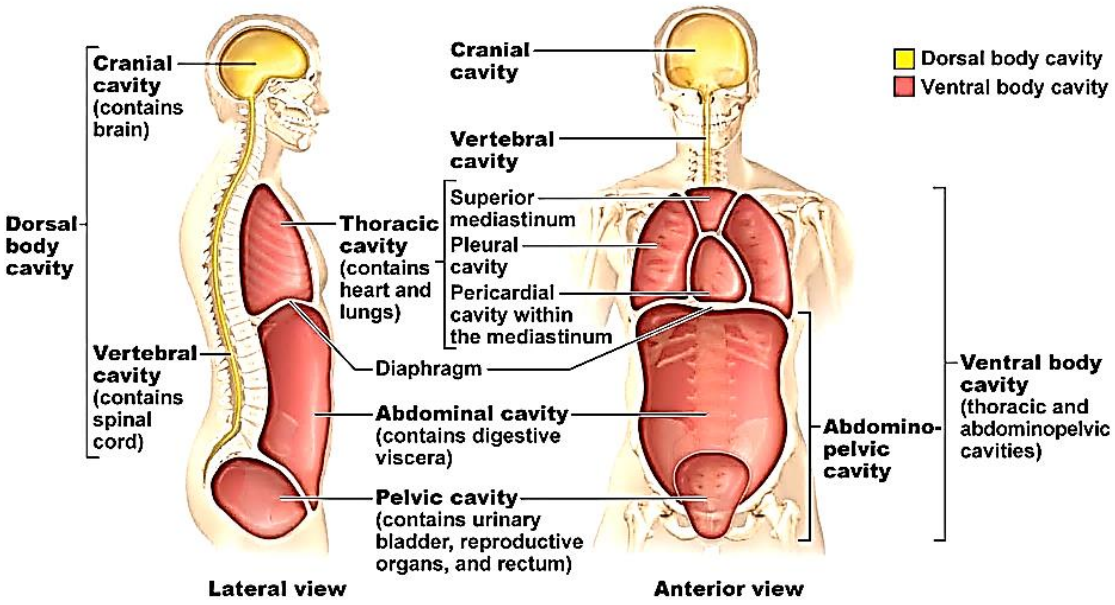
The **ventral cavity** constitutes **the thoracic cavity** (Superior) **and the abdomino-pelvic cavity** (Inferior).

- **Thoracic cavity** houses lung and heart. It is protected by the rib cage & associated musculature and the sternum anteriorly. It consists of the right and left pleural cavities and mediastinum (the portion of tissues and organs that separates the left and right lung).
- **Abdomino-pelvic Cavity** extends from the diaphragm inferior to the floor of the pelvis. It is divided into superior abdominal and inferior pelvic cavity by imaginary line passing at upper pelvis.

Abdominal cavity contains the stomach, intestine, liver, spleen and gallbladder.

The pelvic cavity contains urinary bladder, rectum, and portions of the reproductive organs.

- **The dorsal body cavity:** it constitutes the cephalic cavity containing brain and the vertebral canal containing the spinal cord.



CAVITY	COMMENTS
Cranial cavity	Formed by cranial bones and contains brain.
Vertebral canal	Formed by vertebral column and contains spinal cord and the beginnings of spinal nerves.
Thoracic cavity*	Chest cavity; contains pleural and pericardial cavities and mediastinum.
Pleural cavity	Each surrounds a lung; the serous membrane of each pleural cavity is the pleura.

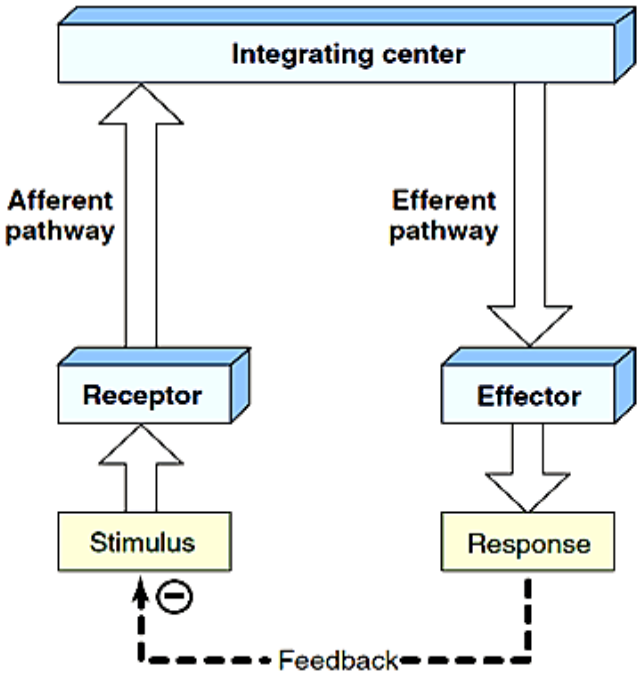
Pericardial cavity	Surrounds the heart; the serous membrane of the pericardial cavity is the pericardium.
Mediastinum	Anatomic region in the central portion of the thoracic cavity between the medial walls of pleural cavities; extends from sternum to vertebral column and from first rib to diaphragm; contains all the structures of the thoracic cavity other than the lungs, including, for example, the heart, thymus, esophagus, trachea, and several large blood vessels.
Abdominopelvic cavity	Subdivided into abdominal and pelvic cavities.
Abdominal cavity	Contains stomach, spleen, liver, gallbladder, small intestine, and most of large intestine; the serous membrane of the abdominal cavity is the peritoneum.
Pelvic cavity	Contains urinary bladder, portions of large intestine, and internal organs of reproduction.

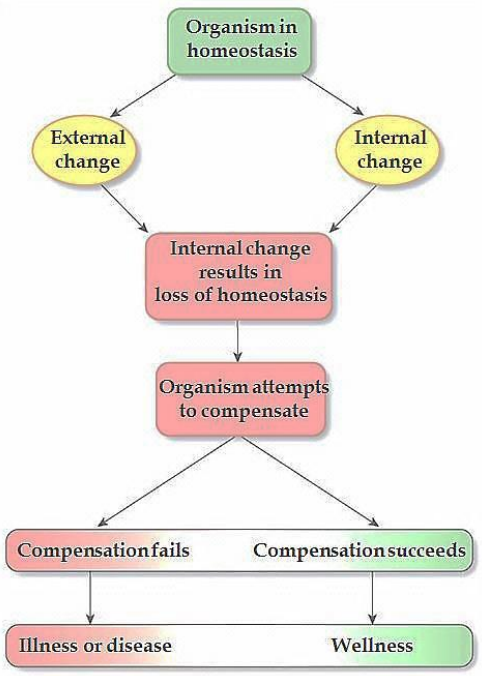
Homeostasis

Homeostasis in a general sense refers to stability, balance or equilibrium. Maintaining a stable internal environment requires constant monitoring and adjustments as conditions change. This adjusting of physiological systems within the body is called homeostatic regulation.

Homeostatic regulation involves three parts or mechanisms:

1. **The receptor** receives information that something in the environment is changing.
2. **The control (integration center** receives and processes information from the receptor.
3. **The effector responds** to the commands of the control center by either opposing or enhancing the stimulus

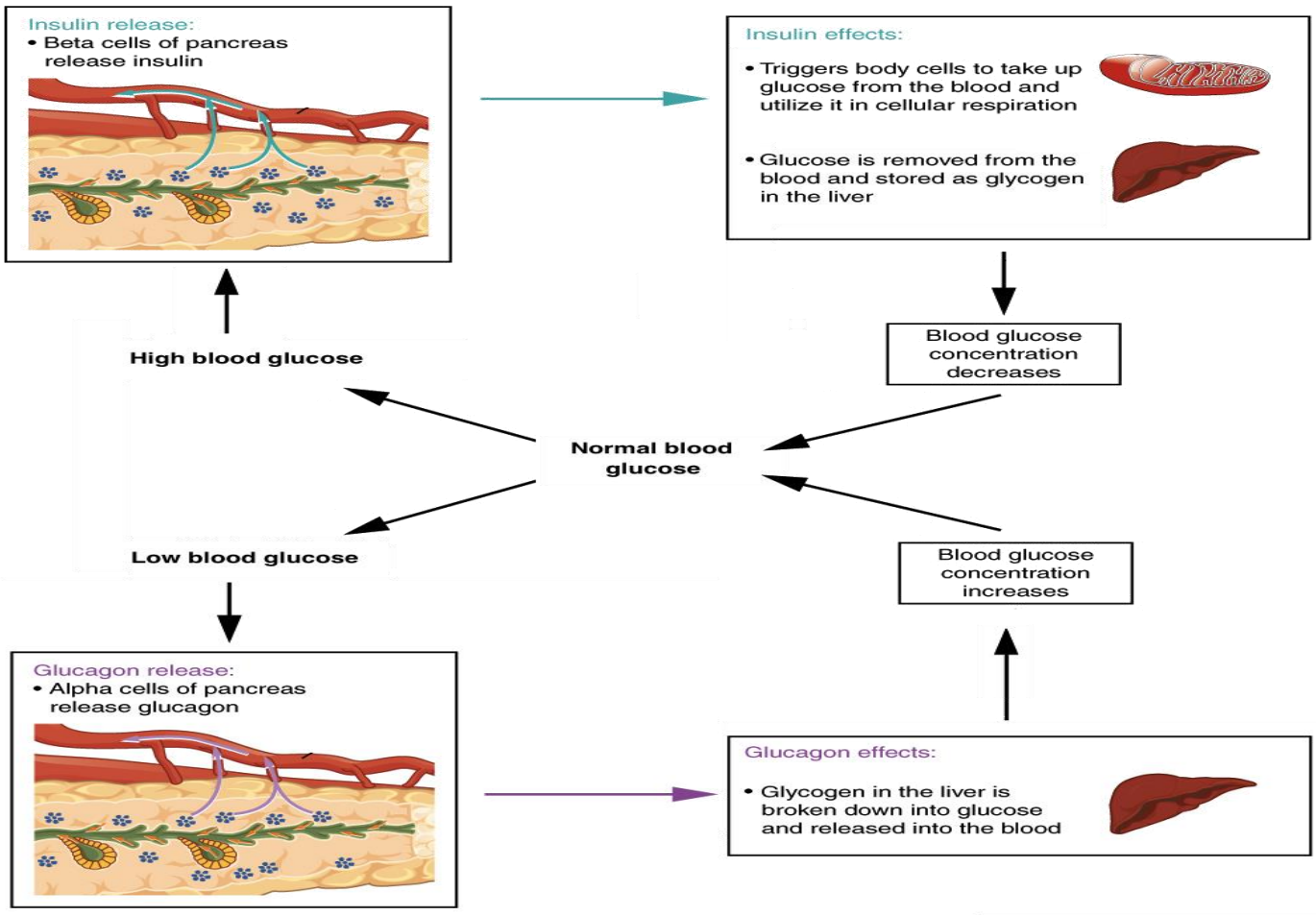




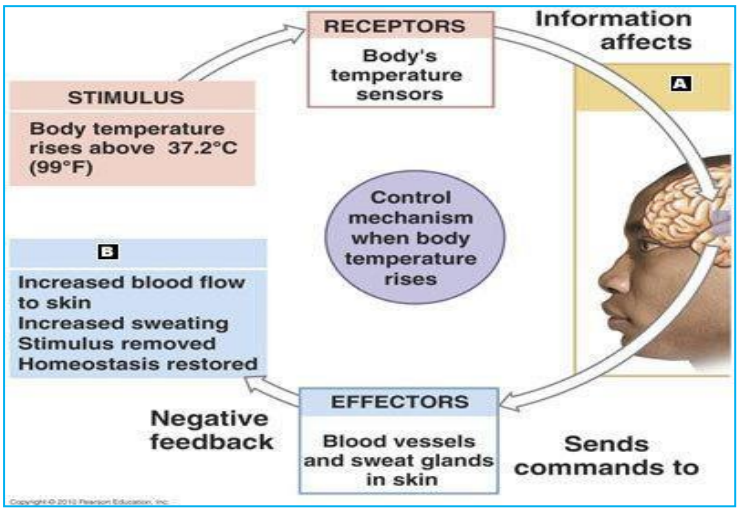
Positive and Negative Feedback

When a change of variable occurs, there are two main types of feedback to which the system reacts:

❖ **Negative feedback:** A negative feedback mechanism happens to lessen the change or stimulus and to restore the system to a stable state. In other words, when the levels are high, the body performs its effort to lower it and in contrast, when the level is low, the body does its action so the level can be raised. **Examples of negative feedback** include regulating the level of glucose in the blood and osmoregulation. Another is thermoregulation. Whenever body temperature deviates, the mechanism starts to work to restore it to normal range. The negative response occurs more often than the positive response in the homeostatic body processes

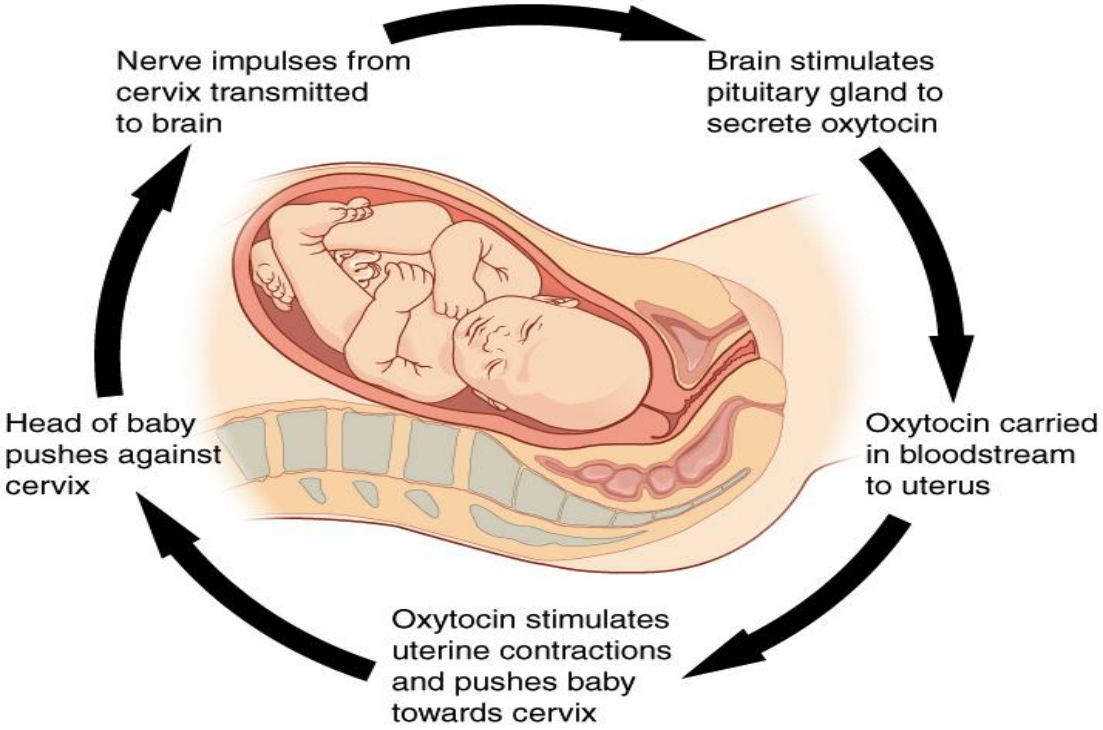


Blood glucose regulation



Thermoregulation

- **Positive feedback** is known as a positive response or a self-reinforcing response to external or internal input. In this, the effector boosts up the stimulus that enhances the product formation for maintaining body stability. It is a response is to amplify the change in the variable.



An example of positive feedback mechanism is that **regulation of childbirth** is seen in the diagram above. When the infant's head bumps up against the cervix, the procedure usually starts. Nerve impulses flow from the cervix to the hypothalamus in the brain as a result of this stimulation. The hypothalamus responds by sending the hormone oxytocin to the pituitary gland, which secretes it into the bloodstream to reach the uterus. Oxytocin causes uterine contractions to increase, pushing the baby closer to the cervix. As a result, the cervix begins to dilate in preparation for the baby's passage. Increased levels of oxytocin, stronger uterine contractions, and wider cervix dilatation continue this cycle of positive feedback until the baby is pushed through the delivery canal and out of the body.

Remember

•**Negative feedback** mechanisms Reverse the original stimulus

•**Positive feedback** mechanisms Reinforce the original stimulus

CHAPTER TWO

CELL PHYSIOLOGY

Cell is the basic living structural and functional unit of the body.

Cytology: It is a branch of science concerned with a study of cells

There are 50-100 trillion cells in the human body.

- **Functions body cells:**
  - Movement
  - Conduction
  - Absorption
  - Secretion
  - Excretion
  - Respiration
  - Reproduction
- **Body Fluids and electrolytes**

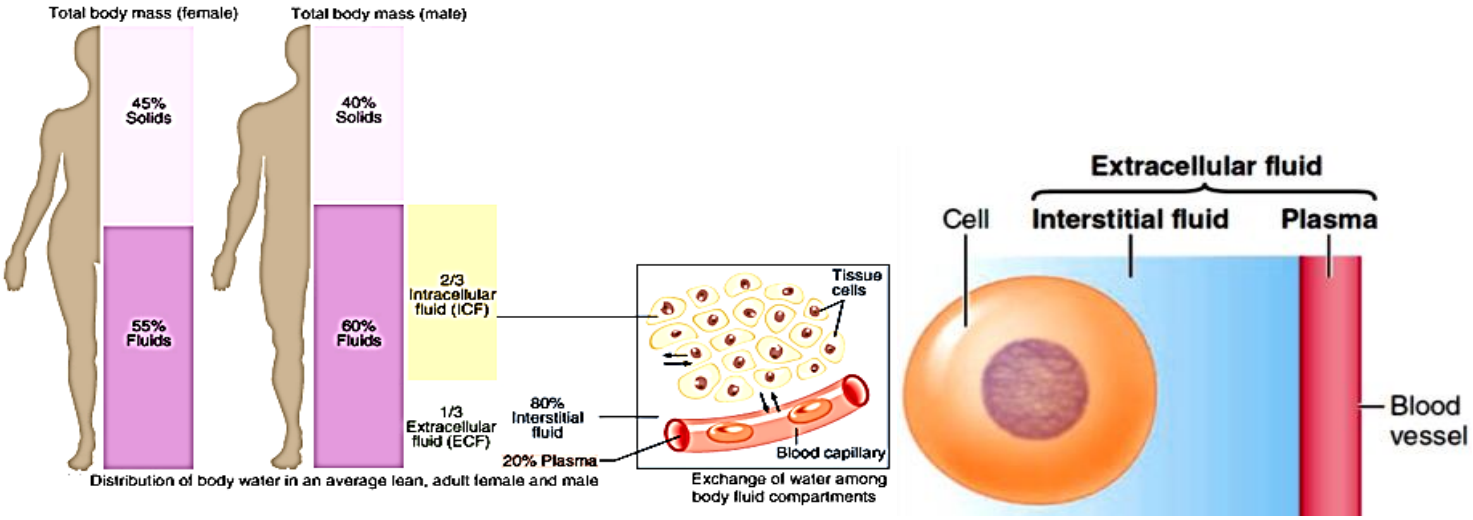
Total body water is distributed as follow:

- **Intracellular fluid (ICF): approximately 2/3 of total of body water**
- **Extracellular fluid (ECF): approximately 1/3 of total body water, includes:**

1. Interstitial fluid (ISF): approximately 2/3 of the extracellular fluid
2. Intravascular fluid (IVF): approximately 1/3 of the extracellular fluid (plasma plus red blood cells).

Human cells or other body fluids contain many dissolved substances (called solutes) such as salts, sugars, acids, and bases.

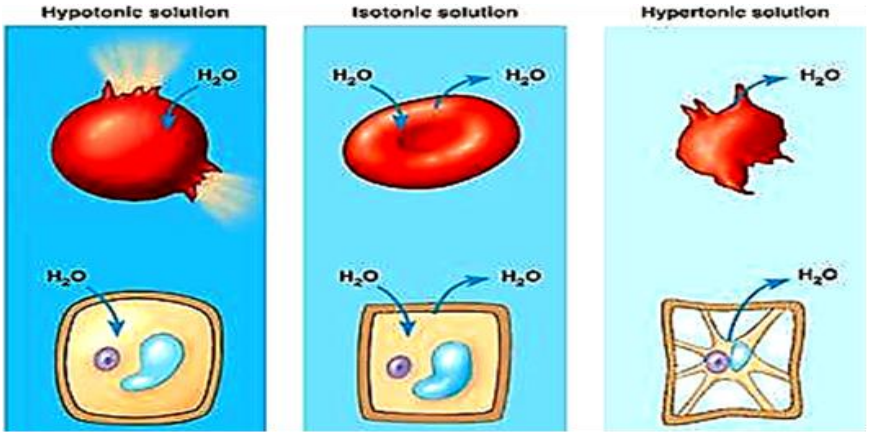
The concentration of solutes in a fluid creates the osmotic pressure of the solution, which in turn determines the movement of water through membranes.



- **Types of crystalloid body solution:**

✓ **Isotonic—** a solution with the same salt concentration as in cells. The blood plasma is isotonic to red blood cells.

- ✓ **Hypotonic –** a solution with a lower salt concentration than in cells. Distilled water (0% salt) is hypotonic to human cells.
- ✓ **Hypertonic –** a solution with a higher salt concentration than in cells. Seawater (3% salt) is hypertonic to human cells.

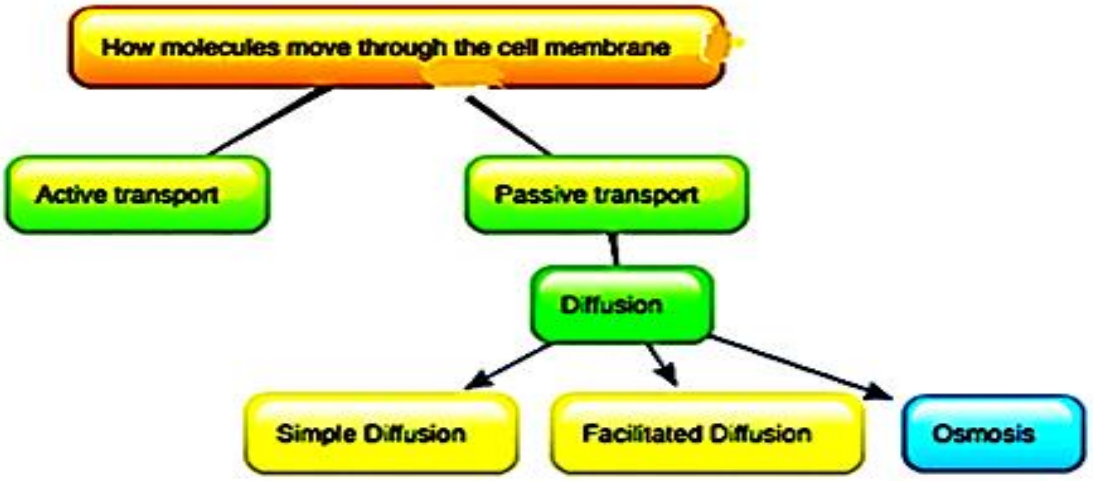


Cellular transport mechanisms

Living cells constantly interact with the blood or tissue fluid around them, taking in some substances and secreting or excreting others.

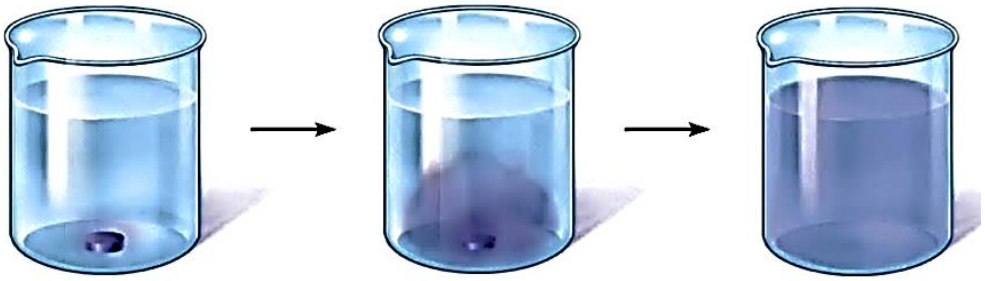
- **Passive transport processes:** don't require any energy or activity of the cell
- **Active transport processes:** require energy, often in the form of ATP.

Both passive and active methods are used to cross the plasma membrane

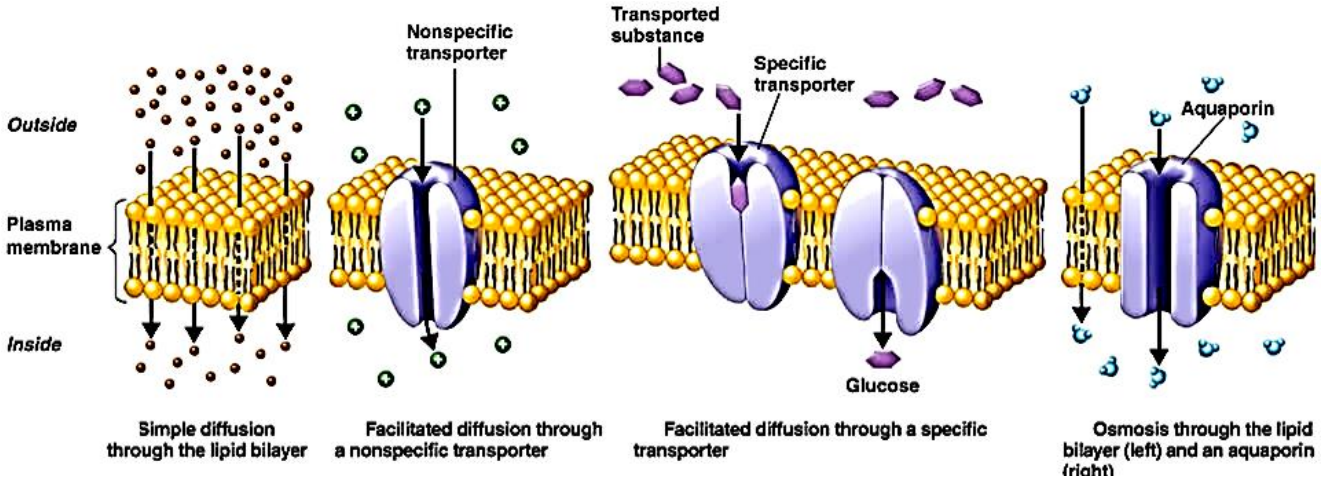


Passive transport process:

Molecules move down concentration gradient from HIGH ⟹ LOW concentration



A. **Simple diffusion:** Movements of molecules from area of high concentration to the area of low concentration. Eg. O2, CO2, fat-soluble vitamins, air in alveoli of lung



B. **Facilitated diffusion:** larger molecules, which are not soluble in lipid need protein channel to pass through the plasma membrane. Transport proteins (carrier or channel proteins) assist molecules across membrane Eg. glucose, amino acids, H2O, ions.

C. **Osmosis**: a special type of diffusion referring to the passage of water through a selectively permeable membrane from an area of high water concentration to lower water concentration. Diffusion of H2O Aquaporins: channel proteins for H2O passage

D. **Filtration**: small molecules pass through selectively permeable membrane in response to force of pressure. Example: filtration in the kidney in the process of urine formation.

- **Special terms:**
  Osmolarity: refers to the number of solute particles per 1 L of solvent
  Osmotic pressure: refer as the minimum pressure that must be applied to a solution to halt the flow of solvent molecules through a semipermeable membrane

Type of passive transport	Definition	Example
Diffusion	Movement of **molecules** from an area of greater concentration to an area of lesser concentration **(along a concentration gradient).**	Exchange of gases in the lungs or body tissues.
Osmosis	Diffusion of **water** through a selectively permeable membrane in the presence of at least one impermeable solute	Absorption of water by the small intestine or kidneys.
Facilitated diffusion	Carrier and transporter enzymes move molecules across cell membranes **(along a concentration gradient).**	Intake of glucose by most cells.
Dialysis	Diffusion of small solute particles but not large particles results in separation of large and small solutes.	During peritoneal dialysis, small solutes diffuse from blood vessels but large do not

Filtration	Movement of water and dissolved substances from an area of higher pressure to an area of lower pressure	Water and small molecules move from the blood to the inside of the kidney tubules.

Active transport process: Energy (ATP) is needed!!

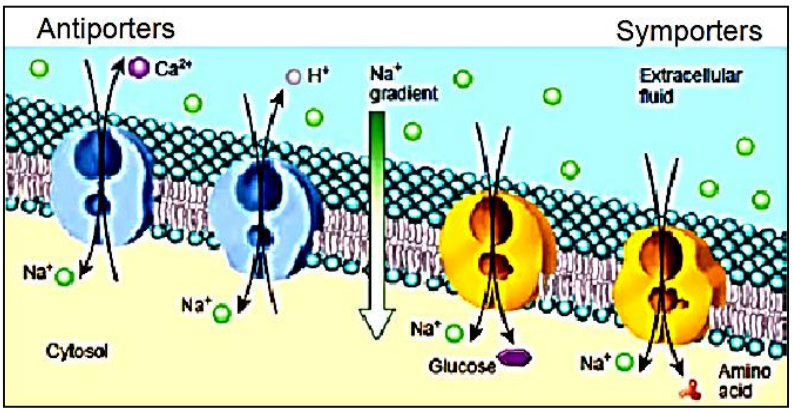
Move molecules against concentration gradient from LOW ⟹ HIGH concentration

Types: Primary and Secondary, Vesicular

A. **Primary Active Tranport**
Directly uses ATP to drive transport Eg. Ca2+ pump, H+ pump, Na+-K+ pump

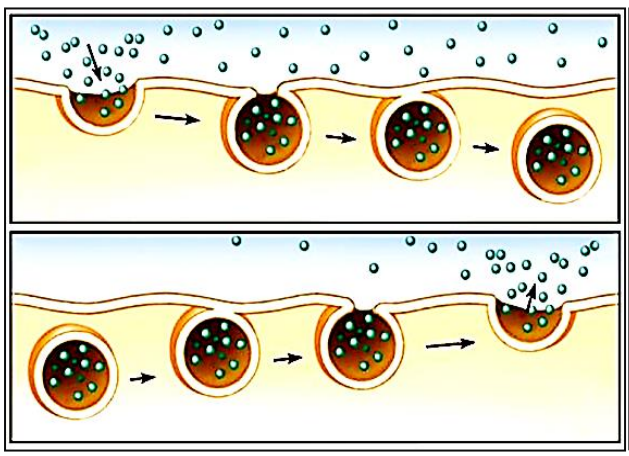
B. **Secondary active transport**
Move more than 1 substance at a time

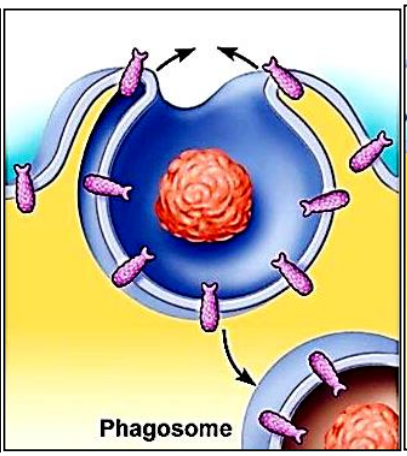
- ✓ Symport: 2 substances moved in same direction
- ✓ Antiport: 2 substances cross in opposite directions Eg. cotransport of sugars, animo acids, ions

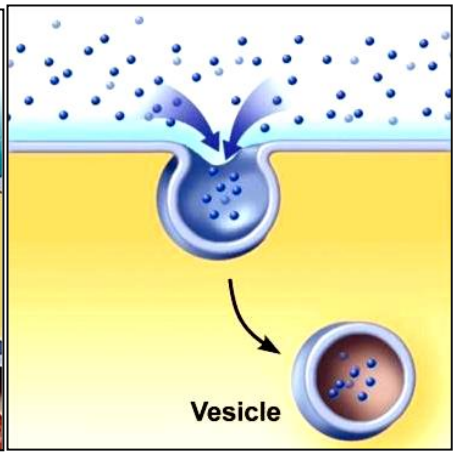


C. **Vesicular transport:** Fluid & large particles transported across membranes in **vesicles** (sacs)
- ✓ **Exocytosis**: "out of cell" – eject substances
- ✓ **Endocytosis:** "within the cell"- ingest substances

1. **Phagocytosis: (cell eating)** – engulf large or solid material eg. WBC engulf bacteria
2. **Pinocytosis: (cell drinking)** – fluid w/dissolved molecules eg. Intestinal cells







CHAPTER THREE
TISSUES & MEMBRANES

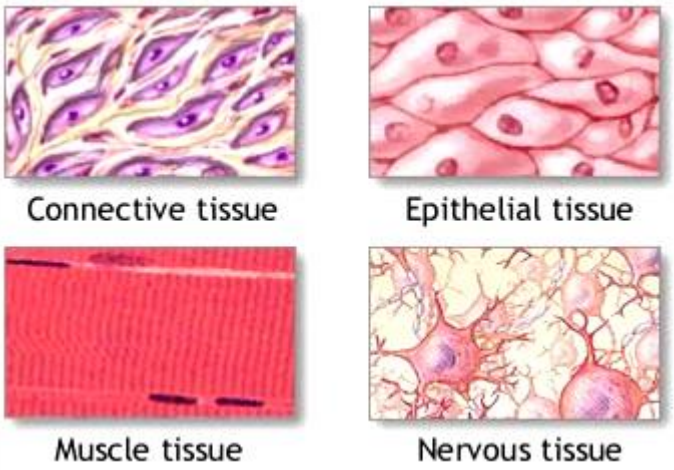
TISSUE: Tissue is a group of similar cells and their intercellular substance that have a similar embryological origin and function together to perform a specialized activity.

- ✓ A science that deals with the study of a tissue is **Histology.**

The various tissues of the body are classified in to four principal parts according to their function & structure.

Types of Tissues:

- ✓ Epithelium (covering)
- ✓ Connective (support)
- ✓ Muscle (movement)
- ✓ Nervous (control)



Type I: Epithelial Tissue

Main characteristics of epithelium:

- ✓ Covers body surfaces
- ✓ Lines body cavities
- ✓ Contains limited amount of matrix
- ✓ Because it covers or line body surfaces, epithelial tissues always have one free surface
- ✓ Avascular: epithelial tissues contain no blood vessels so oxygen and nutrient will diffuse from the underlining capillary.
- ✓ Regeneration.

There are 2 types of epithelial tissues:

- ➢ **Membranous epithelium** that covers the surface or lines body cavities (pleural, pericardial, and peritoneal), blood vessels, respiratory, digestive and genitourinary systems.
- ➢ **Glandular epithelium** is a secretory part, either endocrine or exocrine.

Function:

- ✓ **Protection:** the main function of membranous epithelium as skin
- ✓ **Sensation:** especially in skin, nose, eye and ear
- ✓ **Secretion:** the main function of glandular epithelium including hormones, mucus, digestive juices and sweat
- ✓ **Absorption:** in gastrointestinal epithelium will absorb nutrients and respiratory epithelium will absorb gases
- ✓ **Excretion:** especially in the kidneys

Classifications of epithelial tissue: based on:

- ❖ **Shape of the cell**
- ✓ Squamous – flattened
- ✓ Cuboidal – cube-shaped
- ✓ Columnar – column-like (tall and narrow)

❖ **Number of layers of cells.**

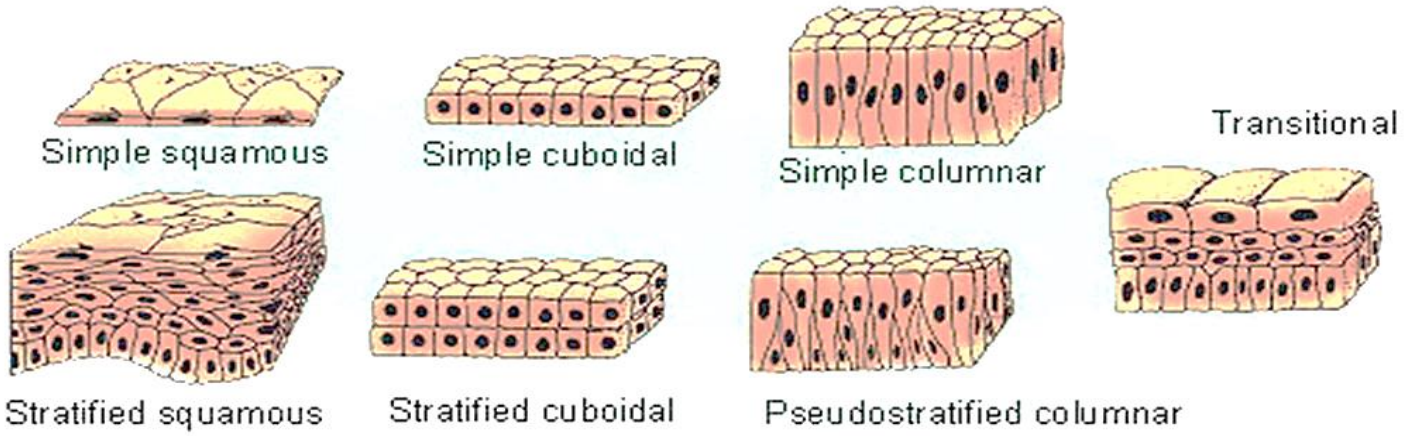
✓ Simple: a single layer of cells

✓ Stratified: many layers of cells are present

✓ Pseudostratified: Single layer, but some cells are shorter than others. Often looks like a double cell layer.

Therefore considering the number of layers and cell shape we can classify covering and lining epithelium into the following groups:

Type	Structure	Location and Function
Simple squamous	One layer of flat cells	▪ **Alveoli of the lungs:** thin to permit diffusion of gases ▪ **Capillaries:** thin to permit exchanges of materials, smooth to prevent abnormal blood clotting
Stratified squamous	Many layers of cells; surface cells flat; lower cells rounded; lower layer undergoes mitosis	▪ **Epidermis:** surface cells are dead; a barrier to pathogens ▪ **Lining of esophagus, vagina:** surface cells are living; a barrier to pathogens
Transitional	Many layers of cells; surface cells change from rounded to flat	▪ **Lining of urinary bladder**: permits expansion without tearing the lining
Cuboidal	One layer of cube- shaped cells	▪ **Thyroid gland:** secretes thyroxine ▪ **Salivary glands:** secrete saliva ▪ **Kidney tubules:** permit reabsorption of useful materials back to the blood
Columnar	One layer of column- shaped cells	▪ **Lining of stomach:** secretes gastric juice ▪ **Lining of small intestine:** secretes enzymes and absorbs end products of digestion (microvilli present)
Ciliated	One layer of columnar cells with cilia on their free surfaces	▪ **Lining of trachea:** sweeps mucus and dust to the pharynx ▪ **Lining of fallopian tube:** sweeps ovum toward uterus



Type II Connective tissue

The main characteristics of connective tissues:

- Found everywhere in the body
- Includes the most abundant and widely distributed tissues
- Connect and hold structures together.
- Provide support and protection to the body.
- It has a large amount of matrix.
- Variations in blood supply
  - Most conn. tissues are well vascularized
  - Tendons and Ligaments have a poor supply
  - Cartilages are avascular

Types of connective tissues:

Connective tissues differ in their fibers present in the matrix. From most rigid to softest, C.T. major classes are (Bone, blood, cartilage, and fibers).

Functions of connective tissues:

- **Connection:** Tissue to tissue, muscle to muscle, muscle to bone, bone to bone
- **Supporting:** it forms supporting frames for the whole body and organs.
- **Transportation:** blood is carrying substances, nutrients, gases ..etc. to every part of the body
- **Defense system:** many cells defend our body against microorganisms

Bone:

- Bone is the **most rigid** of the connective tissues. It has an extremely hard matrix of mineral salts, notably calcium salts, deposited around protein fibers.
- The minerals give bone rigidity, and the protein fibers provide elasticity and strength, much as steel rods do in reinforced concrete.
- The structural unit of the bone tissue is called osteon or Haversian system
- **Cells:** there are 3 forms of bone cells
  - ✓ **Osteocyte:** the mature bone cells.
  - ✓ **Osteoblast:** the mother cell of osteocyte.
  - ✓ **Osteoclast:** the bone eating cell, that dissolves the bone matrix and release the osteocyte free to be a gain active.
- **Matrix** of the bone is formed of:
  - ✓ **Organic Compounds:** proteins, polysaccharides. It represents (35%) of matrix
  - ✓ **Inorganic compounds:** calcium, phosphate sodium. It represents (65%) of matrix
- **Fibers of the bone:** collagen type.

❖ **Functions of bone tissue:**

- ✓ Protection of brain and viscera.
- ✓ Movement.
- ✓ Supporting (skeleton).
- ✓ Mineral store: calcium reservoir

Blood:

- Blood is the unusual type of connective tissue
- The matrix of blood is plasma, which is about 55% of the total blood volume in the body. The water of plasma contains dissolved salts, nutrients, gases, and waste products. The remaining 45% of the blood volume is blood cells and each type of cell has its specific function. Blood cells are:
  - ✓ **Red blood cells (RBCs)** carry oxygen bonded to the iron in their hemoglobin.
  - ✓ **White blood cells (WBCs)** destroy pathogens by phagocytosis, the production of antibodies, or other chemical methods, and provide us with immunity to some diseases.
  - ✓ **Platelets** prevent blood loss; the process of blood clotting involves platelets.

❖ **Functions of blood:**

- ✓ Transports oxygen and carbon dioxide
- ✓ Transports nutrients and hormones
- ✓ Removes wastes from the body
- ✓ Regulates pH and electrolytes
- ✓ Defends against pathogens
- ✓ Stabilizes body temperature

Cartilage:

- Chondrocyte is the mature cartilage cell
- The protein–carbohydrate matrix of cartilage does not contain calcium salts, and also differs from that of bone in that it contains more water, which makes it resilient. It is firm, yet smooth and flexible.
- Unfortunately, because this tissue lacks a direct blood supply (Avascular), it heals very slowly.
- The three types of cartilage are classified according to the type of fiber in the matrix:

Hyaline cartilage:

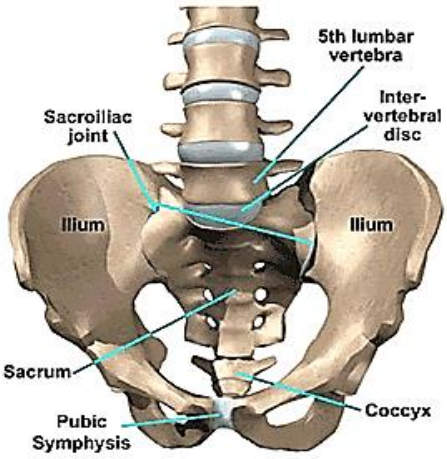
- The most common type of cartilage.
- The matrix, which contains only very fine collagenous fibers, has a glassy, white, opaque appearance.
- This type of cartilage is found in the: Nose, at the ends of the long bones and
- ribs, the supporting rings of the trachea.
- The fetal skeleton is also made of this type of cartilage, although the cartilage is later replaced by bone.

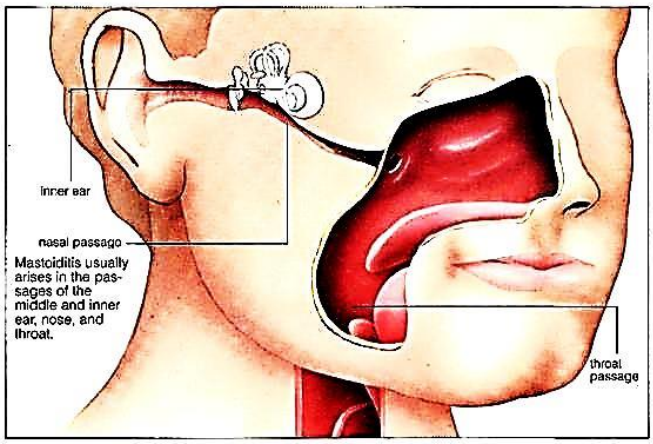
Elastic cartilage:

- It has a matrix containing many elastic fibers, in addition to collagenous fibers. For this reason, elastic cartilage is more flexible than hyaline cartilage.
- Elastic cartilage is found, for example, in the ear pinna and thyroid cartilage.

Fibrocartilage:

- It is the strongest type of cartilage as it has a matrix containing strong collagenous fibers.
- This type of cartilage absorbs shock and reduces friction between joints.
- Fibrocartilage is found in structures that withstand tension and pressure, such as intervertebral disk in the backbone and the wedges in the knee joint.





<u>Fibrous CT:</u> Fibrous connective tissue includes:

Loose (Areolar) connective tissue

- Matrix: soft viscous gel
- Cell: fibroblast (produce collagen), macrophages & WBCs (engulf bacteria and foreign materials)

- Commonly lies between other tissues or between organs, binding them together.
- It fills the spaces between muscle fibers, surrounds blood and lymph vessels, and supports organs in the abdominal cavity.

Adipose tissue

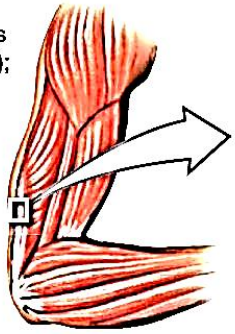
- Contains adipose cell, less fibrous tissue, less macrophages and other cells
- Contains little extracellular matrix
- Functions of adipose tissue are:
  - ✓ Supporting some organs as kidneys
  - ✓ Protections of many parts of the body
  - ✓ Storage of fat which is of high energy
  - ✓ Insulation to prevent heat loss
  - ✓

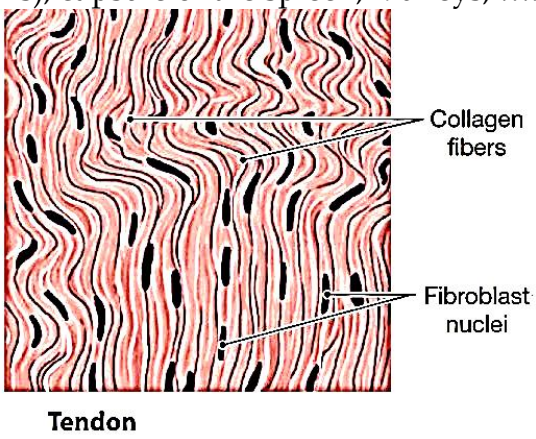
Dense connective tissue.

- It has a matrix produced by fibroblasts that contains bundles of white collagenous fibers.
- In dense regular connective tissue, the bundles are parallel as in tendons (which connect muscles to bones) and ligaments (which connect bones to other bones at joints).
- In dense irregular connective tissue, the bundles run in different directions. This type of tissue is found in the inner portion of the skin (dermis), capsule of the spleen, kidneys, …. etc.

LOCATIONS: Between skeletal muscles and skeleton (tendons and aponeuroses); between bones or stabilizing positions of internal organs (ligaments); covering skeletal muscles; deep fasciae

FUNCTIONS: Provides firm attachment; conducts pull of muscles; reduces friction between muscles; stabilizes relative positions of bones





Tendon

Reticular connective tissue:

- It is formed of cells (reticular cells) that forms the reticular fibers that are fine and branching.
- It forms a framework of many organs as spleen, lymph nodes and bone marrow

Type III Muscle tissue:

- Muscular (contractile) tissue is composed of cells called muscle fibers. Muscle fibers contain actin and myosin, which are protein filaments whose interaction accounts for movement.
- The three types of human muscles:
  - ✓ **Skeletal**
  - ✓ **Smooth**
  - ✓ **Cardiac**

Comparison of Structure and Properties of Muscle Tissue Types

Tissue	Histology	Function	Location
Skeletal	Long cylindrical fiber, striated, many peripherally located nuclei	Voluntary movement, produces heat, protects organs	Attached to bones and around entrance points to body (e.g., mouth, anus)
Cardiac	Short, branched, striated, single central nucleus	Involuntary Contracts to pump blood	Heart
Smooth	Short, spindle-shaped, no evident striation, single nucleus in each fiber	Involuntary movement, moves food, involuntary control of respiration, moves secretions, regulates flow of blood in arteries by contraction	Walls of major organs and passageways

❖ **Functions of muscle tissue:**

- ✓ Body movement (Skeletal muscles attached to bones)
- ✓ Maintenance of posture
- ✓ Respiration (skeletal muscles of the thorax)
- ✓ Production of body heat (when skeletal muscles contract, heat is given off as by- product)
- ✓ Communication (speaking and writing)
- ✓ Contraction of organs and constriction of vessels (contraction of smooth muscles)
- ✓ Heart beat (contraction of cardiac muscles)

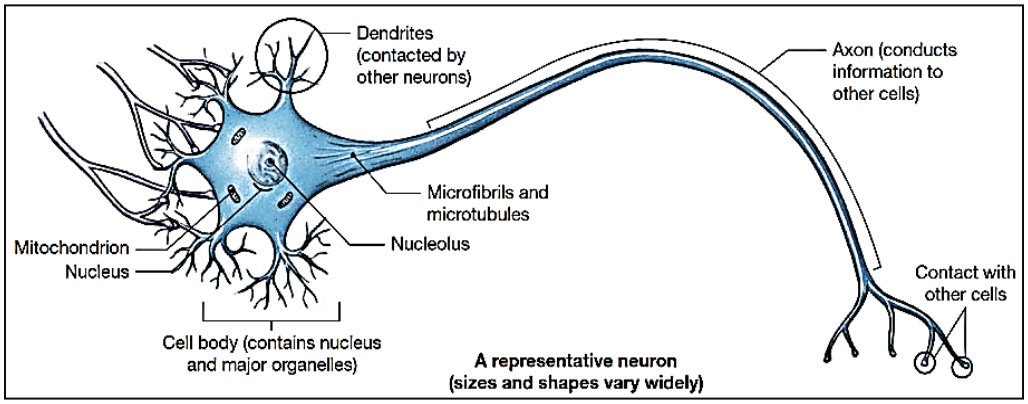
Type IV Nervous tissue:

- Nervous tissue is characterized as being excitable and capable of sending and receiving electrochemical signals that provide the body with information.
- Two main classes of cells make up nervous tissue: the neuron and neuroglia
  - ✓ **Neurons** is the structural and functional unit of the nervous tissue

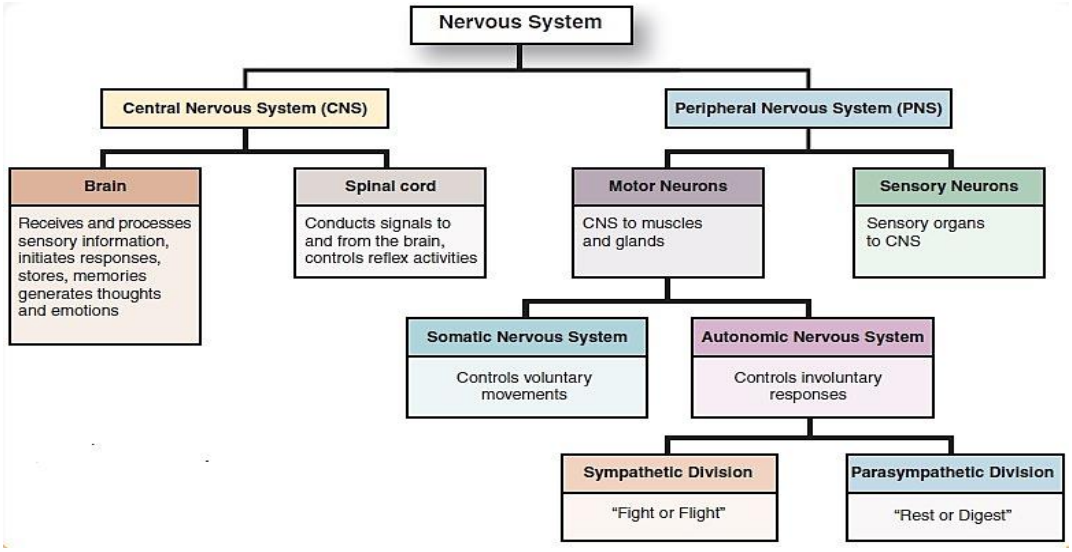
Neurons propagate information via electrochemical impulses, called action potentials, which are biochemically linked to the release of chemical signals.

- ➢ **Neuroglia** are cells that outnumber neurons nine to one and take up more than half the volume of the brain. The primary function of neuroglia is to support and nourish neurons.

- ➢ **Functions of Nervous System**
- ➢ **Sensory (Afferent) function:** to sense changes in the internal and external environment through sensory receptors.



- **Integrative function:** to analyze the sensory information, store some aspects, and make decisions regarding appropriate behaviors.

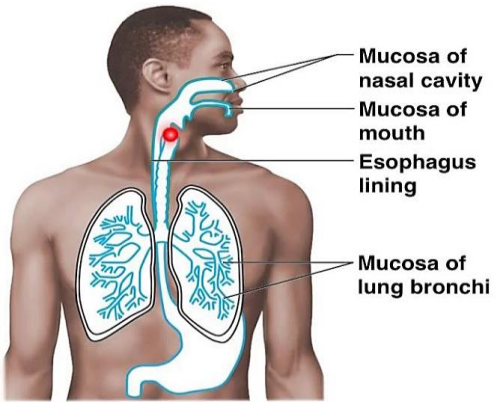


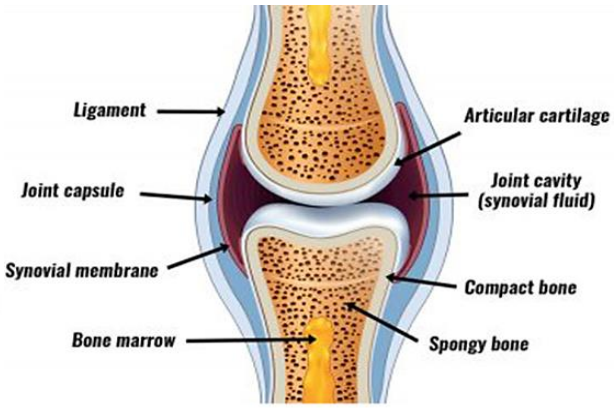
- **Motor (Efferent) function** is to respond to stimuli by initiating action.

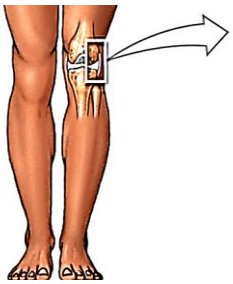
❖ **Body Membranes**

Membranes are thin pliable layers of epithelial and/or connective tissue. They line body cavities, cover surfaces, connect, or separate regions, structures and organs of the body. The **three kinds of membranes are mucous, serous and synovial.**

- **Mucous membranes (mucosa)** lines body cavity that opens directly to the exterior. It is an epithelial layer.
  - Mucous membranes line the entire gastro intestine, respiratory excretory and reproductive tracts and constitute a lining layer of epithelium. The connective tissue layer of mucous membrane is lamina propria. To prevent dry out and to trap particles mucous membranes secret mucous.
- **Serous membrane / serosa:** contains loose connective tissue covered by a layer of mesothelium. It lines body cavity that does not open directly to the exterior. Covers the organs that lie within the cavity. Serosa is composed of parietal layer (pertaining to be outer) and visceral layer (pertaining to be near to the organ). Pleura and pericardium are serous membrane that line thoracic and heart cavity respectively. The epithelial layer of a serious membrane secret a lubricating fluid called serious fluid. The fluid allows organs to glide one another easily.

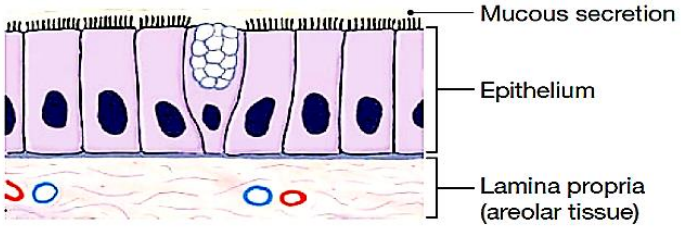






- **Synovial membrane:** Unlike to other membranes this membrane does not contain epithelium. Therefore, it is not epithelial membrane. It lines the cavities of the freely movable joints. Like serious membrane it lines structures that do not open to the exterior. Synovial membranes secret synovial flui that lubricate articular cartilage at the ends of bones as they move at joints.

Mucous membranes are coated with the secretions of mucous glands. These membranes line the digestive, respiratory, urinary, and reproductive tracts.



Serous membranes line the ventral body cavities (the peritoneal, pleural, and pericardial cavities).

CHAPTER FOUR
THE INTEGUMENTARY SYSTEM

▪ The skin is the largest organ in the body which covers the entire surface of the human body. In an adult, the skin has a surface area of about 1.8 square meters
▪ The skin is sometimes called the cutaneous membrane or the integument.
▪ Because the skin has several accessory organs, it is also possible to speak of the integumentary system.

► **Functions of the skin:**

1.Protection

▪ Skin protect the body from dehydration, microorganisms, injury, and ultraviolet radiation/sun damage.

▪ Melanin protect against UV rays of the sun

2. Sensation

▪ The skin allows us to feel and recognize pain, touch/pressure, and temperature heat and cold. It is covered in somatic sensory receptors which relay these sensationsignals to the brain.

2.Temperature Regulation

The skin helps release or preservation of heat.

Sweat glands within the skin release sweat onto the outer layer of skin, which then evaporates to reduce levels of heat.

3.Immunity

The cells within the skin like Langerhans cells (in epidermis), phagocytic cells, and epidermal dendritic cells help with immunity.

4.Permit Movement and Growth

The skin allows for bodily growth and adapts to suit an individual course of movement.

It has elastic and recoil properties on all of its layers, meaning it can adapt for growth and movement.

5.Excretion

The skin can also excrete uric acid, ammonia, urea, and excess water.

6.Endocrine

The skin helps vitamin D synthesis. When the skin is exposed to a nice amount of ultraviolet light, such as the light of the sun, the stratum Basale and stratum spinosum of the skin produce vitamin D3.

This happens because a chemical found within the skin, known as 7- dehydrocholesterol, reacts with the ultraviolet light and converted to vitamin D3.

7. Absorption:

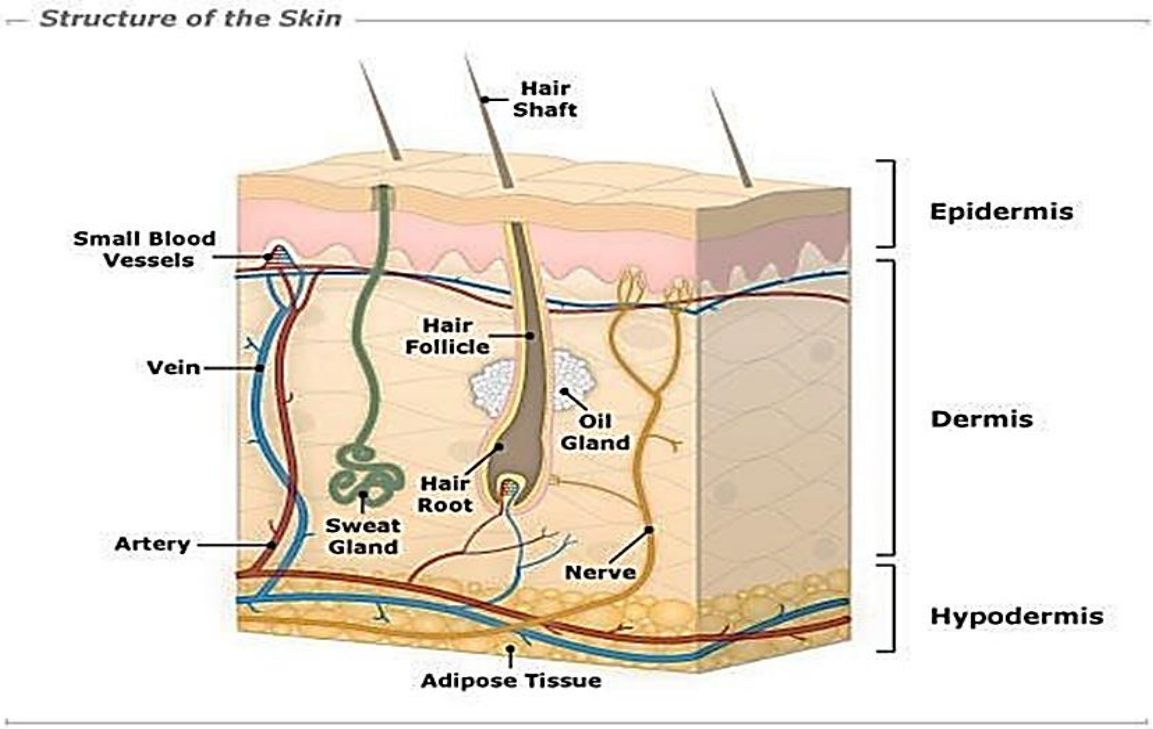
The skin is capable of absorbing necessary substances, such as nitrogen and oxygen.

8. Water resistance:

The skin acts as a barrier, ensuring that all essential minerals and nutrients remain within the body.

Layers of skin:

1. Epidermis – keratinized stratified squamous epithelium
2. Dermis – fibrous connective tissue
3. Hypodermis (subcutaneous fascia)



Epidermis

- The epidermis is the outer and thinner region of the skin.
- The epidermis is composed of keratinized, stratified squamous epithelium.
- The cells that are most abundant in epidermis are called keratinocytes
- Epidermis does not have any blood vessels within it (i.e., it is avascular).
- Epidermis is made of four or five layers of epithelial cells, depending on its location in the body. Two of these layers are of greatest importance: the innermost layer, the stratum germinativum, and the outermost layer, the stratum corneum
- Most of the skin can be classified as thin skin in which epidermis has four layers of cells. Thick skin is found only on the palms of the hands and the soles of the feet.
- Layers of epidermis are ordered from **superficial to deep as following:**

Stratum corneum: (Horny layer)

- The most superficial layer of the epidermis.
- The stratum corneum consists of many layers of dead squamous cells filled with keratin. The protein keratin is relatively waterproof.
- **Functions:** 1. Water barrier2. Microorganism barrier

Stratum lucidum: (Clear layer)

- Absent in thin skin (eye lid) and more apparent in thick skin (palms and soles).
- **Function:** prevent water loss and penetration

Stratum granulosum: (granular layer):

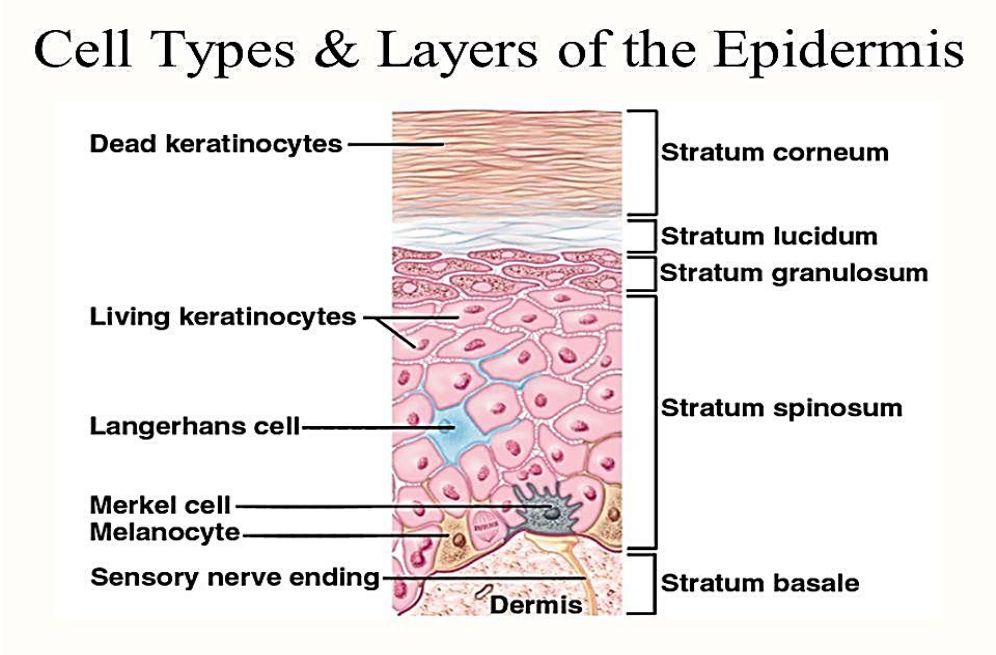
- As the cells move up into this layer, they start to lose their nuclei and cytoplasmic organelles, and turn into the keratinized squamous of the next layer.
- The granules contain a lipid rich secretion, which acts as a water sealant.

Stratum spinosum (spin layer)

- As the name suggests, the stratum spinosum is spiny in appearance due to the protruding cell processes that join the cells via a structure called a desmosome.
- Contains dendritic (Langerhans) cells that help protect body against pathogens by "presenting" them to the immune system
- It is the site of cancer skin.

Stratum Basale (base layer) .

- The stratum basale is a single layer of cells primarily made of basal cells.
- A basal cell is a cuboidal-shaped stem cell that is a precursor of the keratinocytes of the epidermis.
- All of the keratinocytes are produced from this single layer of cells, which are constantly going through mitosis to produce new cells. As new cells are formed, the existing cells are pushed superficially away from the stratum basale.
- Two other cell types are found in the stratum basale:
  - ✓ **Merkel cell,** which functions as a receptor and is responsible for stimulating sensory nerves that the brain perceives as touch. These cells are especially abundant on the surfaces of the hands and feet.
  - ✓ **Melanocyte,** a cell that produces the pigment melanin. Melanin gives hair and skin its color, and also helps protect the living cells of the epidermis from ultraviolet (UV) radiation damage.

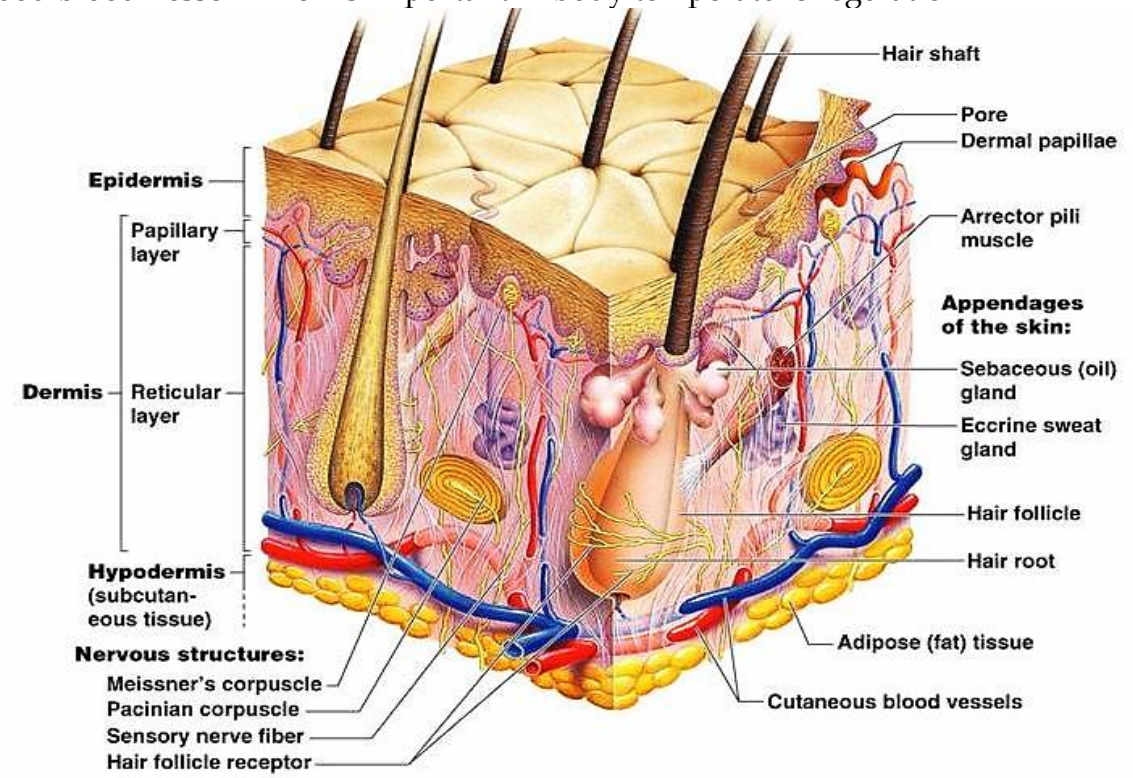


Dermis:

- The dermis, a deeper and thicker region than the epidermis, is composed of dense irregular connective tissue.
- It contains blood and lymph vessels, nerves, and other structures, such as hair follicles and sweat glands.
- Capillaries are abundant here to nourish not only the dermis but also the stratum germinativum. The epidermis has no capillaries of its own, and the lower, living cells depend on the blood supply in the dermis for oxygen and nutrients.
- The dermis is made of two layers of connective tissue that compose an interconnected mesh of elastin and collagenous fibers, produced by **fibroblasts.**
- The **collagenous fibers** are flexible but offer great resistance to overstretching; they prevent the skin from being torn.

- The **elastic fibers** stretch to allow movement of underlying muscles and joints, but they maintain normal skin tension.
- The upper layer of the dermis has fingerlike projections called **dermal papillae**. Dermal papillae project into and anchor the epidermis. In the overlying epidermis, dermal papillae cause ridges, resulting in spiral and concentric patterns commonly known as "fingerprints". Because they are unique to each person, fingerprints and footprints can be used for identification purposes.
- **Functions of dermis:**

✓ Provide the skin with elasticity and stretch
✓ Contains sweat glands (3000/in2) which is important in controlling body temperature and excretion of wastes
✓ Contains sebaceous (oil) glands (2 per hair) which release sebum that keep epidermis and hair flexible and water proof
✓ The sensory area for pain, touch, temperature and pressure
✓ It has a good blood vessel which is important in body temperature regulation



Hypoderms (subcutaneous layer)

- It is found beneath the dermis. It is a subcutaneous layer (under the skin).
- Hypodermis is composed of loose, fibrous connective tissue, which is richly supplied with lymphatic and blood vessels and nerves.
- Hypodermis is much thicker than dermis. Within it coils of ducts of sudoriferous (sweat) glands, and the base of hair follicles.

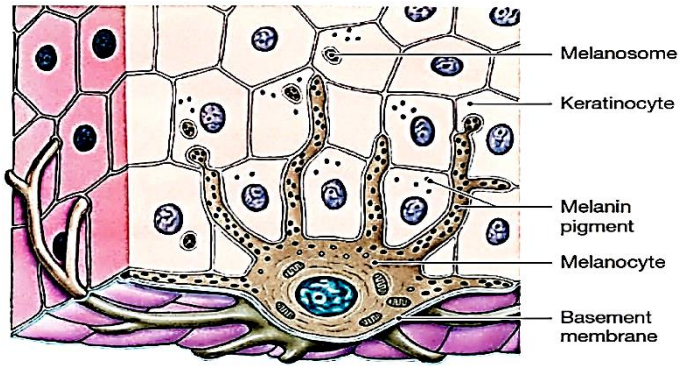
Color of the skin

Skin's color is determined by 3 factors:

1. The presence of melanin a dark pigment produced by specialized cell called melanocyte
2. The accumulation of yellow pigment carotene.
3. The color of blood reflected through the epidermis

- The main function of melanin is to screen out excessive ultraviolet rays.

- All races have some melanin in their skins although the darker races have slightly more melanocyte. The person who is genetically unable to produce any melanin is an albino.



Accessory Structures of the Skin

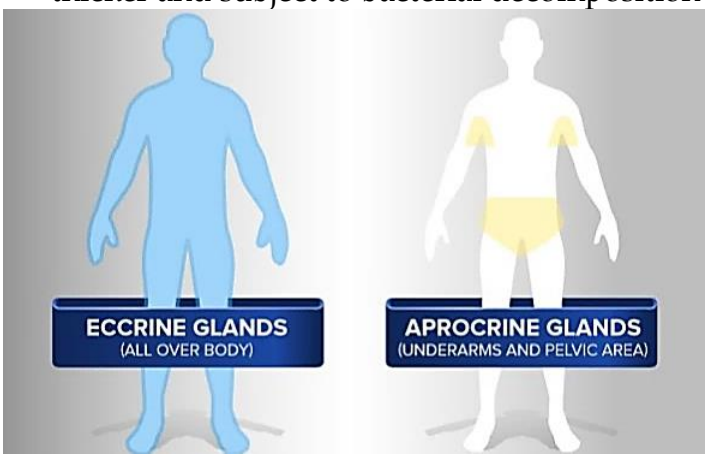
- Hair
- Nails
- Skin glands
  - ✓ Sweat glands
  - ✓ Sebaceous glands.
  - ✓ Ceruminous glands

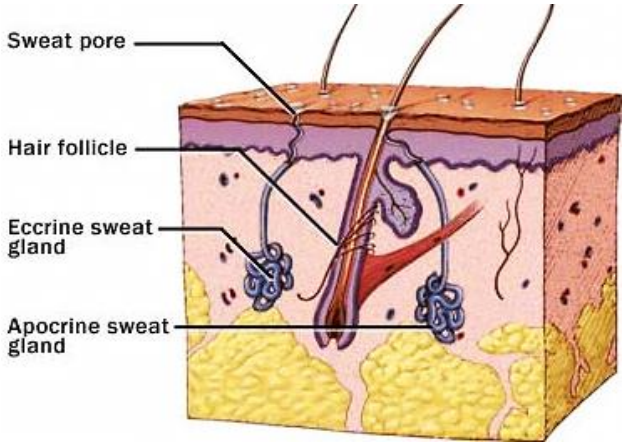
Sweat glands:

There are two types of sweat glands:

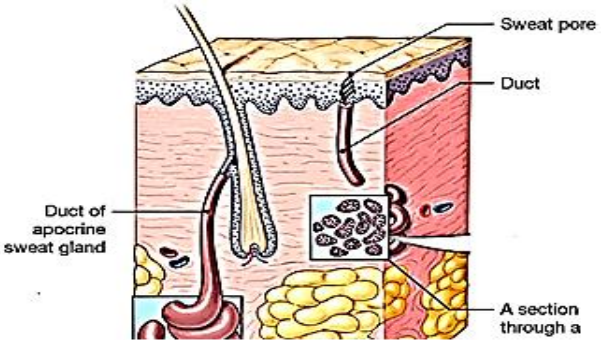
- **Eccrine glands:**

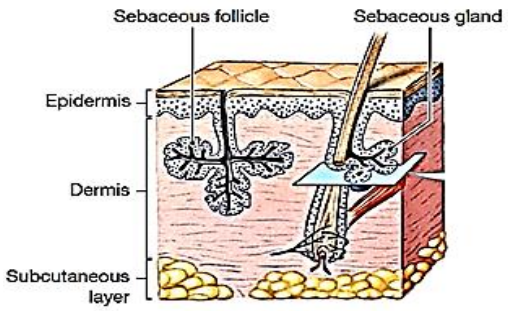
- Most numerous sweat glands (3000/in2) but are especially abundant on the palms of the hand, the soles of the feet, and the forehead
- They are quite small and coiled glands lying deep in the dermis, open onto the surface of the skin.
- They become active when a person is hot, helping to lower body temperature as sweat evaporates.
- The sweat (perspiration) produced by these glands is mostly water, but it also contains salts and some urea, a waste substance.
- **Function:** Regulation of body temperature.
  - **Apocrine glands:**
    - Apocrine sweat glands are larger than eccrine sweat glands
    - Lie deeper in the dermis, sometimes even reaching the hypodermis
    - Duct normally emptying into the hair follicle.
    - Apocrine glands open into hair follicles in the anal region, groin and armpits.
    - These glands begin to secrete at puberty
    - In addition to water and salts, apocrine sweat includes organic compounds that make the sweat thicker and subject to bacterial decomposition and subsequent smell.



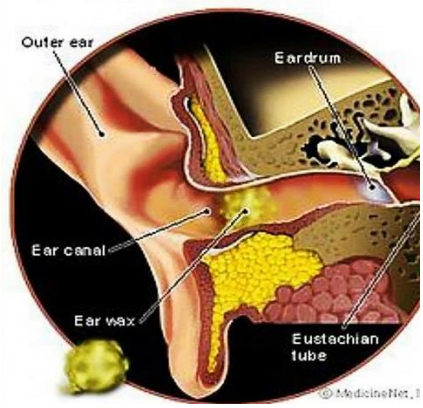


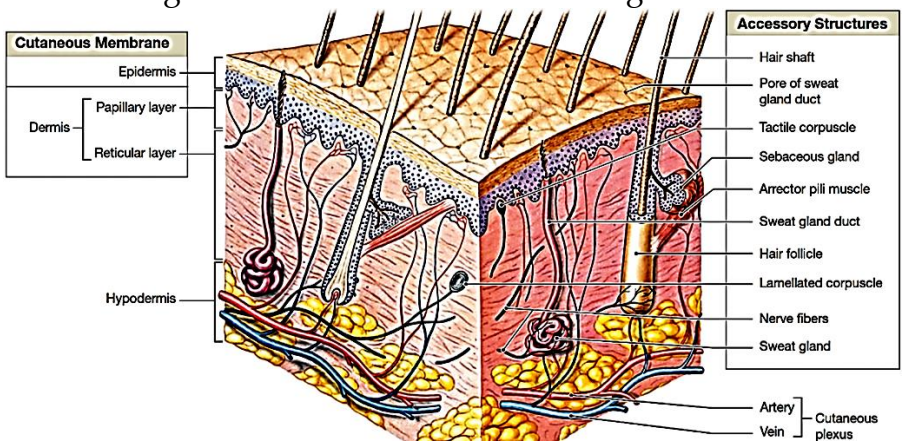
- **Sebaceous glands:**
  - A sebaceous gland is a type of oil gland that is found all over the body (except palms and soles) and helps to lubricate and waterproof the skin and hair.
  - Most sebaceous glands are associated with hair follicles.
  - These glands secrete an oily substance called sebum that flows into the follicle and then out onto the skin surface.
  - Sebum lubricates the hair and skin, and helps waterproof them.





- **Ceruminous glands:**
  - Present only in the external ear canal alone or with sebaceous glands.
  - Mixed secretions of sebaceous and Ceruminous glands called cerumen (wax)
  - Function of cerumen to protect area from dehydration.
    - Excess secretion can cause blockage of ear canal and loss of hearing.





Hair

Hair is composed of Keratinized threads of cells, which develops from the epidermis. Because it arises from the skin, it is considered an appendage of the skin. It covers the entire body except the palms, soles, lips, tip of penis, inner lips of vulva and nipples.

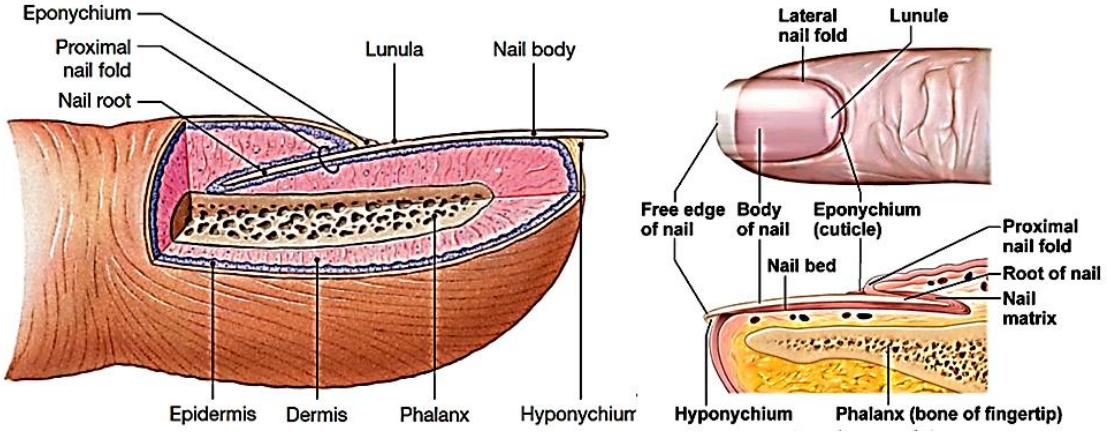
Function of Hair:

- ✓ Insulation against cold in scalp
- ✓ Against glare in eye brows
- ✓ Screen against foreign particles (eye lashes(
- ✓ In the nostrils trap dust particles in the inhaled air
- ✓ Protect openings from foreign particles.

Nail

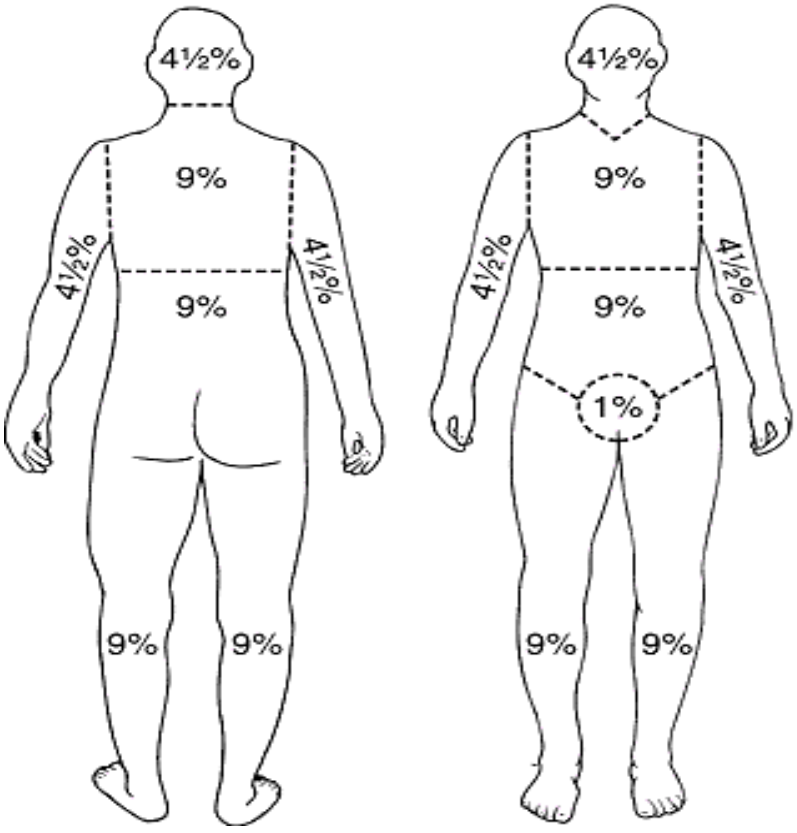
- Nails, like hair are modifications of the epidermis. They are made of hard keratin.
- Nails are composed of flat, cornified plates on the dorsal surface of the distal segment of the fingers and toe.
- The proximal part of nail is lunula, which is white in its color because of the capillary's underneath are covered by thick epithelium.

- Nail has body and root. The body is the exposed part and the root is hidden under the skin. The nail ends with a free edge that overhangs the tip of the fingers. Epithelial layer covering underneath of the fore-hang nail is hyponychium.
- The nail rests on an epithelial layer of skin called nail bed. The thicker layer of skin beneath the nail root is the matrix, where new cells are generated. Nail grows 0.5 m.m a week. Thin layers of epidermis called eponychium originally cover the growing nail. Our nail protects our fingers and toes. It also allows picking up and grasping objects as well we use them to scratch.



Estimating body surface area

- **Rule of palm:** the palm size of burn represents 1% of body surface.
- **Rule of nine**
  - ✓ Head and Neck represents: 9% of total body surface.
  - ✓ Each upper limb represents: 9% of total body surface.
  - ✓ Each lower limb represents: 18% of total body surface.
  - ✓ Anterior surface of the trunk represents: 18% of body surface.
  - ✓ Posterior surface of the trunk represents: 18% of body surface.
  - ✓ External genitalia represent: 1% of body surface.



CHAPTER FIVE
THE SKELETAL SYSTEM

The word skeleton comes from the Greek word skeleton meaning "dried up". It is strong yet light adapted for its function of body protection and motion.

The skeletal system includes bones, joints, cartilages and ligaments.

The joint give the body flexibility and allow movements to occur. But from structural point of view, the human skeletal system consists of two main types of supportive connective tissue, bone and cartilage.

Functions of the skeletal system:

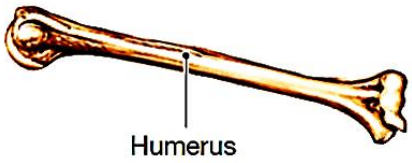
1. **Support:** it forms the internal framework that supports and anchors all soft organs.
2. **Protection:** bones protect soft body organs.
3. **Movement:** skeletal muscles attached to the skeletal system use the bone to levers to move the body and its part.
4. **Storage:** fat is stored in the internal cavities of bones. Bone it self-serves as a storehouse of minerals. The most important being calcium and phosphors.
5. **Blood cell formation:** it occurs within the marrow cavities of certain bones.

Bone

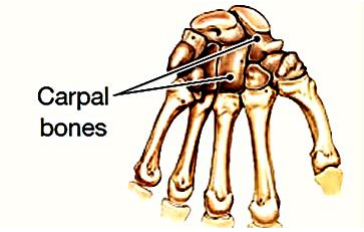
Bone (osseous) is specialized connective tissue that has the strength of cast iron and lightness of pinewood. Living bone is not dry, brittle or dead. It is a moist changing, productive tissue that is continually resorbed, reformed and remodeled.

Types of bone

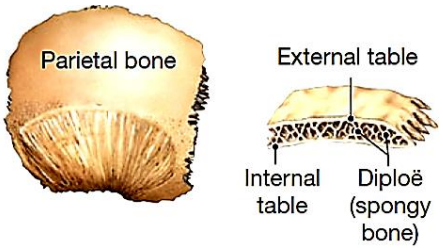
► **Long bone,** are called long as its length is greater than its width. The most obvious long bones are in the arm and leg. They act as levers that pulled by contraction of muscles.



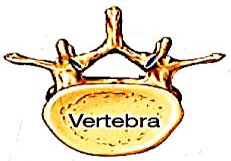
► **Short bones** are about equal in length, width and thickness, which are shaped with regular orientation. They occur in the wrist and ankle.



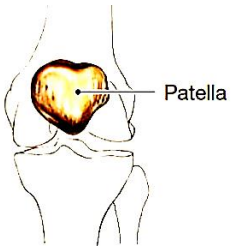
► **Flat bones** are thin or curved more often they are flat. This includes ribs, scapulae, sternum and bone of cranium.



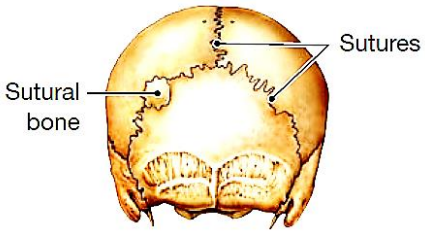
► **Irregular bones**, they do not fit neatly into any other category. Examples are the vertebral, facial, and hipbone.



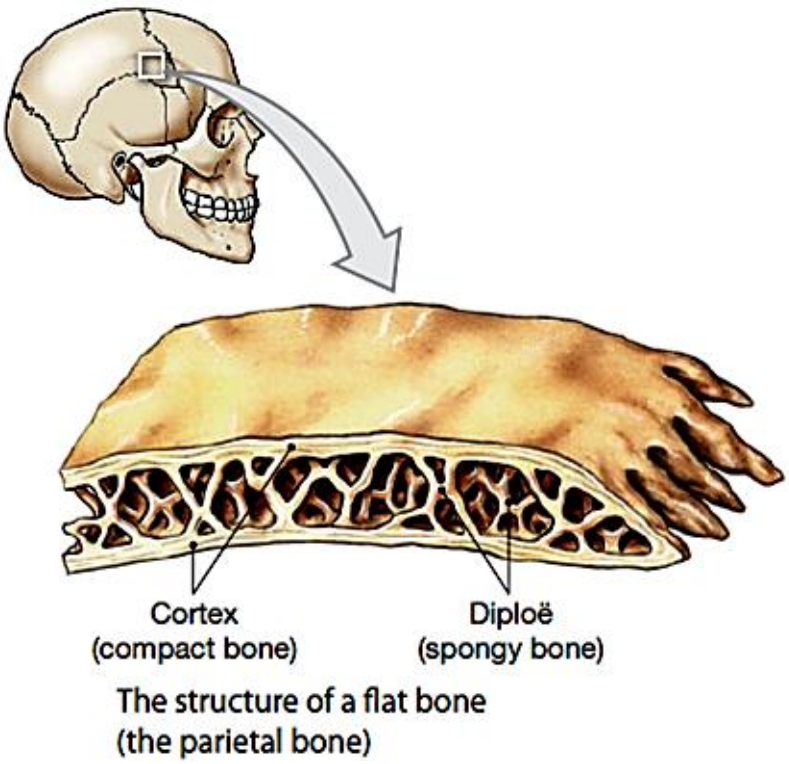
► **Sesamoid bones** are small bones embedded within certain tendons, the fibrous cord that connects muscle to bones.
Typical sesamoid bones are patella and pisiform carpal bone, which are in the tendon of quadriceps femuris and flexor carp ulnaris muscle respectively.



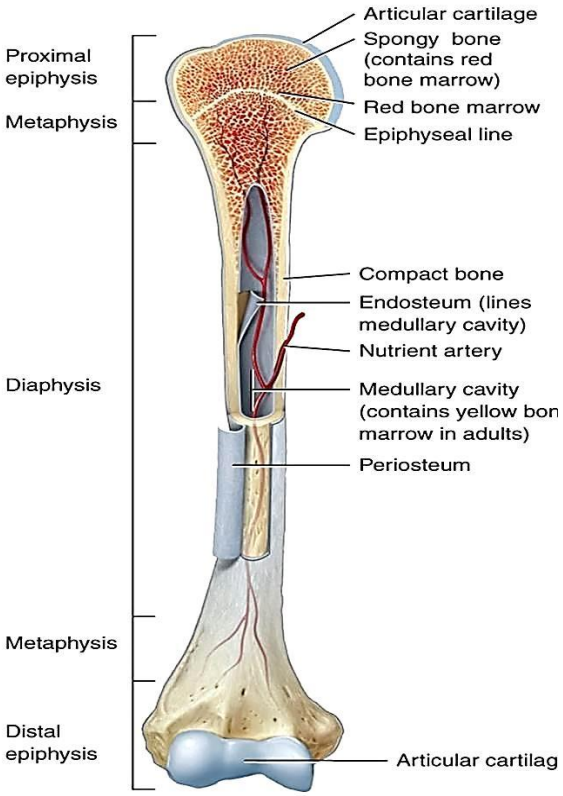
► **Accessory bones** are most commonly found in the feet. They usually occur in the developing bone and do not fuse completely. They look like extra bones or broken on X-ray.
Sutural (wormian) bones are examples of accessory bones.



Gross anatomy of a typical long bone: Eg. Tibia (in the leg one of the longest bones in the body.
In adults it have:
Diaphysis, the tubular shaft, hallow cylindrical with walls of compact bone tissue.
The center of the cylinder is the medullary cavity, which is filled with marrow.
Epiphysis is roughly spherical end of the bone. It is wider than the shaft. Flat and irregular bones of the trunk and limbs have many epiphysis and the long bones of the finger and toe have only one epiphysis.
Metaphysis is the part separating diaphysis from epiphysis. It is made up of epiphyseal plate and adjacent bony trabeculae of cancellous bone tissue.



The structure of a flat bone (the parietal bone)



Epiphyseal plate is a thick plate of hyaline cartilage, which provides the framework of synthesis of the cancellous bone tissue within metaphysis.

•The medullary cavity running through the length of the diaphysis contains Yellow marrow.

•The porous latticework of the spongy epiphyses is filled with red bone marrow. The red marrow also known as myeloid tissue

Endosteum is the lining the medullary cavity of compact bone tissue and covering the trabeculae of spongy bone tissue.

Periosteum: it is covering the outer surface of the bone. It is absent at joints and replaced by articular cartilage.

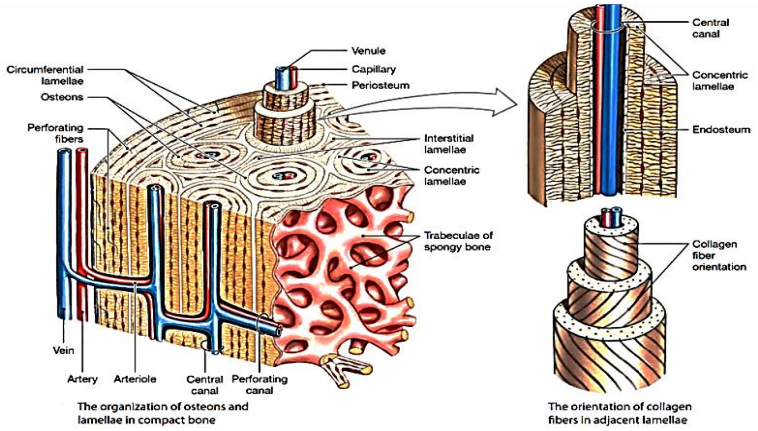
Bone (Osseous) Tissue

Bone tissue is composed of cells embedded in a matrix of ground substances and fibers. It is more rigid than other tissues because it contains inorganic salts mainly calcium phosphate & calcium carbonate. A network of collagenous fibers in the matrix gives bone tissue its strength and flexibility. Most bones have an outer sheet of compact bone tissue enclosing an interior spongy bone tissue.

Compact bone tissue forms the outer sheet of a bone. It is very hard and dense. It appears to naked eye to be solid but not. Compact bone tissue contains cylinders of calcified bone known as osteons (Haversian system).

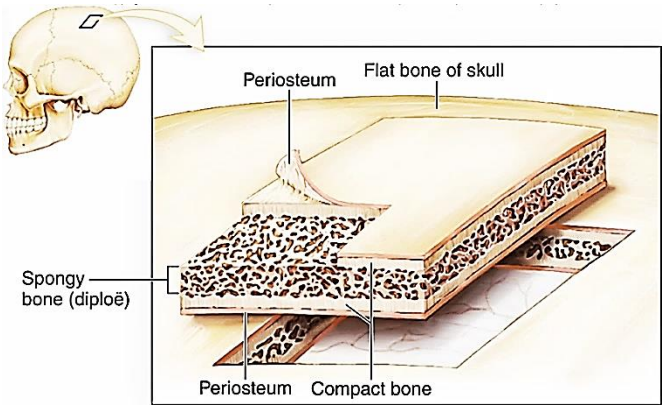
Osteons are made up of concentric layers called lamellae, which are arranged seemingly in wider and wider drinking straws. In the center of the osteons are central canals (haversion canal) , which are longitudinal canals that contains blood vessels, nerves and lymphatic vessels. Central canals, usually have branches called perforating canals /valkmann's canal that run at right angle to central canal extending the system of nerves and vessels out ward to periosteum and to endosteum. Lacunae (Little spaces) that houses osteocytes (bone cells) are contained in lamella. Radiating from each lacuna are tiny canaliculi containing the slender extensions of the osteocytes
where nutrients and wastes can pass to and from central canal.

Spongy (cancellous) Bone tissue Is in the form of an open interlaced pattern that with stands maximum stress and supports in shifting stress. Trabeculae are tiny spikes of bone tissue surrounded by bone matrix that has calcified.



The organization of osteons and lamellae in compact bone

The orientation of collagen fibers in adjacent lamellae

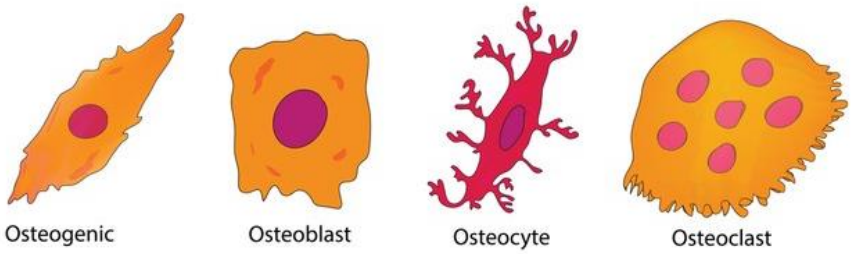


Bone Cells

Bone contain five types of cells

a) Osteogenic (osteoprogenitor) cells: these are small spindle shaped cell. They found mostly in the deepest layer of periosteum and endosteum. They have high mitotic potential and can be transformed into bone forming cells (osteoblasts).

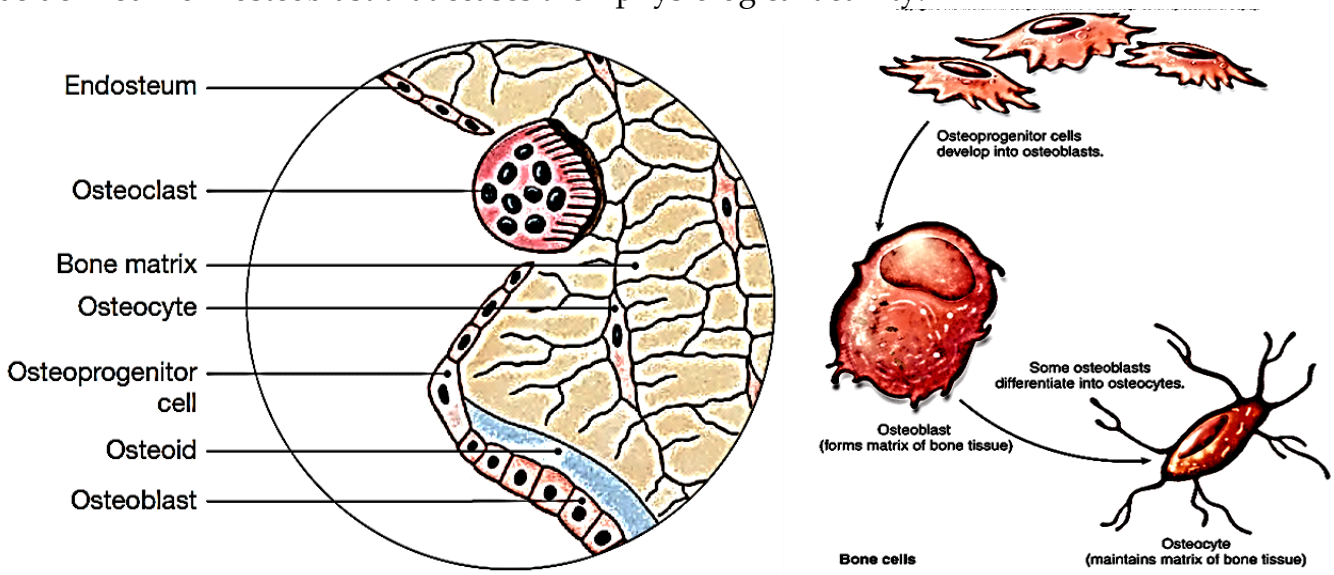
b) Osteoblasts are found in the growing portion of bone including periosteum. They are able to synthesize and secrete un-mineralized ground substance, act as pump cell to move calcium and phosphate in and out of bone tissue.



c) Osteocytes are the main cell of fully developed bones. They have a cell body that occupies a lacuna. Osteocytes are derived from osteoblasts. They together with osteoclasts play an important role of homeostasis by helping to release calcium.

d) Osteoclasts are multinuclear giant cell, which are found where bone is resorbed during its normal growth. Osteoclasts are derived from white blood cells called monocytes.

e) Bone - lining cells are found on the surface of most bones in the adult skeleton. They are believed to be derived from osteoblast that ceases their physiological activity.



Developmental Anatomy and Growth of Bones

Bones develop through a process known as Ossification.

Bone in embryo develops in two ways:

Intra-membranous ossification, If bone develops directly from mesenchymal tissue. Examples are vault of the skull, flat bones and part of the clavicle. In this type of ossification development continues rapidly from the center.

Endochondral Ossification, when bone tissue develops by replacing hyaline cartilage.

The cartilage itself do not converted into bone but the cartilage is replaced by bone through the process. Endochondral ossification produces long bones and all other bones not formed by intra-membranous ossification.

Function of bone

• **Supportive and protection** of internal organs.

Bones provide structural support and serve as a framework for the entire body.

Bones protect many delicate tissues and organs from injury and trauma.

• **The store house** and main supply of reserve calcium and phosphate.

More than 90% of the body's reserves of the minerals calcium and phosphate are stored and released by bone .

Calcium: needed for

- muscle contraction
- blood clotting
- nerve impulse transmission .

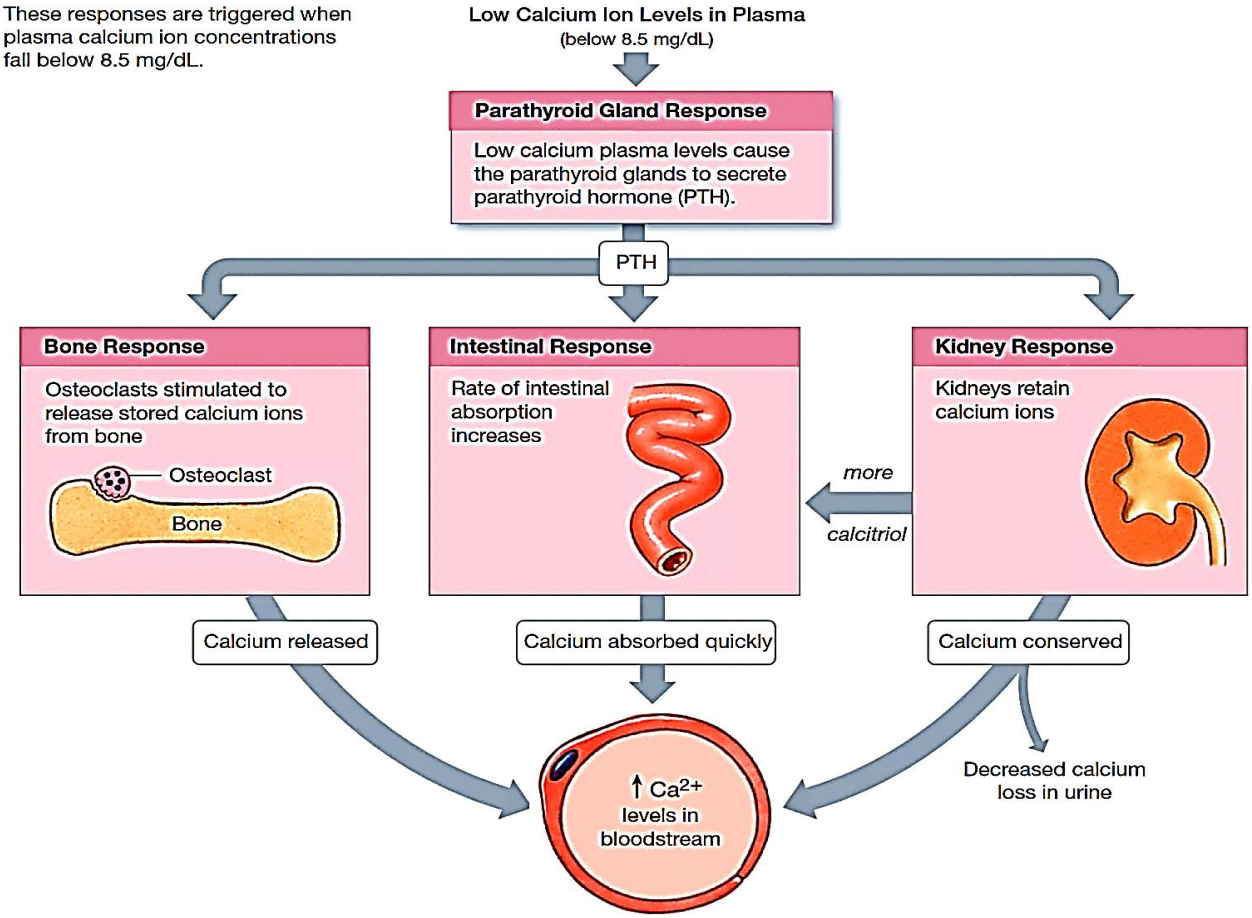
Phosphate: needed for

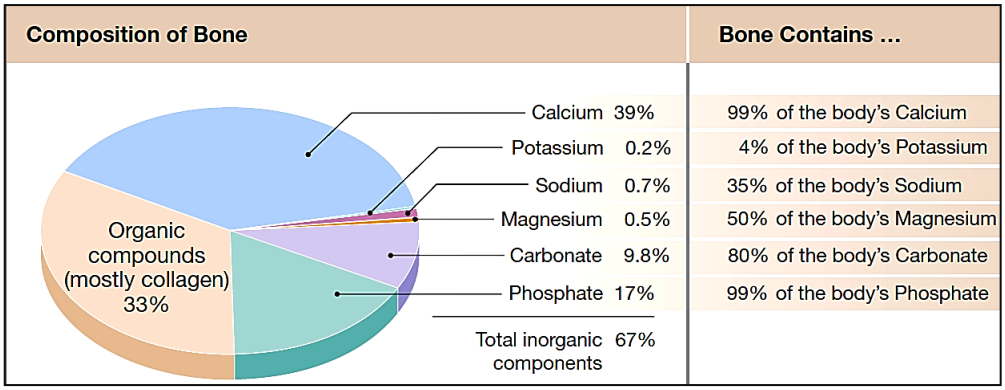
- ATP utilization
- structure of nucleic acids (DNA, RNA)

The manufacture of red and white blood cell.

Blood cell production in red bone marrow located in some spongy bone.

Red bone marrow contains stem cells form all of the blood cell types.





Effects of Vitamins

Vitamin A: activates osteoblasts

Vitamin C: normal synthesis of collagen

Vitamin D: absorption and transport of calcium and phosphate

The Skeleton

❖ **General feature's and surface markings**

Looking to the bone reveals the surface is not smooth but scarred with bumps, holes and ridges. These are surface markings where muscles, tendons and ligaments attached, blood & lymph vessels and nerves pass.

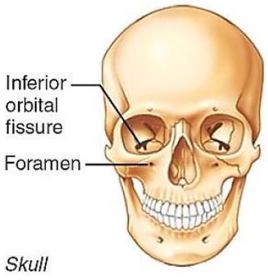
Depression and openings

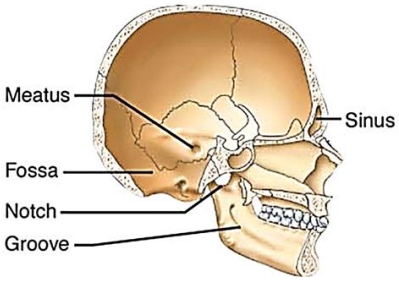
Fissure narrow, cleft like opening between adjacent parts of bone. Example: Supra of orbital fissure.

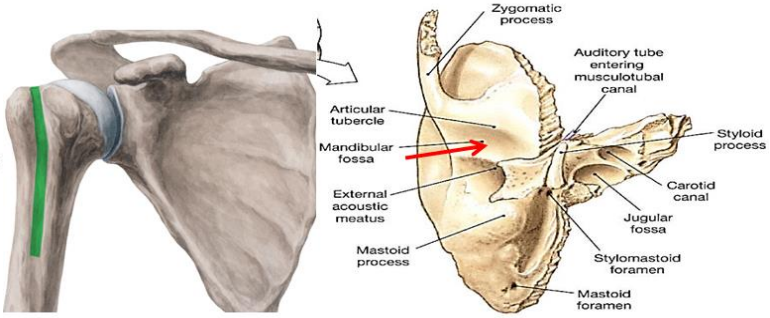
Foramen, a bigger, round opening. Example: Foramen magnum.

Meatus: a relatively narrow tubular canal. Example: External auditory meatus

Groves and sulcus: are deep furrow on the surface of a bone or other structure. Example: Inter-vertebral and radial groves of humers.







Fossa: shallow depressed area. Example: Mandibular fossa.

❖ **Processes that form joints**

Condyle / knuckle like process/ concave or convex. Example Medial condyle of femur Head, expanded, rounded surface at proximal end of a bone often joined to shaft by a narrowed neck. Example: Head of femur

Facet: small, flat surface. Example: Articular facet of ribs.

❖ **Process to which tendons, ligaments and other Connective tissue attach**

Tubercle: it is a knob like process. Example: Greater tubercle of humerus.

Tuberosity: it is large, round roughened process. Example: ischeal tuberosity.

Trochanter: it is a large, blunt projection found only on femur Crest is a prominent ridge. Example: Iliac crest.

Line: it is a less prominent ridge than a crest.

Spinous process (spine) is a sharp, slender process. Example Ischeal spin

Epicondyle is a prominence above condyle. Example medial Epicondyle of Femur

Division of the skeletal system

The Adult human skeletons have 206 named bones that are grouped in to two principal parts.

☼ **These are the axial and appendicular skeleton.**

- The Axial skeleton consist bones that lie around the axis. And the appendicular skeleton consists bones of the body out of the axial group.
- These are appendages. Upper & lower extremities and bones of girdles are grouped under appendicular skeleton.

The number of skull bones is sometimes listed as 22, when the Ossicles of the ears (6 bones) and the single hyoid bone is counted separately. Technically, the hyoid bone is not part of the skull.

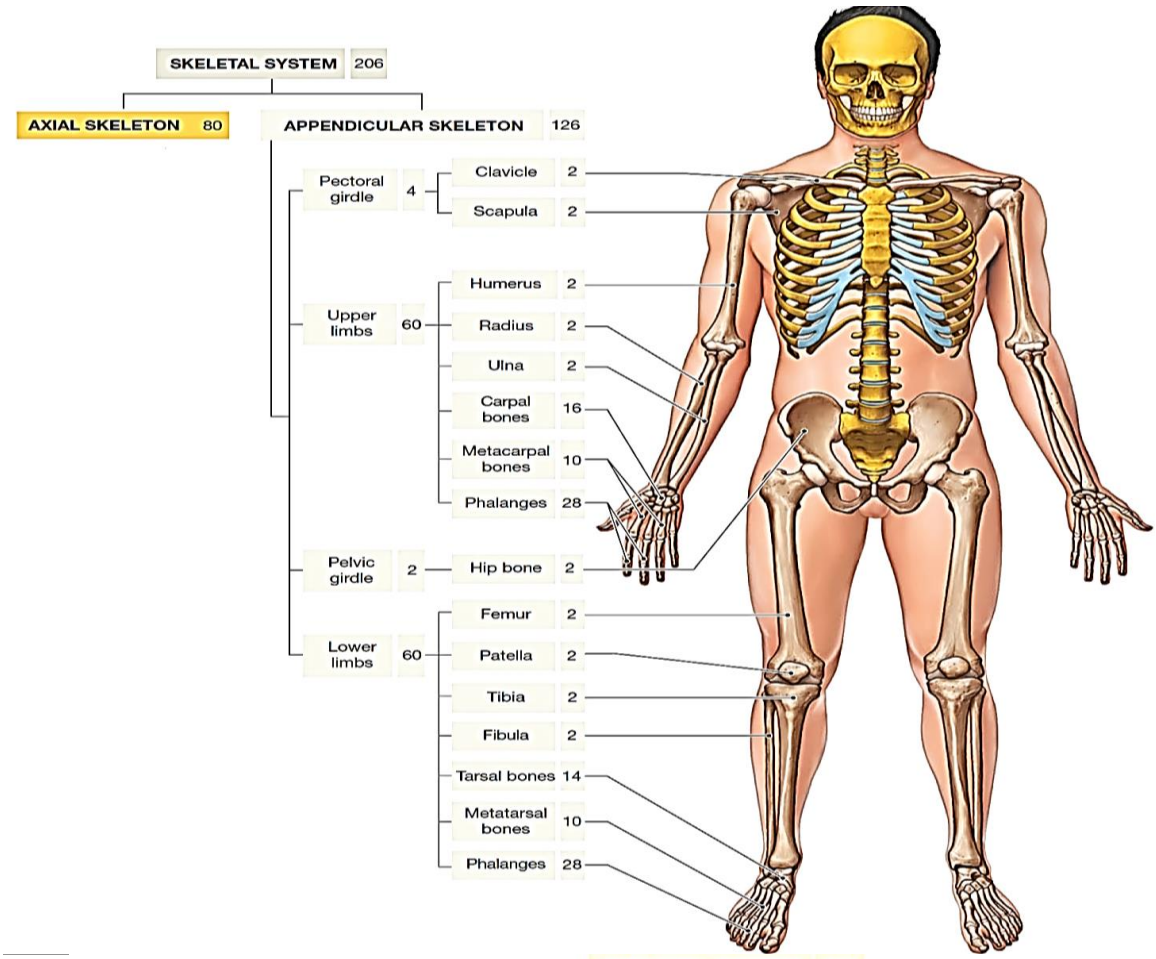
The thoracic vertebrae are sometimes included in this category.

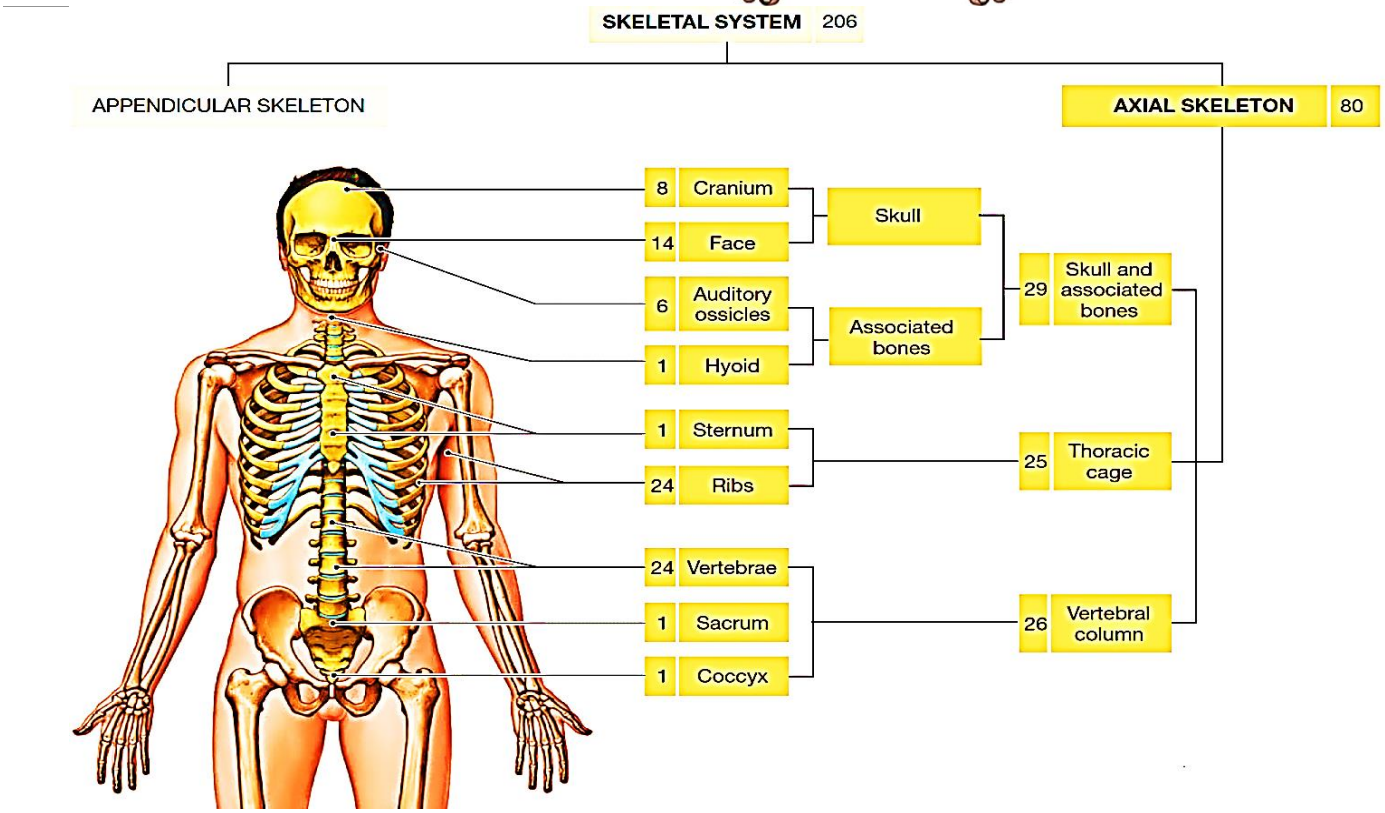
#Technically, the term arm refers to the upper extremity between the shoulder and elbow; the forearm is between the elbow and wrist. The upper part of the lower extremity, between the pelvis and knee, is the thigh; the leg is between the knees an ankle.

The Axial skeleton

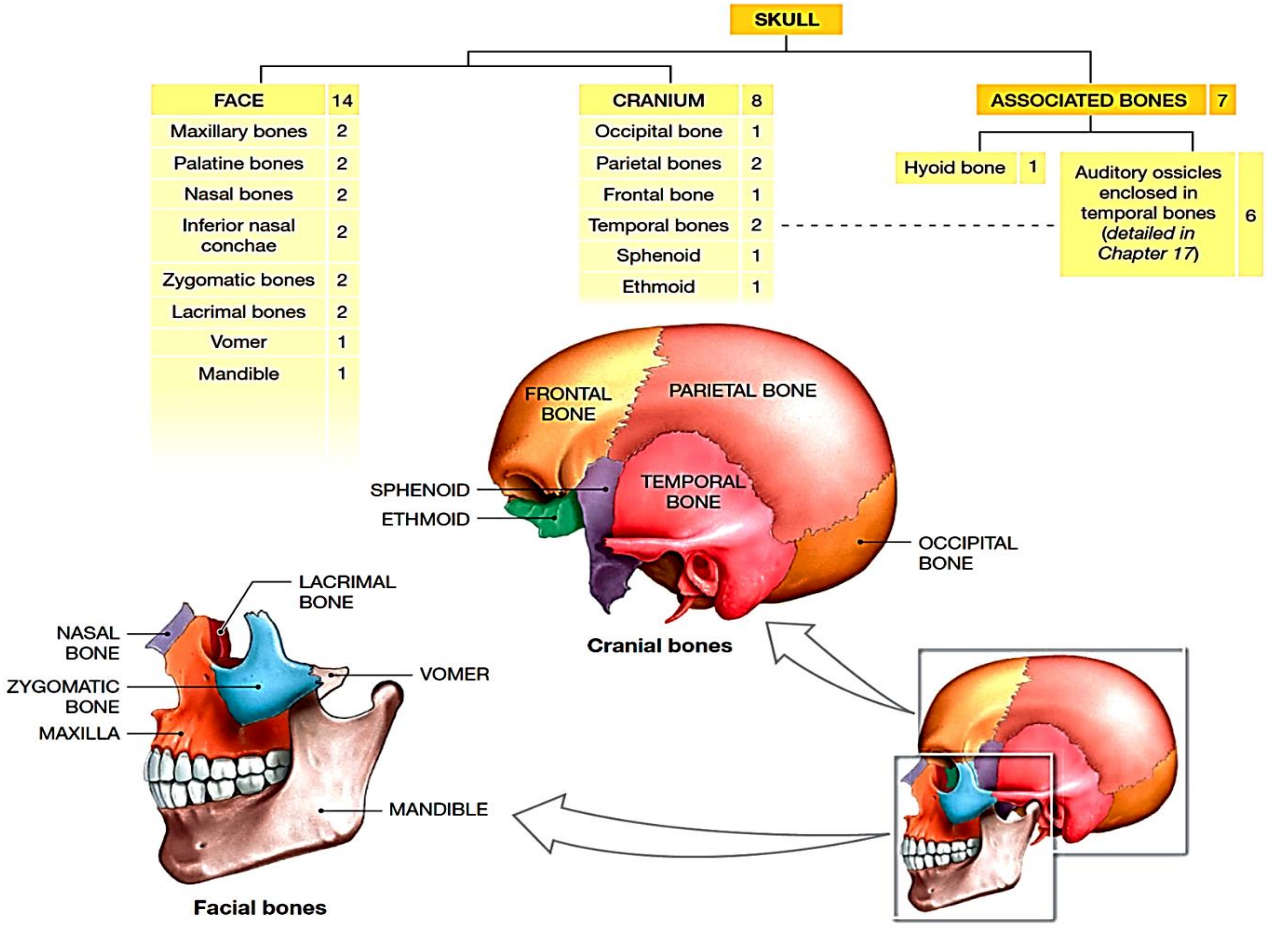
The skull

It Contain 22 bones. The skull rests on the superior of vertebral column. It is composed of cranial and facial bones.

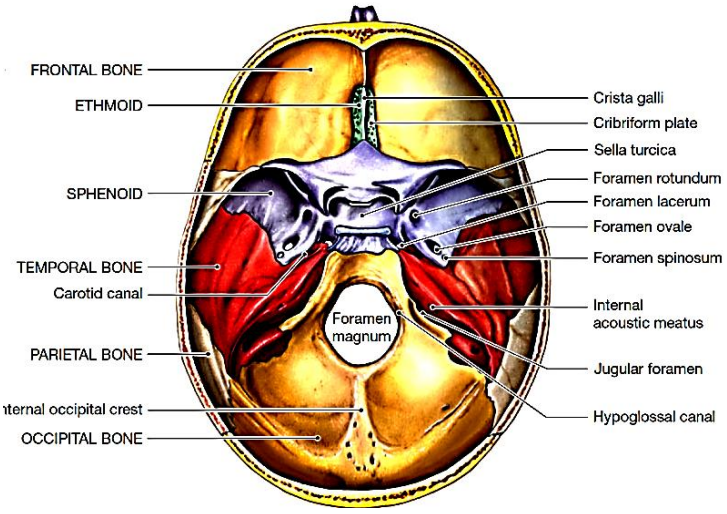


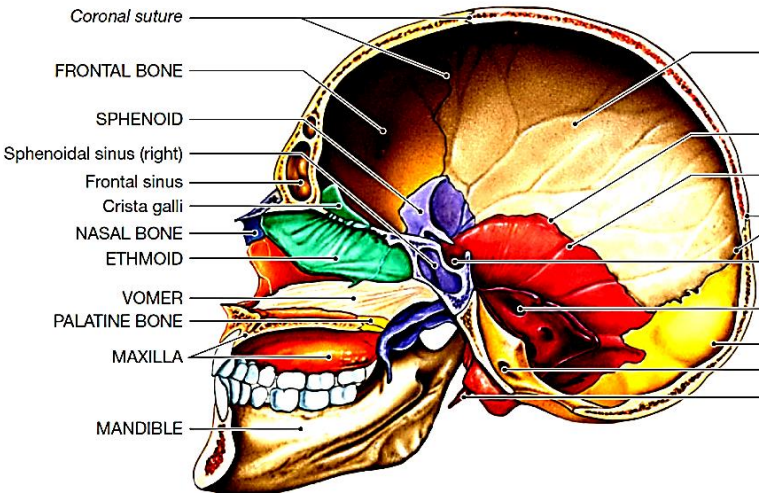


Description and function of Cranial Bones



Ethmoid (1) Base of cranium, anterior to body of sphenoid. Made up of horizontal, cribriform plate, median perpendicular plate, paired lateral masses; contains ethmoidal sinuses, crista galli, superior and middle conchae. Forms roof of nasal cavity and septum, part of cranium floor; site of attachment for membranes covering brain.

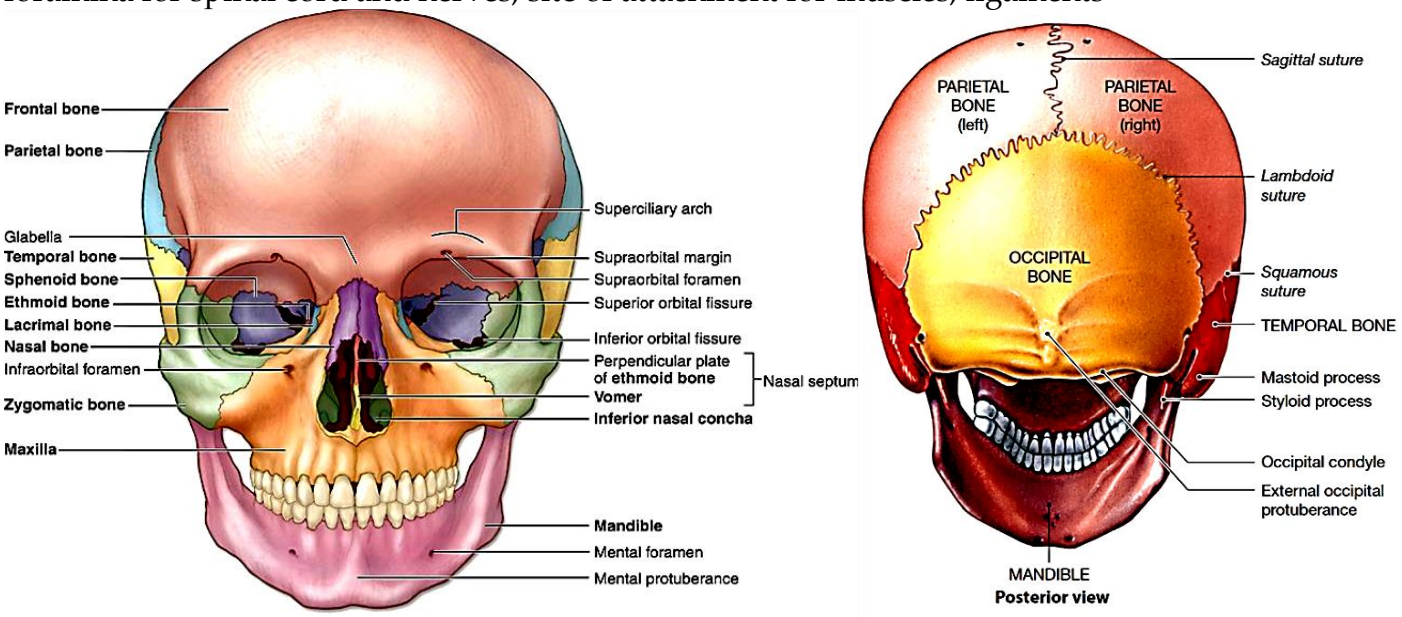




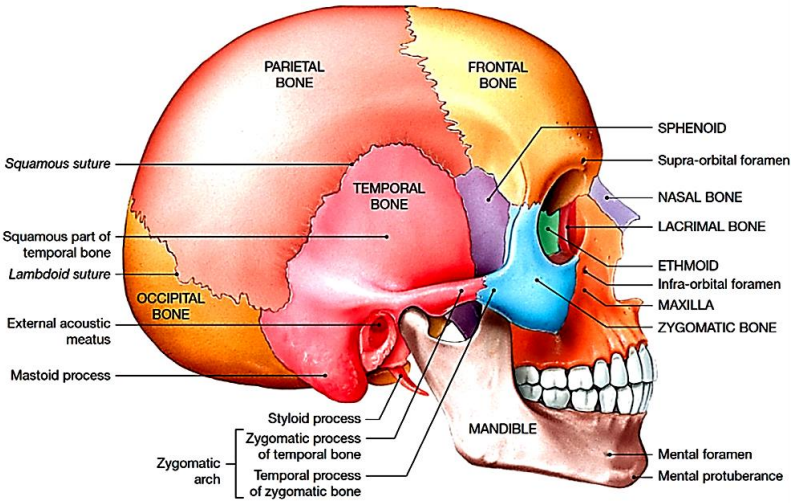
❖ **Frontal (1)** Anterior and superior parts of cranium, forehead, brow areas. Shaped like large scoop; frontal squama forms forehead; orbital plate forms roof of orbit; supraorbital ridge forms brow ridge; contains frontal sinuses, supraorbital foramen. Protects front of brain; contains passageway for nerves, blood vessels.

❖ **Occipital (1)** Posterior part of cranium, including base. Slightly curved plate, With turned- up edges; made up of squamous, base, and two lateral parts; contains foramen magnum, occipital condyles, hypo-gloss

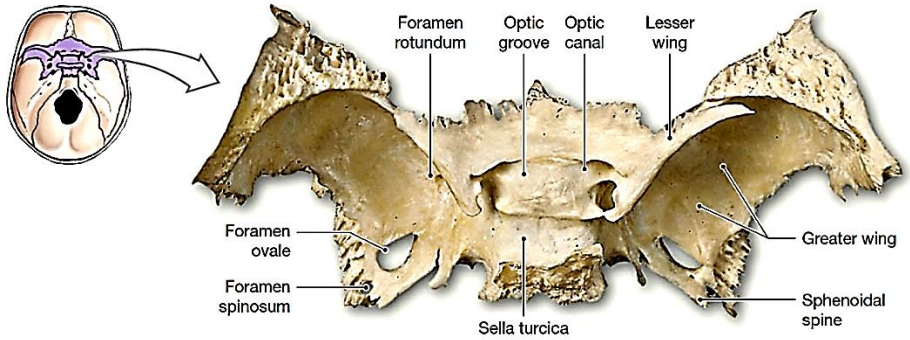
canals, atlanto-occipital joint, external occipital crest and protuberance. Protects posterior part of brain; forms foramina for spinal cord and nerves; site of attachment for muscles, ligaments



❖ **Parietal (2)** Superior sides and roof of cranium, between frontal and occipital bones. Broad, slightly convex plates; smooth exteriors and internal depressions. Protect top, sides of brain, passageway for blood vessels.



❖ **Sphenoid (1)** Base of cranium, anterior to occipital and temporal bones. Wedge-shaped; made up of body, greater and lesser lateral wings, pterygoid processes; contains sphenoidal sinuses, sella turcica, optic foramen, superior orbital fissure, foramenovale, foramen rotundum, foramen spinosum Forms anterior part of base of cranium; houses pituitary gland; contains foramina for cranial nerves, meningeal artery to brain.



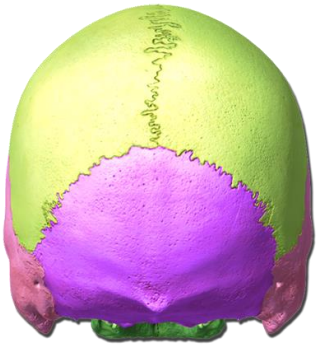
❖ **Temporal (2)** Sides and base of cranium at temples. Made up of squamous, petrous, tympanic, mastoid areas; contain zygomatic process, mandibular fossa, ear Ossicles, mastoid sinuses. Form temples, part of cheekbones; articulate with lower jaw; protect ear ossicles; site of attachments for neck muscles.

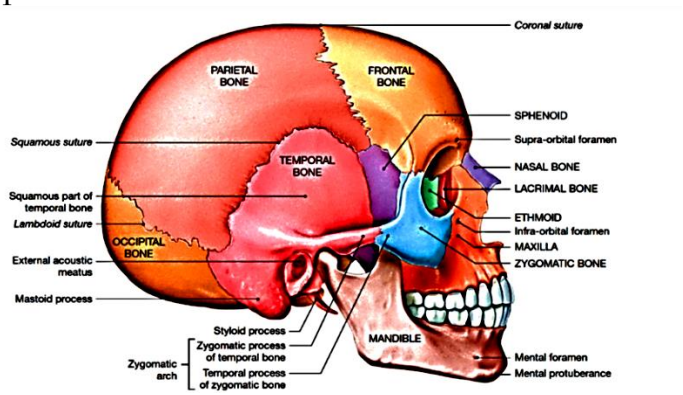
Sutures

Meaning to stitch, are immovable joint found between skull bones.

☼ There are four main sutures in the skull.

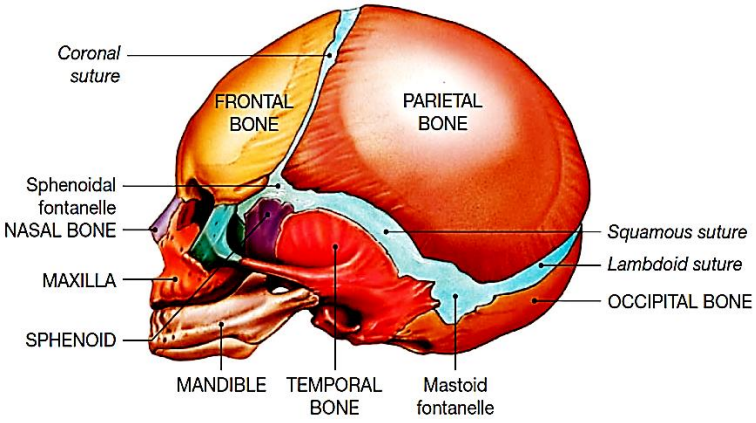
a) **Coronal suture:** between the frontal & the two-parital bone.
b) **Sagittal suture:** between the two parietal bones.
c) **Lambdoidal suture:** between parietal & occiputal bone.
d) **Squamosal suture:** between parietal bone and temporal bone.

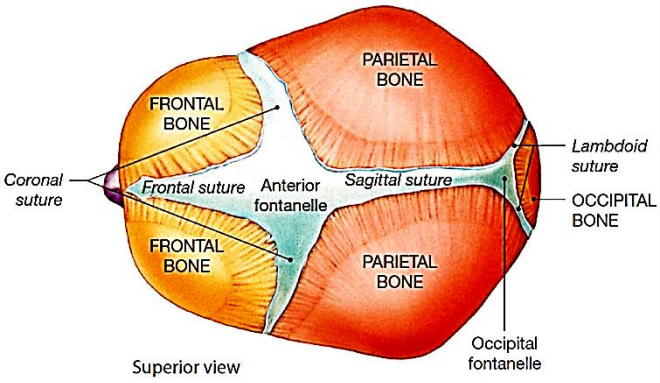




Fontanels

The skeleton of a newly formed embryo consists cartilage or fibrous membrane structures, which gradually replaced by bone the process is called ossification. At birth membrane filled spaces on the skull are called fontanel. They are found between cranial bones.





Function

- They enable skull of the fetus to compress as it passes through the birth canal
- Permit rapid growth of brain during infancy
- Serves as a landmark (anterior fontanel) for withdrawal of blood from the superior sagittal sinus
- Aid in determination of fetal position prior to birth.

In the skull of the fetus there are 6 prominent fontanels:

a) **The Anterior (frontal) fontanel**, between angle of two parietal bones & segment of the frontal bone. It is diamond shaped and is the largest fontanel. It closes 18 to 24 months after birth.
b) **The posterior (occipital) fontanel**, between parietal& occipital bone. It is also diamond shaped but smaller than the anterior fontanel. It closes 2 months after birth.
c) **The Anterolateral (sphenoidal) fontanel**, they are pair, one in each side. Found at the junction of frontal, parietal, temporal & sphenoidal bone. They are small & irregular in shape and closes at 3rd month after birth.

d) **The posterolateral (mastoid) fontanel**, Paired one in each side. Found at the junction of parietal, occipital and temporal bones. They are irregular in shape and begin to close at 1 or 2 months after birth and completed by 12 months.

Description and function of Facial bones

- **Inferior nasal**: Lateral walls of nasal cavities, below superior and middle
- **Conchii (2)** conchae of Ethmoid bone. Thin, cancellous shaped like curved leaves.,
- **Lacrimal (2)** Medial wall of orbit, behind frontal process of maxilla. Small, thin, rectangular; contains depression for lacrimal sacs, nasolacrimal tear duct.

Mandible (1) Lower jaw, extending from chin to mandibular fossa of temporal bone.
Largest, strongest facial bone; horseshoe-shaped horizontal bony with two perpendicular rami; contains tooth socket coronoid, condylar, alveolar processes, mental foramina. forms lower jaw, part of temporomandibular joint; site of attachment for muscles.

Maxillae (2) Upper jaw and anterior part of hard palate. Made up of zygomatic, frontal, palatine, alveolar processes; contain infraorbital foramina, maxillary sinuses, tooth sockets. Form upper jaw, front of hard palate, part of eye sockets.

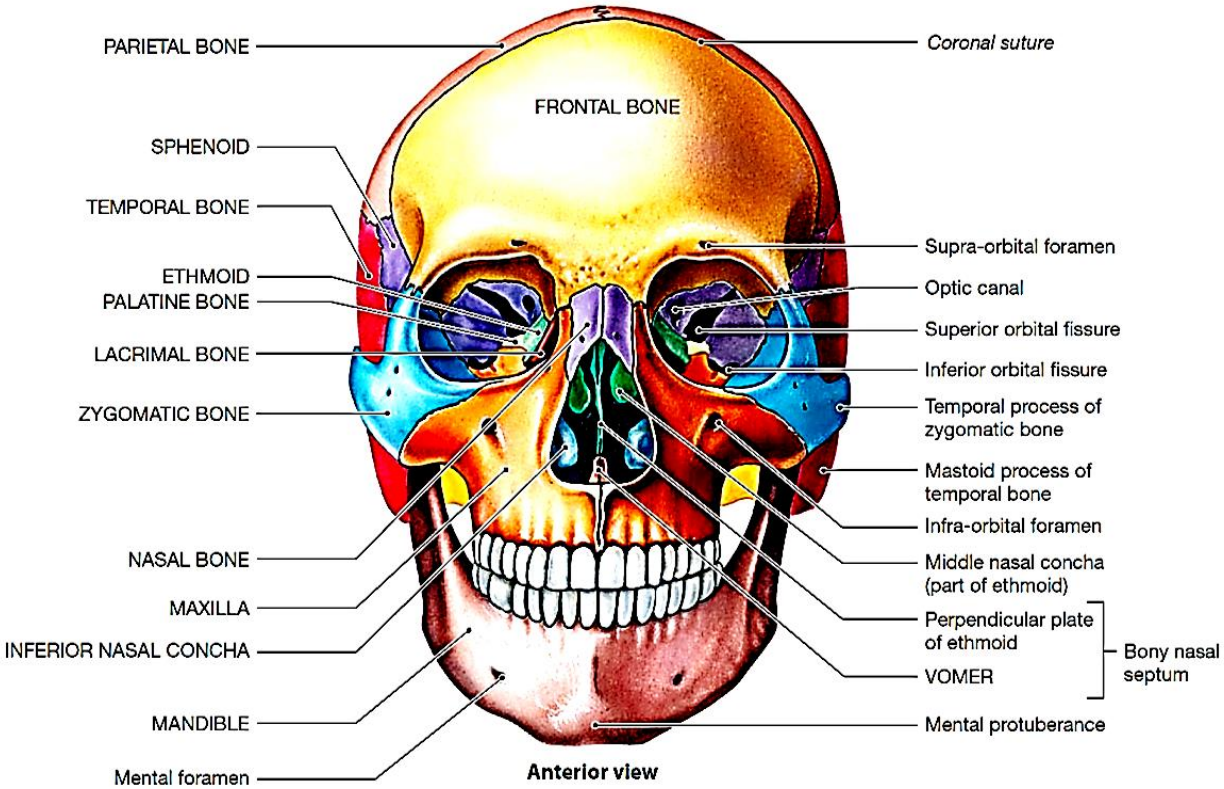
Nasal (2) Upper bridge of nose between frontal processes of maxillae. Small, oblong; attached to a nasal cartilage. Form supports for bridge of upper nose.

Palatine (2) Posterior part of hard palate, floor of nasal cavity and orbit; posterior to maxillae. L-shaped, with horizontal and vertical plates; contain greater and lesser palatine foramina. **Horizontal plate** forms posterior part of hard palate; vertical plate forms part of wall of nasal cavity, floor of orbit.

Vomer (1) Posterior and inferior part of nasal septum. Thin, shaped like plowshare. Forms posterior and inferior nasal septum dividing nasal cavities.

Zygomatic (2) Cheekbones below and lateral to orbit. Curved lateral part of (molar) cheekbones; made up of temporal process, zygomatic arch; contain zygomatico-facial and zygomatico-temporal foramina. form cheekbones, outer part of eye sockets.

Hyoid (1) Below root of tongue, above larynx. U-shaped, suspended from styloid process of temporal bone; site of attachment for some muscles used in speaking, swallowing.



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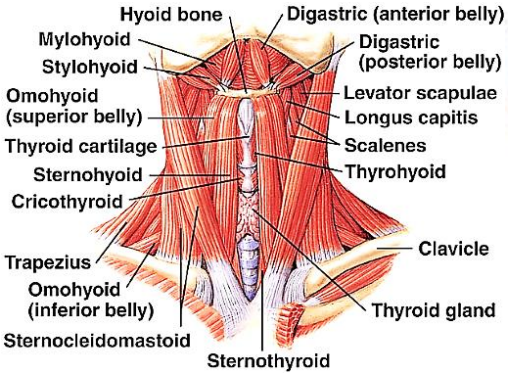
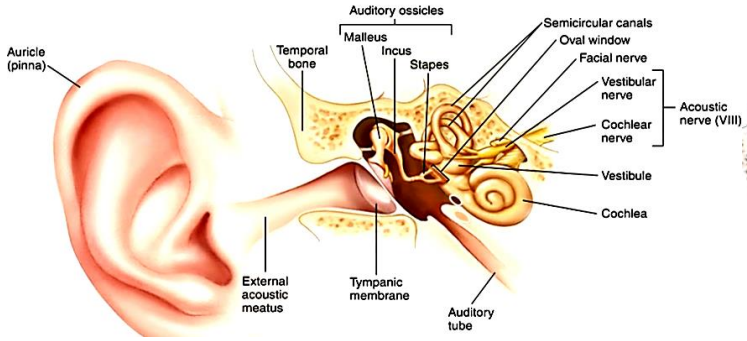
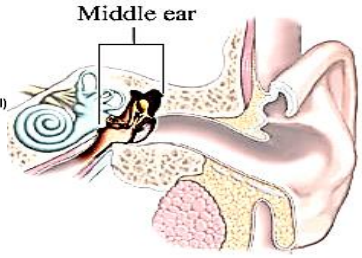
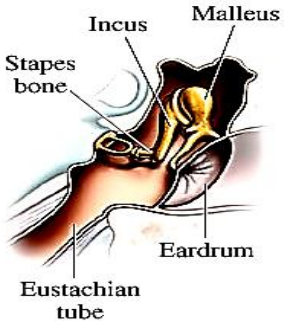


Ossicles of ear (6) Inside cavity of petrous portion of temporal bone.
Tiny bones Incus(2) shaped like anvil, hammer, stirrup, articulating with one another malleus (2) and attached to tympanic membrane. Convey sound vibrations stapes (2) from eardrum to oval window



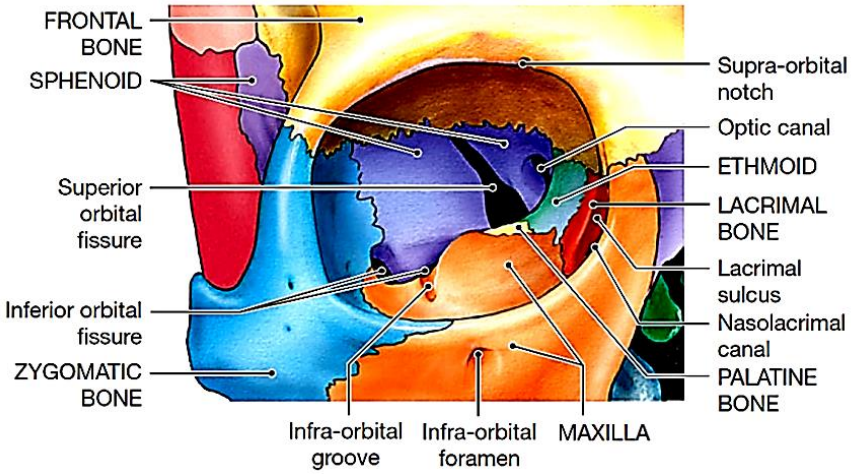


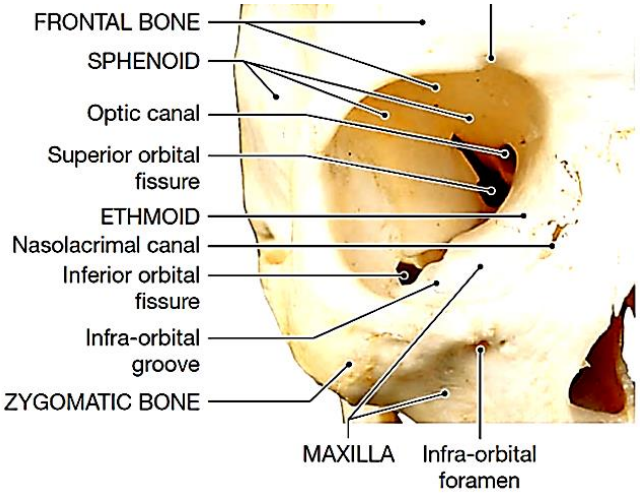


Orbits: A pyramid shaped space that contains the eyeball & associate structures. It is formed by bones of the skull. **Orbit has four walls and apex:**

- The roof of the orbit consists of parts of the frontal& sphenoid bone.
- The lateral wall is formed by portions of zygomatic and sphenoid bone.
- The floor of the orbit is formed by parts of the maxilla, zygomatic and palatine bone.
- The medial wall is formed by portion of the maxilla, lacrimal, ethimoid & sphenoid bone.

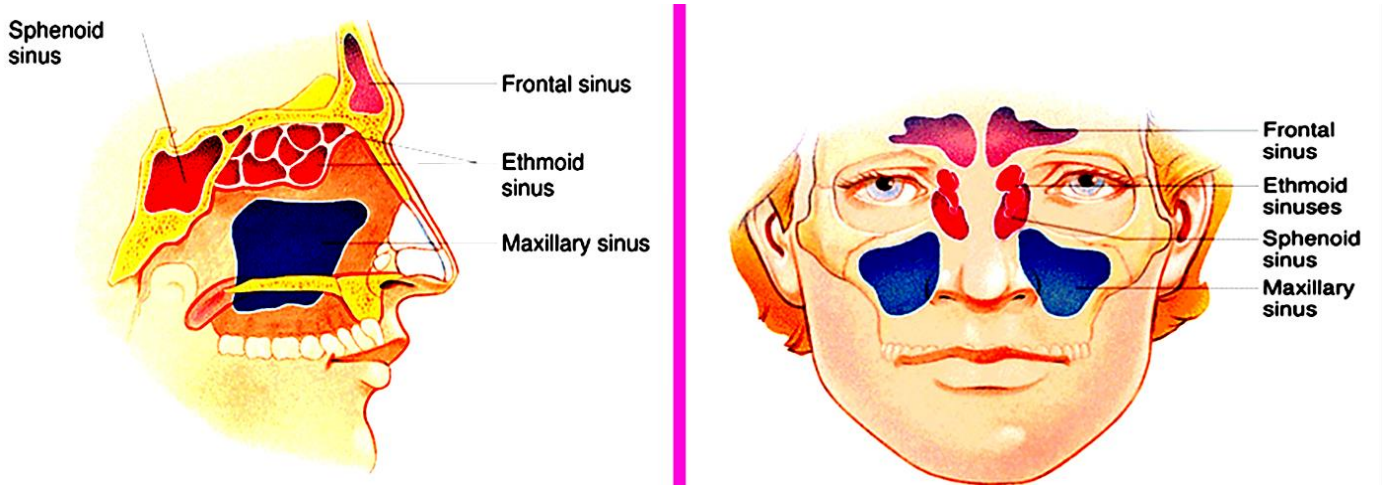
In the orbit there are openings that pass structures. Some of the principal openings and And the structures passing through are:
Optic foramen (canal) passes optic nerve Superior orbital fissure passes supra orbit nerve and artery. Inferior orbital fissure passes maxillary branch of trigeminal and zygomatic nerve and infra orbital vessel. Supra orbital foramen (notch) passes oculomotor, trochlear, ophthalmic branch of trigeminal and abducent nerves. Canal for Naso-lacrimal duct passes Naso-lacrimal duct.





Facial cavities (sinuses)

- Sphenoid sinus
- Frontal sinus
- Ethmoid sinus
- Maxillary sinuses

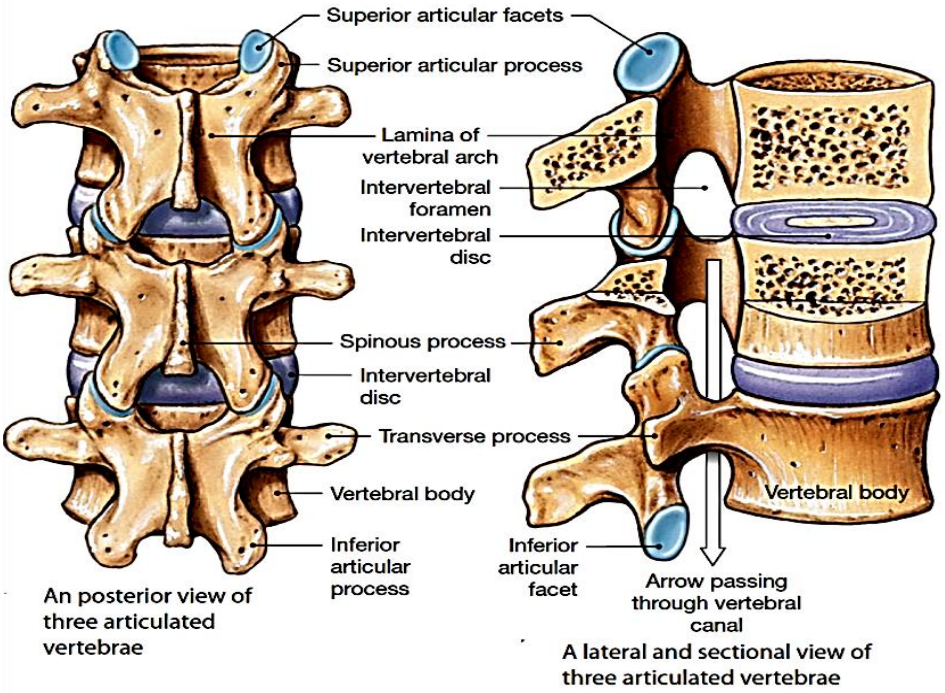


The vertebral column

The vertebral column together with the sternum & ribs constitutes the skeleton of the trunk of the body. It composes2/5th of the height of the body and has average length in male of 71 c.m. and in female 61 c.m. The adult vertebral column contains 26 vertebras. Prior to fusion of sacral & coccygeal vertebrae the total number is 33. It is a strong and flexible to either direction & rotated on it self. Encloses & protect spinal cord, supports the head and serves as a point of attachment for the ribs & muscles of the back.

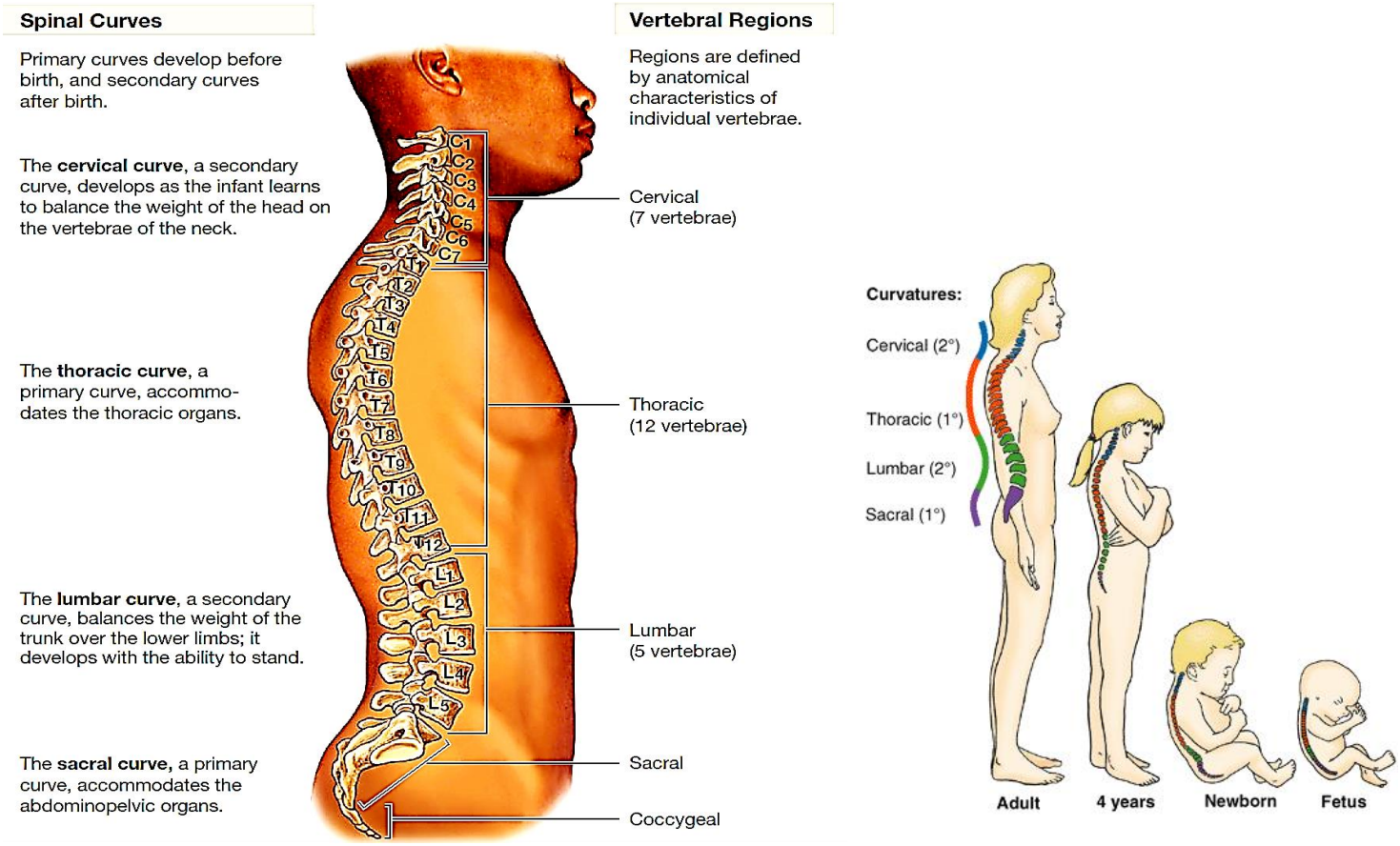
Inter vertebral discs

Between adjacent vertebrae from 1st to sacrum there are inter vertebral discs. They are fibro-cartilaginous. Each disc is composed of the outer fibrous ring consisting fibro-cartilage called annulus fibrosis and the inner soft, pulpy highly elastic structure called the nucleus pulpous. The disc permits various movement of the vertebral column, absorb shock and form a strong joint.



Normal curves

The vertebral column when viewed from side is not a straight line rather have bandings. These are normal curves of the vertebral column. There are 4 normal curves formed by vertebras, two are concave and the other two are convex.



The presences of the curve have several functions; these are absorption of shock, maintenance of balance, protection of column from fracture and increasing the strength of the column.

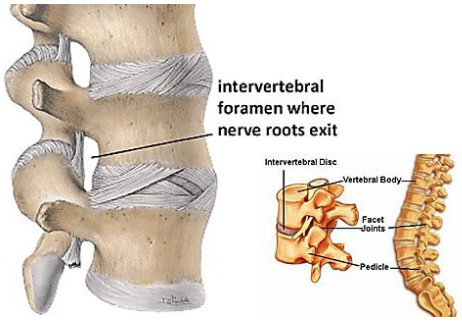
In the age of the fetus there is only a single anterior concave curve, but approximately the third post natal month, when the child begins to hold head erect, the cervical curve develops.

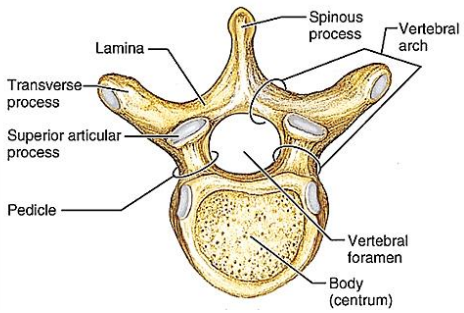
Later when the child sits up, stands and walks the lumbar curve develops. The cervical & lumbar curves are an anteriorly convex and because they are modification of the fetal position they are called secondary curves. The thoracic and sacral curves are anteriorly concave, since they retain the anterior concavity of the fetal curve they are referred primary curves.

Typical vertebra

Although there are variations in size and shape, the vertebras of the different region have basically similar structure. It consists the body, vertebral arch and seven processes.

The body (Centrum), thick, disc shaped, anterior part. It has superior and inferior roughened area for attachment with intervertebral discs.

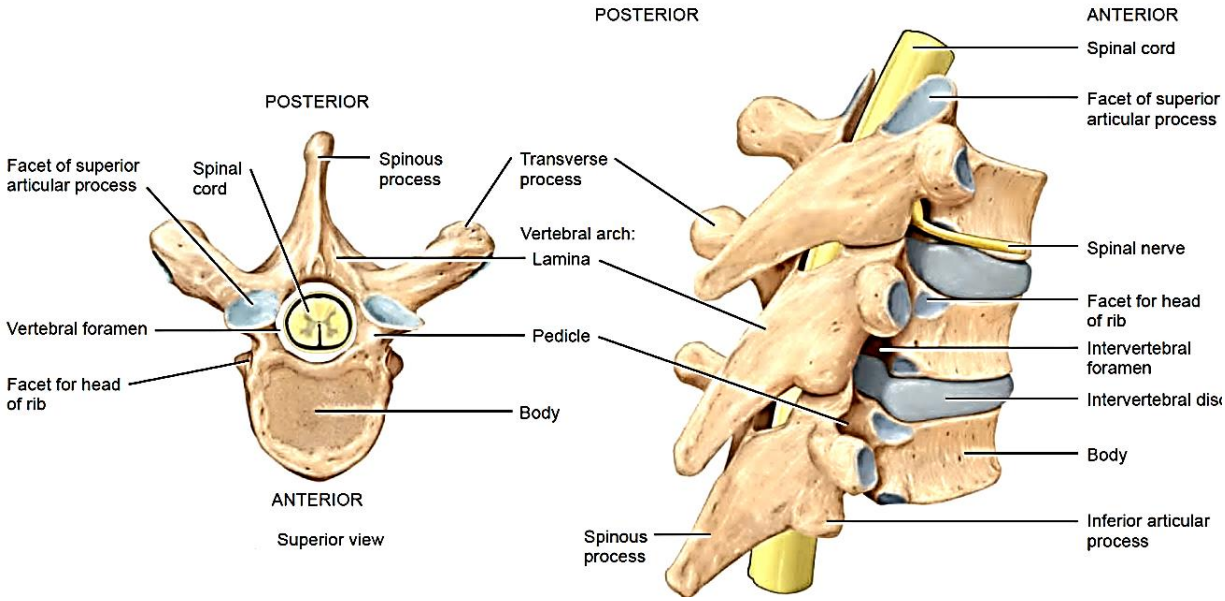




The vertebral (neural) arch extends posterior from the body of the vertebrae. With the body it surrounds the spinal cord. It is formed by two short, thick process called pedicles. It projects posteriorly to meet at laminae. The laminae are flat parts that join to form the posterior portion of the vertebral arch. The space that lies between the vertebral arch and body contains the spinal cord called vertebral foramina. The vertebral foramina of all vertebras together form the vertebral (spinal)
canal.

The pedicles are notched superiorly & inferiorly to form an opening between vertebrae on each side of the column called Intervertebral foramen. Intervertebral foramen is an opening between the vertebras that serves as passage of nerves that come out of spinal cord to supply the various body parts.

There are seven processes that arise from the vertebral arch at the point where the lamina and pedicle joins.



•The transverse processes on both side extends laterally.

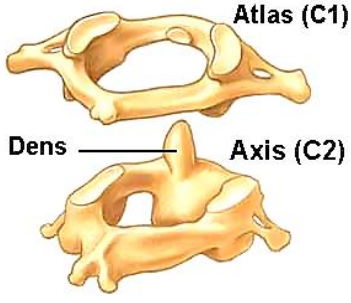
•The Spinous processes extends posteriorly & inferiorly from the junction of the laminae.

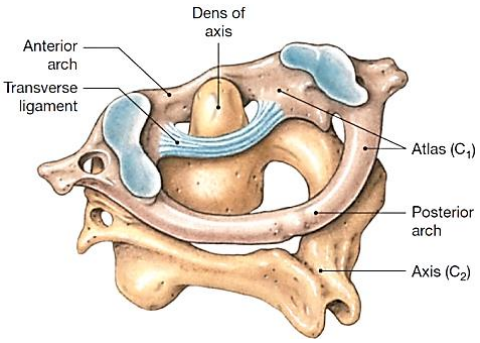
•Both the transverse & spinous processes are muscle attachments. The remaining four processes form joints with another vertebra. Two of them articulate with the immediate superior vertebra. And the other two articulate with the immediate inferior vertebra.

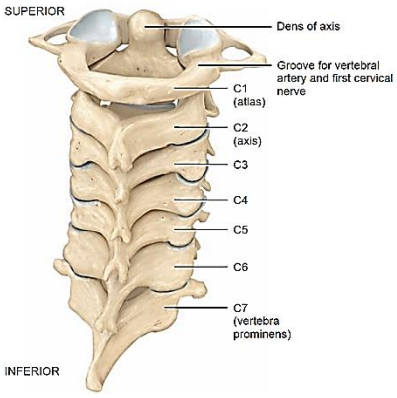
Description and function of bones of the vertebral column

Cervical vertebrae (7) First (atlas), second (axis), and seventh vertebrae are C1-C7 modified; third through sixth are typical; all contain transverse formina.

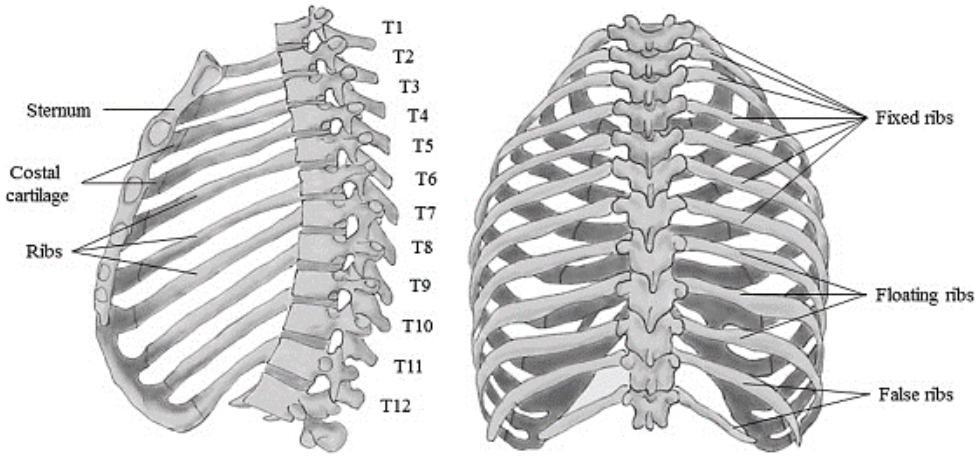
Atlas supports head, permits "yes" motion of head at joint between skull and atlas; axis
Permits "no" motion at joint between axis and atlas.



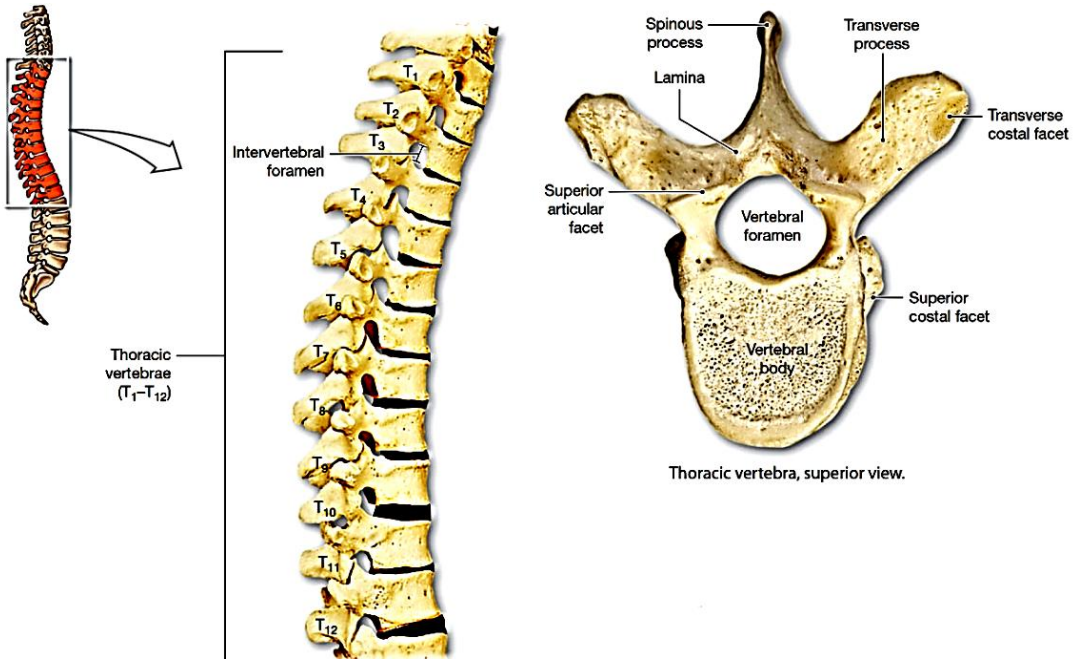




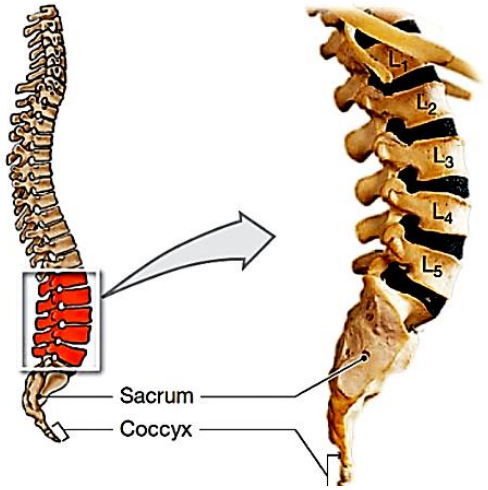
Thoracic vertebrae (12) Bodies and transverse processes have facets that articulate T1-T12 with ribs; laminae are short, thick, and broad. Articulate with ribs; allow some movement of spine in thoracic area.



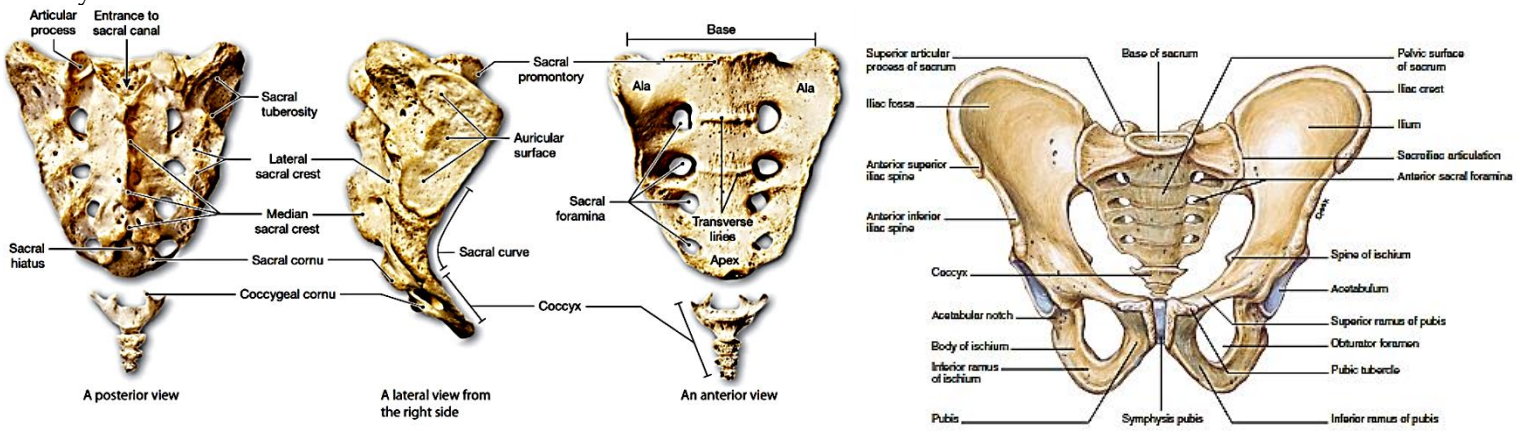
Lumbar vertebrae (5) Largest, strongest vertebrae; adapted for attachment of back L1-L5 muscles. Support back muscles; allow forward and backward bending of spine.



Sacrum Wedge-shaped, made up of five fused bodies united by four (5 fused bones) intervertebral disks. Support vertebral column; give strength and stability to pelvis.



Coccyx Triangular tailbone, united with sacrum by intervertebral (3 to 5 fused bones) disk. Vestige of an embryonic tail.



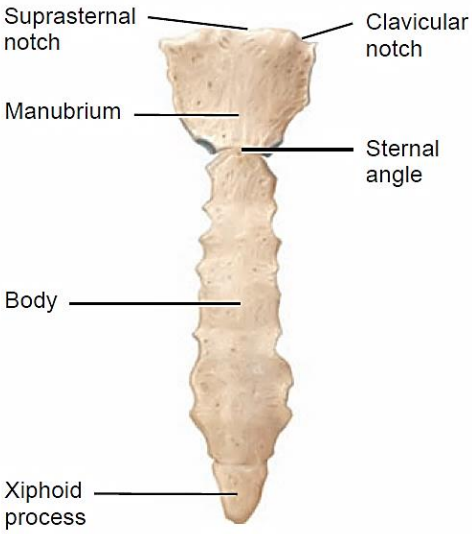
•In a child there are 33 separate vertebrae, the 9 in the sacrum and coccyx not yet being fused.

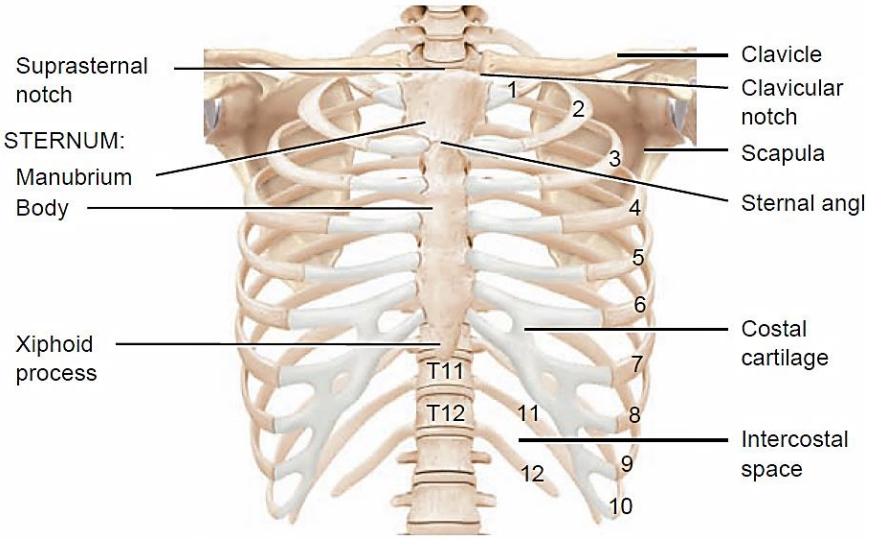
The Thorax

Refers to the chest. Thorax is a bony cage formed by sternum (breast bone), costal cartilage, ribs and bodies of the thoracic vertebra.

Sternum (breast bone)

Sternum is flat, narrow bone measuring about 15 c.m. (6 inch) located in the median line of anterior thoracic wall. It consists 3 basic portions: the manubrium (superior portion), the body (middle & largest portion) and the xiphoid process (inferior & smallest portion). The junction of the manubrium and the body forms the sternal angle. The manubrium on its superior portion has a depression called jugular (supra sternal) notch.



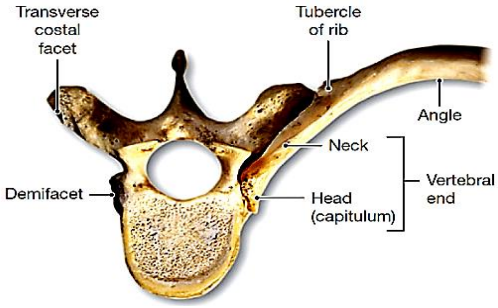


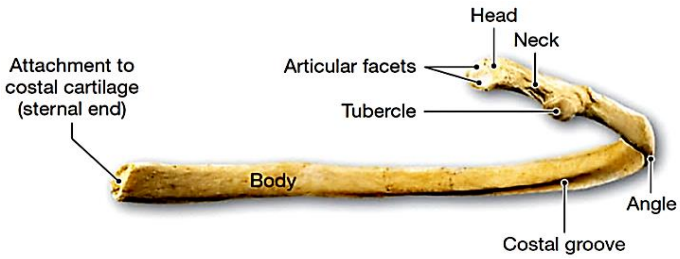
On each side of the jugular notch are clavicular notch that articulates with medial end of clavicle. The manubrium also articulates with the 1st and 2nd rib. The body of the sternum articulates directly or indirectly with 2nd to10th rib. The xiphoid (ZI-foyd) process consists hyaline cartilage during infancy and child hood and do not ossify completely up to the age of 40 The xiphoid process is used as a palpable landmark during the administration of cardiopulmonary resuscitation (CPR), and CPR training strongly emphasizes proper hand positioning to reduce the chances of breaking ribs or the xiphoid process.

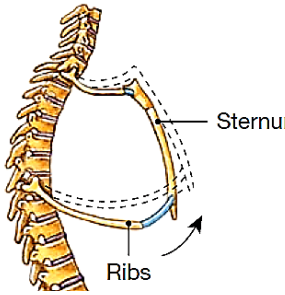
Ribs

Human being contains 12 Pair of ribs that make up the side of thoracic cavity. Ribs increase in length from 1st through 7th and they decrease in length through 12th. Each ribs posteriorly articulates with the body of its corresponding thoracic vertebra.

Anteriorly the 1st seven ribs have direct attachment to sternum by costal cartilage hence they are called true (vertebro – sternal) ribs.
The 8th – 10th ribs, which are groups of the **false ribs** are called vertebra-chondral ribs because their cartilage attach one another and then attaches to the cartilage of the 7th rib.
The 11th & 12th ribs are designated as **floating ribs** because their anterior part even doesn't attach indirectly to sternum. Although there is variation when we examine a typical rib (3rd to 9th) contains a head, neck and body parts.





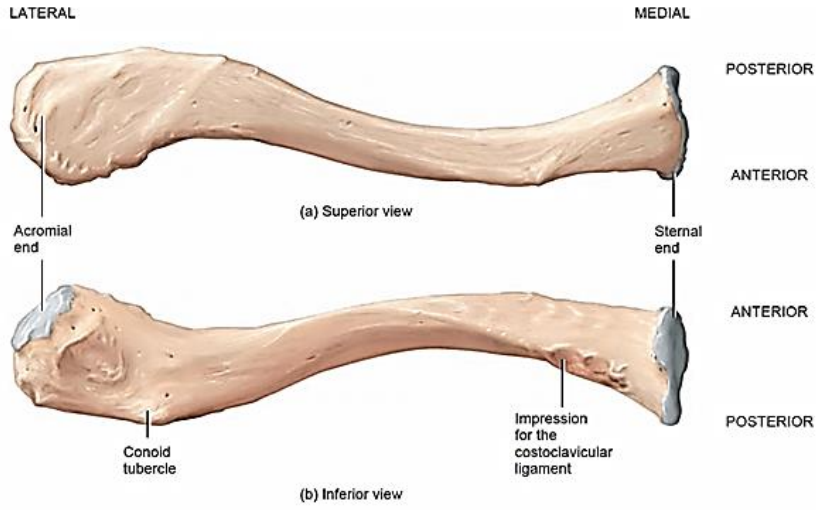


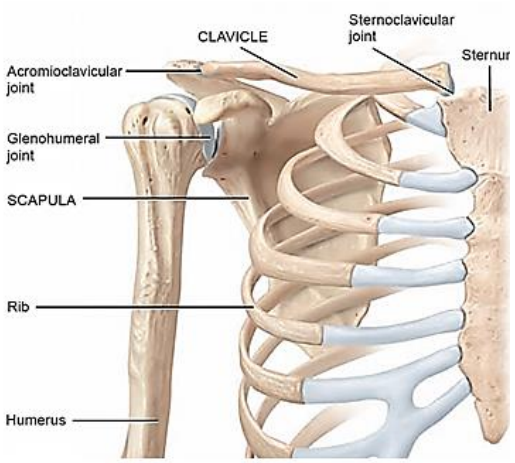
The Head is a projection at posterior end of the rib. It consists one or two facet that articulate with facet of the vertebra.
The neck is constricted portion just lateral to the head. One or two knobs like structures on the posterior end where the neck joins the body is the tubercles, which articulate with the transverse process of the vertebra and to attach with muscles of the trunk.
The body (shaft) is main part of the rib. . The costal angle is the site where the rib changes its direction. The inner side of the costal angle is costal grove. Where thoracic nerves and blood vessels are protected.
The Appendicular skeleton

The upper extremities (limbs)

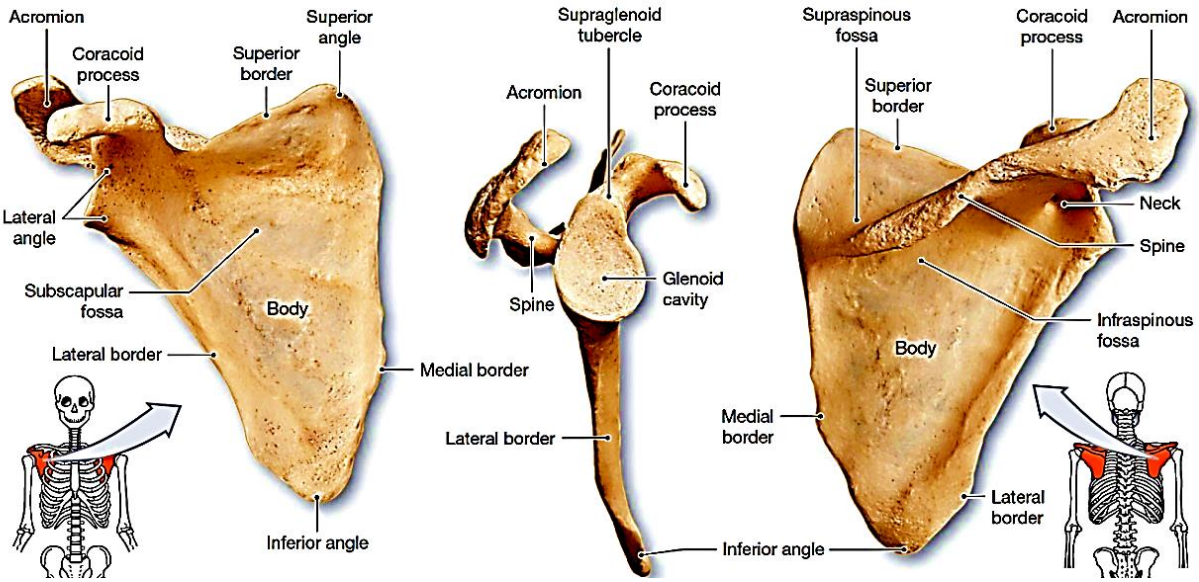
The upper extremities consist of 64 bones. Connected and supported by the axial skeleton with only shoulder joint and many muscles from a complex of suspension bands from the vertebral column, ribs and sternum to the shoulder girdle.
Description and function of bones of the upper extremity
Shoulder (Pectoral) Girdle

- **CLAVICLE (2)** Collarbone; S-shaped bones, double-curved, long bone with rounded medial end and flattened lateral end; held in place by ligaments. Holds shoulder joint and arm away from thorax so upper limb can swing freely.



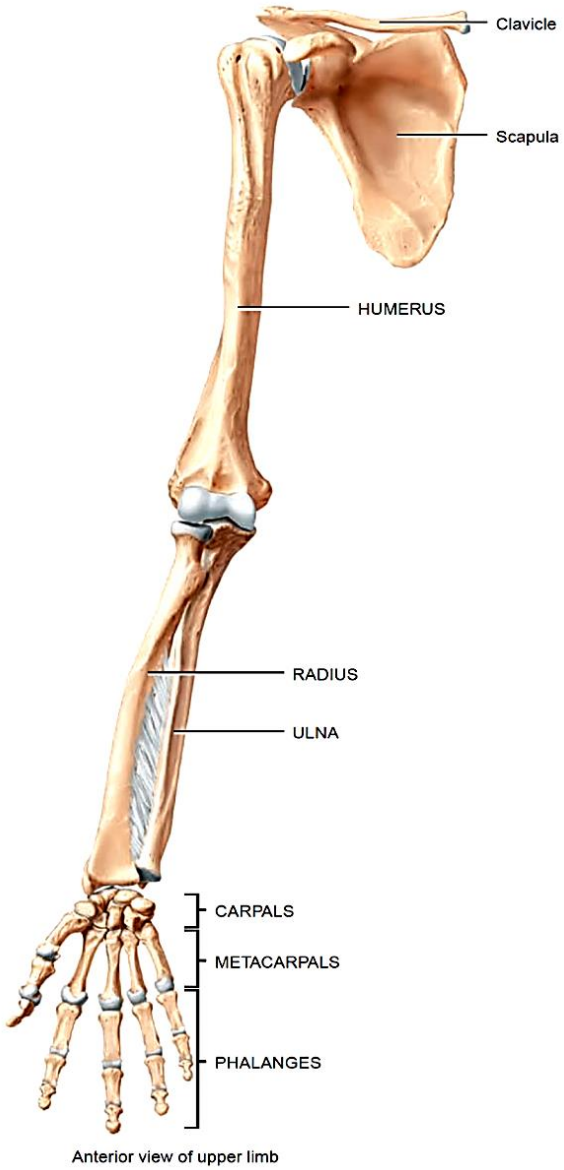


- **Scapula (2)** Shoulder blade; flat, triangular bone with horizontal spine separating fossae. Site of attachment for muscles of arm and chest.



Arm (Brachium)

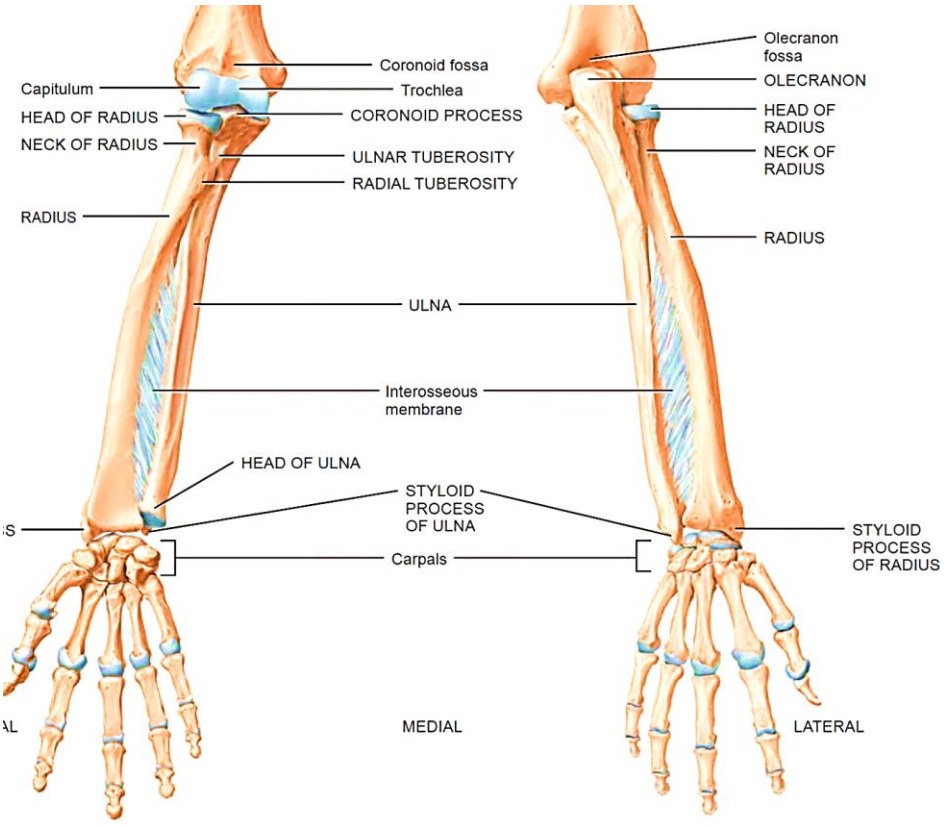
- **Humerus (2)** Longest, largest bone of upper limb; forms ball and socket joint with glenoid fossa of scapula. Site of attachment for muscles of shoulder and arm, permitting arm to flex and extend at elbow.



Anterior view of upper limb

Forearm (Antebrachium)

- **Radius (2)** Larger of two bones in forearm; large proximal end consists of olecranon process (prominence of elbow). Forms hinge joint at elbow.

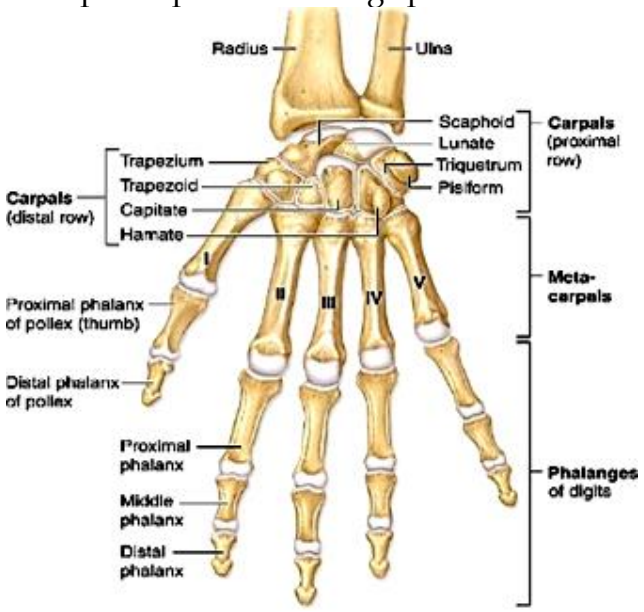


Wrist

- **Carpals (16)** Small short bones; in each wrist, 8 carpals in 2 transverse rows of 4. With attached ligaments, allow slight gliding movement.

Hands and Fingers

- **Metacarpals (10)** Five miniature long bones in each hand in fanlike arrangement; articulate with finge at metacarpophalangeal joint (the Knuckle). Aid opposition movement of thumb; enable cupping of hand.
- **Phalanges (28)** Miniature long bones, 2 in each thumb, 3 in each finger; articulate with each other at interphalangeal joint. Allow fingers to participate in stable grips.



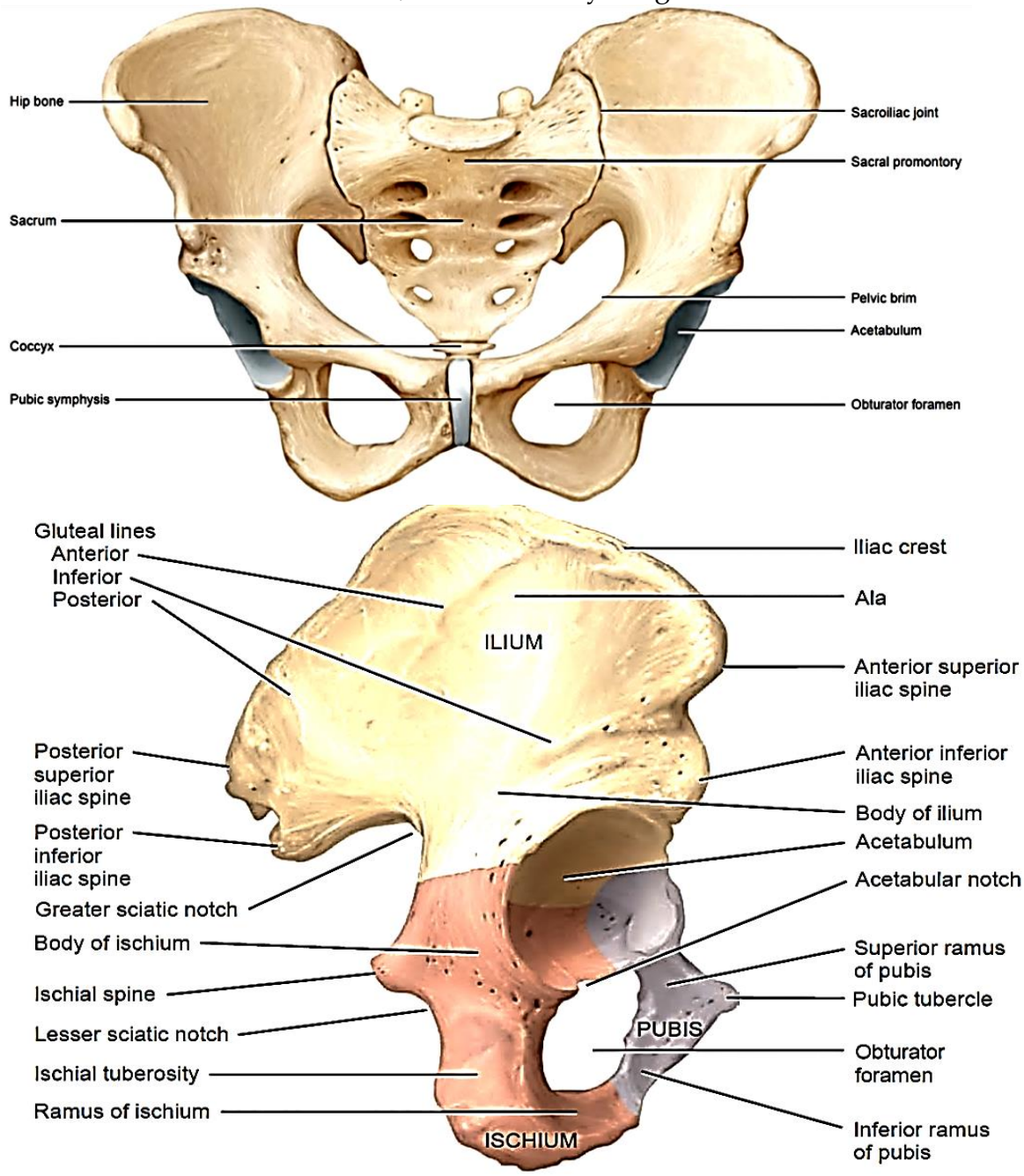
The lower extremity

It consists 62 bones. The lower extremity is connected to the axial skeleton with the hip girdle.

Description and function of the lower extremity

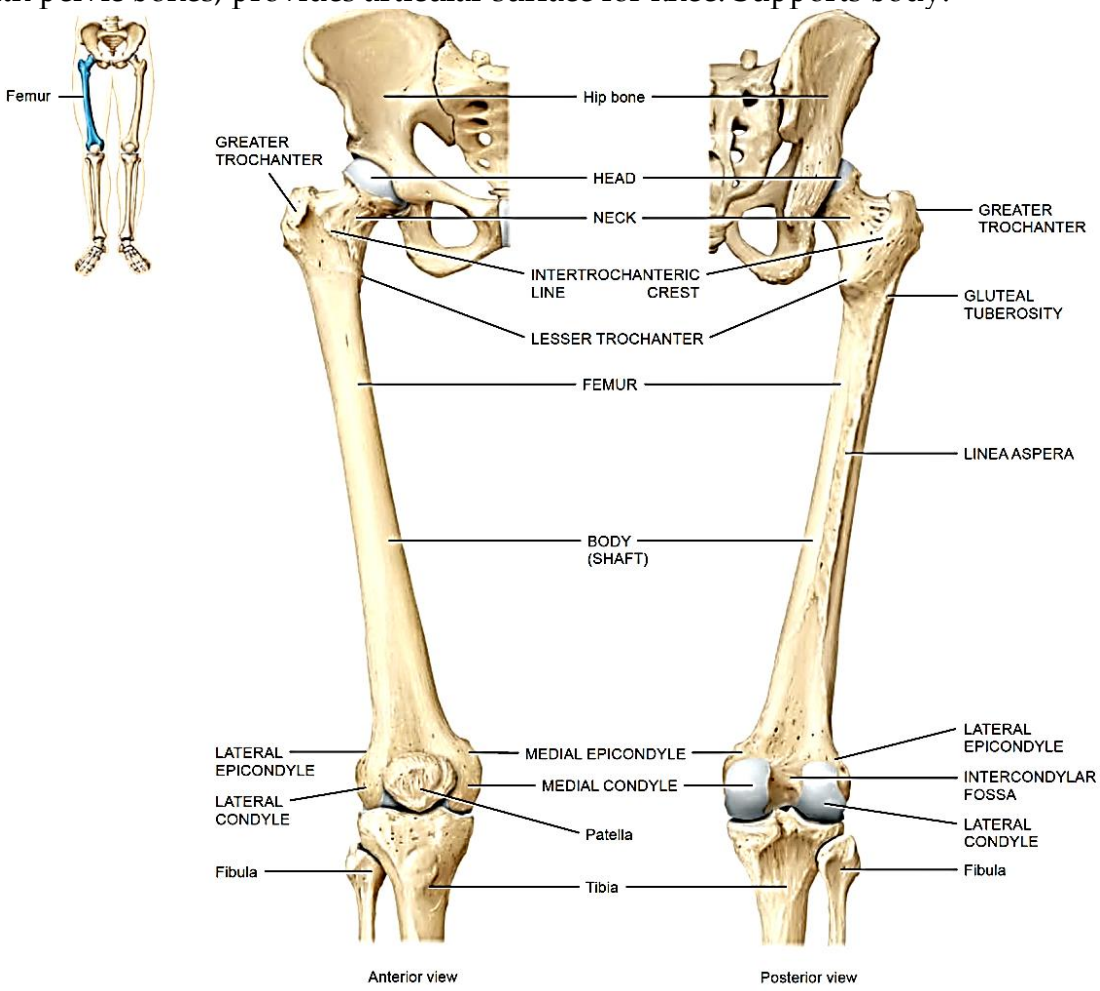
Pelvic Girdle

- **Hipbone** Irregular bone formed by fusion of ilium, ischium, pubis; with (Coxal) (2) sacrum and coccyx forms pelvis; forms socket of ball-and- socket joint with femur. Site of attachment for trunk and lower limb muscles; transmits body weight to femur.

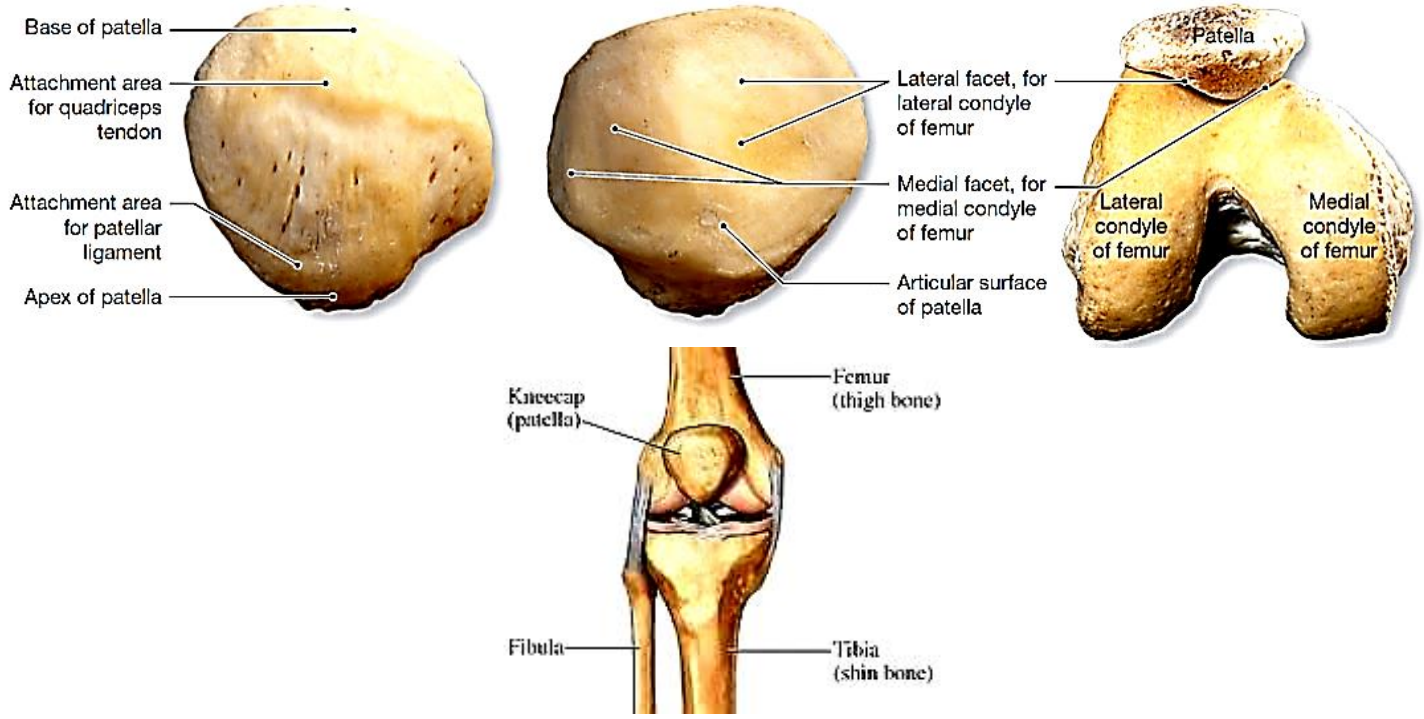


Thigh

- **Femur (2)** Thighbone; typical long bone; longest, strongest, heaviest bone; forms ball of ball-and-socket joint with pelvic bones; provides articular surface for knee. Supports body.

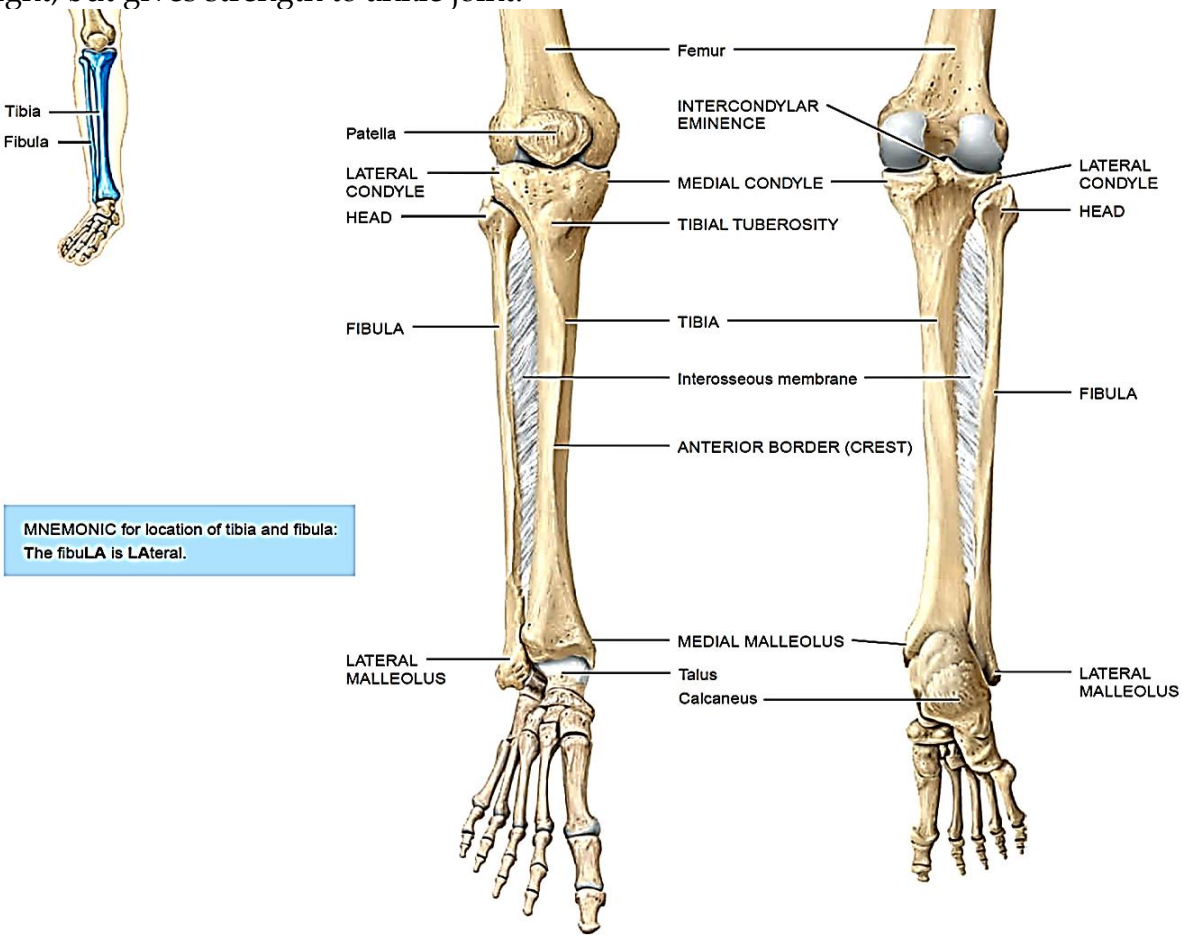


- **Patella (2) Kneecap**; sesamoid bone within quadriceps femuris tendon. Increases leverage for quadriceps muscle by keeping tendon Away from axis of rotation.



Leg

- **Fibula (2)** Smaller long bone of lower leg; articulates proximally with tibia and distally with talus. Bears little body weight, but gives strength to ankle joint.



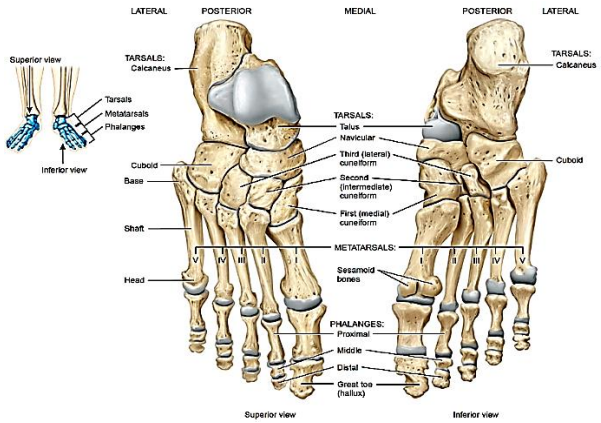
- **Tibia** (2) Larger long bone of lower leg; articulates with femur fibula, talus. Supports body weight, transmitting it from femur to talus.

Ankle

- **Tarsals (14)** Ankle, heel bones; short bones; 7 in each ankle including talus, calcaneus, cuboid, navicular, 3 cuneiforms; with metatarsals, form arches of foot. Bear body weight; raise body and transmit thrust during running and walking.

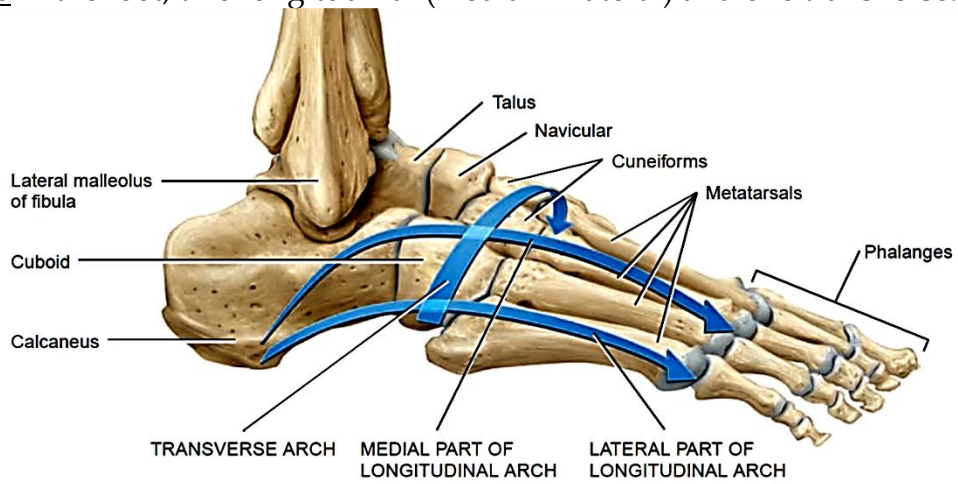
Foot and Toes

- **Metatarsals (10)** Miniature long bones; 5 in each foot; form sole; with tarsal, form arches of feet. Improve stability while standing; absorb shocks; bear weight; aid in locomotion.
- **Phalange (28)** Toes; miniature long bones; 2 in each big toe, 3 in each other toe; arranged as in hand. Provide stability during locomotion.



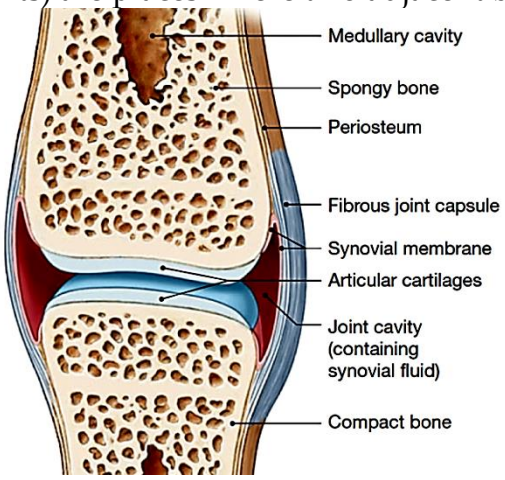
Arches of the foot

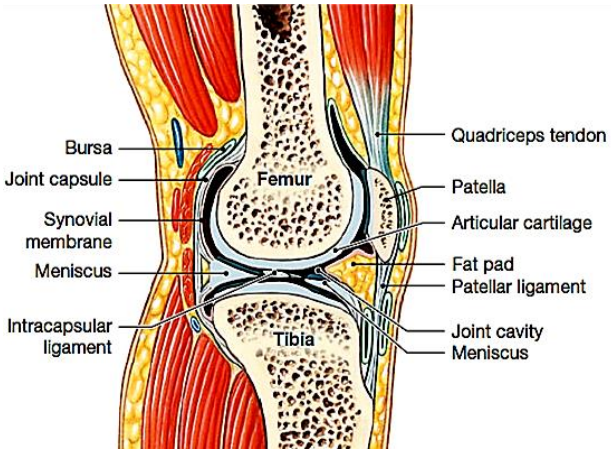
The sole of your foot is arched for the same reason that your spine is curved. Beside its function of absorbing shock it prevents nerves and blood vessels in the sole of the foot from being crushed.
There are three arches in the foot, two longitudinal (medial& lateral) and one transverse.



Articulations

Bones being structural framework, muscles give it power. But movable joints provide the mechanism that allows the body to move.
Articulations (joints) are places where two adjacent bones or cartilages meet.





Classifications Joints are classified by two methods

- By function-degree of movement
- By structure – presence of cavity.

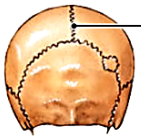
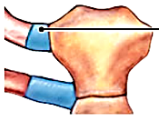
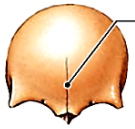
According to functional classification joints may be :

- immovable(synartherosis),
- slightly movable (amphiartherosis) and
- freely movable (diarthrosis).

According to structure joints can be classified into,

- fibrous,
- cartilaginous &
- synovial.

At a synarthrosis, the bony edges are quite close together and may even interlock. These extremely strong joints are located where movement between the bones must be prevented.

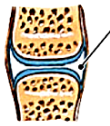
Fibrous	**Suture**		A **suture** (*sutura*, a sewing together) is a synarthrotic joint located only between the bones of the skull. The edges of the bones are interlocked and bound together at the suture by dense fibrous connective tissue.
Fibrous	**Gomphosis**		A **gomphosis** (gom-FŌ-sis; *gomphosis*, a bolting together) is a synarthrosis that binds the teeth to bony sockets in the maxillae and mandible. The fibrous connection between a tooth and its socket is a *periodontal* (per-ē-ō-DON-tal) *ligament* (*peri*, around + *odontos*, tooth).
Cartilaginous	**Synchondrosis**		A **synchondrosis** (sin-kon-DRŌ-sis; *syn*, together + *chondros*, cartilage) is a rigid, cartilaginous bridge between two articulating bones. The cartilaginous connection between the ends of the first pair of vertebrosternal ribs and the sternum is a synchondrosis. Another example is the epiphyseal cartilage, which connects the diaphysis to the epiphysis in a growing long bone.
Bony fusion	**Synotosis**		A **synostosis** (sin-os-TŌ-sis) is a totally rigid, immovable joint created when two bones fuse and the boundary between them disappears. The coronal suture of the frontal bone and the epiphyseal lines of mature long bones are synostoses.

AMPHIARTHROSIS
(little movement)

An amphiarthrosis permits more movement than a synarthrosis, but is much stronger than a freely movable joint. The articulating bones are connected by collagen fibers or cartilage.

Fibrous	**Syndesmosis**		At a **syndesmosis** (sin-dez-MŌ-sis; *syndesmos*, ligament), bones are connected by a ligament. One example is the distal articulation between the tibia and fibula.
Cartilaginous	**Symphysis**		At a **symphysis**, the articulating bones are separated by a wedge or pad of fibrocartilage. The articulation between the two pubic bones (the *pubic symphysis*) is an example of a symphysis.

DIARTHROSIS
(free movement)



Diarthroses, or **synovial** (si-NŌ-ve-ul) **joints**, permit a wider range of motion than do other types of joints. They are typically located at the ends of long bones, such as those of the upper and lower limbs.

Synovial	**Planes of Movement**
• **Monaxial**	Movement in one plane; elbow, ankle
• **Biaxial**	Movement in two planes; ribs and wrist
• **Triaxial**	Movement in three planes; shoulder, hip

CHAPTER SIX

THE MUSCULAR SYSTEM

THERE ARE OVER 600 SKELETAL MUSCLES IN THE BODY -150 IN THE HEAD AND NECK.

There are about 60 muscles in the face.

Smiling is easier than frowning.

It takes 20 muscles to smile and over 40 to frown.

Smile and make someone happy.

The muscular system

The term muscle tissue refers to all the contractile tissues of the body: **skeletal, cardiac, and smooth muscle.** Muscles are machines for converting stored chemical energy into mechanical energy (work) and heat . Muscles constitute 50% of the body weight (40% skeletal muscles and 10% smooth muscles and cardiac muscle).

There are three types of muscles ;

-Skeletal muscles ,

-Smooth muscles and

-Cardiac muscle.

The muscular system, however, refers to the skeletal muscle system: the skeletal muscle tissue and connective tissues that makeup individual muscle organs, such as the biceps brachii muscle.

► Cardiac muscle tissue is located in the heart and is therefore considered part of the cardiovascular system.

► Smooth muscle tissue of the intestines is part of the digestive system, whereas smooth muscle tissue of the urinary bladder is part of the urinary system and so on.

Functions of muscle tissue

Through sustained contraction or alternating contraction and relaxation, muscle tissue has three key functions: producing motion, providing stabilization, and generating heat.

1. Motion: Motion is obvious in movements such as walking and running, and in localized movements, such as grasping a pencil or nodding the head.

These movements rely on the integrated functioning of bones, joints, and skeletal muscles.

2. Stabilizing body positions and regulating the volume of cavities in the body: Besides producing movements, skeletal muscle contractions maintain the body in stable positions, such as standing or sitting. Postural muscles display sustained contractions when a person is awake, for example, partially contracted neck muscles hold the head upright. In addition, the volumes of the body cavities are regulated through the contractions of skeletal muscles.

For example, muscles of respiration regulate the volume of the thoracic cavity during the process of breathing.

3. Thermo genesis (generation of heat). As skeletal muscle contracts to perform work, a by-product is heat.

Much of the heat released by muscle is used to maintain normal body temperature. Muscle contractions are thought to generate as much as 85% of all body heat.

Physiologic Characteristics of muscle tissue:

Muscle tissue has four principal characteristics that enable it to carry out its functions and thus contribute to homeostasis.

1. **Excitability (irritability),** a property of both muscle and nerve cells (neurons), is the ability to respond to certain stimuli by producing electrical signal called action potentials (impulses). For example, the stimuli that trigger action potentials are chemicals-neurotransmitters, released by neurons, hormones distributed by the blood.

2. **Contractility** is the ability of muscle tissue to shorten and thicken (contract), thus generating force to do work. Muscles contract in response to one or more muscle action potentials.

3. **Extensibility** means that the muscle can be extended (stretched) without damaging the tissue. Most skeletal muscles are arranged in opposing pairs. While one is contracting, the other not only relaxed but also usually is being stretched.

4. **Elasticity** means that muscle tissue tends to return to its original shape after contraction or extension.

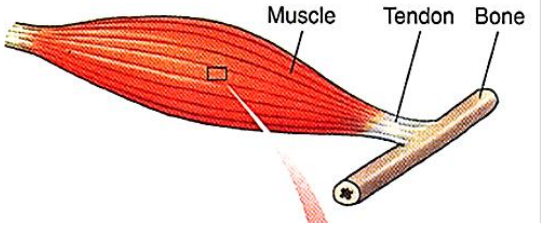
Muscle structure

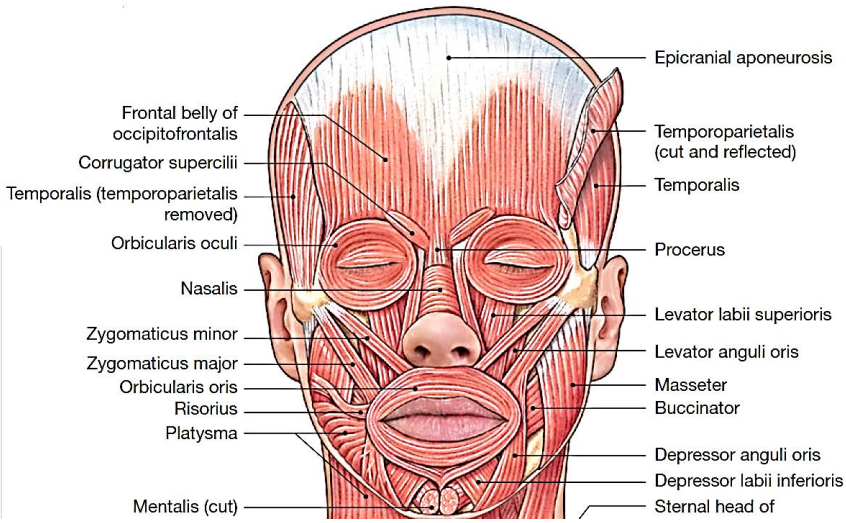
A. Connective Tissue Component

A skeletal muscle is an organ composed mainly of striated muscle cells and connective tissue. Each skeletal muscle has two parts; the connective tissue sheath that extend to form specialized structures that aid in attaching the muscle to bone and the fleshy part the belly or gaster.

The extended specialized structure may take the form of a cord, called a tendon; alternatively, a broad sheet called an aponeurosis may attach muscles to bones or to other muscles, as in the abdomen or across the top of the skull.

A connective tissue sheath called facia surrounds and separates muscles .





Connective tissue also extends into the muscle and divides it into numerous muscle bundles (fascicles).

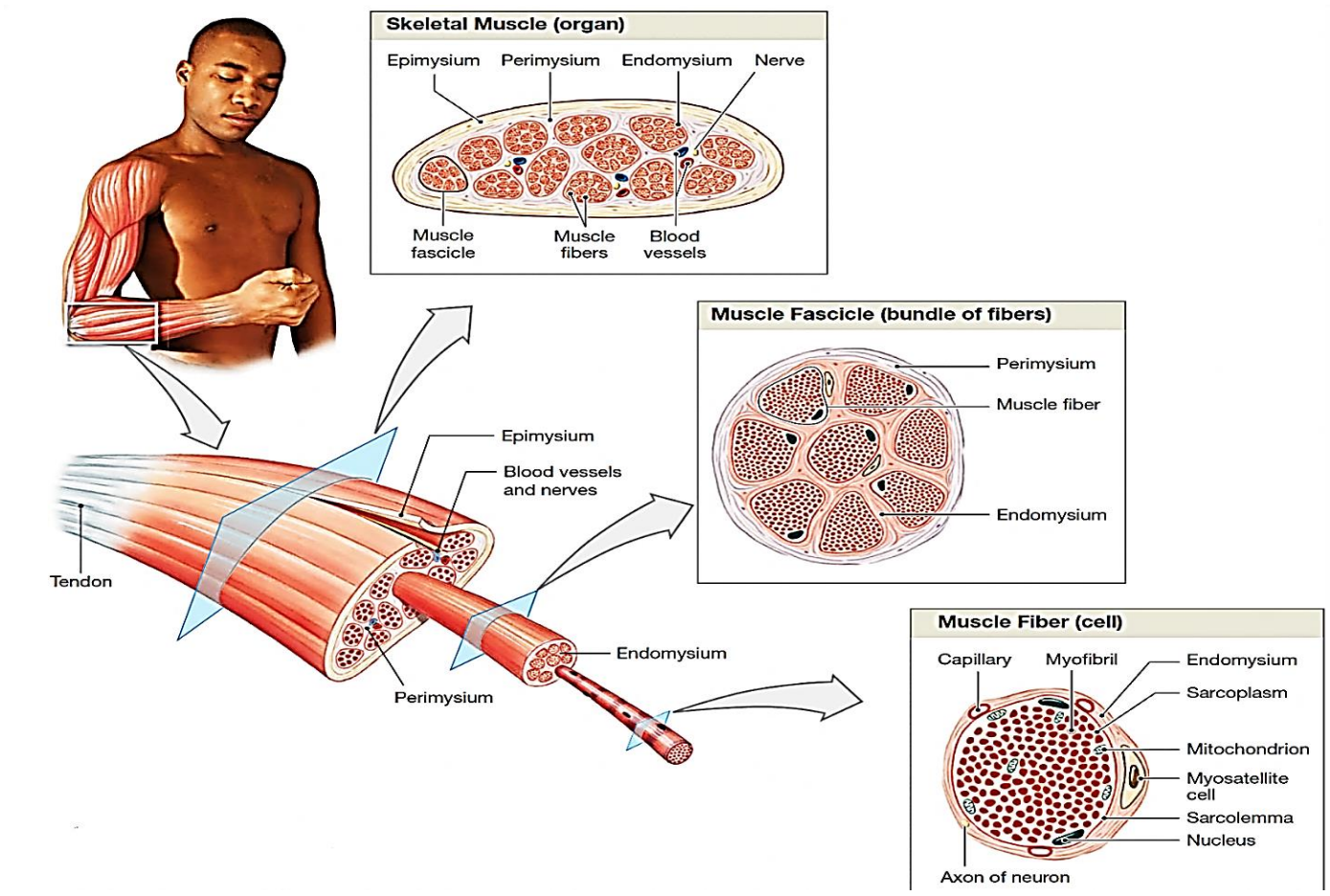
There are three connective tissue components that cover a skeletal muscle tissue.

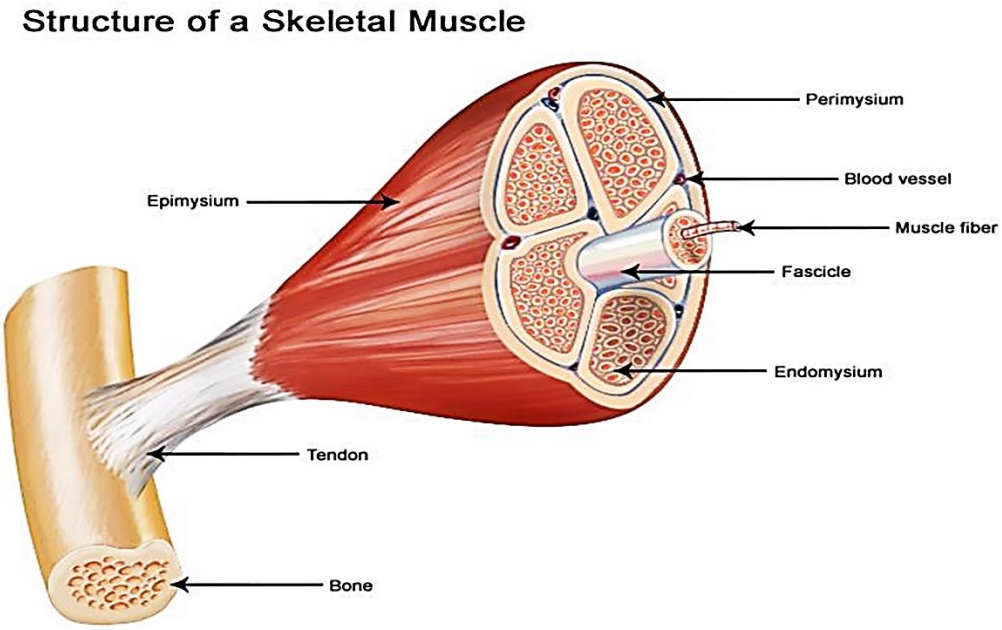
These are:

1.Epimysium— a connective tissue sheath that surrounds and separates muscle.

2.Perimysium— a connective tissue that surrounds and holds fascicles together.

3.Endomysium— a connective tissue that surrounds each muscle fiber.

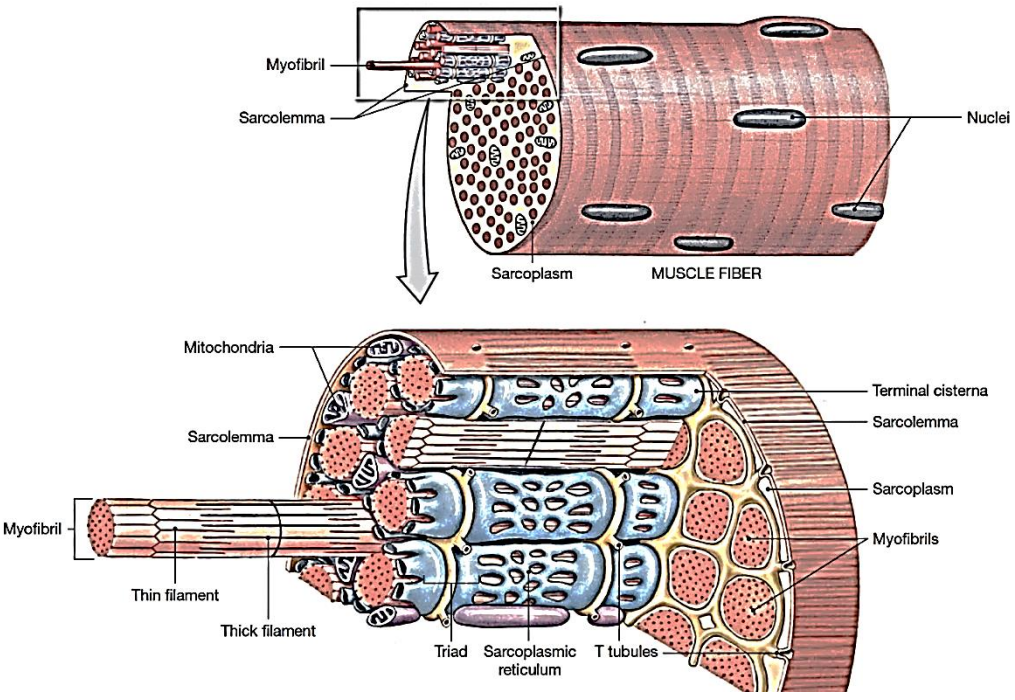




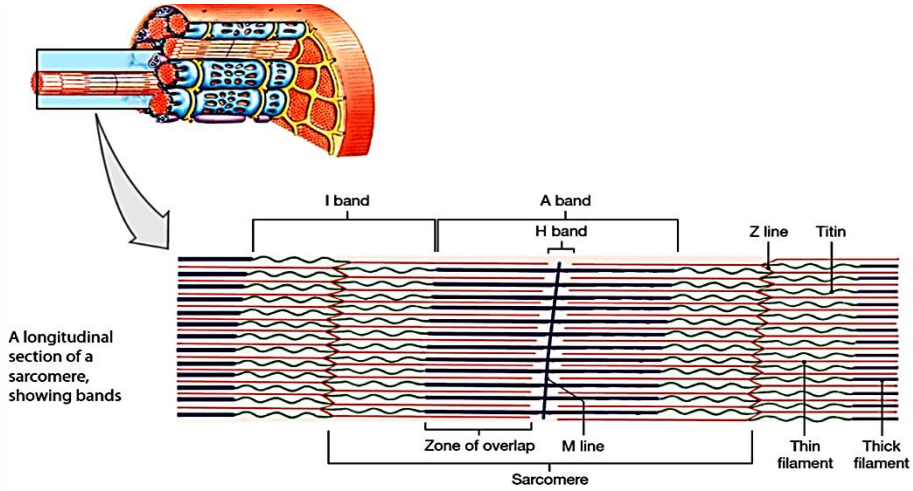
B. Microscopic structures

The muscle bundles are composed of many elongated muscle cells called muscle fibres. Each muscle fiber is a cylindrical cell containing several nuclei located immediately beneath the cell membrane (sarcolemma). The cytoplasm of each muscle fiber (sarcoplasm) is filled with myofibrils. Each myofibril is a thread-like structure that extends from one end of the muscle fiber to the other. Myofibrils consist of two major kinds of protein fibers: actins or thin myofilaments, and myosin or thick myofilaments.

The actins and myosin myofilaments form highly ordered units called sarcomers, which are joined end-to-end to form the myofibrils (see Figure).

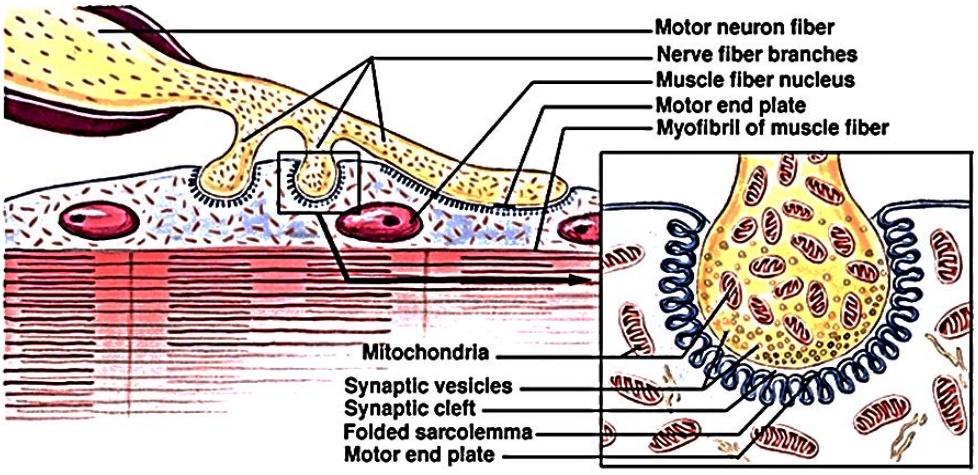


Sarcomere is a structural and functional unit of muscle tissue. The ends of a sarcomere are a network of protein fibers, which form the Z-lines when the sarcomere is viewed from side. The Z-lines form an attachment site for actins myofilaments. The arrangement of the actin and myosin myofilaments in a sarcomere gives the myofibril a banded appearance because the myofibril appears darker where the actin and myosin myofilaments overlap. The alternating light (I-band) and dark (A-band) areas of the sarcomers are responsible for striation (banding pattern) seen in skeletal muscle cells observed through the microscope.



The sarcolemma of a muscle fibre forms blind-ended sacs or T-tubules which penetrate the cell and lie in the spaces between the myofibrils. The T-tubules contain interstitial fluid and do not open into the interior of the muscle fibre. Within the sarcoplasm of the muscle fibre there is an extensive network of branching and anastomosing channels, which forms the sarcoplasmic reticulum (this structure is a modified endoplasmic reticulum). The channels of the sarcoplasmic reticulum lay in close contact around the ends of T-tubules, and contain stores of calcium.

Physiologic Anatomy of the Neuromuscular Junction – The Motor End Plate.



The nerve fiber forms a complex of branching nerve terminals that invaginate into the surface of the muscle fiber but lie outside the muscle fiber plasma membrane.

The entire structure is called the motor end plate.

In the axon terminal are many mitochondria that supply ATP, the energy source that is used for synthesis of an excitatory ***transmitter acetylcholine .***

The acetylcholine in turn excites the muscle fiber membrane. Acetylcholine is synthesized in the cytoplasm of the terminal, but it is absorbed rapidly into many small synaptic vesicles, about 300,000 of which are normally in the terminals of a single end plate.

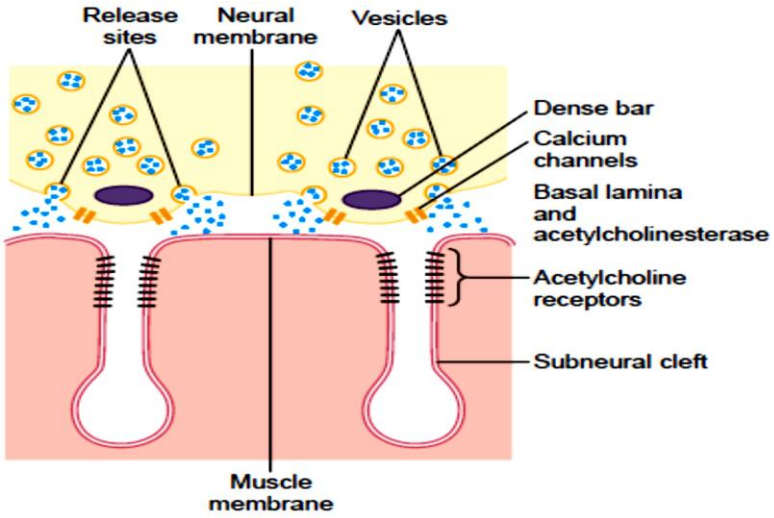
In the synaptic space are large quantities of the enzyme ***acetylcholinesterase,*** which destroys acetylcholine a few milliseconds after it has been released from the synaptic vesicles.

Secretion of Acetylcholine by the Nerve Terminals

When a nerve impulse reaches the neuromuscular junction, about 125 vesicles of acetylcholine are released from the terminals into the synaptic space.

On the inside surface of the neural membrane are linear dense bars ,

To each side of each dense bar are protein particles that penetrate the neural membrane; these are voltage gated calcium channels .



When an action potential spreads over the terminal, these channels open and allow calcium ions to diffuse from the synaptic space to the interior of the nerve terminal.

The calcium ions, in turn, are believed to exert an attractive influence on the acetylcholine vesicles, drawing them to the neural membrane adjacent to the dense bars .

The vesicles then fuse with the neural membrane and empty their acetylcholine into the synaptic space by the process of exocytosis.

☼ **Effect of Acetylcholine on the Postsynaptic Muscle Fiber Membrane to Open Ion Channels.**

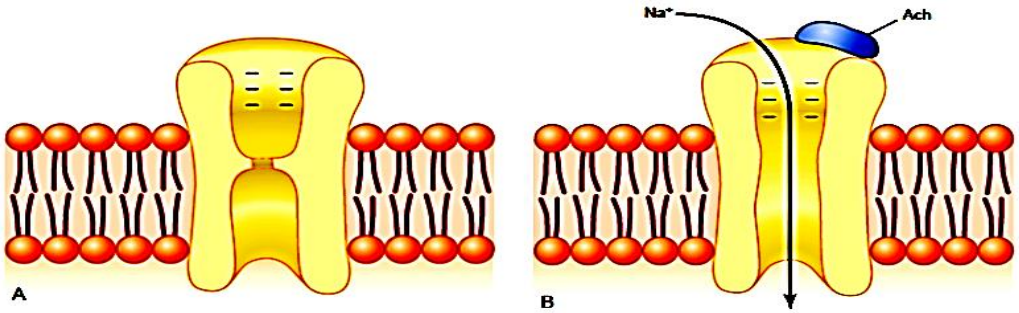
Many very small acetylcholine receptors in the muscle fiber membrane; these are acetylcholine-gated ion channels, and they are located almost entirely near the mouths of the subneural clefts lying immediately

below the dense bar areas, where the acetylcholine is emptied into the synaptic space. Each receptor is a protein complex.

The complex is composed of five subunit proteins, two alpha proteins and one each of beta, delta, and gamma proteins.

The channel remains constricted, as shown in section A of the figure, until two acetylcholine molecules attach respectively to the two alpha subunit proteins .

This causes a conformational change that opens the channel, as shown in section B of the figure.



The opened acetylcholine channel has a diameter of about 0.65 nanometer, which is large enough to allow the important positive ions—(Na+), (K+), and (Ca++)— to move easily through the opening.

Conversely, negative ions, such as chloride ions, do not pass through because of strong negative charges in the mouth of the channel that repel these negative ions.

The principal effect of opening the acetylcholine-gated channels is to allow large numbers of Na+ to pour to the inside of the fiber, carrying with them large numbers of positive charges .

This creates a local positive potential change inside the muscle fiber membrane, called the end plate potential.

Destruction of the Released Acetylcholine by Acetylcholinesterase

It is removed rapidly by two means:

(1) Most of the acetylcholine is destroyed by the enzyme acetylcholinesterase.

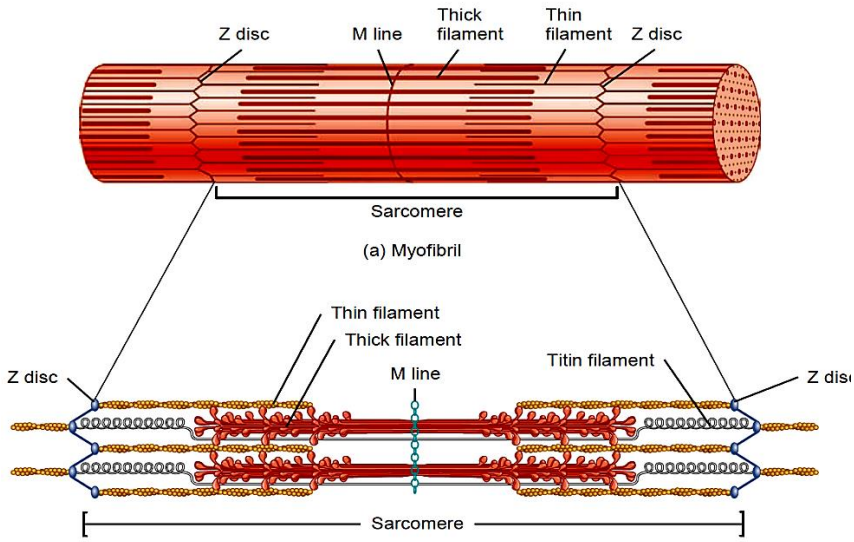
(2) A small amount of acetylcholine diffuses out of the synaptic space and is then no longer available to act on the muscle fiber membrane.

❖ **Muscle contractions**

The thick myofilaments are composed of a protein called myosin. Each myosin filament has small regular projections known as crossbridges. The crossbridges lie in a radial fashion around the long axis of the myofilament. The rounded heads of the crossbridges lie in apposition to the thin myofilaments.

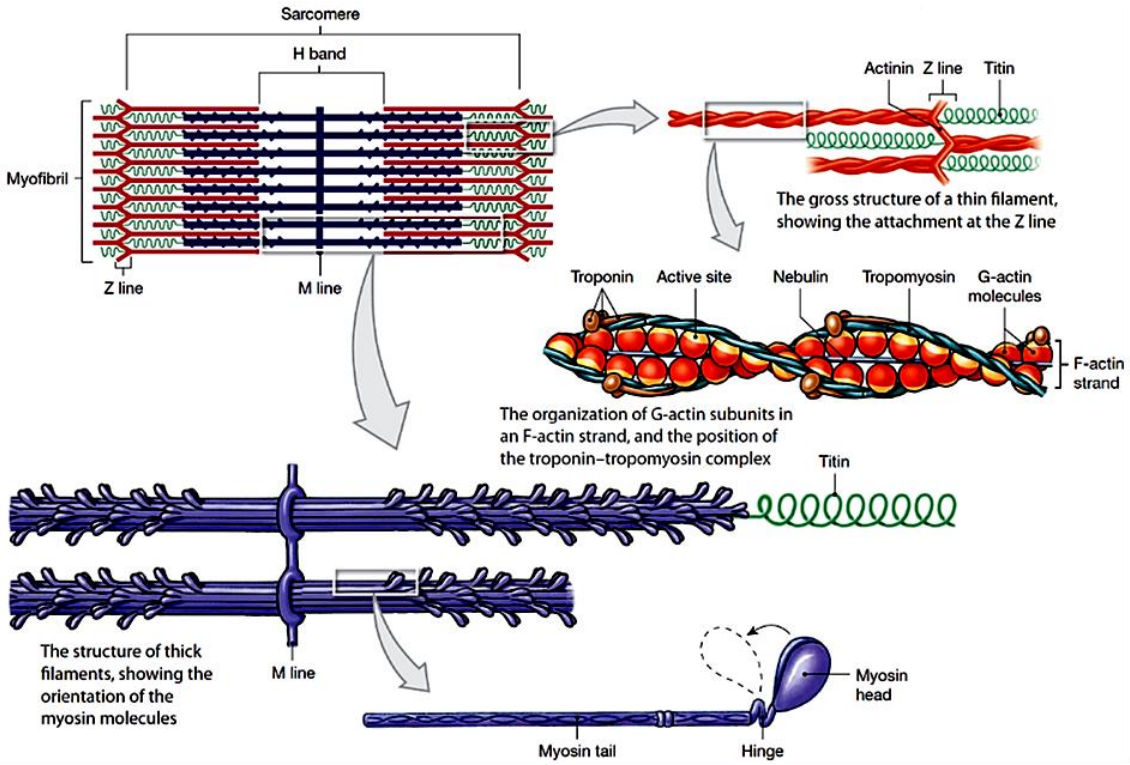
The thin myofilaments are composed of a complex protein called actin, arranged in a double stranded coil. The actin filaments also contain two additional proteins called troponin and tropomysin.

Sarcomere represent the Contractile unit of a muscle fiber



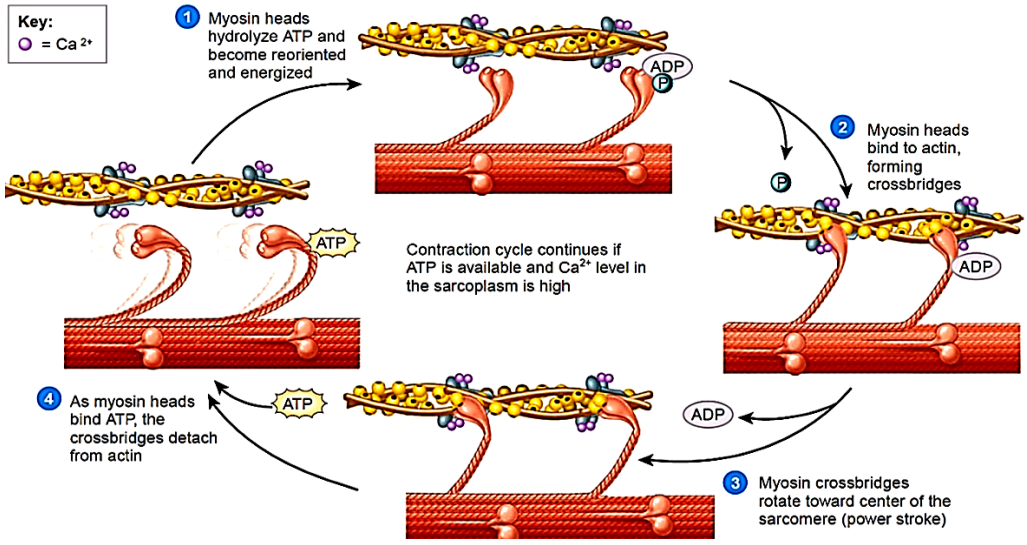
In a resting muscle fibre the myosin crossbridges are prevented from combining with the actin filaments by the presence of troponin and tropomysin.

When a nerve impulse reaches a muscle fibre it is conducted over the sarcolemma and in to the T-tubules, then to the sarcoplasmic reticulum.

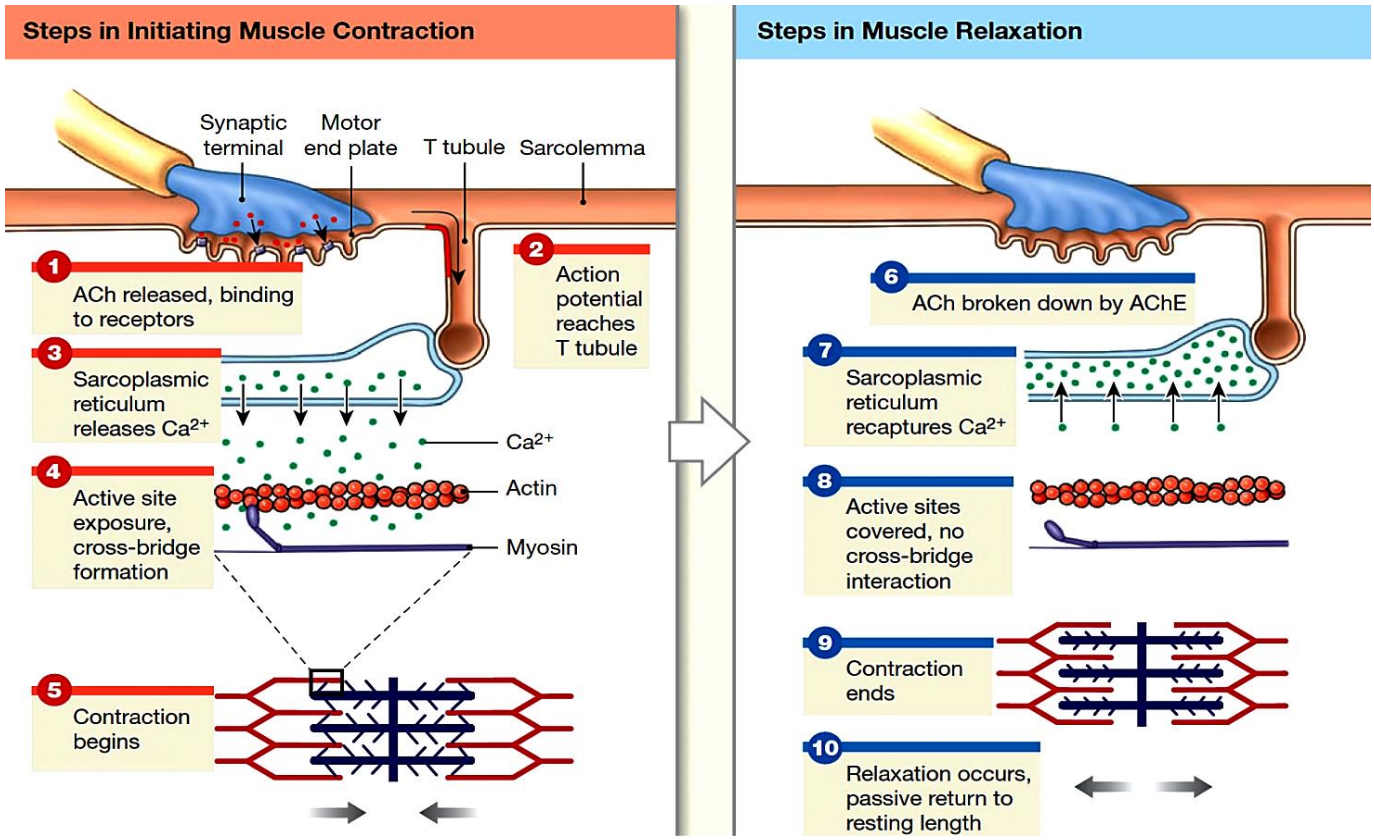


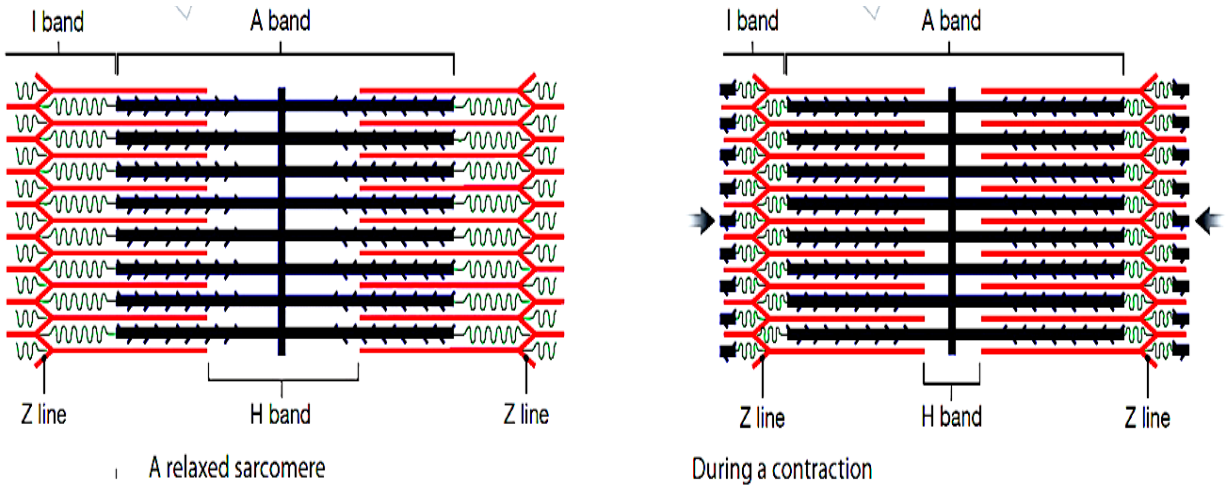
The sarcoplasmic reticulum releases calcium ions into the sacrcoplasm. The liberated calcium ions combine with troponin causing it to push tropomysin away from the receptor sites on the actins filaments. The myosin crossbridges interact with the actin receptor sites and pull the actins myofilaments toward the

center (H-zone) of each sarcomere. The bond between the myosin crossbridges and actin breaks down under the influence of enzymes and the crossbridges are then free to rejoin with other actin receptor sites. The actin filaments do not shorten but slide past the myosin filaments overlapping them so that the Z lines are drawn toward each other, shortening the sarcomere. As each sarcomere shortens the whole muscle fibre contracts.



❖ **<u>Relaxation of the muscle fibers</u>** occurs when the calcium ions are actively reabsorbed by the sarcoplasmic reticulum thus allowing troponin and tropomysin to again inhibit the interaction of the actins and myosin filaments.





❖ **Energy Requirements for Muscle Contraction**

Contraction o skeletal muscle requires adenosine triphosphate (ATP). The ATP releases energy when it breaks down to adenosine diphosphate (ADP) and a phosphate (P), some of the energy is used to move the crossbridges and some of the energy is released as heat.

ATP→ADP + P + Energy (for crossbridge movement) + Heat

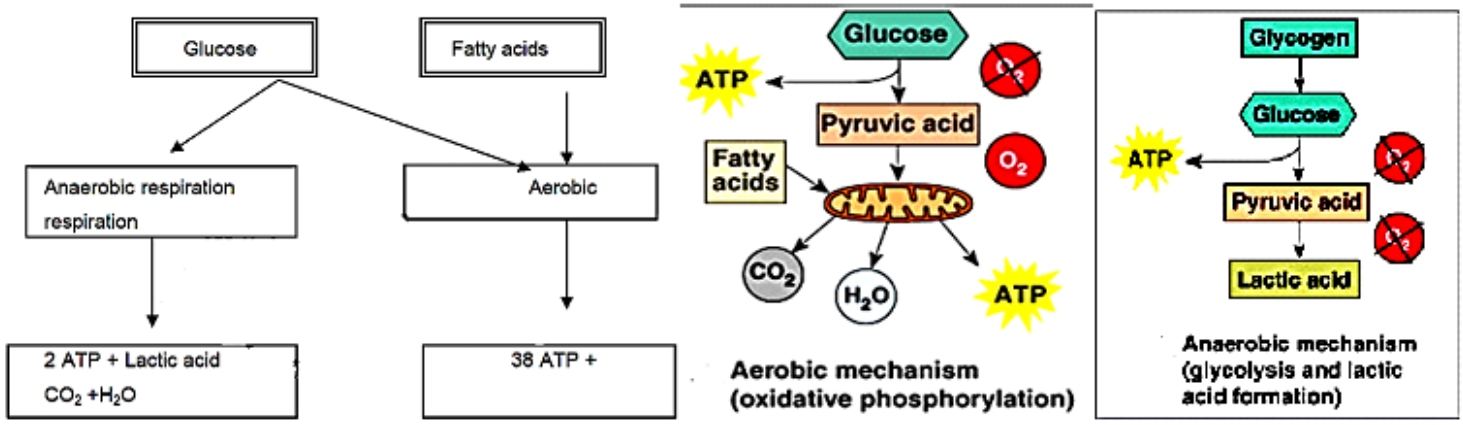
The ATP required for muscle contraction is produced primarily in numerous mitochondria located within the muscle fibers.

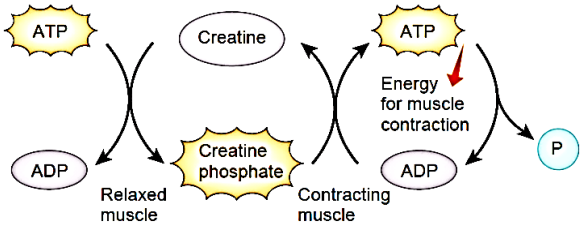
Because ATP is a very short-lived molecule and rapidly degenerates to the more stable ADP, it is necessary for muscle cells to constantly produce ATP.

ATP is produced by anaerobic or aerobic respiration.

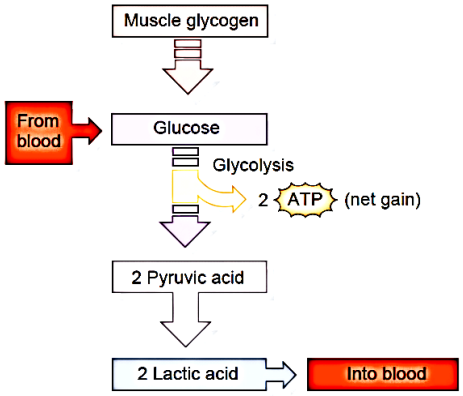
Anaerobic respiration, which occurs in the absence of oxygen, results in the breakdown of glucose to yield ATP and lactic acid.

Aerobic respiration requires oxygen and breaks down glucose to produce ATP, carbon dioxide, and water. Compared with anaerobic respiration, aerobic respiration is much more efficient. The breakdown of glucose molecule by aerobic respiration theoretically can produce 19 times as much ATP as is produced by anaerobic respiration.

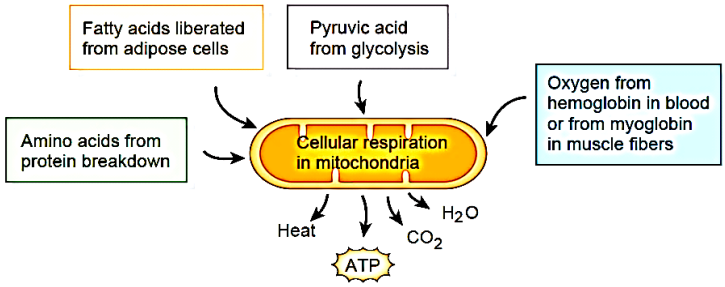




ATP from creatine phosphate



ATP from anaerobic glycolysis



In addition, aerobic respiration can utilize a greater variety of nutrient molecules to produce ATP than can anaerobic respiration. For example, aerobic respiration can use fatty acids to generate ATP. Although anaerobic respiration is less efficient than aerobic respiration, it can produce ATP when lack of oxygen limits aerobic respiration. By utilizing many glucose molecules, anaerobic respiration can rapidly produce much ATP, but only for a short period.

Resting muscles or muscles undergoing long-term exercise such as a long-distance bicycling on level ground depend primarily on aerobic respiration for ATP synthesis. Although some glucose is used as an energy source, fatty acids are a more important energy source during sustained exercise as well as during resting conditions. On the other hand, during intense exercise such as riding a bicycle up a steep hill, anaerobic respiration provides enough ATP to support intense muscle

Contractions for approximately 1 to 2 minutes. Anaerobic respiration is ultimately limited by depletion of glucose and a build up of lactic acid within the muscle fiber. Lactic acid can also irritate muscle fibers, causing short-term muscle pain.

Muscle pain that lasts for a couple of days following exercise, however, results from damage to connective tissue and muscle fibers within the muscle.

❖ **Muscle fatigue** results when ATP is used during muscle contraction faster than it can be produced in the muscle cells, and lactic acid builds up faster than it can be removed. As a consequence, ATP levels are too low to sustain crossbridge movement and the contractions become weaker and weaker.

For most of us, however, complete muscle fatigue is rarely the reason we stop exercising. Instead, we stop because of psychological fatigue, the feeling that the muscles have tired.

A burst of activity in a tired athlete as a result of encouragement from spectators is an example of how psychological fatigue can be overcome.

After intense exercise, the respiration rate remains elevated for a period. Even though oxygen is not needed for anaerobic production of ATP molecules for contraction, oxygen is needed to convert the lactic acid produced by anaerobic respiration back to glucose.

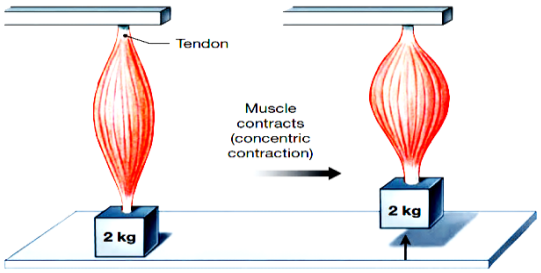
The increased amount of oxygen needed in chemical reactions to convert lactic acid to glucose is the oxygen debt. After the oxygen debt is paid, respiration rate returns to normal.

Types of muscle contraction

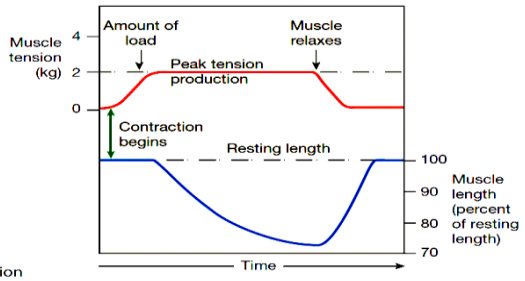
Muscle contractions are classified as either isotonic or isometric.

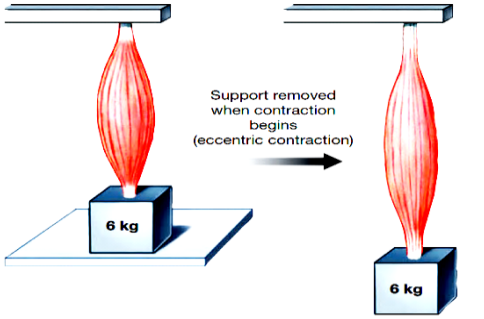
In isotonic contractions, the amount of tension produced by the muscle is constant during contraction, but the length of the muscle changes; for example, movement of the fingers to make fist. In isometric contractions, the length of the muscle does not change, but the amount of tension increases during the contraction process. Clenching the fist harder and harder is an example. Most movements are a combination of isometric and isotonic contractions. For example, when shaking hands, the muscles shorten some distance (isotonic contractions) and the degree of tension increases (isometric contractions).

Isometric contractions are also responsible for muscle tone, the constant tension produced by muscles of the body for long periods. Muscle tone is responsible for posture; for example, keeping the back and legs straight, the head held in upright position, and the abdomen from bulging.



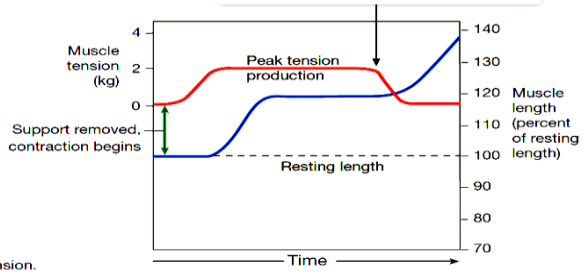
In this experiment, a muscle is attached to a weight less than its peak tension capabilities. On stimulation, it develops enough tension to lift the weight. Tension remains constant for the duration of the contraction, although the length of the muscle changes. This is an example of isotonic contraction.

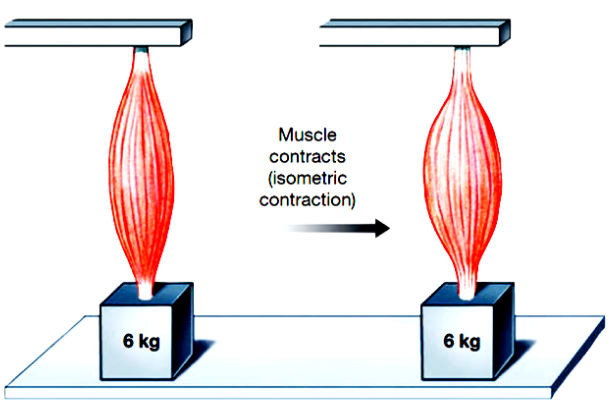




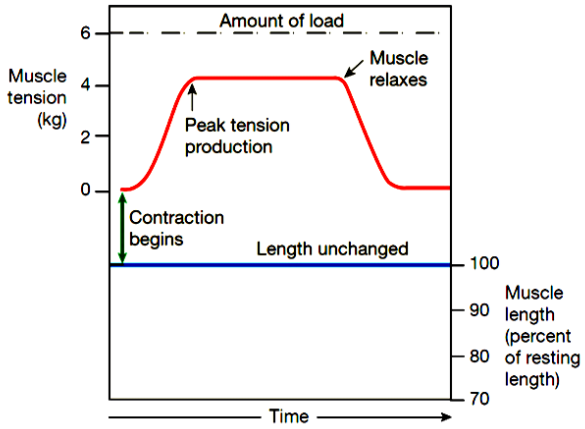
In this eccentric contraction, the muscle elongates as it generates tension.

When the eccentric contraction ends, the unopposed load stretches the muscle until either the muscle tears, a tendon breaks, or the elastic recoil of the skeletal muscle is sufficient to oppose the load.





The same muscle is attached to a weight that exceeds its peak tension capabilities. On stimulation, tension will rise to a peak, but the muscle as a whole cannot shorten. This is an isometric contraction.



Muscle attachments

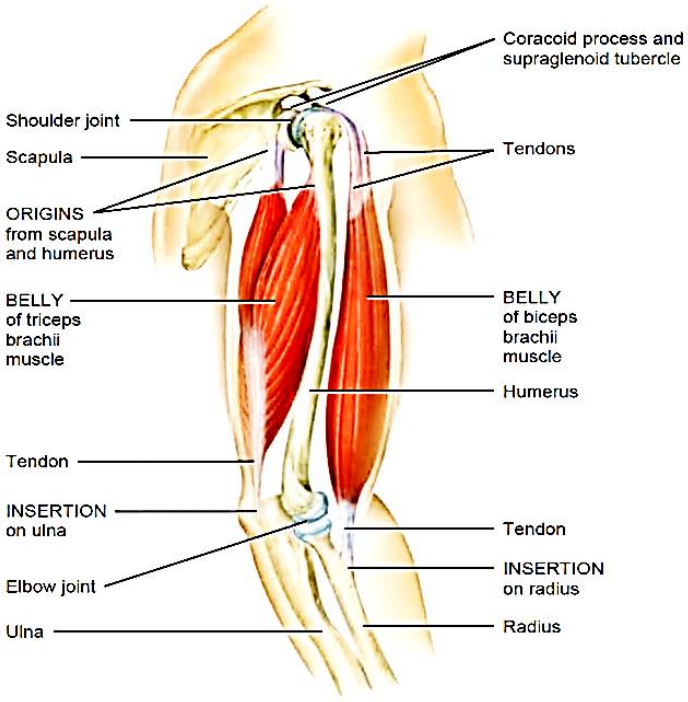
Most muscles extend from one bone to another and cross at least one movable joint.

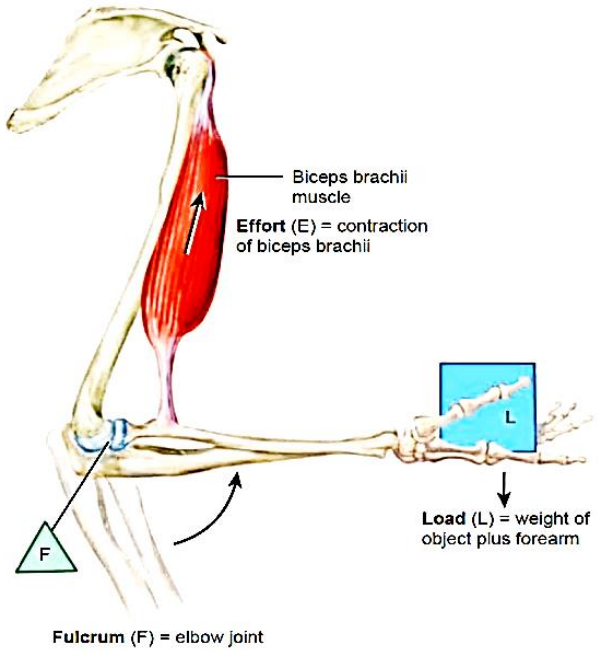
Muscle contraction causes most body movements by pulling one of the bones towards the other across the movable joint. Some muscles are not attached to bone at both ends.

For example, some facial muscles attach to the skin, which moves as the muscles contract.

The points of attachment of each muscle are its origin and insertion (Figure). At these attachments points, the muscle is connected to the bone by a tendon. -

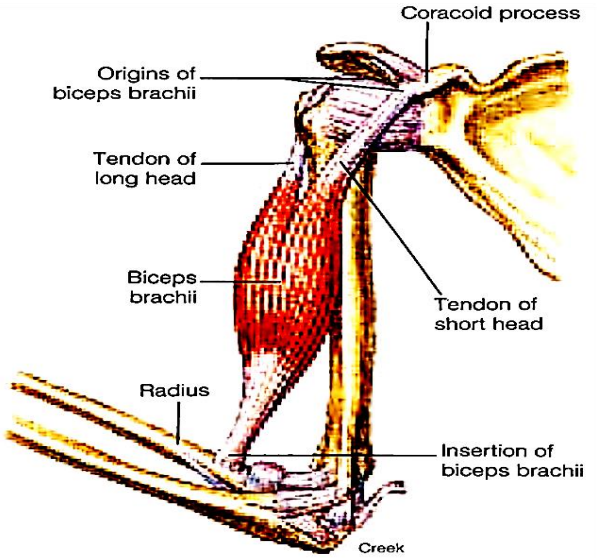
The origin is the most stationary end of the muscle and the insertion is the end of the muscle attached to the bone undergoing the greatest movement.





- Some muscles have more than one origin, but the principle is the same the origin act to anchor or hold the muscle so that the force of contraction causes the insertion to move.

- For example, the biceps brachii causes the radius to move, resulting in flexion of the forearm. The triceps brachii muscle has three origins; two on the humerus and one on the scapula. The insertion of the triceps brachii is on the ulna and contraction results in extension of the forearm.



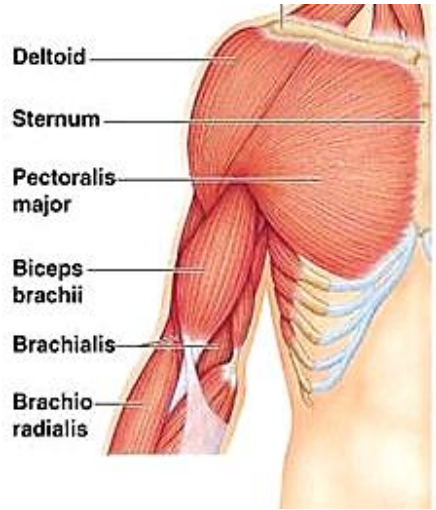
Several muscles contract while others relax to produce almost any movement you can imagine. Of all the muscles contracting simultaneously, the one mainly responsible for producing a particular movement is called the prime mover for that movement. The other muscles that help in producing the movement are called synergists. As prime movers and synergist muscles at a joint contract, other muscles called antagonists, relax. When those antagonist muscles contract, they produce a movement opposite to that of those prime movers and their synergist muscles.

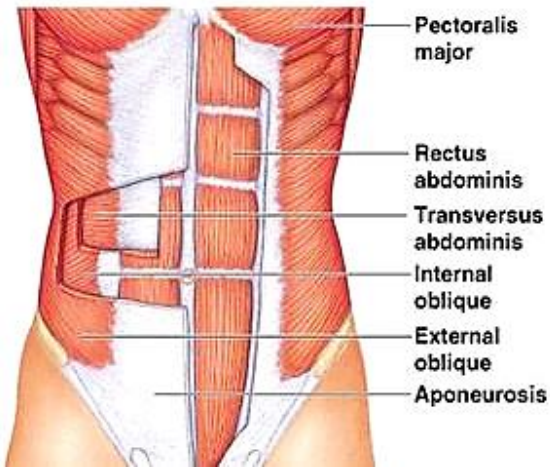
Naming skeletal muscles

Most of the skeletal muscles are named according to one or more of the following basis:

1. Direction of muscle fibers relative to the midline of the body or longitudinal axis of a structure

Rectus means the fibers run parallel to the midline of the body or longitudinal axis of a structure. Example, rectus abdominis





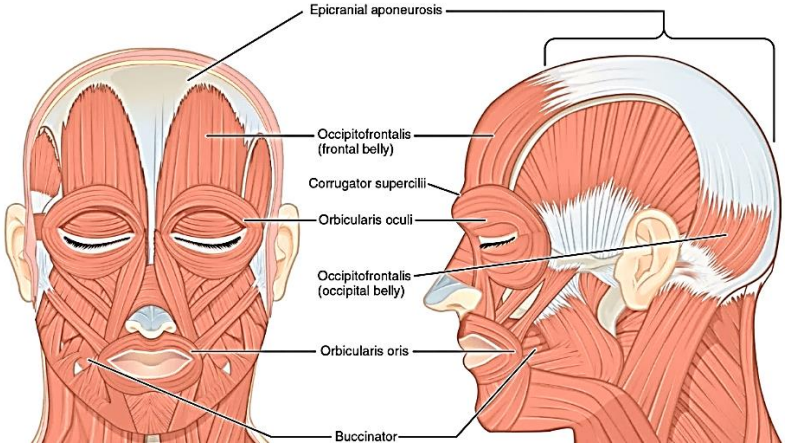
Transverse means the fibres run perpendicular to the midline longitudinal axis of a structure. Example, transverse abdominis

Oblique means the fibres run diagonally to the midline longitudinal axis of a structure.

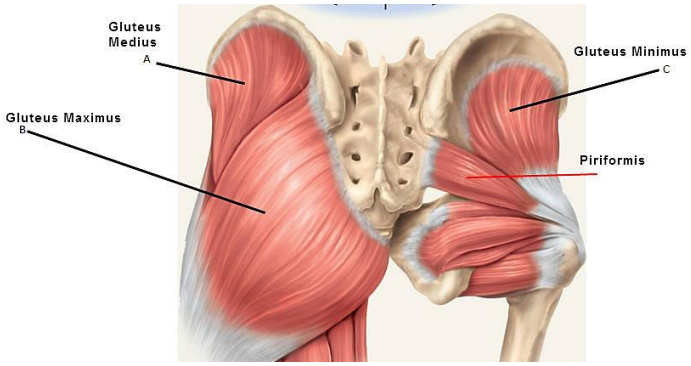
Example, external oblique

2. Location−structure to which a muscle is found closely related

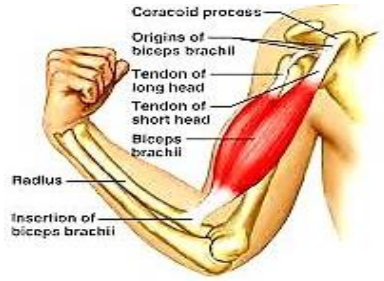
Example: Frontal, a muscle near the frontal bone Tibialis anterior, a muscle near the front of tibia



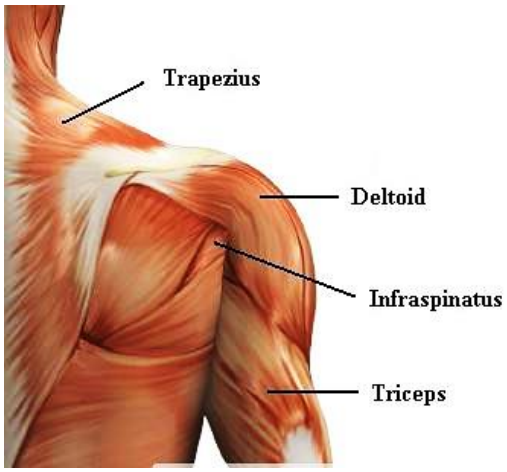
3. Size−relative size of the muscle Maximus means largest. Example, gluteus maximus Minimus means smallest. Example, gluteus minimus Longus means longest. Example, Adductor longus Brevis means short. Example, Peroneous brevis



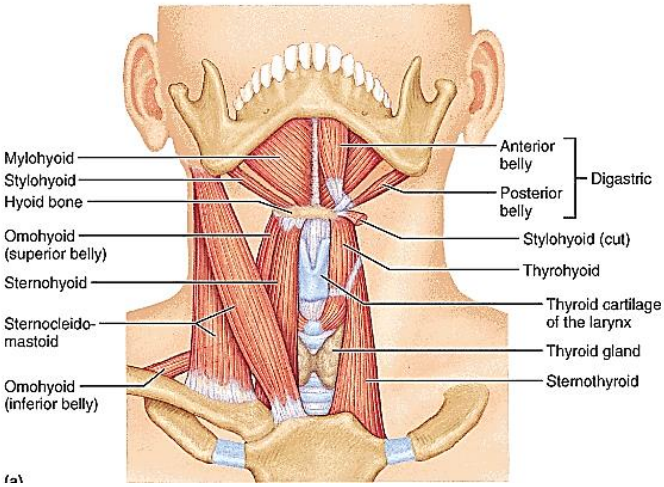
4. Number of origins−number of tendons of origin Biceps means two origins. Example, biceps brachii Triceps means three origins. Example, triceps brachii Quadriceps means four origins. Example, quadriceps femoris



5. Shape −relative shape of the muscle Deltoid means triangular. Example, deltoid Trapezius means trapezoid. Example, trapezius Serratus means saw-toothed. Example, serratus anterior Rhomboideus means rhomboid or diamond shape. Example, Rhomboideus major



6. Origin and insertion−sites where muscles originates and inserts Example, sternocleidomastoid−originates on sternum and clavicle and inserts on mastoid process of temporal bone.



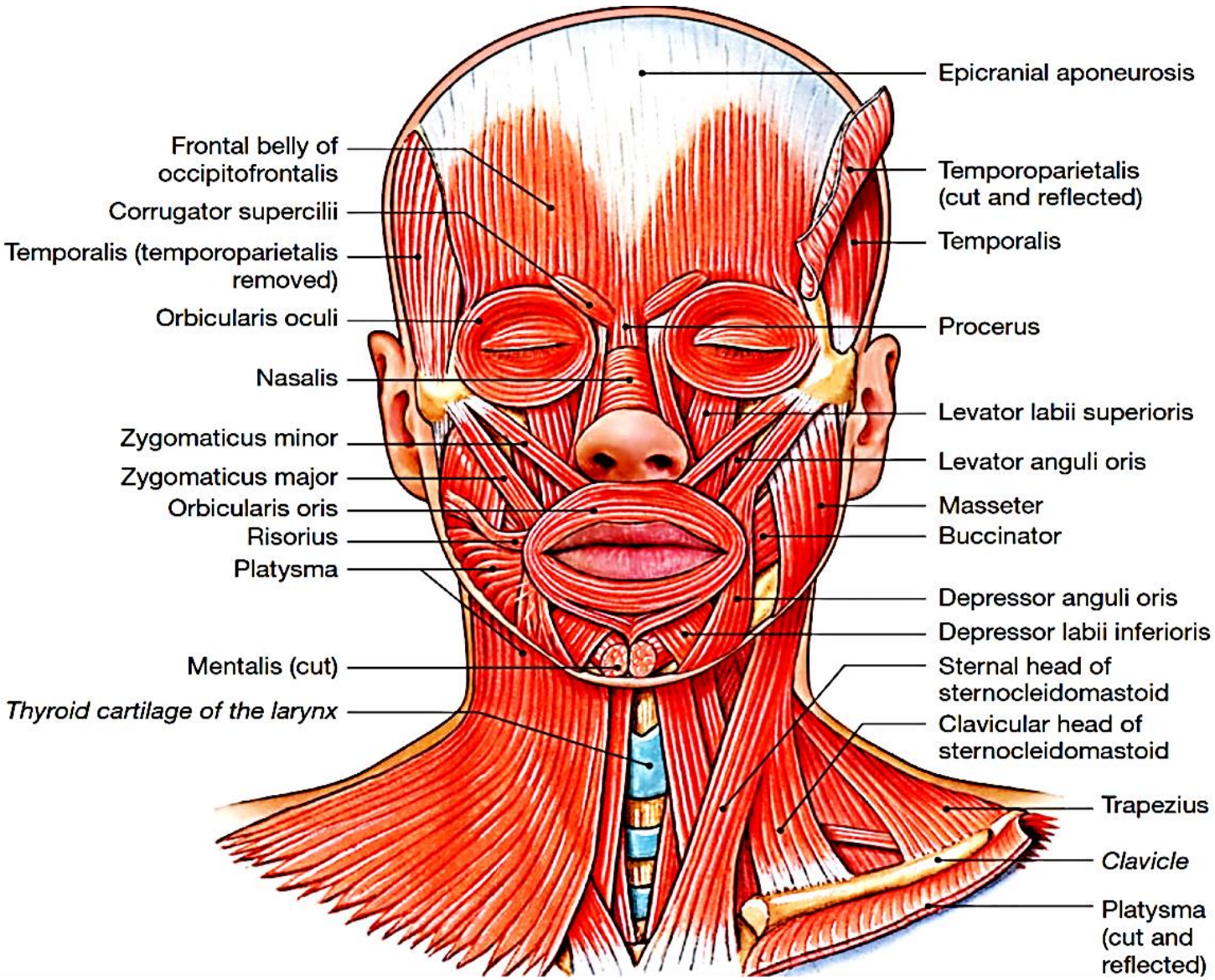
7. Action−principal action of the muscle

- **Flexor:** decrease the angle at a joint. Example, flexor carpiradialis
- **Extensor**: increases the angle at a joint. Example, extensor carpiulnaris
- **Abductor**: moves a bone away from the midline. Example, abductor policis brevis
- **Adductor:** moves a bone closer to the midline. Example, adductor longus
- **Levator:** produces an upward movement. Example, levator labii superioris
- **Depressor**: produces a downward movement. Example, depressor labii inferioris
- **Supinator:** turns the palm upward or anteriorly. Example, supinator
- **Pronator:** turns the palm downward or posteriorly. Example, pronator teres
- **Sphincter**: control the size of an opening. Example, external anal sphincter
- **Tensor:** makes a body part more rigid. Example, tensor fasciae latae

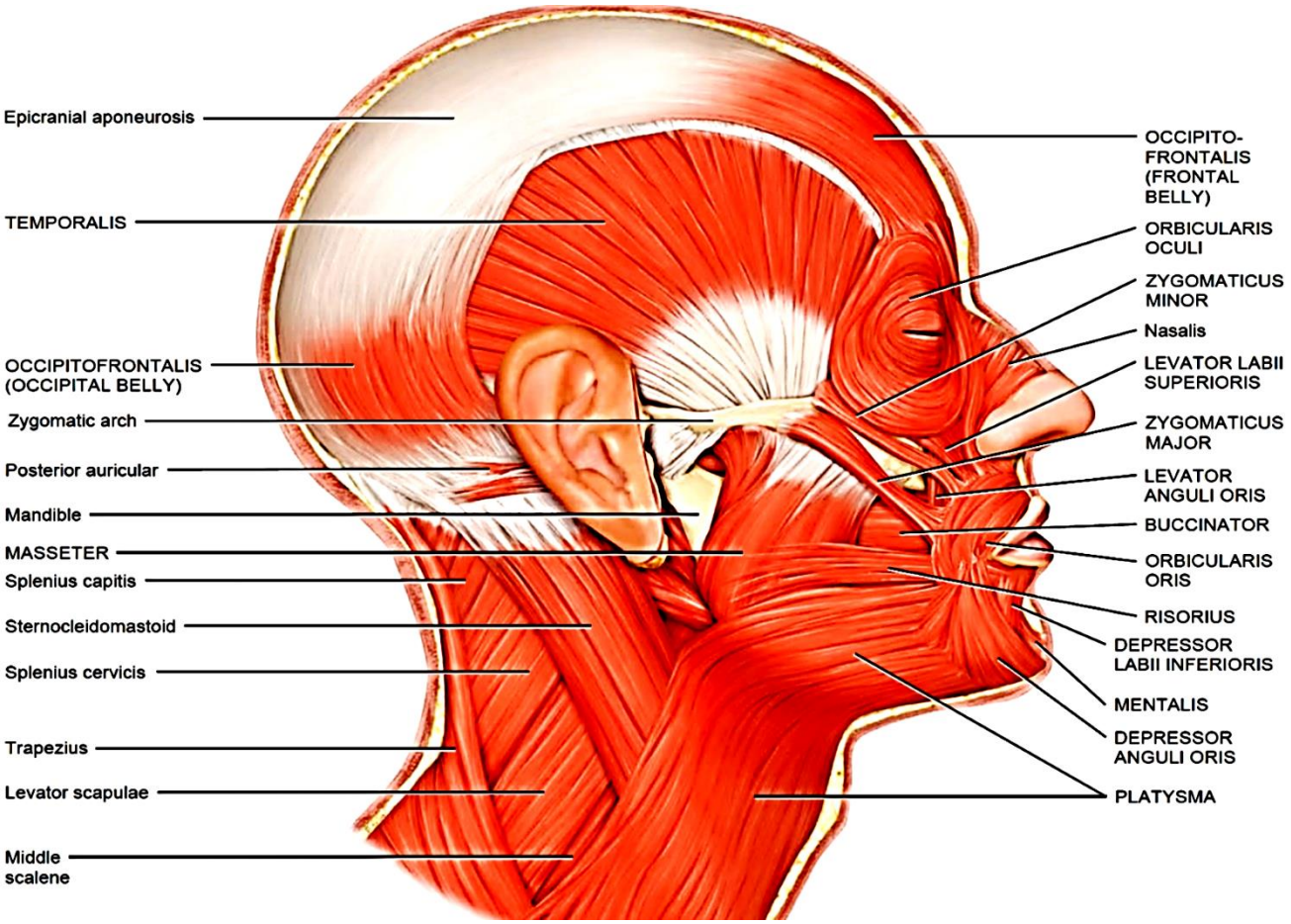
• **Rotator**: moves a bone around its longitudinal axis. Example, obturator externus

Principal skeletal muscles Although there are over 700 individual skeletal muscles in the human body, an appreciation and understanding of skeletal muscles can be accomplished by concentrating on the large superficial muscles and muscle groups.

Head and neck muscles

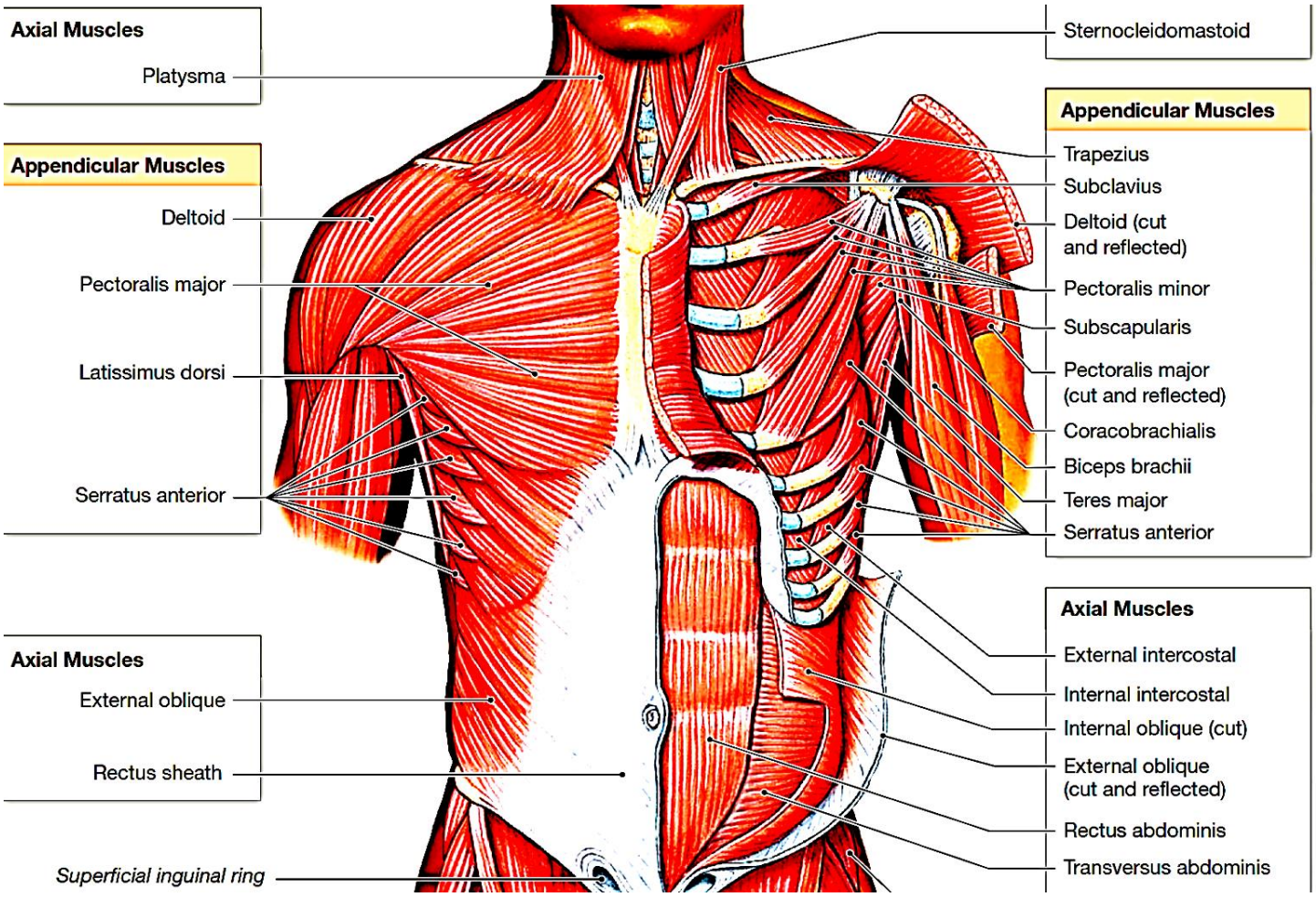


Muscle	Origin	Insertion	Action
Muscles of facial expression			
Occipitofrontalis	Occipital bone	Skin of eye brow	Elevates eye brows
orbicularis oculi	Maxilla & frontal	Skin around the eye	Closes eye
Orbicularis oris	Maxilla & mandible	Skin around the lips	Closes lip
Buccinator	Mandible & maxilla	Corner of mouth	Flattens cheeks
Zygomaticus muscles	Zygomatic bone	Corner of mouth	Elevates corner of mouth
Levator labii superioris	Maxilla	Upper lip	Elevates upper lip
Corrugator supercilli	Frontal bone	Skin of eye brow	Lowers and draws together eye brows
Depressor anguli oris	Mandible	Lower lip near corner of mouth	Depresses corner of the mouth
Muscles of mastication Temporalis	Temporal region on side of the skull	Mandible	Closes jaw
Massetor	Zygomatich arch	Mandible	Closes jaw
Muscles that move the head rapezius	Occipital bone and vertebrae	Scapula and Clavicle	Extends head and neck
Sternocleidomastoid	Sternum & clavicle	Mastoid process of temporal bone	Rotates head and flexes neck



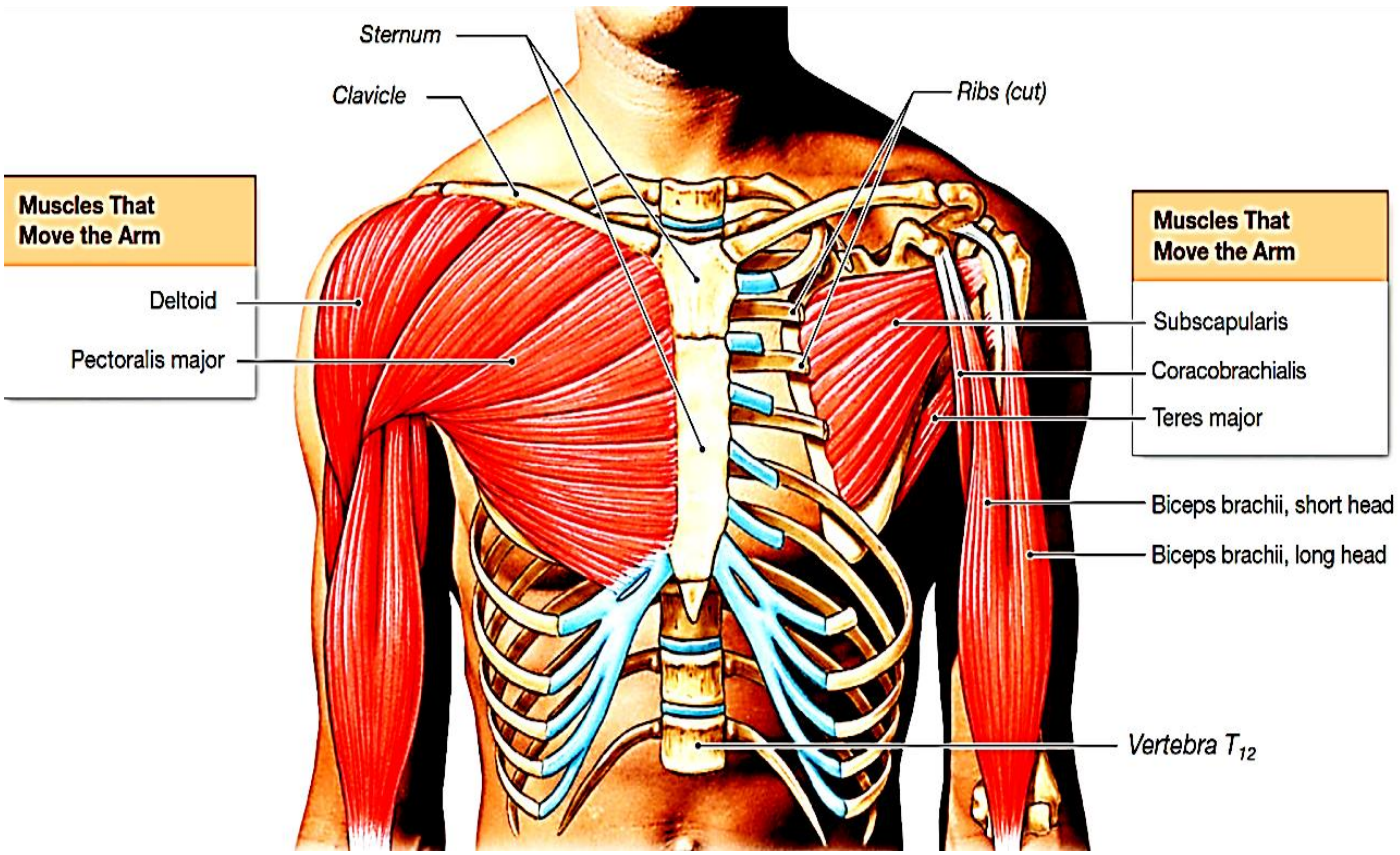
Trunk muscles

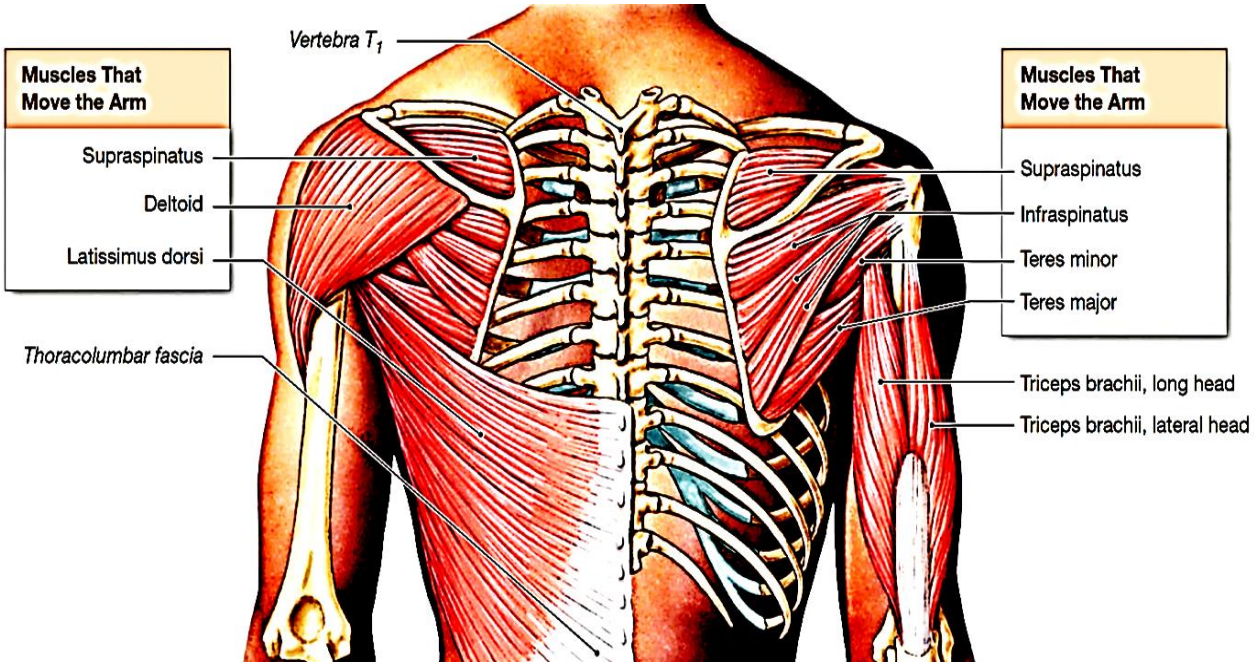
Muscle	Origin	Insertion	Action
Muscles that move the vertebral column			
Erector spinae	Ilium, sacrum, vertebrae	Superior vertebrae Ribs	Extend, abduct, and rotate vertebrae
Deep back muscles	Vertebrae	Vertebrae	Extend. abduct, and rotate vertebrae
Rectus abdominis	Pubis	Xiphoid process of ste- rnum & lower ribs	Flexes vertebrae; compress abdomen
External abdominal oblique	Rib cage	Iliac crest & facia of rectus abdominis	Flexes & rotates vertebral column; compress abdomen
Internal abdominal oblique	Iliac crest and vertebrae	Lower ribs and facia of rectus abdominis	Flexes & rotates vertebral column; compress abdomen
Transversus abdominis	Ribcage, vertebrae and	Xiphoid process of sternum, facia of	Compress abdomen



Upper limb muscles

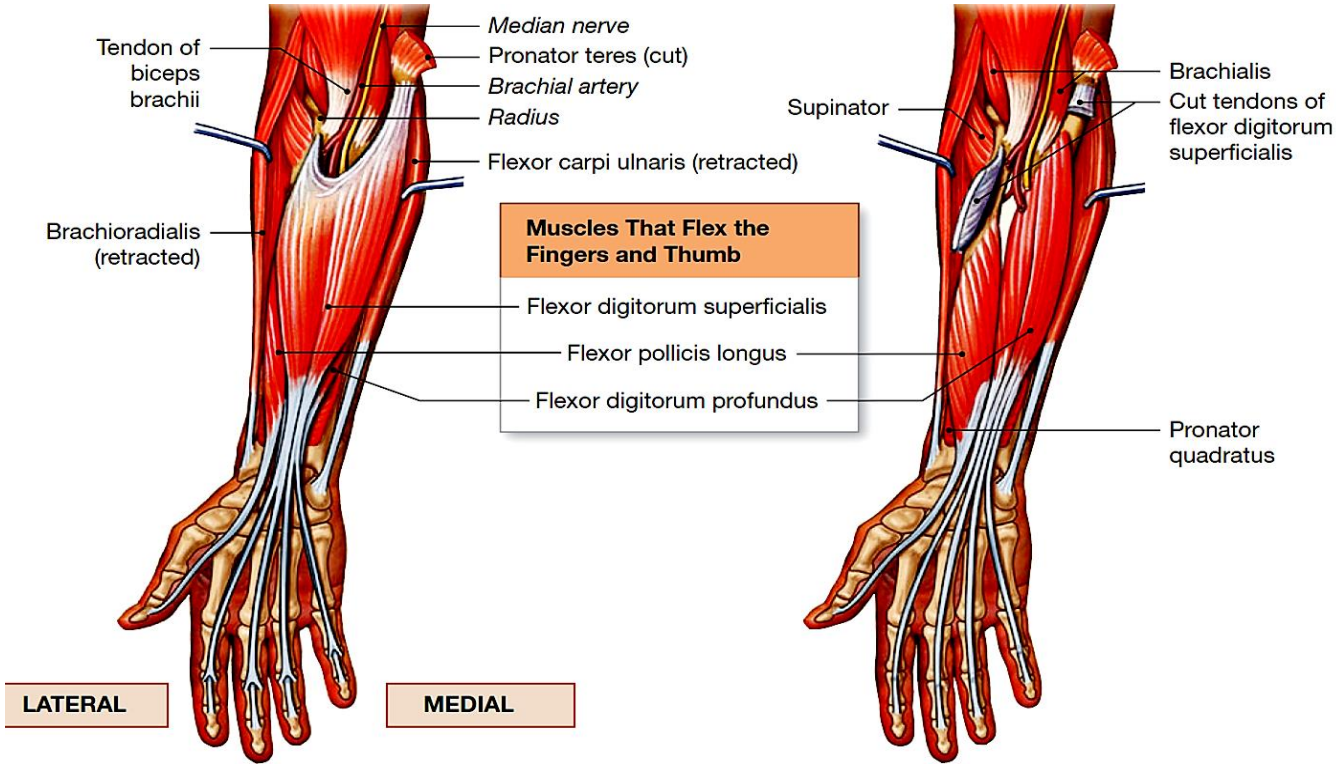
Muscle	Origin	Insertion	Action
Muscles that move the scapula			
Trapezius	Occipital bone and vertebrae	Scapula clavicle	holds scapula in place rotates scapula
Serratus anterior	Ribs	Medial border of the scapula	Rotates scapula and pulls anteriorly
Muscles that move the arm			
Pectoralis major	Sternum, ribs, and clavicle	Tubercle of humerus	Adducts and flexes arm
Lattismus dorsi	Vertebrae	Tubercle of humerus	Adducts and extends arm
Deltoid	Scapula and clavicle	Shaft of humerus	Abducts, flexes, and extends arm
Teres major	Scapula	Tubercle of humerus	Adducts and extends arm
Infraspinalis	Scapula	Tubercle of humerus	Extends arm
Muscles that move the forearm			
Brachilis	Shaft of humerus	Coracoids process of ulna	Flexes and forearm
Biceps brachii	Coracoids process of Supinates scapula	Radial tuberosity	
Triceps brachii	Shaft of humerus and lateral border of scapula	Olecranon process of ulna	Extends forearm

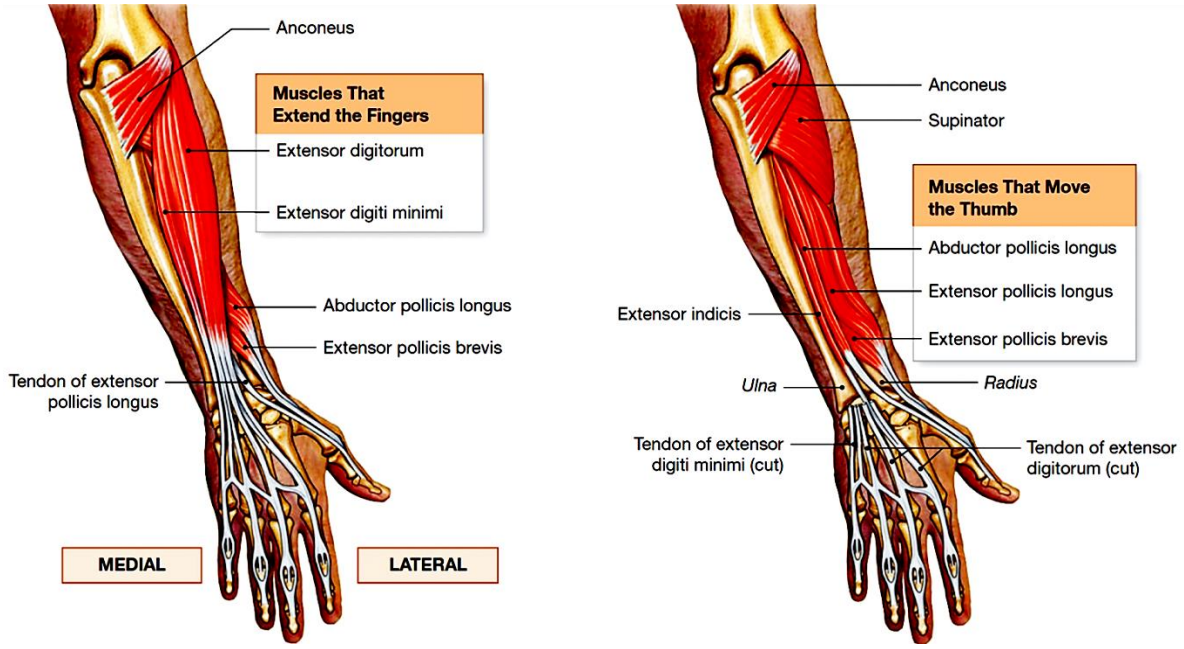




❖ Muscles that move the wrist and fingers

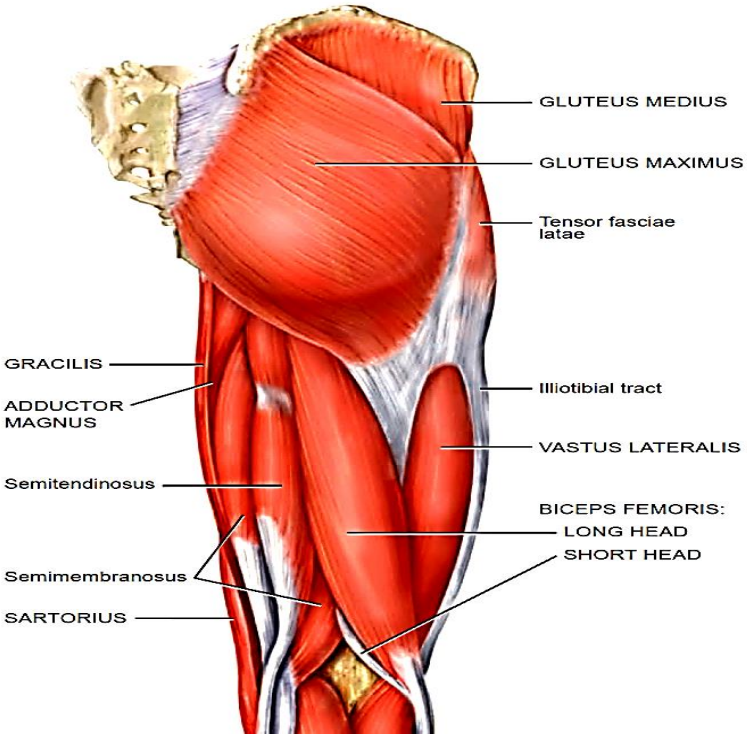
Muscle	Origin	Insertion	Action
Anterior fore arm muscles	Medial epicondyle	Carpals,metacarpals, and phalanges	Flex wrist, fingers and thumb; pronate forearm
Posterior forearm muscles	Lateral epicondyle	Carpals,metacarpals, and phalanges	Extend wrist, fingers and thumb; supinate forearm
Intrinsic hand muscles	Carpals metacarpals	Phalanges	Abduct, adduct, flex, and extend fingers and thumb

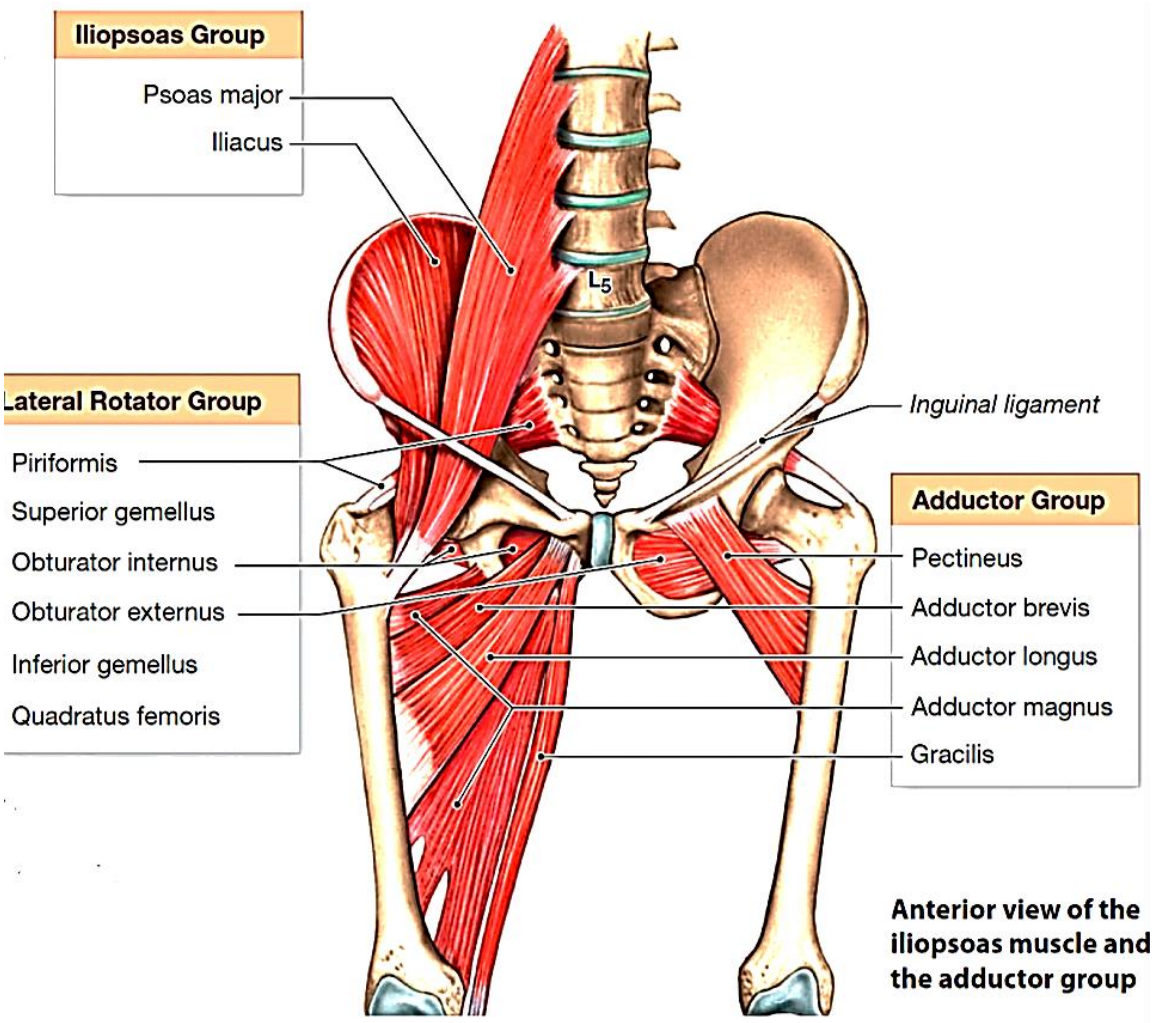




❖ Lower limb muscles

Muscle	Origin	Insertion	Action
Muscles that move the thigh			
Iliopsoas	Ilium and vertebrae	Trochanter of femur	Flexes thigh
Tensor fascia latae	Anterior superior iliac spine	Lateral condyle of tibia	Abducts thigh
Gluteus maximus	Ilium, sacrum, and coccyx	Lateral side of femur	Extends and abducts thigh
Gluteus medius	Ilium	Trochanter of femur	Abducts thigh
Adducter muscles of thigh	Pubis	Femur	Adduct thigh

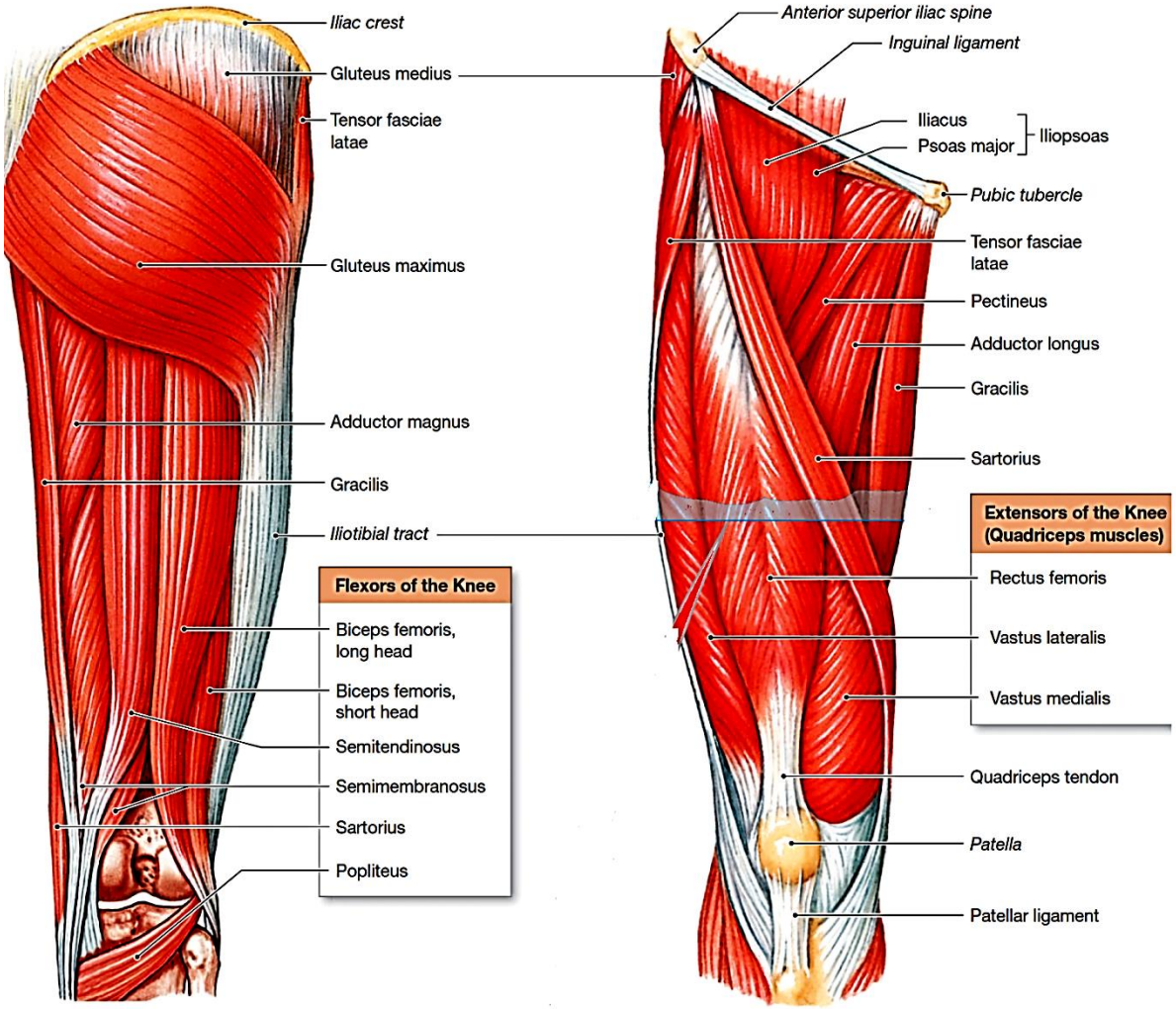




Anterior view of the iliopsoas muscle and the adductor group

Muscles that move the leg

Quadriceps femoris Rectus femoris	Anterior superior iliac spine	Tibial tuberosity	Extends leg and flexes thigh
Vastus lateralis	Femur	Tibial tuberosity	Extends leg
Vastus medialis	Femur	Tibial tuberosity	Extends leg
Vastus intermedius (not shown in illustration)	Femur	Tibial tuberosity	Extends leg
Sartorius	Anterior superior iliac spine	Tibia	Flexes leg and thigh
Hamstring muscles Biceps femoris	Ischium and femur	Fibula	Flexes leg and extends thigh
Semimembranosus	Ischium	Tibia	Flexes leg and extends thigh
Semitendinosus	Ischium	Tibia	Flexes leg and extends thigh

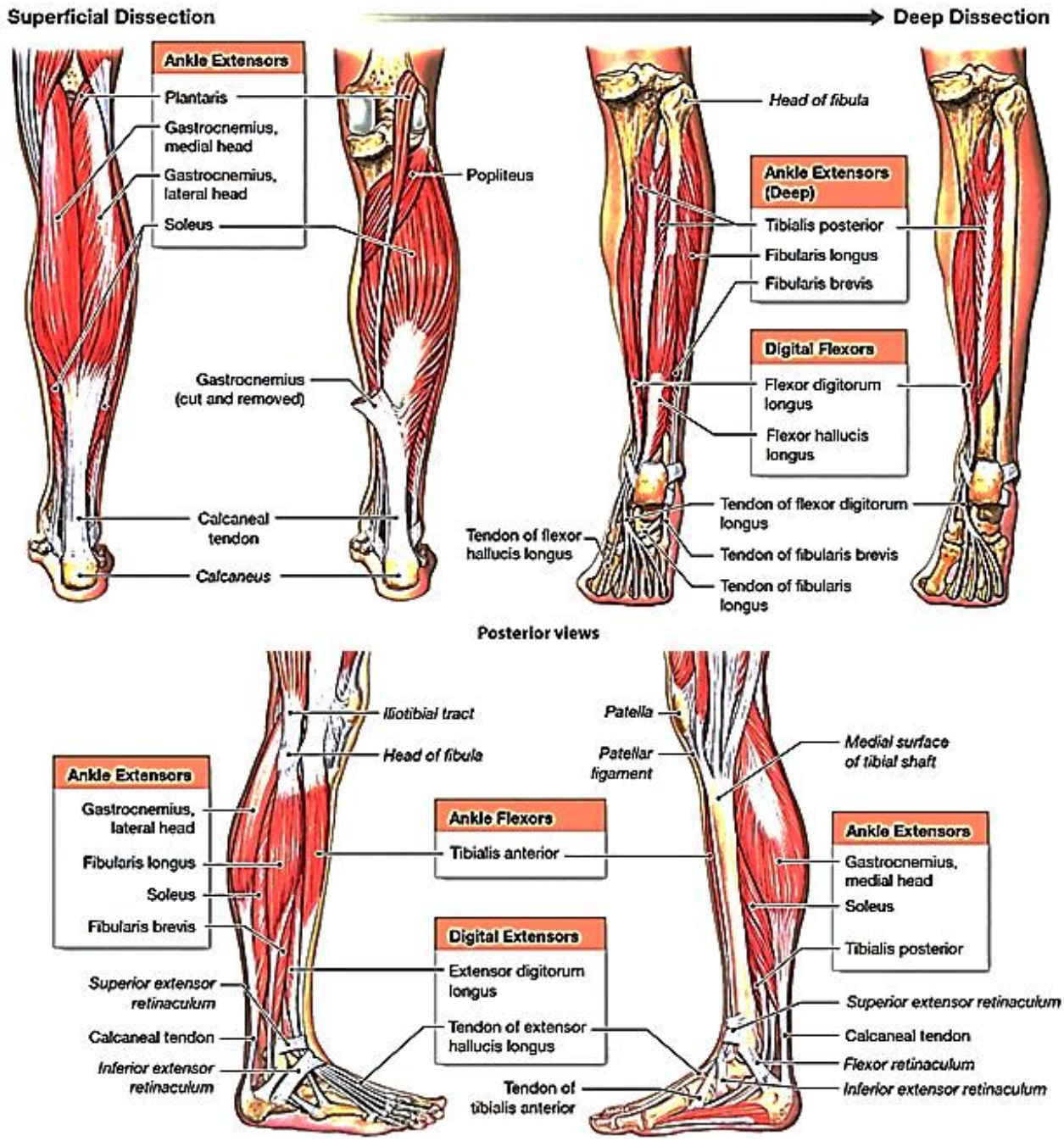


Hip and thigh, posterior view

Quadriceps and thigh muscles, anterior view

❖ Muscles that move the ankle and toes

Tibialis anterior	Tibia	Tarsal and first metatarsal	Dorsiflexes foot
Deep anterior leg muscles	Tibia or fibula	Phalanges, metatarsals, tarsals	Extend toes
Gastrocnemius	Medial and lateral epicondyle of femur	Calcaneus	Plantar flexes foot
Soleus	Tibia and fibula	Calcaneus	Plantar flexes foot
Deep posterior leg muscles	Tibia of fibula	Phalanges, metatarsals, tarsals	Evert foot
Peroneus muscle Intrinsic foot muscles	Fibula and tibia Tarsals or metatarsals	Tarsals and metatarsals Phalanges	Evert foot Abduct, adduct, flex, and extend toes



CHAPTER SEVEN

THE NERVOUS SYSTEM

With a mass of only 2 kg, about 3% of total body weight, the nervous system is one of the smallest and yet the most complex of the 11 body systems.

General Function

None of the body system is capable of functioning alone. All are interdependent and work together as one unit so that normal conditions within the body may prevail.

Control of the body's billions of cells is accomplished mainly by two communication systems:

- **The Nervous System And**
- **The Endocrine System.**

Both systems transmit information from one part of the body to another, but they do it in different ways. The nervous system transmits information very rapidly by nerve impulses conducted from one body area to another.

The endocrine system transmits information more slowly by chemicals secreted by ductless glands into blood steam and circulated from glands to other parts of the body.

The nervous system serves as the chief coordinating agency. Conditions both within and outside the body are constantly changing; the purpose of the nervous system is to respond to these internal and external changes (known as stimuli) and so cause the body to adapt to new conditions. It is through the nerve impulse sent to the various organs by the nervous system that a person's internal harmony and the balance between the person and the environment are maintained.

The nervous system has been compared to a telephone exchange, in that the brain and the spinal cord act as switching centres and the nerve trunks act as cables for carrying messages to and from these centres.

The body's primary communication and control system. Can be divided according to:

- **Structural categorie.**
- **Functional categories.**

► Structural subdivisions of the nervous system:

A. According to the location:

a. Central nervous system (C.N.S.) formed of brain and spinal cord.

b. Peripheral nervous system (P.N.S.): formed of :

1. **Cranial nerves:** 12 nerves their cell bodies are present in the brain.
2. **Spinal nerves:** their cell bodies are present in spinal cord.

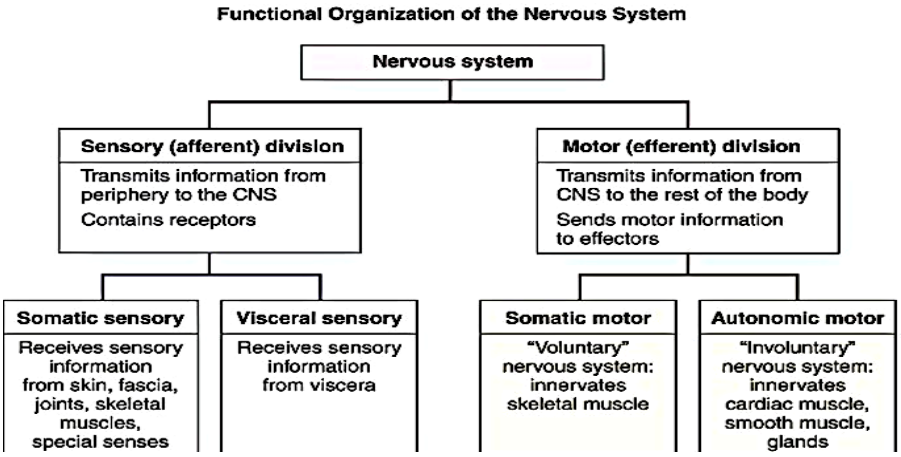
B. According to the function:

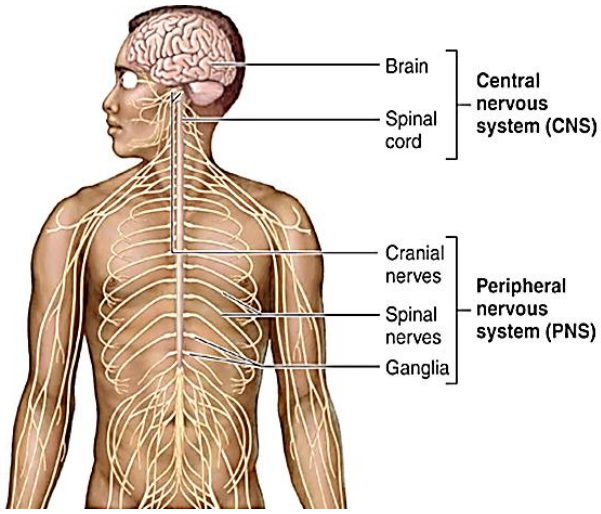
a. Somatic nervous system: controls the skeletal muscles. There are **<u>two types:</u>**

1. **Sensory (afferent)**: receives sensory information (input) from receptors transmits this information to the CNS.
2. **Motor (efferent):** transmits motor impulses (output) from the CNS to muscles or glands (effector organs).

b. Autonomic nervous system: is the system of viscera and formed **of <u>2 divisions:</u>**

1. **Sympathetic N.S.** is the system of fight and flight.
2. **Parasympathetic N.S.** is the system of rest and digest





The different function of the body organs and systems are regulated by **two main regulatory mechanisms:**

<u>1- Chemical regulation;</u> by hormones, vitamins, enzymes, ions, gases and chemical transmitters. Chemical regulation is characterized by having a slow onset and prolonged duration of action; chemical regulation controls mainly the metabolic functions of the body.

<u>2- Nervous regulation;</u> by the nervous system.

It is characterized by having a rapid onset and a short duration .

So, nervous regulation controls the rapid activities of the body.

The two mechanisms are complementary and can therefore influence each other e.g. certain chemical substances can either stimulate or inhibit the nervous system, and in turn influence the nervous regulation . On the other hand, stimulation of the nervous system leads to release of chemical substances e.g. hormones which influence the chemical regulation.

<u>The nervous system performs the following functions:</u>

1- Sensory functions: The sensory system receives sensory information from the skin, deeper tissues (subcutaneous tissues, muscles, joints and bones) viscera and special sense organs (eye, nose, ear and tongue).

The sensory stimuli give the brain information about the external and internal environment which helps in control of the body functions .

When this sensory information reaches the (CNS), they produce the following:

Immediate reflex action by sending motor impulses to the muscles or glands.

Perception of the specific sensation.

Stored in the brain in the form of memory (if important) or ignored.

2- Motor functions: The nervous system controls:

Voluntary movements, muscle tone and body posture.

Activity of smooth muscles and cardiac muscle.

Secretion of all exocrine glands as well as most of the endocrine glands.

Arterial blood pressure, respiration and body temperature.

3- Higher mental functions: As memory, learning, thinking, emotions, personality character etc.

Cells of the nervous system and their functions

The two types of cells found in the nervous system are called:

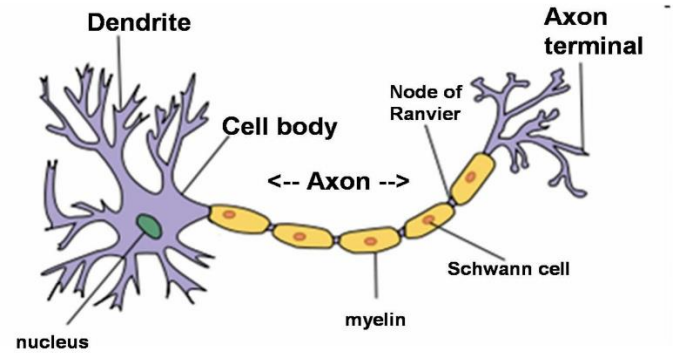
- Neurons or nerve cells and
- Neuroglia, which are specialized connective tissue cells.

Neurons conduct impulses, whereas neuroglia supports neurons.

► **Neurons**

Each neuron consists of four parts:

- A main part called the neuron cell body,
- One or more branching projections called dendrites, and
- One elongated projection known as an axon.
- Axon terminalis



Dendrites are the processes or projections that transmit impulses to the neuron cell bodies, and **Axons** are the processes that transmit impulses away from the neuron cell bodies.

Neurons can be classified structurally and functionally.

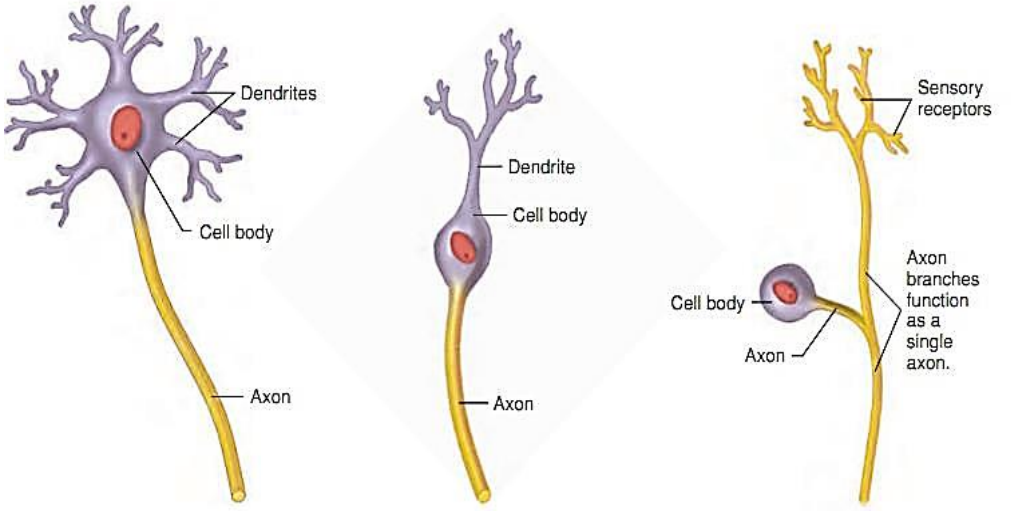
- **Classification of neuron:**

Structural: According to the number of extensions from the cell body.

Multipolar neuron: has one axon and several dendrites. Most of neurons are multipolar especially motor neurons.

Bipolar neuron: has only one axon and one dendrite. Located in sensory organs e.g. retina, nasal cavity.

Unipolar neuron: has only one process from the cell body then divides into two branches one to the C.N.S. and the other to the P.N.S. usually this type is: sensory.

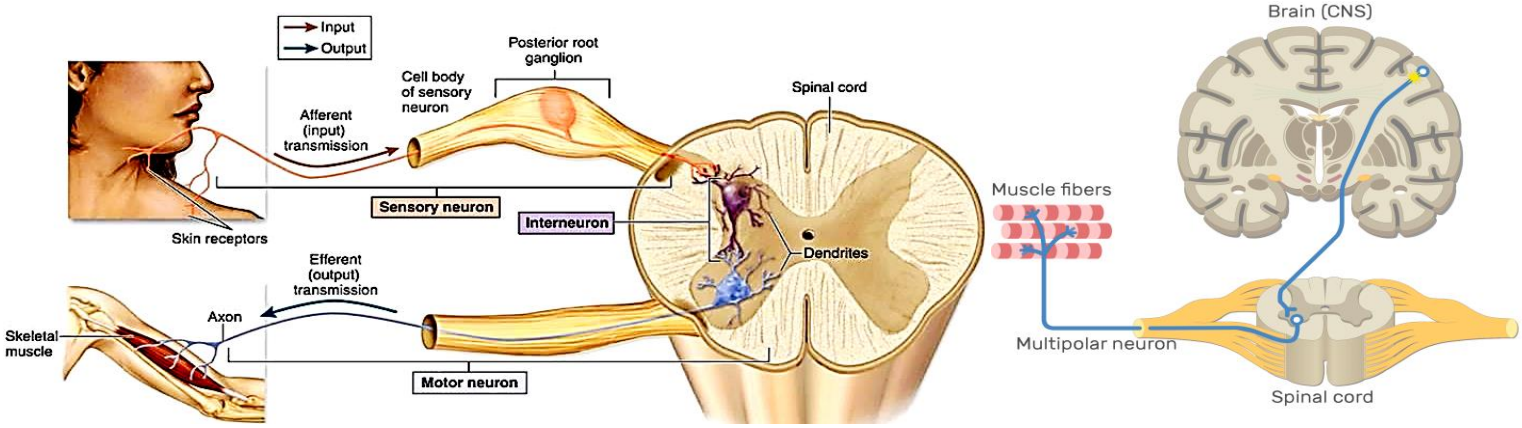


(a) A **multipolar neuron** has many dendrites and an axon.

(b) A **bipolar neuron** has a dendrite and an axon.

(C) **unipolar neuron** appears to have an axon and no dendrites.

The four types of functional classification of neurons are according to the direction in which they transmit impulses. These are: - **sensory neurons, motor neurons, and interneurons .neurosecretory .**



Sensory neurons transmit impulses to the spinal cord and brain from all parts of the body. Motor neurons transmit impulses in the opposite direction-away from the brain and spinal cord.

They do not conduct impulses to all parts of the body but only to two kinds of tissue-muscle and glandular epithelial tissue.

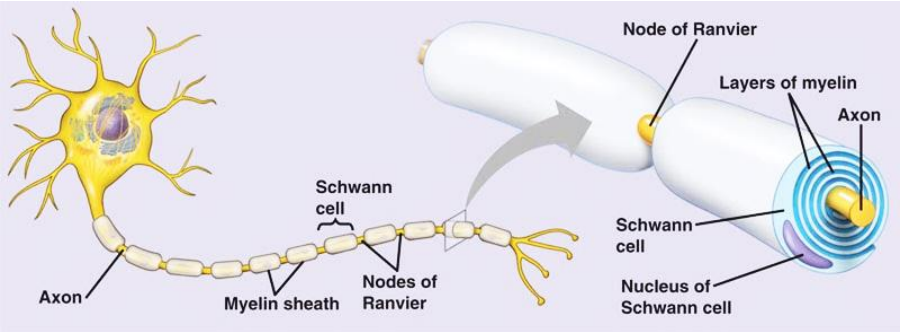
Interneurons conduct impulses from sensory neurons to motor neurons.

Sensory neurons are also called afferent neurons; motor neurons are called efferent neurons, and interneurons are called central or connecting neurons.

The axon shown in Figure is surrounded by a segmented wrapping of a material called myelin sheath.

Myelin sheath is a white, fatty substance formed by Schwann cells that wrap around some axons outside the central nervous system. Such fibers are called Myelinated fibers.

Nodes of Ranvier are indentions between adjacent Schwann cells.

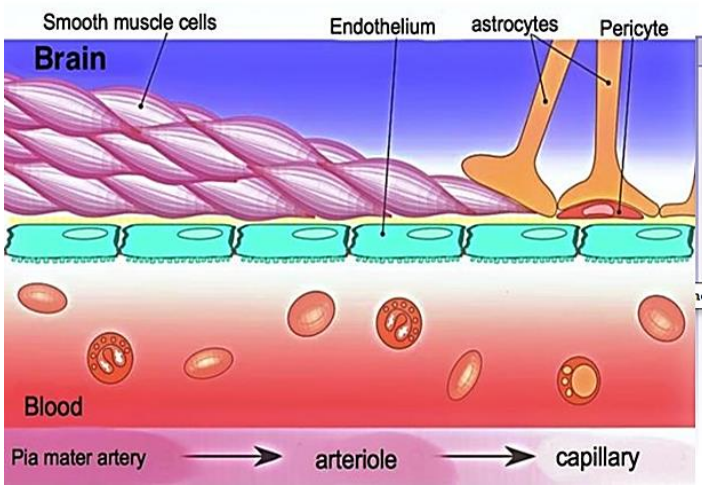


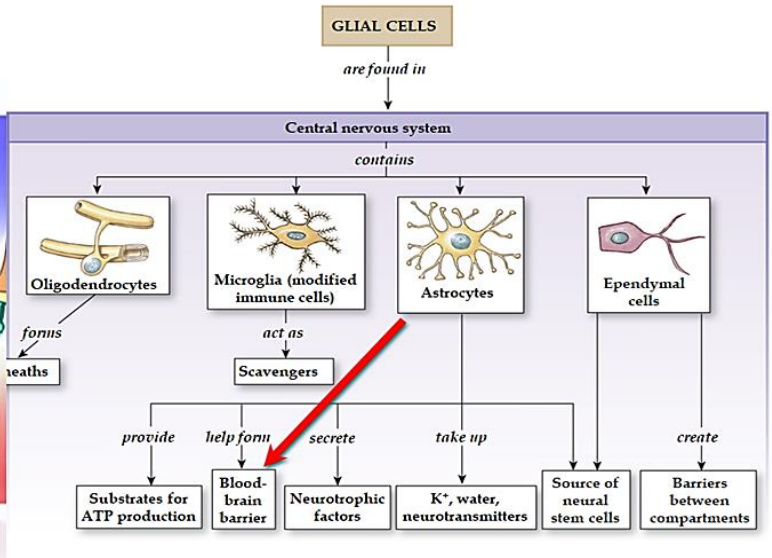
The outer cell membrane of a Schwann cell is called the neurilemma. The fact that axons in the brain and cord have no neurilemma is clinically significant because it plays an essential part in the regeneration of cut and injured axons.

Therefore the potential for regeneration in the brain and spinal cord is far less than it is in the peripheral nervous system.

► **Neuroglia**

Neuroglia does not specialize in transmitting impulses. Instead, they are special types of connective tissue cells.





Their name is appropriate because it is derived from Greek word glia meaning "glue." One function of neuroglia cells is to hold the functioning neurons together and protect them.

MAIN PROPERTIES OF NERVOUS TISSUE

1.Excitability

2. Conductivity

1.Excitability: Ability to produce and conduct electrical impulses.

Excitability is an inherent property of nervous tissue .

It possesses the ability to enter into an active state, what is known as a state of excitation, in response to stimulation.

The excitation of nervous tissue occurs as a result of stimulation of sense organs or sensory nerve endings known as receptors, by any sort of stimuli acoustic, optic, thermal, gustatory, etc.

The conduction of excitation along nerve fiber is accompanied by appearance of bio-electric phenomena, called action potentials (or nerve impulses) in nervous tissue.

Excitation is conducted in the nerves in isolation (separately) along each nerve fiber and is never transmitted from one nerve fiber to an adjacent nerve fiber.

It explains that excitation may either be transmitted along all the fibers forming a nerve, or just along some of them.

This makes possible the contraction of individual muscle fibers and certain muscles, and

not necessarily the entire group of muscles innervated by a particular nerve.

2.Conductivity : Nervous tissue is characterized by the fact that excitation does not remain at the site of its origin but is transmitted along nerve fibers. This property of nervous tissue to transmit excitation is called conductivity.

STIMULUS

A stimulus can be defined as a sudden change in the environment (external or internal) which is strong enough to excite the nerve or muscle or organism as a whole.

If the stimulus is capable to bring a change or to excite a given tissue, it is called adequate stimulus or threshold stimulus.

If the stimulus is not capable to excite or fails to elicit any response, it is called inadequate stimulus .

There are many types of stimuli that may excite the tissues; the most important ones

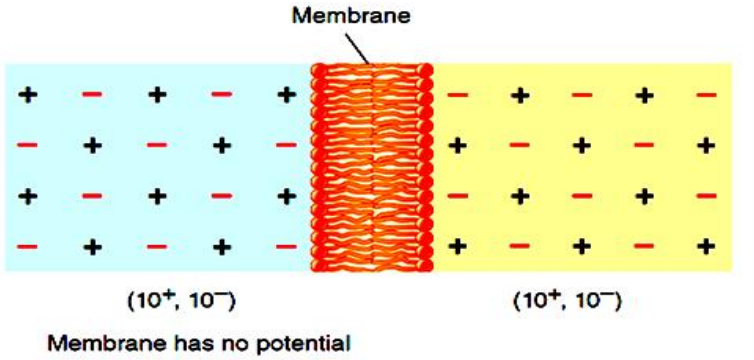
are the following :

1. **Mechanical stimuli:** When a given tissue is excited by mechanical means, such stimuli are known as mechanical stimuli.
2. **Physical stimuli**: Heat, cold and humidity serve as physical stimuli for the living tissues.
3. **Chemical stimuli:** Various chemicals including acids and bases are referred to as chemical stimuli for the living tissues.
4. **Electrical stimuli:** Electricity functions as electrical stimuli and it is capable to excite the living tissues.

Impulse Generation and Conduction

► The Nerve Impulse

The cell membrane of an unstimulated (resting) neuron carries an electric charge. Because of positive and negative ions concentrated on either side of the membrane, the inside of the membrane at rest is negative as compared with the outside.



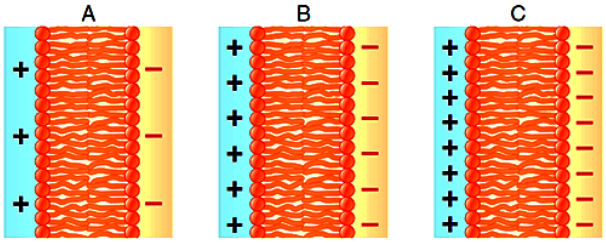
The membrane itself is not charged.

The term membrane potential refers to the difference in charge between the wafer-thin regions of ICF and ECF lying next to the inside and outside of the membrane, respectively .

The magnitude of the potential depends on the number of opposite charges separated:

The greater the number of charges separated, the larger the potential.

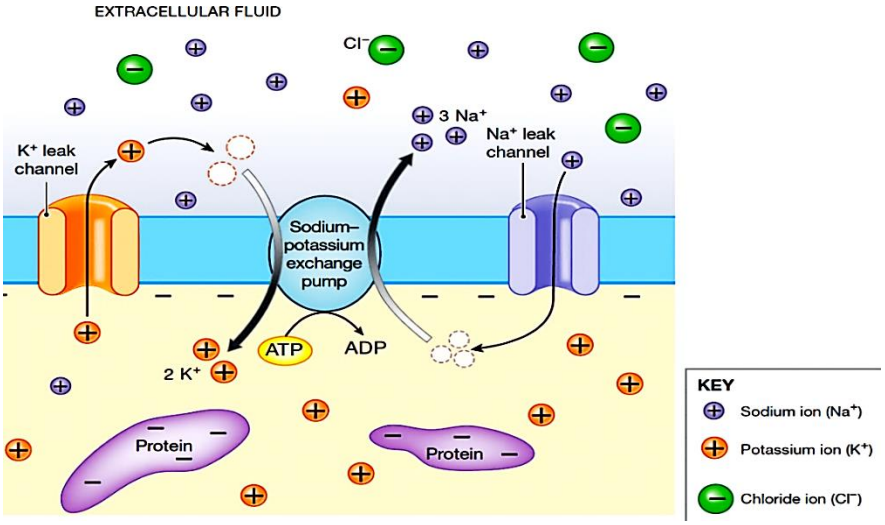
Membrane B has more potential than A and less potential than C.



A nerve impulse is a local reversal in the charge on the nerve cell membrane that then spreads along the membrane like an electric current. This sudden electrical change in the membrane is called an action potential. A stimulus, then, is any force that can start an action potential.

This electric change results from rapid shifts in sodium and potassium ions across the cell membrane. The reversal occurs very rapidly (in less than one thousandth of a second) and is followed by a rapid return of the membrane to its original state so that it can be stimulated again.

A myelinated nerve fiber conducts impulses more rapidly than an unmyelinated fiber of the same size because the electrical impulse "jumps" from node (space) to node in the myelin sheath instead of traveling continuously along the fiber.

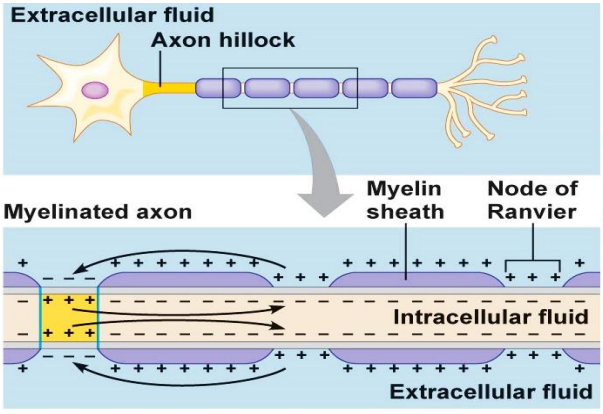


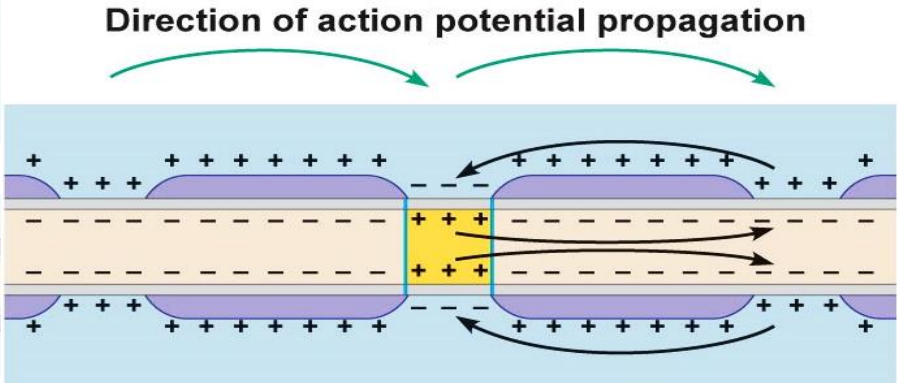
Properties of nerve impulse:

1. When a nerve is subjected to a threshold stimulus it undergoes excitation or it is stimulated. Threshold stimulus is roughly constant for a given type of nerve fiber, but is higher in small fibers than in large.
2. Normally nerve impulses pass in one direction only.
3. Generally, impulses flowing in a nerve fiber do not spread with the neighboring fibers because nerves or nerve fibers are generally covered by an insulating nerve sheath called the medullary sheath.
4. After one impulse has passed along the nerve fiber, there is a refractory period during which the nerve fiber is neither excitable nor conductive. During this period a second stimulus, however, of large strength, remains ineffective. In this period the fiber recovers and after recovery it again responds to its maximum. If an inadequate stimulus is followed by another one within a very brief period, say 0.5 ms, there may be a response, so that there can be summation of two inadequate stimuli. The absolute refractory period is followed by a relatively refractory one, during which the threshold is increased.
5. The nerve impulses follow the "all or none law", i.e., if any stimulus is an adequately effective one that is either of threshold strength or more, the nerve fiber responds to a great extent with an impulse of maximum intensity but if it is below threshold strength it is incapable to elicit any response whatsoever. Any stimulus bringing out a response in the nerve, the response cannot be increased further by increasing the strength of the stimulus.
6. The impulse takes a certain time to travel along the nerve fiber. The larger the fibers the more rapidly they conduct, and medullated fibers conduct more rapidly than non-medullated. Thus the speed or the

velocity of nerve impulse differs in different animals. In frog it is about 30 meters per second at the room temperature and for the human motor and sensory nerves it is about120 meters per 'second. While crossing from one fiber to another there is some resistance at the synapse which breaks the speed of the impulse.

7. In a nerve fiber the impulse is always conducted at a constant amplitude and velocity.
8. For regular conduction of nerve impulses a continuous supply of Oxygen is required in order to liberate the energy and generally it is believed that nerves do not undergo fatigue, however, the continued stimulation in the absence of oxygen may result in certain disorders or irregularities.
9. Normally the nerve fibers are capable in sending enormous number of impulses.
10. The nerve impulses travel in a nerve in a quick sequence. Each nerve impulse is of the same nature and is similar in all the nerves.





Condition affecting the passage of a nerve impulse

1. Repeated stimulation does not lessen the excitability and conductivity of a nerve fiber or nerve but if the interval between two stimuli is shorter than the relative refractory period, the second and subsequent stimuli produce progressively smaller impulses than the first.

2. Temperature also plays an important role in the passage of a nerve impulse. Below a certain temperature the propagation of the excitatory process in the nerve does not take place.

3. pH value also regulates the conduction and propagation of nerve impulses. Actually pH value does not affect the nerve impulses but alters the excitability and after potentials.

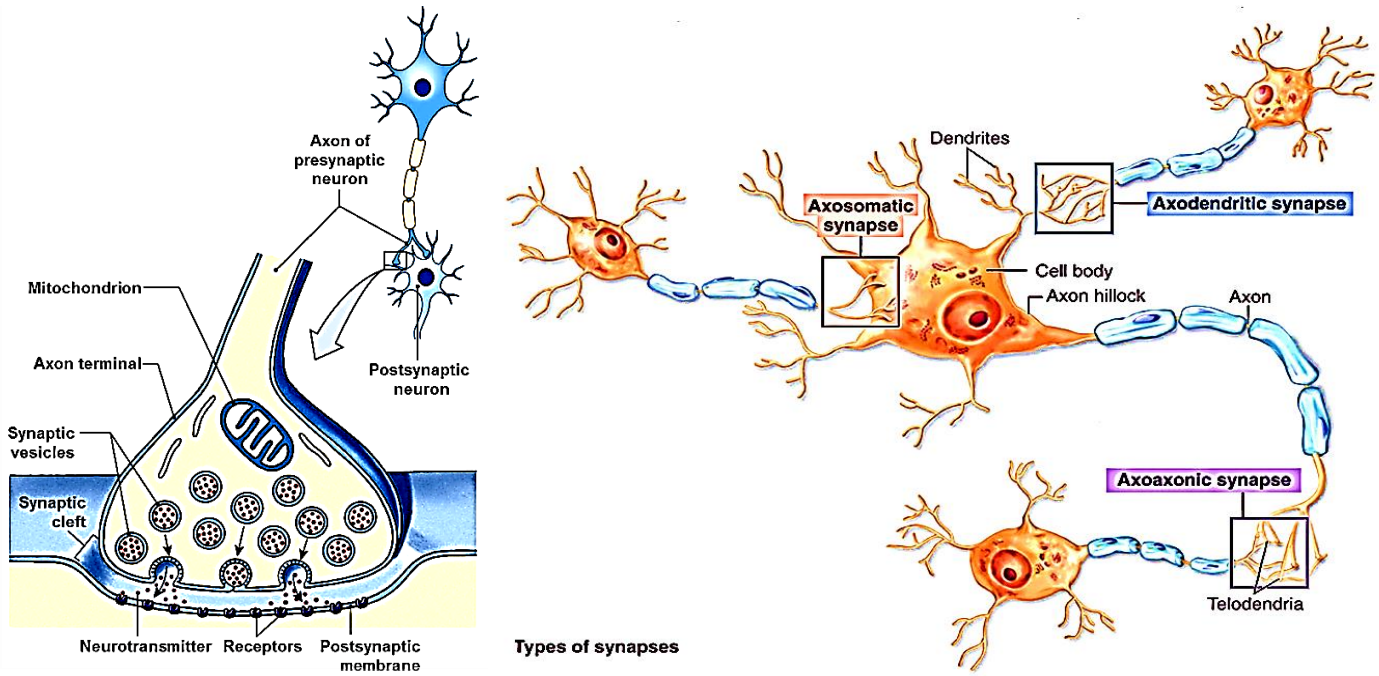
► **The Synapse**

Each neuron is a separate unit, and there is no anatomic connection between neurons.

✓ How then is it possible for neurons to communicate?

✓ In other words, how does the axon of one neuron make functional contact with the membrane of another neuron?

This is accomplished by the synapse, from a Greek word meaning "to clasp." Synapses are points of junction for the transmission of nerve impulses (Fig).



Types of synapses

Within the branching endings of the axon are small bubbles(vesicles) containing a type of chemical known as a neurotransmitter. When stimulated, the axon releases its neurotransmitter into the narrow gap, the synaptic cleft, between the cells.

The neurotransmitter then acts as a chemical signal to stimulate the next cell, described as the postsynaptic cell. On the receiving membrane, usually that of a dendrite, sometimes another part of the cell, there are special sites, or receptors, ready to pick up and respond to specific neurotransmitters. Receptors in the cell membrane influence how or if that cell will respond to a given neurotransmitter.

Although there are many known neurotransmitters, the main ones are epinephrine, also called adrenaline; a related compound, norepinephrine, or noradrenaline; and acetylcholine.

Acetylcholine (Ach) is the neurotransmitter released at the neuromuscular junction, the synapse between a neuron and a muscle cell.

All three of the above neurotransmitters function in the autonomic nervous system.

Functions:

The characteristic functions of synapses can be summarized as follows:

1. It transmits impulses from one neuron to other neuron or to glands, muscle cells or sonic other cells. 2. It can transmit impulses only in one direction.

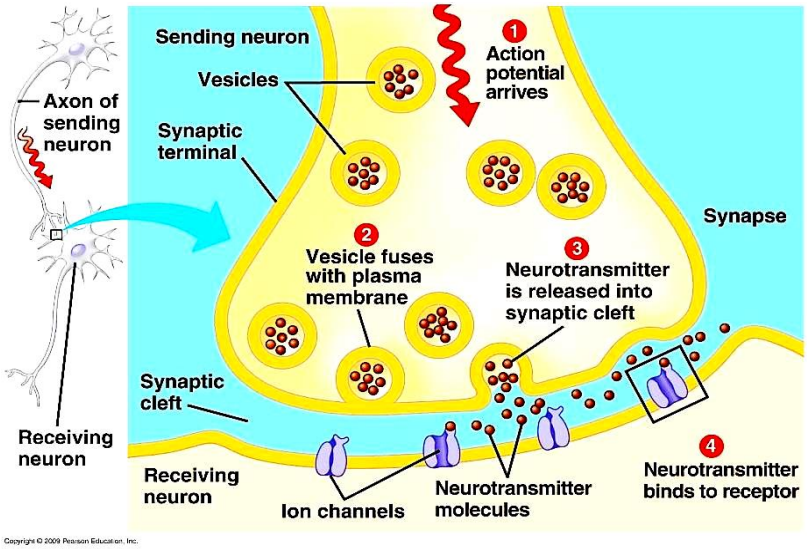
3. It retards the passage of an impulse more so than the nerve fiber does.

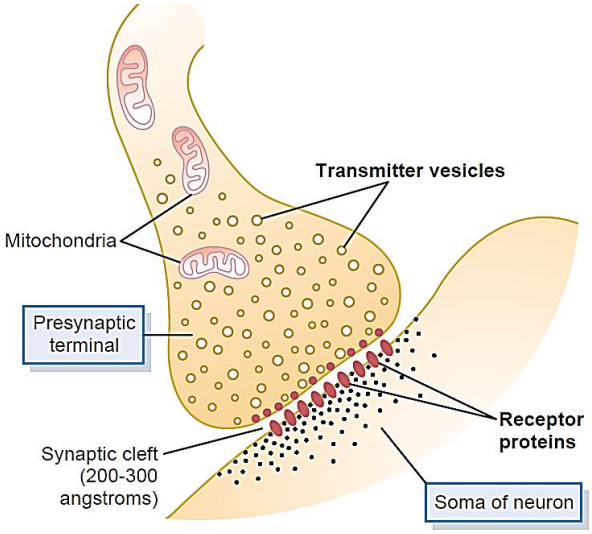
Consequently, there is a synaptic delay in the passage of an impulse from the pre-synaptic to the post-synaptic neuron.

4. It is a site where repetitive discharge of a neuron may occur.

5. It is very vulnerable to the action of many chemical compounds, some of which increase its responsiveness (strychnine), whereas others decrease it (anesthetics).

6. It has the characteristics of summating and of inhibiting impulses. One axon may receive the excitatory or inhibitory effects of impulses from several other neurons. Similarly one axon may transmit impulses through its nerve endings to several other neurons.





The Reflex Arc

As the nervous system functions, both external and internal stimuli are received, interpreted, and acted on. A complete pathway through the nervous system from stimulus to response is termed a reflex arc . This is the basic functional pathway of the nervous system.

❖ **The parts of a typical reflex arc are:**

1. Receptor-the end of a dendrite or some specialized receptor cell, as in a special sense organ, that detects a stimuli.

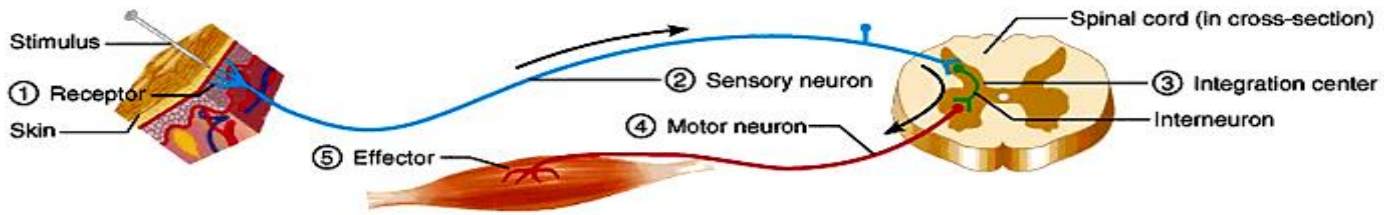
2. Sensory neuron, or afferent neuron-a cell that transmits impulses toward the CNS.

3. Central neuron-a cell or cells within the CNS.

These neurons may carry impulses to and from the brain, may function within the brain, or may distribute impulses to different regions of the spinal cord.

4. Motor neuron, or efferent neuron-a cell that carries impulses away from the CNS.

5. Effector-a muscle or a gland outside the CNS that carries out a response.



At its simplest, a reflex arc can involve just two neurons, one sensory and one motor, with a synapse in the CNS. There are very few reflex arcs that require only this minimal number of neurons.

The knee jerk reflex is one of the few examples in humans. Most reflex arcs involve many more, even hundreds, of connecting neurons within the central nervous system.

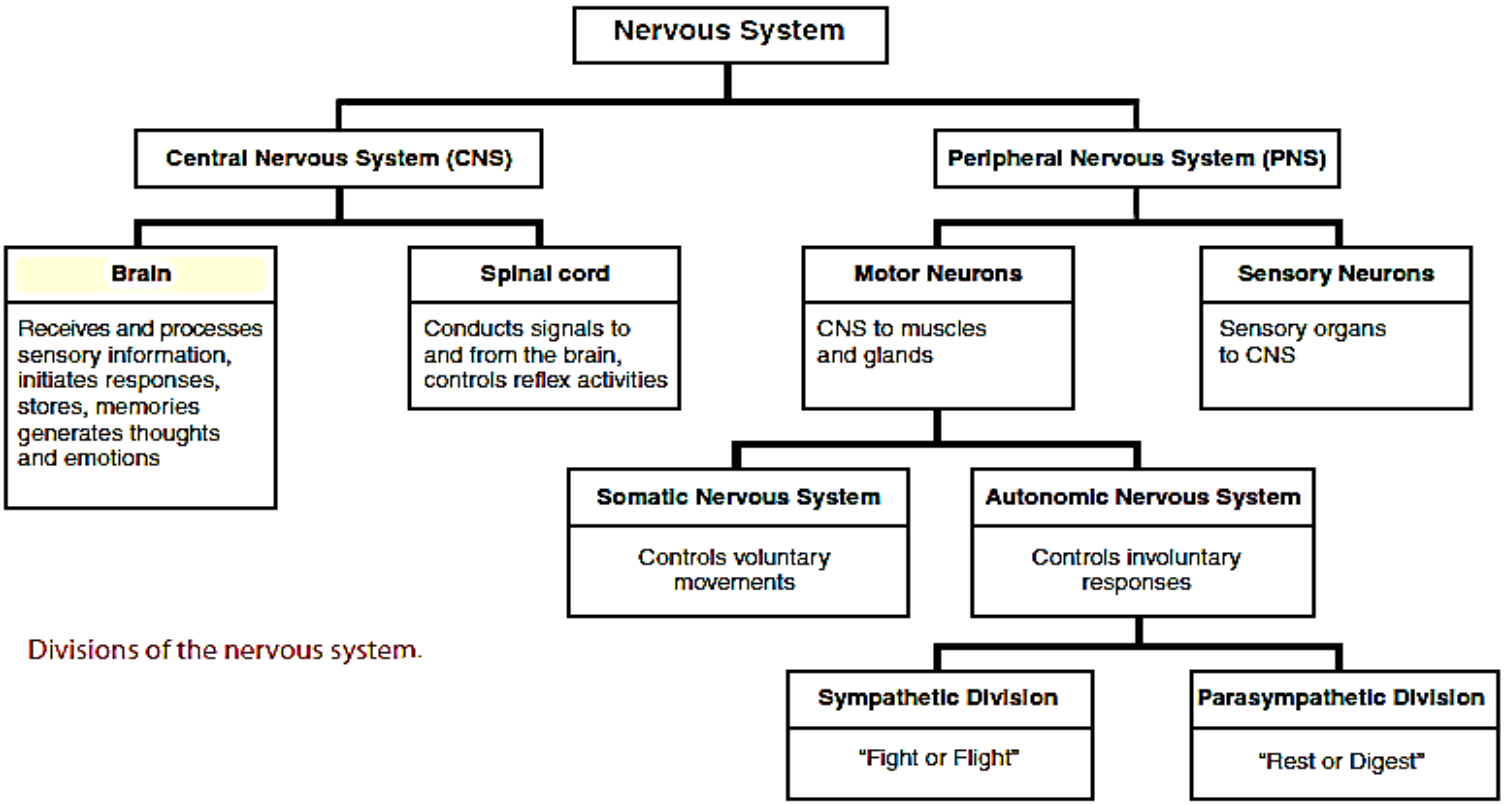
Division of the Nervous System

- The nervous system as a whole consists of two principal divisions called the **central nervous system and peripheral nervous system.**

Because the brain and spinal cord occupy a midline or central location in the body, they are together called the central nervous system or CNS.

Similarly, the usual designation for the nerves of the body is the peripheral nervous system or PNS. Use of the term peripheral is appropriate because nerves extend to outlying or peripheral parts of the body.

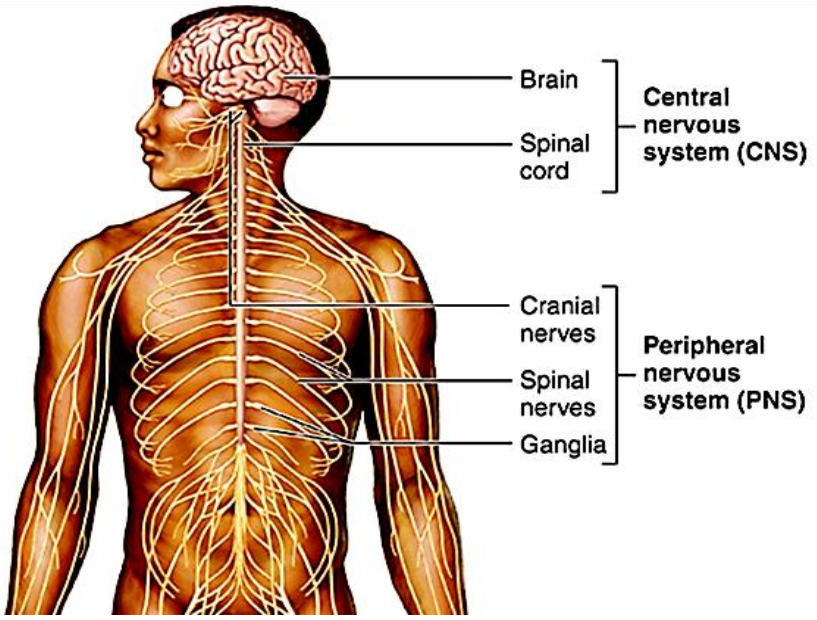
- A subdivision of the peripheral nervous system called the autonomic nervous system (ANS) consists of structures that regulate the body's autonomic or involuntary functions (for example, the heart rate, the contractions of the stomach, and intestines, and the secretion of chemical compounds by glands).



Divisions of the nervous system.

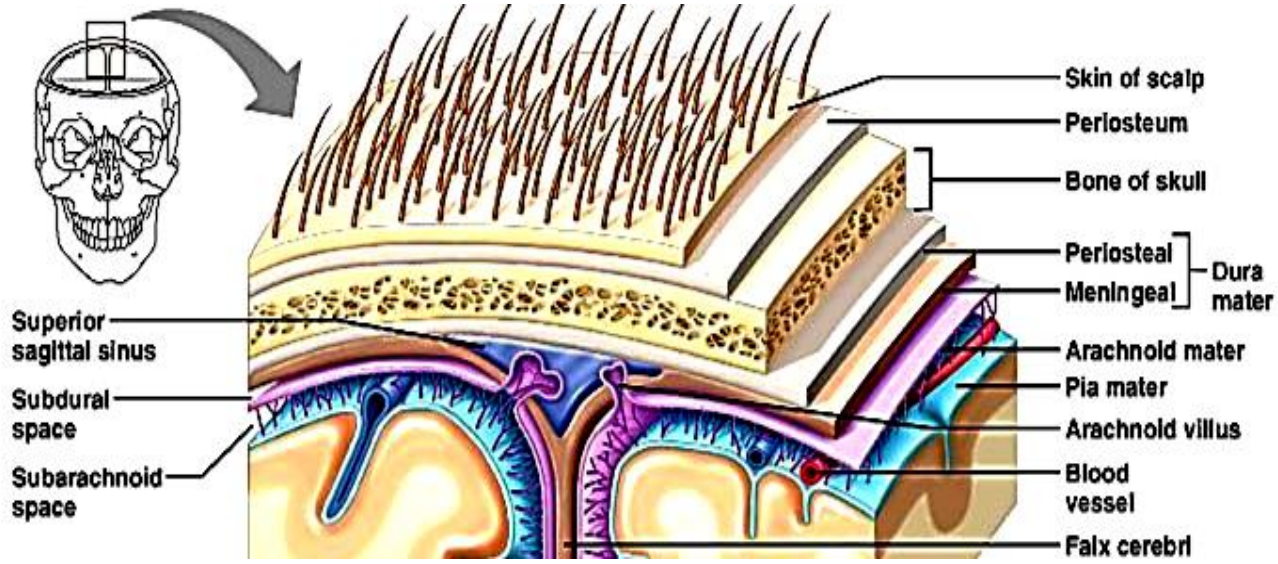
Central Nervous System

The CNS as its name implies, is centrally located. Its two major structures, the brain and spinal cord, are found along the midsagittal plane of the body.



The brain is protected in the cranial cavity of the skull, and the spinal cord is surrounded in the spinal column.

In addition, protective membranes called meninges cover the brain and spinal cord.



Covering and Fluid Spaces Of the Brain and Spinal Cord

Nervous tissue is not a sturdy tissue. Even moderate pressure can kill nerve cells, so nature safeguards the chief organs made of this tissue-the spinal cord and the brain-by surrounding them with three fluid-containing membranes called the meninges.

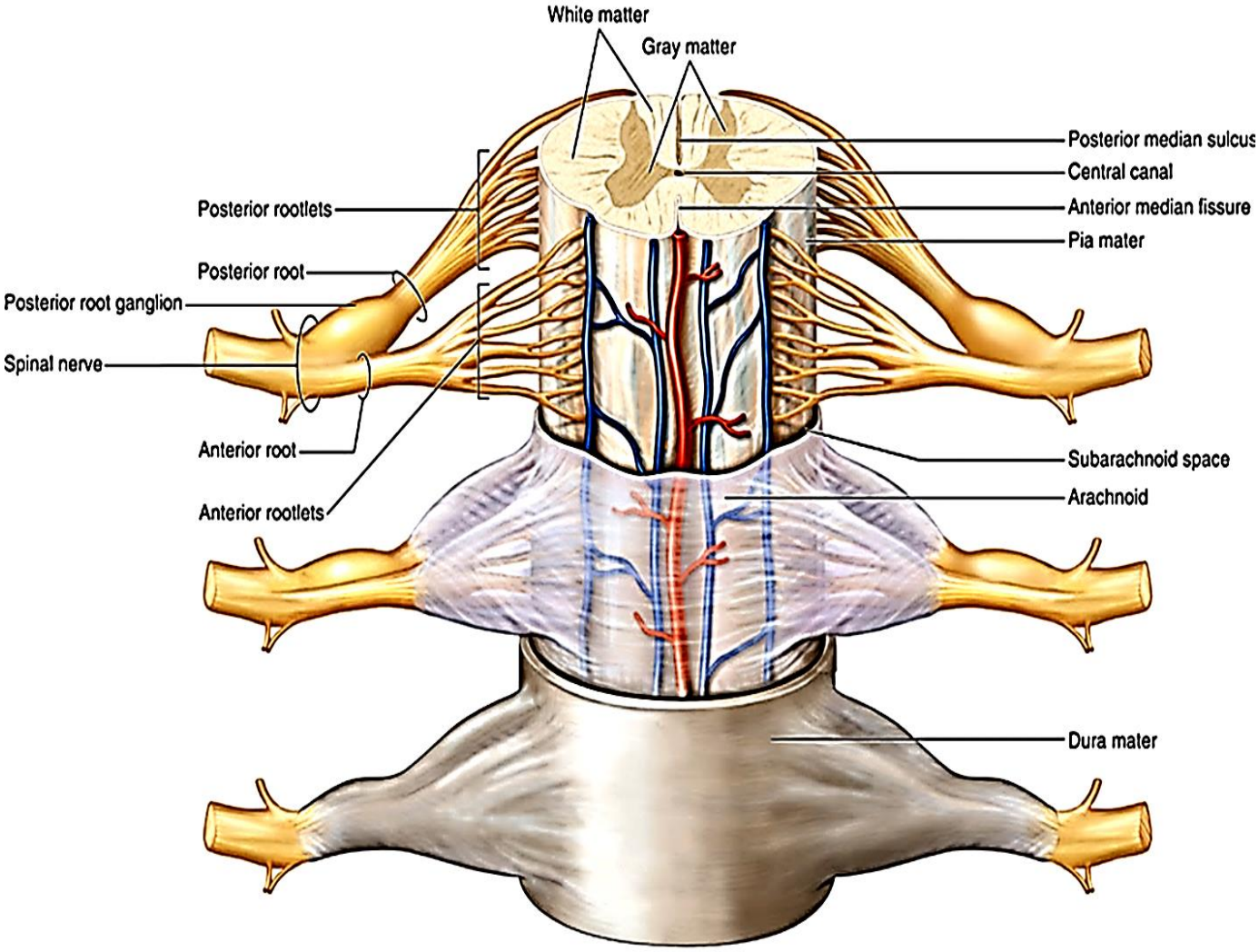
The meninges are then surrounded by bone. The spinal meninges form tube like covering around the spinal cord and line the bony vertebral foramen of the vertebrae that surround the cord.

Look at Figure , and you can identify the three layers of the spinal meninges.

They are the dura mater, which is the tough outer layer that lines the vertebral canal, the pia mater, which is the inner most membrane covering the spinal cord itself, and the arachnoid, which is the membrane between the dura and pia mater.

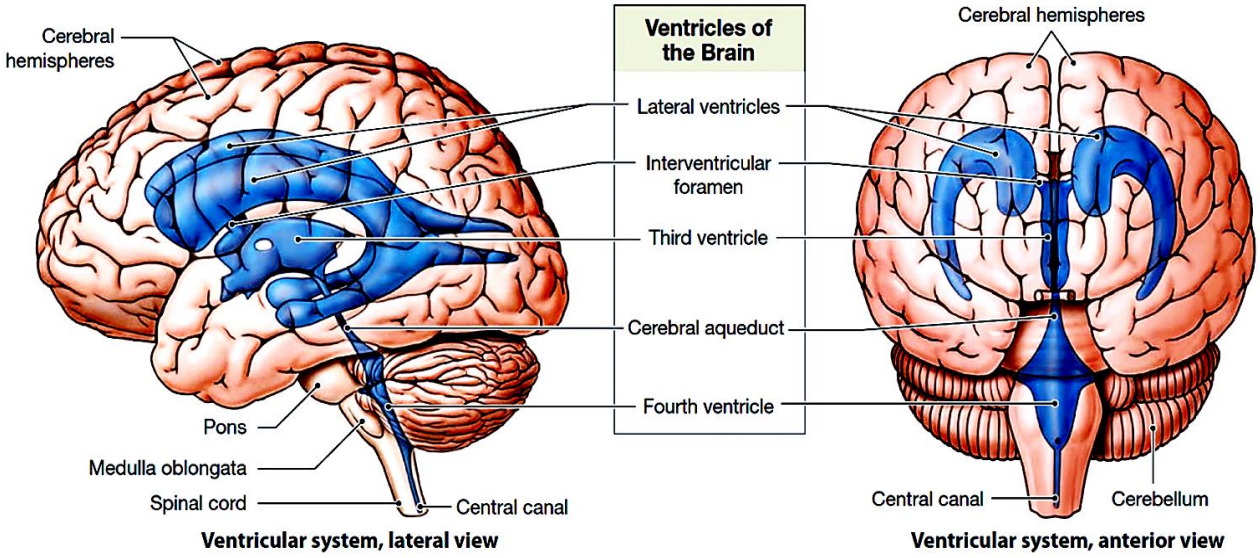
The term arachnoid means "cobweblike." It comes from arachne, which is the Greek word for spider. The meninges that form the protective covering around the spinal cord also extend up and around the brain to enclose it completely.

Fluid fills the subarachnoid spaces between the pia mater and arachnoid in the brain and spinal cord. This fluid is called cerebrospinal fluid (CSF). Cerebrospinal fluid also fills spaces in the brain called cerebral ventricles.



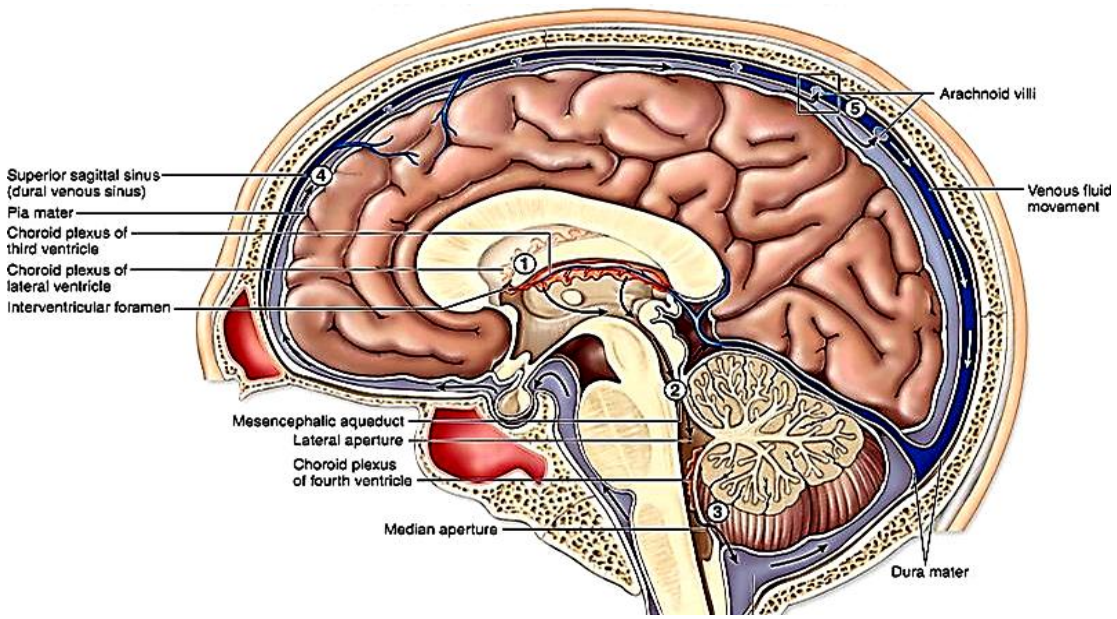
In Figure, you can see the irregular shapes of the ventricles of the brain.

These illustrations visualize the location of the ventricles if you remember that these large spaces lie deep inside the brain and that there are two lateral ventricles.



One lies inside the right half of the cerebrum (the largest part of the human brain), and the other lies inside the left half of the cerbrum. CSF is one of the body's circulating fluids. It forms continually from fluid filtering out of the blood in a network of brain capillaries known as the choroid plexus and into the ventricles.

CSF : A clear, colorless liquid called cerebrospinal fluid(CSF) fills the ventricles and canals of the CNS and bathes its external surface. The brain produces about 500 mL of CSF per day, but the fluid is constantly reabsorbed at the same rate and only 100 to 160 mL are present at one time. About 40% of it is formed in the subarachnoid space external to the brain, 30% by the general ependymal lining of the brain ventricles, and 30% by the choroid plexuses. CSF forms partly by the filtration of blood plasma through the choroid plexuses and other capillaries of the brain. CSF has more sodium and chloride than the blood plasma, but less potassium, calcium, and glucose and very little protein.CSF seeps from the lateral ventricles into the third ventricle and flows down through the cerebral aqueduct into the fourth ventricle.



It moves from the fourth ventricle into the small, tube like central canal of the cord and out into the subarachnoid spaces.

Then it moves leisurely down and around the cord and up and around the brain (in the subarachnoid spaces of their meninges) and returns to the blood (in the veins of the brain).

Cerebrospinal fluid serves three purposes:

1. Buoyancy :

Because the brain and CSF are very similar in density, the brain neither sinks nor floats in the CSF but remains suspended in it—that is, the brain has neutral buoyancy. Neutral buoyancy allows the brain to attain considerable size without being impaired by its own weight.

If the brain rested heavily on the floor of the cranium, the pressure would kill the nervous tissue.

2. Protection.

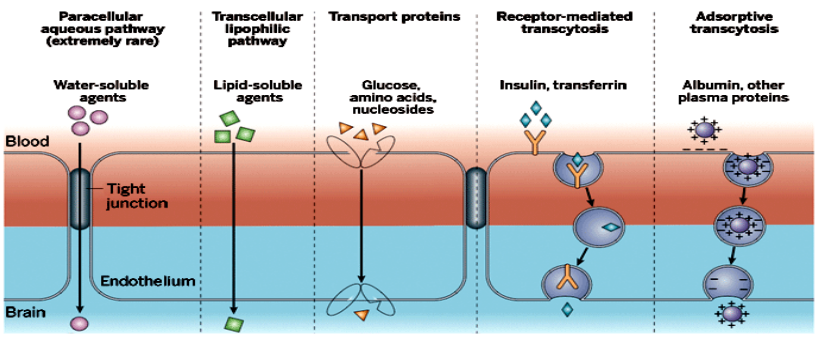
CSF also protects the brain from striking the cranium when the head is jolted. If the jolt is severe, however, the brain still may strike the inside of the cranium or suffer shearing injury from contact with the angular surfaces of the cranial floor. This is one of the common findings in child abuse (shaken child syndrome) and in head injuries (concussions) from auto accidents, boxing, and the like.

3. Chemical stability. CSF is secreted into each ventricle of the brain and is ultimately absorbed into the bloodstream. It provides a means of rinsing metabolic wastes from the CNS and homeostatically regulating its chemical environment. Slight changes in its composition can cause malfunctions of the nervous system. For example, a high glycine concentration disrupts temperature and blood pressure control, and a high pH causes dizziness and fainting.

Blood-Brain Barrier

Nervous tissue is protected from the general circulation by the blood-brain barrier.

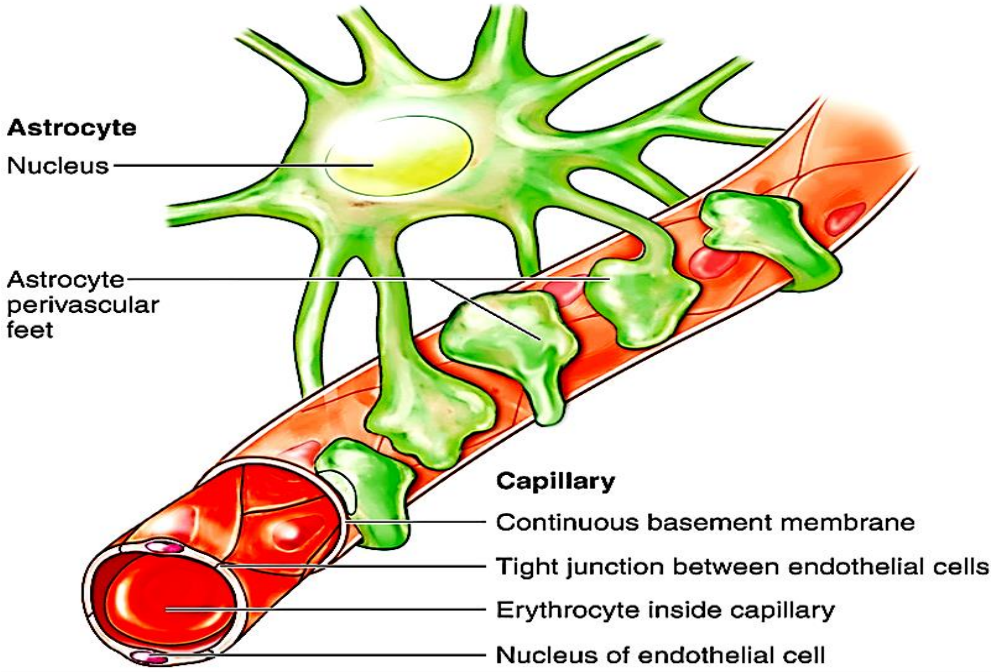
Strictly regulates what substances can enter the interstitial fluid of the brain. Prevents exposure of neurons in the brain to drugs, waste products in the blood, and variations in levels of normal substances (ions, hormones) that could adversely affect brain function.



Tight junctions prevent materials from diffusing across the capillary wall. Astrocytes act as "gatekeepers" that permit materials to pass to the neurons after leaving the capillaries. Is markedly reduced or missing in three distinct locations in the CNS: the choroid plexus, hypothalamus, and pineal gland.

Molecules within brain capillaries moved selectively through endothelial cells by:

Diffusion. Active transport. Endocytosis. Exocytosis



☼ **In general**, only lipid-soluble compounds (including carbon dioxide; oxygen; ammonia; lipids, such as steroids or prostaglandins; and small alcohols) can diffuse across the membranes of endothelial cells into the interstitial fluid of the brain and spinal cord. Water and ions must pass through channels in the apical and basement plasma membranes. Larger, water-soluble compounds can cross the capillary walls only by active or passive transport.

Divisions of the Brain

The brain, one of our largest organs, consists of the following major divisions, named in ascending order beginning with most inferior part:

1. **Brain stem:** (Medulla oblongata , Pons , Midbrain)
2. **Cerebellum.**
3. **Diencephalon:** (Thalamus, hypothalamus, Epithalamus)
4. **Cerebrum.**

The major parts in the brain is also divided into:

1. **Forebrain**

   Cerebrum.

   Diencephalon.

2. **Midbrain.**

3. **Hindbrain.**

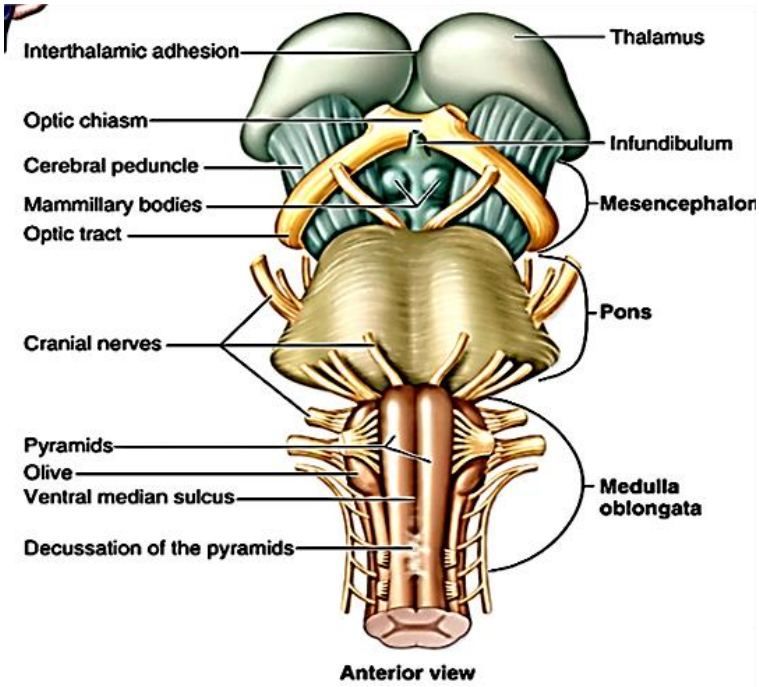
   Pons .

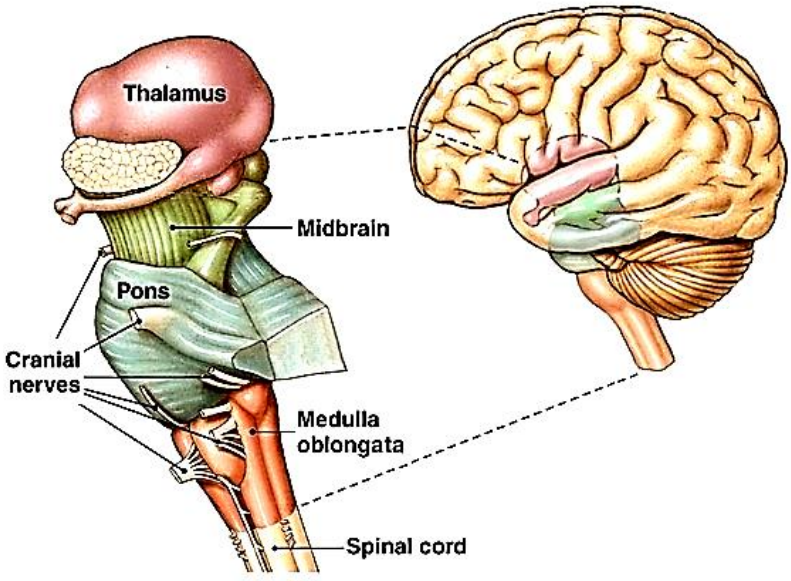
   Medulla oblongata.

   Cerebellum.

Brain Stem

The lowest part of the brain stem is the medulla oblongata. Immediately superior to the medulla lies the pons and superior to that the midbrain. Together these three structures are called the brain sterm.





The medulla oblongata is an enlarged, upward extension of the spinal cord , about 3 cm long. It lies just inside the cranial cavity superior to the large hole in the occipital bone called the foramen magnum. Like the spinal cord, the medulla consists of gray and white matter, but their arrangement differs in the two organs. In the medulla, bits of gray matter mix closely and intricately with white matter to form the reticular formation (reticular means "netlike"). In the spinal cord, gray and white matter does not intermingle; gray matter forms the interior core of the cord, and white matter surrounds it.

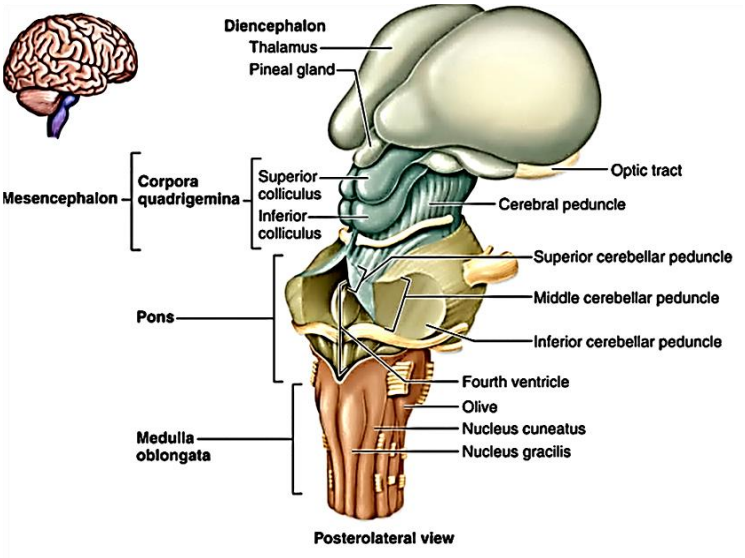
❖ **The core of the medulla contains:**

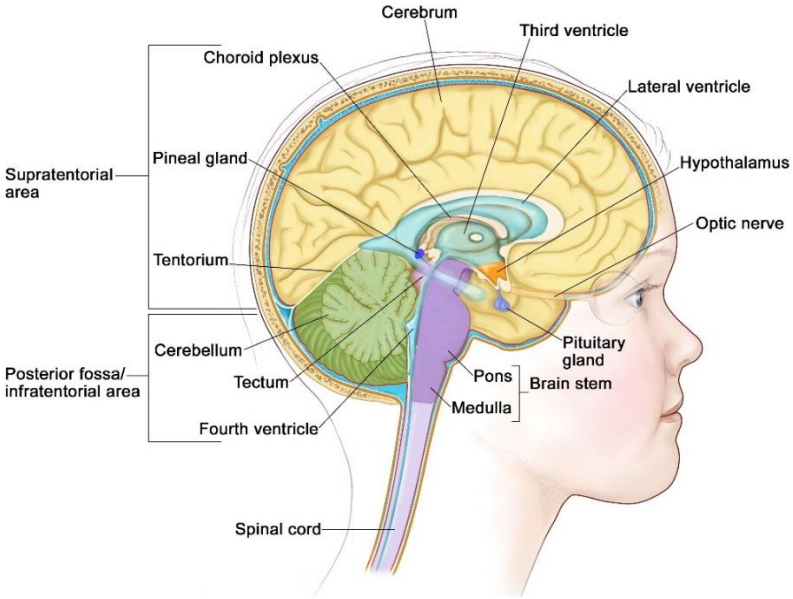
Much of the reticular formation

- Nuclei influence autonomic functions

Visceral centers of the reticular formation include:

- Cardiac center
- Vasomotor center
- The medullary respiratory center
- Centers for hiccupping, sneezing, swallowing, and coughing

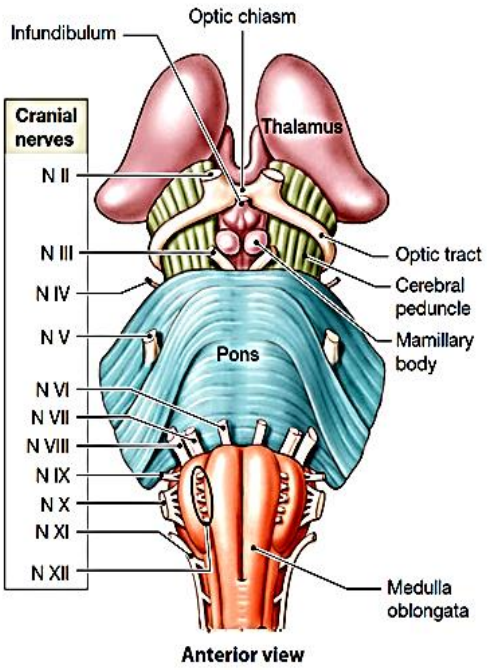




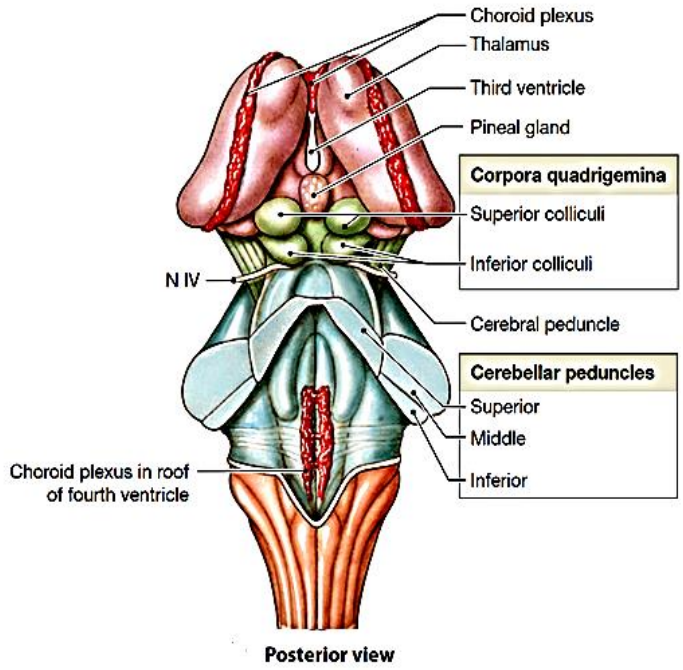
The pons and midbrain, like the medulla, consist of white matter and scattered bits of gray matter.

Rounded bulge on underside of brainstem Between medulla oblongata and midbrain Helps regulate rate and depth of breathing Relays nerve impulses to and from medulla oblongata and cerebellum

All three parts of the brain stem function as two-way conduction paths. Sensory fibers conduct impulses up from the cord to other parts of the brain, and motor fibers conduct impulses down from the brain to the cord. The cardiac, respiratory, and vasomotor centers (collectively called the vital centers) are located in the medulla. Impulses from these centers control heartbeat, respirations and blood vessel diameter (which is important in regulating blood pressure). In addition, many, important reflex centers lie in the brain stem.



Anterior view

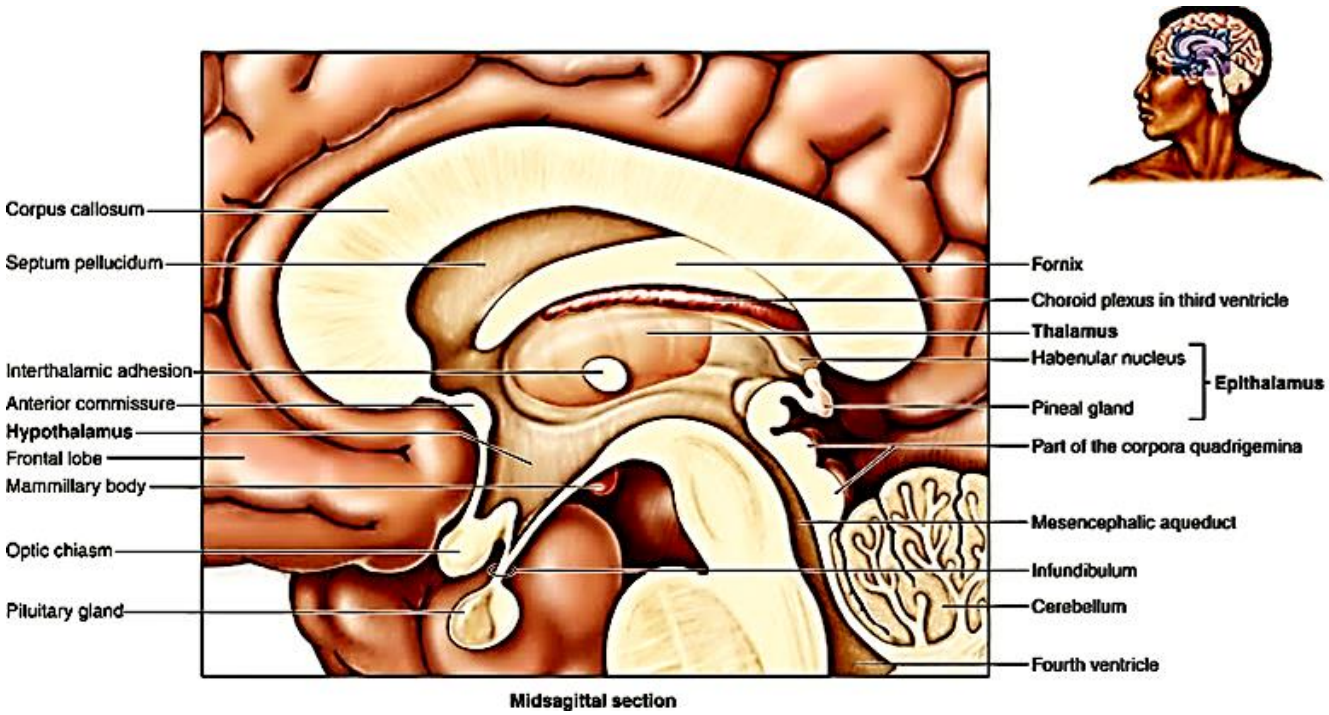


Posterior view

Diencephalon

The diencephalon is a small but important part of the brain located between the midbrain inferiorly and the cerebrum superiorly. Composed of three paired structures: **Hypothalamus Thalamus, and Epithalamus.** The ventricles of the diencephalons is the 3rd ventricle.

Hypothalamus. The hypothalamus, as its name suggests, is located inferior to the thalamus. The posterior pituitary gland, the stalk that attaches it to the undersurface of the brain, and areas of gray matter located in the side walls of a fluid-filled space called the third ventricle are extensions of the hypothalamus.



Midsagittal section

Measured by size, it is one of the least significant parts of the brain, but measured its contribution to healthy survival; it is one of the most important brain structures.

Impulses from neurons whose dendrites and cell bodies lie in the hypothalamus are conducted by their axons to neurons located in the spinal cord, and many of these impulses are then relayed to muscles and glands all over the body.

Thus the hypothalamus exerts a major control over virtually all-internal organs. Among the vital functions that it helps control are the heartbeat, constriction and dilation of blood vessels, and contractions of the stomach and intestines.

Some neurons in the hypothalamus function in a surprising way; they make the hormones that the posterior pituitary gland secretes into the blood. Because of one of these hormones (called antidiuretic hormone or ADH) affects the volume of urine excreted, the hypothalamus plays an essential role in maintaining the body's water balance.

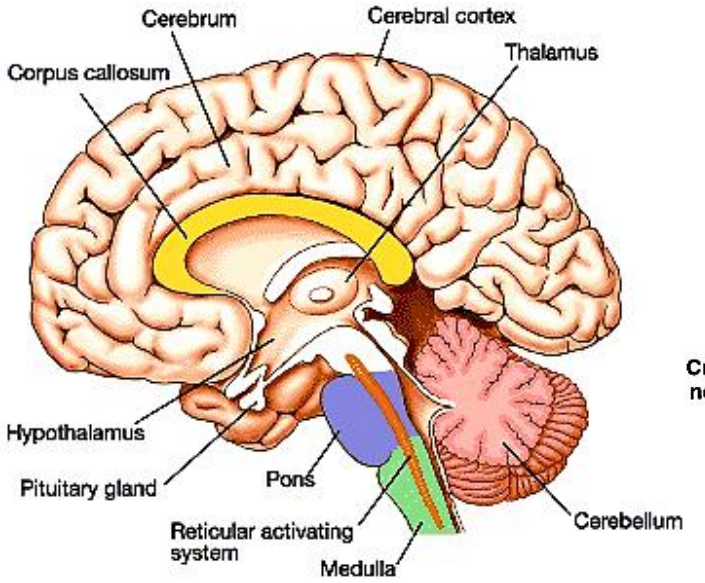
Some of the neurons in the hypothalamus function as endocrine glands. Their axons secrete chemicals called releasing hormones into the blood, which then carries them to the anterior pituitary gland. Releasing hormones, as their name suggests, control the release of certain anterior pituitary hormones.

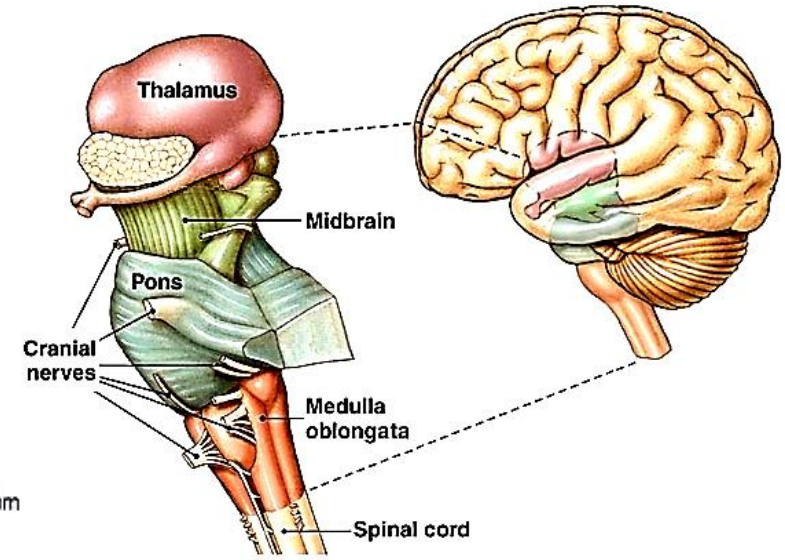
These in turn influence the hormone secretion of other endocrine glands.

Thus, the hypothalamus indirectly helps control the functioning of every cell in the body.

The hypothalamus is a crucial part of the mechanism for maintaining body temperature. Therefore, a marked elevation in body temperature in the absence of disease frequently characterizes injuries or other abnormalities of the hypothalamus. In addition, this important center is involved in functions such as the regulation of water balance; sleep cycles, and the control of appetite and many emotions involved in pleasure, fear, anger, sexual arousal, and pain.

Thalamus. Just superior to the hypothalamus is a dumbbell shaped section or largely gray matter called the thalamus.





Each enlarged end of the dumbbell lies in a lateral wall of the third ventricle.

The thin center section of the thalamus passes from left to right through the third ventricle.

The thalamus is composed chiefly of dendrites and cell bodies of neurons that have axons extending up to the sensory areas of the cerebrum. It performs the following functions :

1. It helps produce sensations. Its neurons relay impulses to the cerebral cortex from the sense organ of the body.

2. It associates sensations with emotions. Almost all sensations are accompanied by a feeling of some degree of pleasantness or unpleasantness. The way that these pleasant and unpleasant feelings are produced is unknown except that they seem to be associated with the arrival of sensory impulses in thalamus.

3. It plays a part in the so -called arousal or alerting mechanism.

4. It contains important nuclei such as medial geniculate which is responsible for auditory sense and lateral geniculate which is responsible for vision.

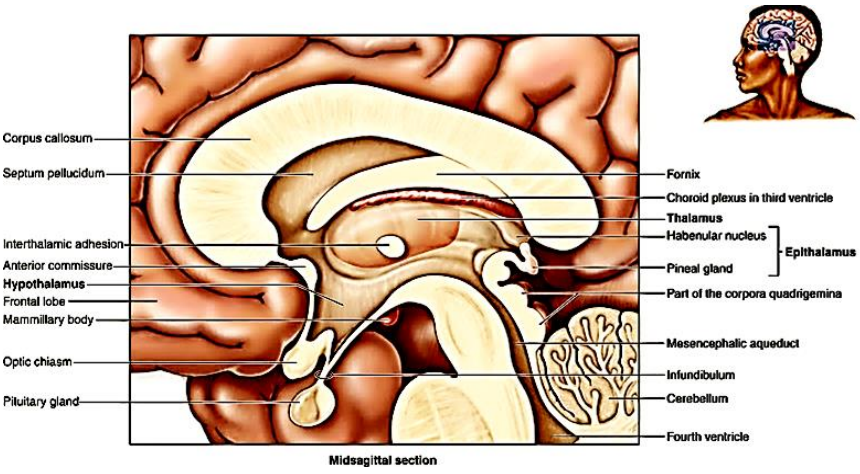
The Epithalamus:

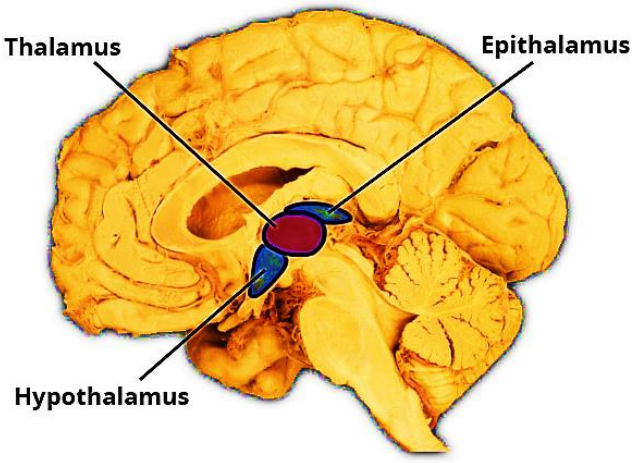
Forms part of the "roof" of the third ventricle

Consists of a tiny group of nuclei

Includes the pineal gland (pineal body)

- Secretes the hormone melatonin
- Under influence of the hypothalamus





Cerebellum

Structure. The cerebellum is the second largest part of the human brain. It lies under the occipital lobe of the cerebrum. In the cerebellum, gray matter composes the outer layer, and white matter composes the bulk of the interior. It consists of right and left by a narrow bridgelike vermis.

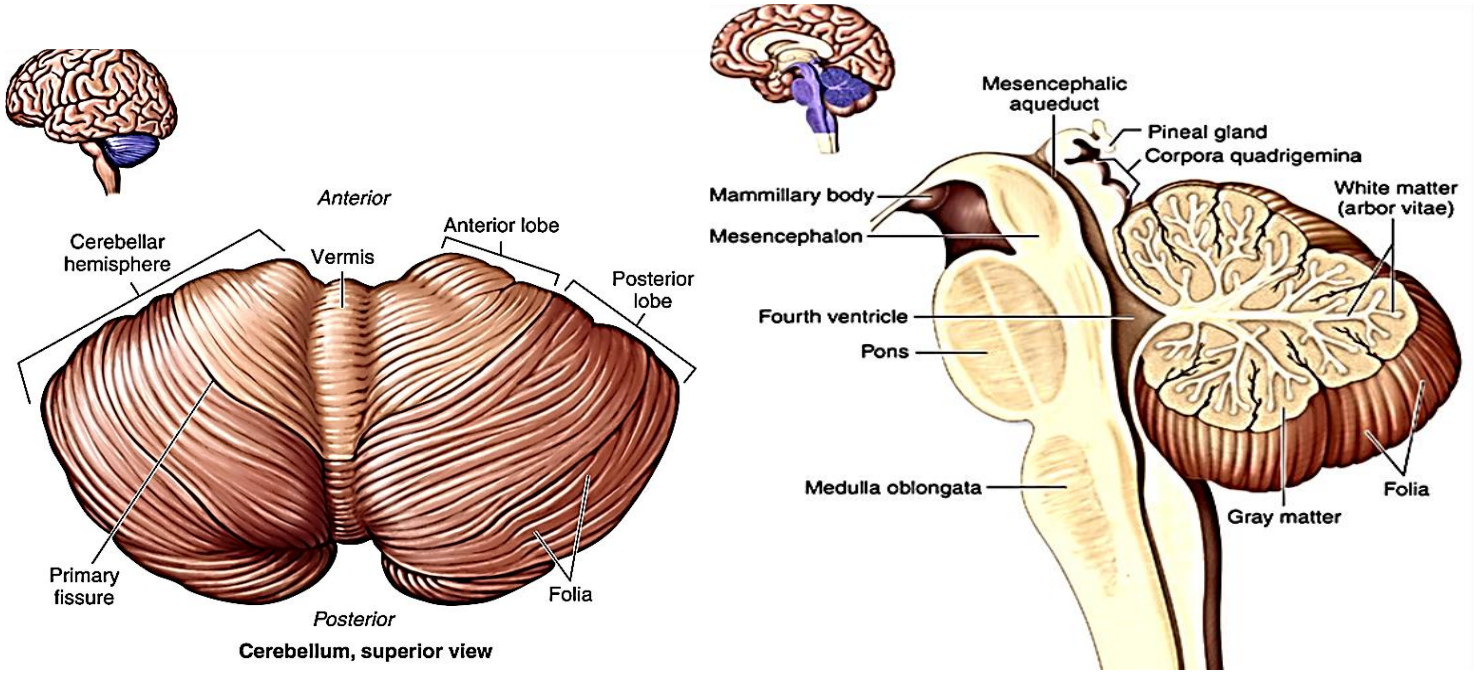
Consists of two cerebellar hemispheres Surface folded into ridges called folia Separated by fissures

Hemispheres each subdivided into:

- **Anterior lobe**
- **Posterior lobe**

The cerebellum (SER-eh-BEL-um) lies inferior to the cerebrum and occupies the posterior cranial fossa. It is also marked by gyri, sulci, and fissures.

The cerebellum is the second-largest region of the brain; it constitutes about 10% of its volume but contains over 50% of its neurons connect the cerebellum to the brainstem.



Cerebellum, superior view

Function. Most of our knowledge about cerebellar functions has come from observing patients who have some sort of disease of the cerebellum and from animals who have had the cerebellum removed. From such observations, we know that the cerebellum plays an essential part in the production of normal movements. Perhaps a few examples will make this clear.

A patient who has a tumor of the cerebellum frequently loses balance and may topple over and reel like a drunken person when walking.

It may be impossible to coordinate muscles normally. Frequent complaints about being clumsy and unable to even drive a nail or draw a straight line are typical.

With the loss of normal cerebellar functioning, the ability to make precise movements is lost.

The general functions of the cerebellum, then, are to produce smooth coordinated movements, maintain equilibrium, and sustain normal postures.

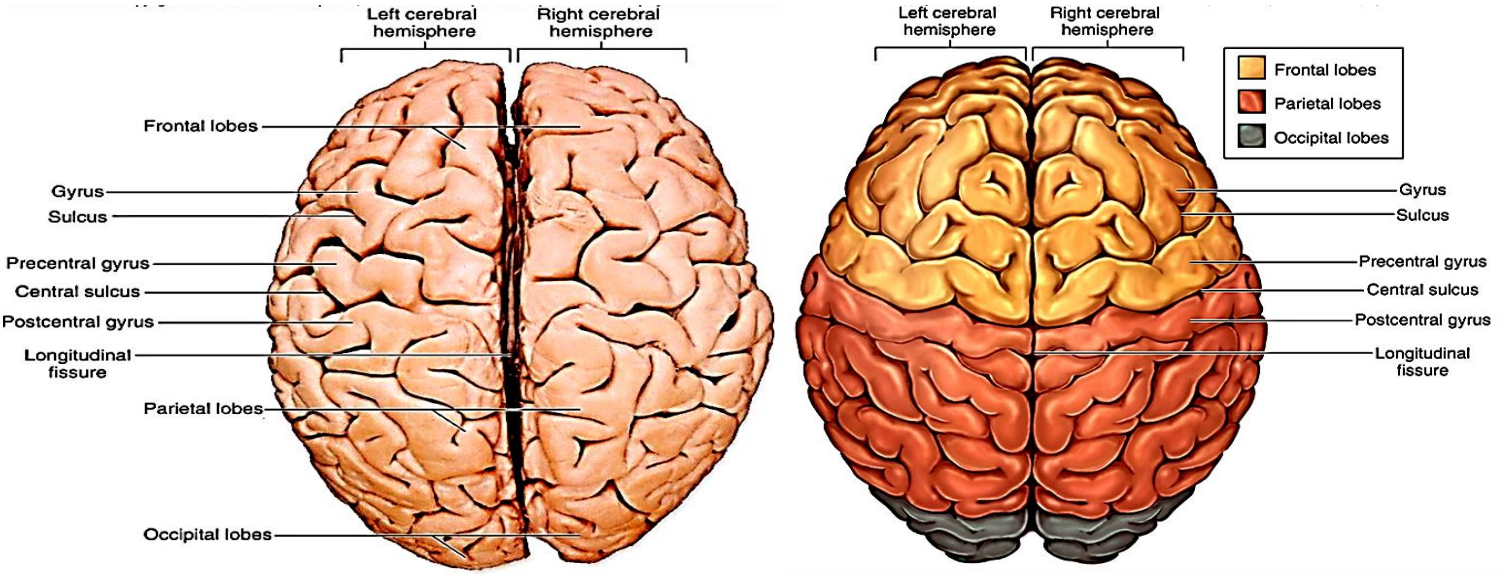
Cerebrum

The cerebrum is the largest and upper most part of the brain. The average adult brain weighs about 1,600 g in men and 1,450 g in women. Its size is proportional to body size, not intelligence.

1-The cerebrum SER-eh-brum or seh-REE-brum) constitutes about 83% of its volume and consists of a pair of cerebral hemispheres. Each hemisphere is marked by thick folds called gyri (JYrye; singular, gyrus) separated by shallow grooves called sulci (SUL-sye; singular, sulcus).

A very deep groove, the longitudinal fissure, separates the right and left hemispheres from each other .

At the bottom of this fissure, the hemispheres are connected by a thick bundle of nerve fibers called the corpus callosum a prominent landmark for anatomical description.



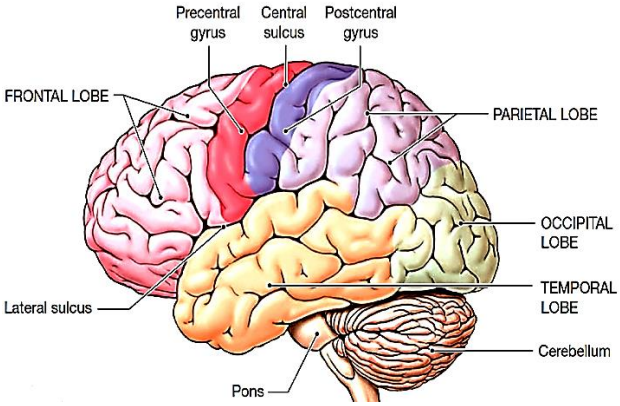
The ridges are called convolutions or gyri, and the grooves are called sulci. The deepest sulci are called fissures; the longitudinal fissure divides the cerebrum into right and left halves or hemispheres. These halves are almost separate structures except for their lower mid portions, which are connected by a structure called the corpus callosum .

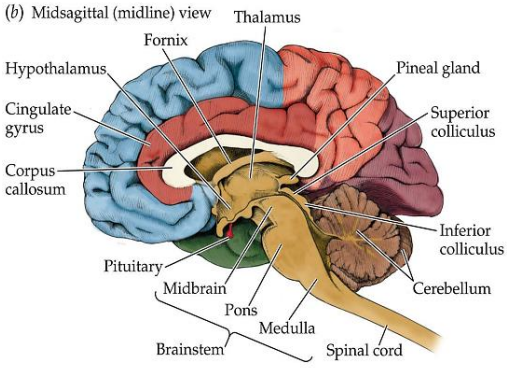
Two deep sulci subdivide each cerebral hemisphere into four major lobes and each lobe into numerous convolutions. The lobes are named for the bones that lie over them: the frontal lobe, the parietal lobe, the temporal lobe, the occipital lobe and insula. A thin layer of gray matter, made up of neuron dendrites and cell bodies, composes the surface of the cerebrum.

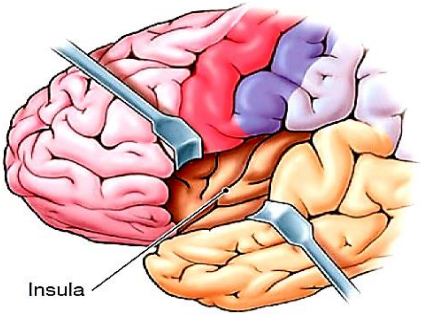
Each hemisphere divided into 5 lobes:

1. **Frontal**
2. **Parietal**
3. **Occipital**

4. **Temporal**
5. **Insula**



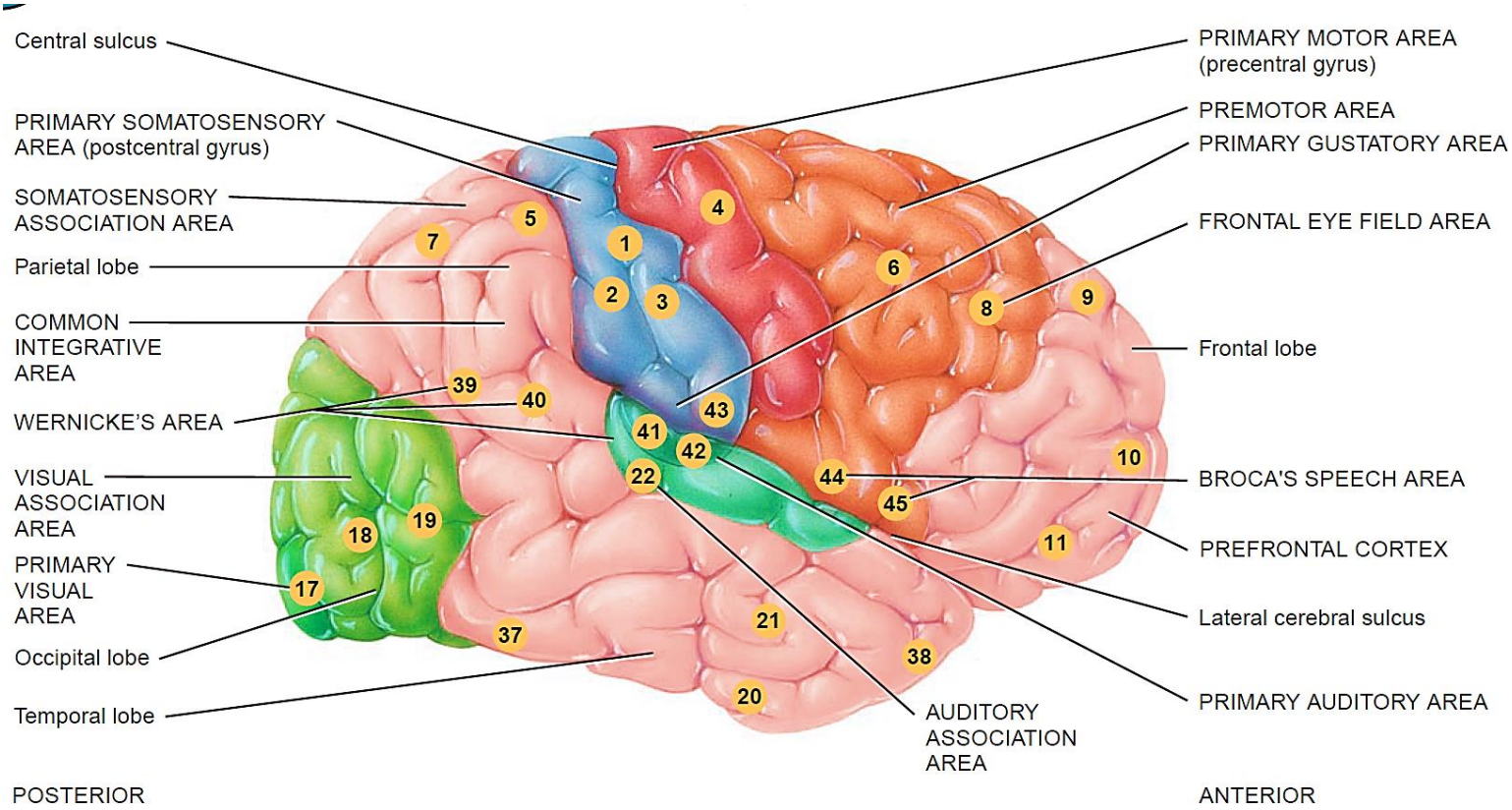




Its name is the cerebral cortex. White matter made up of bundles of neuronal fibers (tracts), composes most of the interior of the cerebrum. Within this white matter, however, are a few islands of gray matter known as the basal ganglia, whose functioning is essential for producing automatic movements and postures.

☼ **Cerebral cortex:**

Composed of gray matter Neuronal cell bodies, dendrites, and short axons
Folds in cortex – triples its size
Approximately 40% of brain's mass
Brodmann areas – 52 structurally distinct areas



1.The frontal lobe: lies immediately behind the frontal bone, superior to the orbits.

Its posterior boundary is the central sulcus .

The frontal lobe is chiefly concerned with voluntary motor functions, motivation, foresight, planning, memory, mood, emotion, social judgment, and aggression.The nerve fibres of the white matter of the cerebral hemispheres are of three groups: commissural, association and projection.

2. The parietal: lobe forms the upper most part of the brain and underlies the parietal bone.

Its rostral boundary is the central sulcus and its caudal boundary is the parieto-occipital sulcus, visible on the medial surface of each hemisphere

This lobe is concerned with the sensory reception and integration of some sthetic,taste, and some visual information.

3. The occipital lobe is at the rear of the head underlying the occipital bone. It is the principal visual center of the brain.

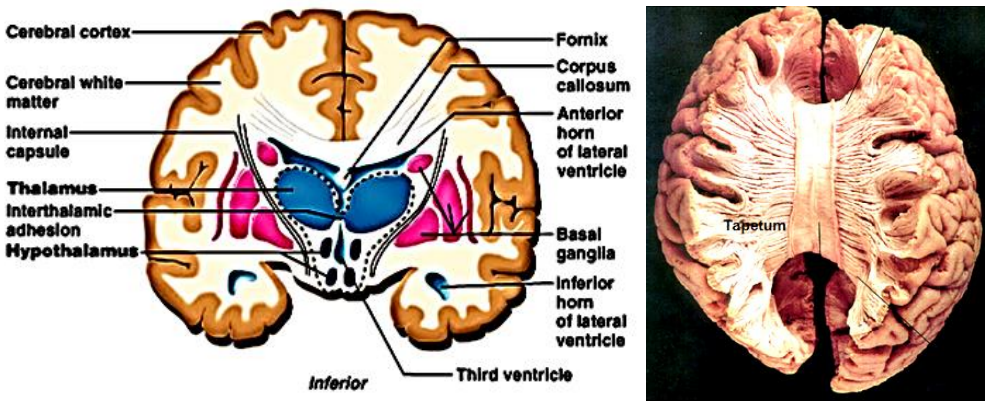
4. The temporal lobe is a lateral, horizontal lobe deep to the temporal bone, separated from the

parietal lobe above it by a deep lateral sulcus. It is concerned with hearing, smell, learning, memory, visual recognition, and emotional behavior

5. The insula is a small mass of cortex deep to the lateral sulcus, made visible only by retracting or cutting away some of the overlying cerebrum, It is not as accessible to study in living people as other parts of the cortex and is still little-known territory. It apparently plays roles in understanding spoken language, in the sense of taste, and in integrating sensory information from visceral receptors.

Commissural fibres unite corresponding areas of the cortex of the two hemispheres across the midline. They comprise the corpus callosum, and the anterior and fornix commissures.

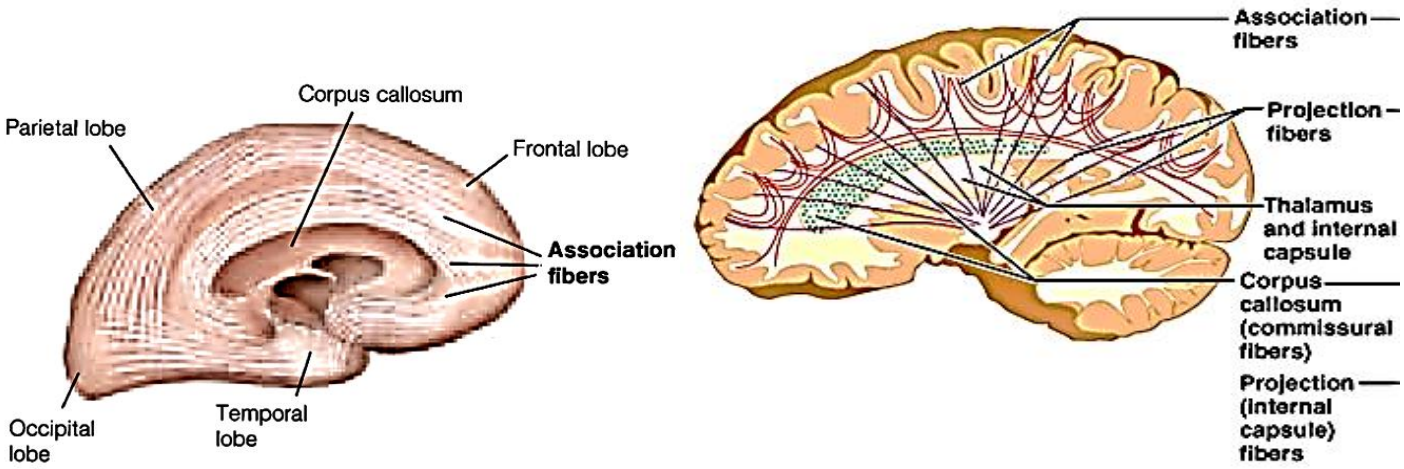
The corpus callosum is a broad band of fibres passing between corresponding cortical areas of the two hemispheres.

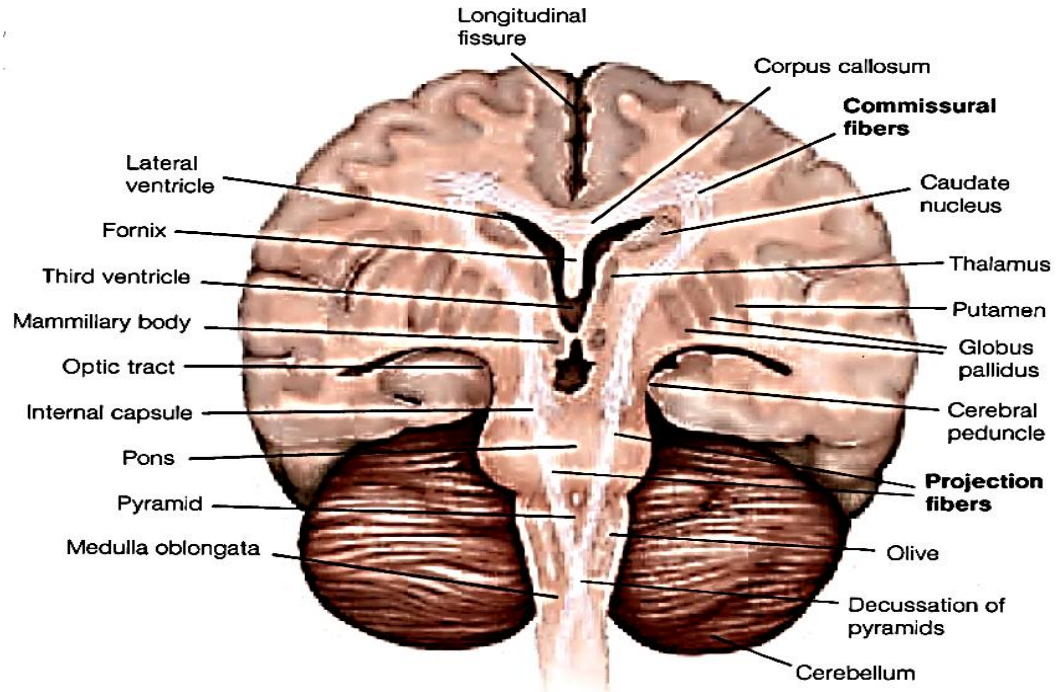


It lies at the base of the median longitudinal fissure and above the diencephalons and midbrain. In midsagital section it is the shape of a hook lying horizontally with its bend anteriorly and its point downwards. The pointed portion is known as the rostrum, the bend as genu, the horizontal part as the body and the expanded posterior end as the splenium.

The callosum extends laterally into each hemisphere; the anterior fibres pass forwards into the frontal pole and are known as the forceps major, passes backwards into the occipital poles. The rostrum of the corpus callosum fuses inferiorly with the lamina terminalis. A bundle of fibres within the lamina , the anterior commissure , unites the piriform areas and the olfactory tracts of the two sides.

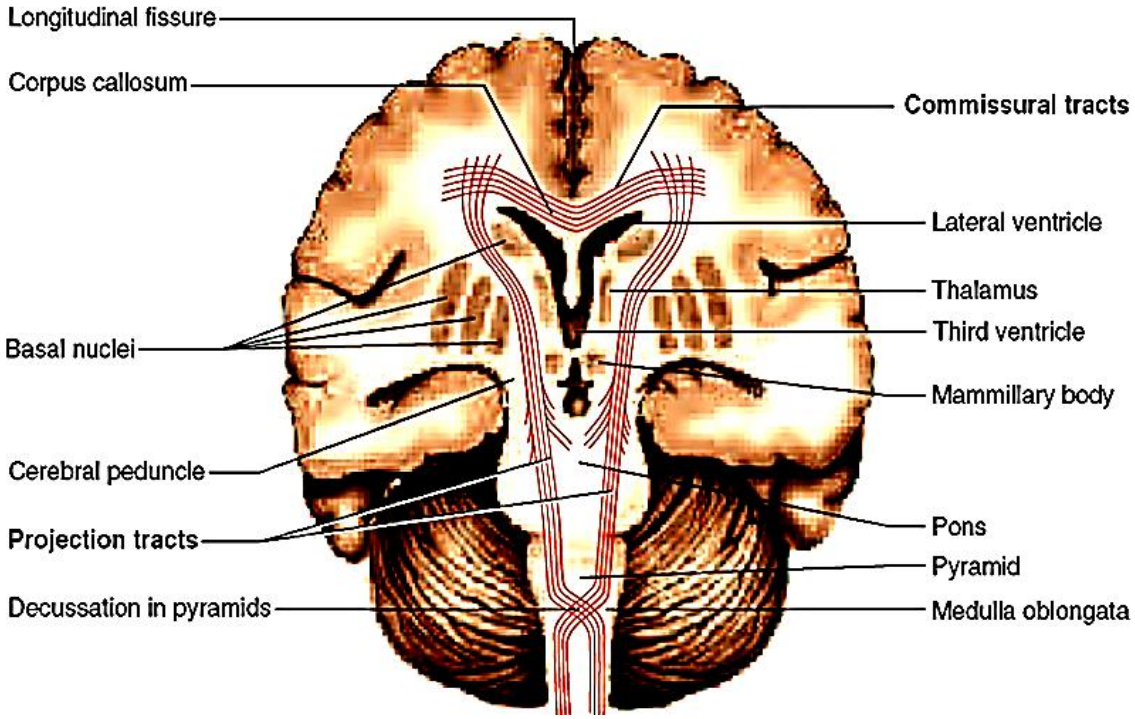
The fornix (hippocampal) commissure is found on the undersurface of the corpus callosum where the two crura meet and form the fornix. Fibres here pass across the midline between the two hippocampi.





Association fibres may be long or short, uniting adjacent or widely separated gyri of the same hemisphere.

Projection fibres ascend from or descend to lower lying parts of the central nervous system. Many form a well defined layer, the internal capsule, between the lentiform nucleus laterally and the thalamus and caudate nucleus medially.



Superiorly its fibres fan out as the corona radiate interdigitating with the fibres of the corpus callosum. In horizontal section the capsule is V-shaped. It possesses an anterior limb (between the caudate nucleus and the lentiform nucleus and crossed by fibres and grey matter uniting the two structures), an apex (the genu) pointing medially, and a posterior limb lying between the thalamus and the lentiform nucleus.

The anterior limb carries:

(a) frontopontine fibres from the frontal lobe to the pons, and

(b) fibres from the thalamus (medial and ventro-anterior nuclei) to the frontal lobe.

The posterior limb carries from before backwards,

a) Pyramidal fibres from the motor cortex which pass to the cranial nerve nuclei (corticospinal fibres),

b) Somatosensory fibres passing from thalamus (ventroposterior nucleus) to the postcentral (somatosensory) cortex,

c) Temporopontine fibres from the temporal lobe to the pons,

d) The auditory radiations passing from the medial geniculate body under the lentfiform nucleus, to the superior temporal gyrus,

e) The visual radiations passing from the lateral geniculate body around the lateral aspect of the posterior horn of the lateral ventricle to the visual cortex. The course of the fibres is such that many cross the midline (decusate) and end on the opposite (contralteral) side.

Some fibres however end on the same (ipsilateral) side. The motor areas of each hemisphere control the voluntary muscles of the contralateral side of the body and the sensory areas receive information from the contralateral side. What functions does the cerebrum perform? This is a hard

question to answer briefly because the neurons of the cerebrum do not function alone. They function with many other neurons m many other parts of the brain and in the spinal cord. Neurons of these structures continually bring impulses to cerebral neurons and continually transmit impulses away from them. If all other neurons were functioning normally and only cerebral neurons were not functioning, here are some of the things that you could not do. You could not think or use your will. You could not remember anything that has ever happened to you. You could not decide to make the smallest movement, nor could you make it. You would not see or hear. You could not experience any of the sensations that make life so rich and varied. Nothing would anger or frighten you, and nothing would bring you joy or sorrow. You would, in short, be unconscious. These terms, then sum up cerebral functions: Consciousness, thinking, memory, sensations, emotions, and willed movements. Figure7- 6, B, shows the areas of the cerebral cortex essential for willed movements, general sensations, vision, hearing, and normal speech.

It is important to understand that very specific areas of the cortex have very specific functions. For example, the temporal lobe's auditory areas interpret incoming nervous signals from the ear as very specific sounds. The visual area of the cortex in the occipital lobe helps you identify and understand specific images. Localized areas of the cortex are directly related to specific functions, as shown in Figure 7-6, B. This explains the very specific symptoms associated with an injury to localized areas of the cerebral cortex after a stroke or traumatic injury to the head.

Spinal Cord

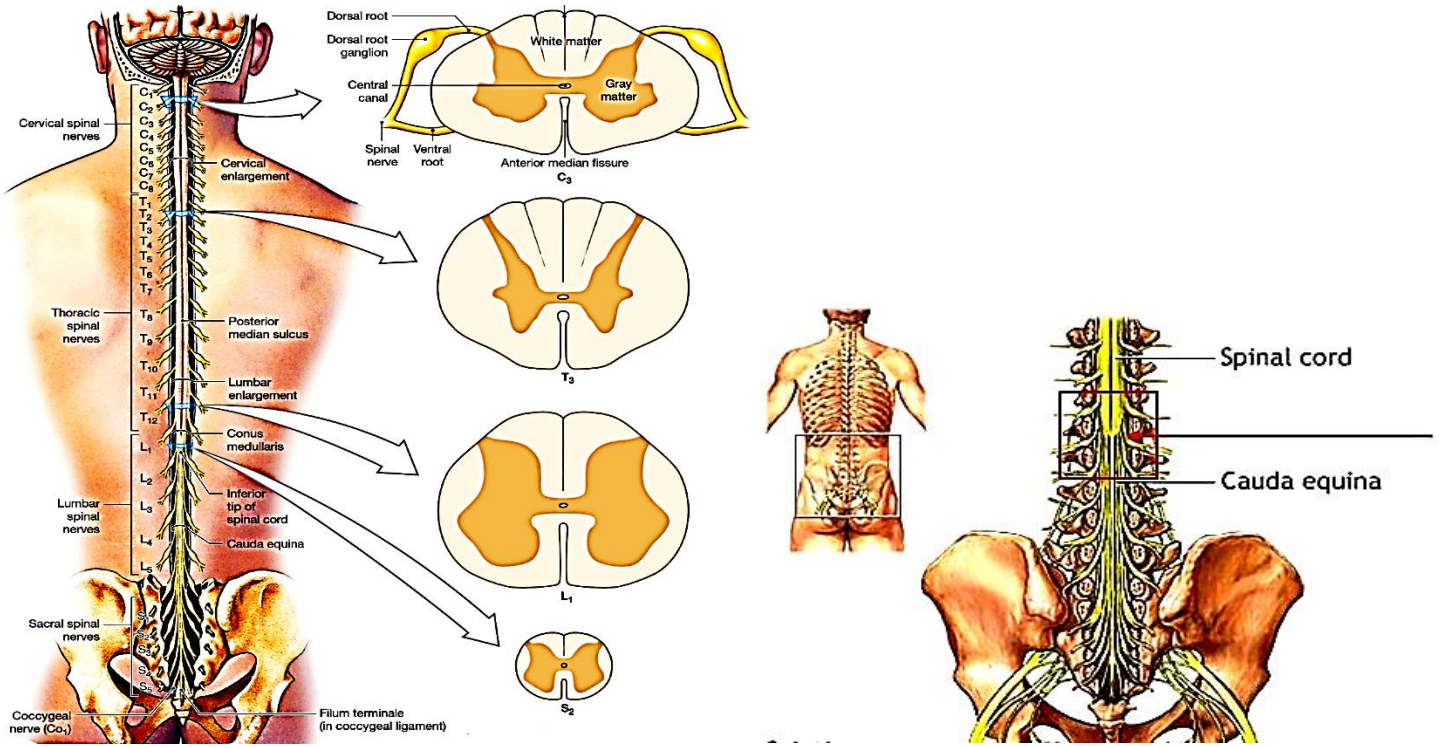
Location of the Spinal Cord

In the embryo, the spinal cord occupies the entire spinal canal and so extends down into the tail portion of the vertebral column. However, the column of bone grows much more rapidly than the nerve tissue of the cord, so that eventually the end of the cord no longer reaches the lower part of the spinal canal. This disparity in growth continues to increase; in the adult the cord ends in the region just below the area to which the last rib attaches (between the first and the second lumbar vertebrae.

Structure of the Spinal Cord

The spinal cord lies within the vertebral canal and extends from the foramen magnum to the level of the second lumbar vertebrae after which a fibrous remnant, the filum terminale, descends to be attached to the back of the coccyx. The cord is about 45 cm long. It is cylindrical in shape, flattened slightly

anteroposteriorly, and has cervical and lumbar enlargements where the nerves supplying the upper and lower limb originate the enlargements lie opposite the lower cervical and lower thoracic vertebrae. Since the spinal cord is shorter than the vertebral canal, the nerves descend with increasing obliquity before leaving the canal through the intervertebral foramina.



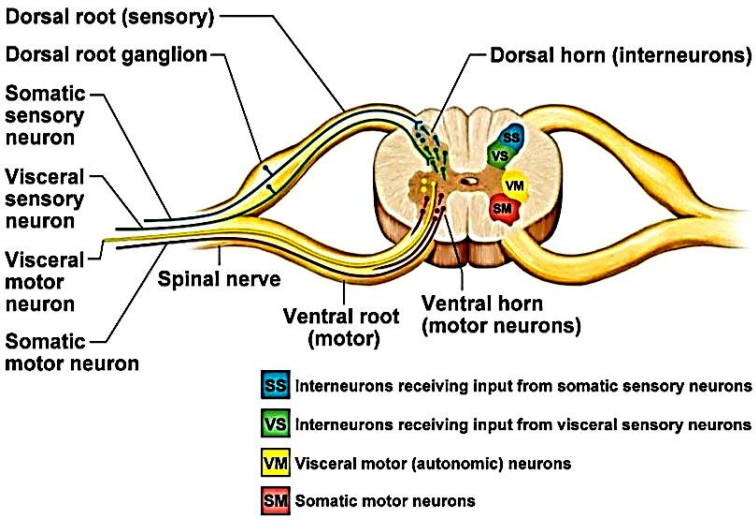
The collection of lower lumbar, sacral and coccygeal nerves below the spinal cord, with the filum terminale, is known as the cauda equina.

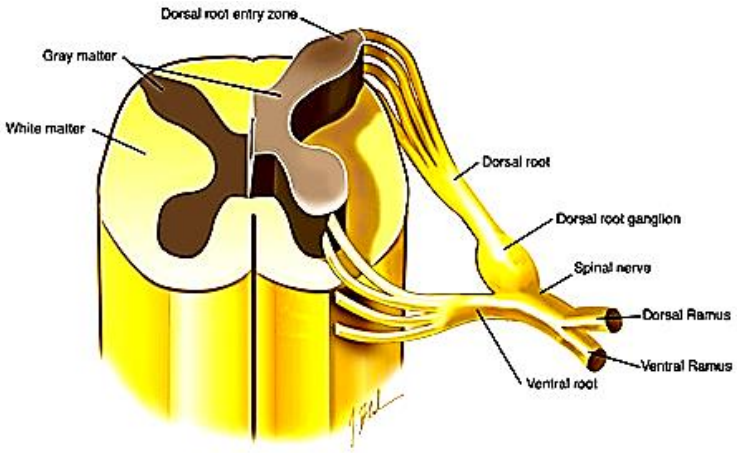
The cord has an anterior median fissure and a posterior median sulcus. On its sides the rootlets of the spinal nerves emerge from anterolateral and posterolateral sulci.

The spinal cord has a small, irregular shaped internal section that consists of gray matter (nerve cell bodies) and a larger area surrounding this gray part that consists of white matter (nerve cell fibers).

- **The gray matter** is so arranged that a column of cells extend up and down dorsally, one on each side; another column is found in the ventral region on each side.

  These two pairs of columns, called the dorsal and ventral horns, give the gray matter an H-shaped appearance in cross section. In the center of the gray matter is a small channel, central canal that contains cerebrospinal fluid, the liquid that circulates around the brain and spinal cord.





- **The white matter** consists of thousands of nerve cell fibers arranged in three areas external to the gray matter on each side.

Functions of the Spinal Cord

The spinal cord is the link between the spinal nerves and the brain. It is also a place where simple responses, known as reflexes can be coordinated even without involving the brain.

The functions of the spinal cord may be divided into three categories:

1. Conduction of sensory impulses upward through ascending tracts to the brain

2. Conduction of motor impulses from the brain down through descending tracts to the efferent neurons that supply muscles or glands

3. Reflex activities. A reflex is a simple, rapid, and automatic response involving very few neurons. When you fling out an arm or leg to catch your balance, withdraw from a painful stimulus, or blink to avoid an object approaching your eyes, you are experiencing reflex behaviour. A reflex pathway that passes through the spinal cord alone and does not involve the brain is termed a spinal reflex.

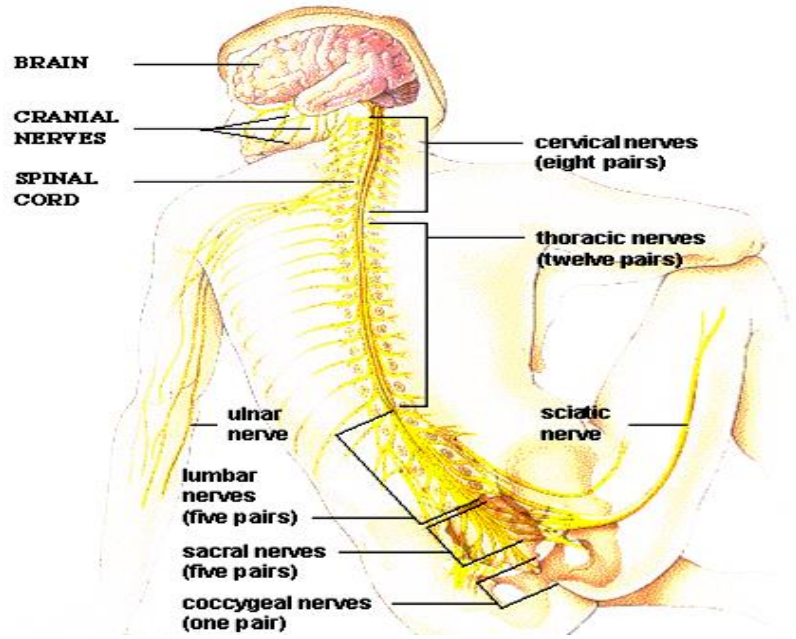
The stretch reflex, in which a muscle is stretched and responds by contracting, is one example. If you tap the tendon below the kneecap (the patellar tendon), the muscles of the anterior thigh (quadriceps femoris) contracts, eliciting the knee jerk. Such stretch reflexes may be evoked by appropriate tapping of most large muscles (such as the triceps brachii in the arm and the gastrocnemius in the calf of the leg). Because reflexes occur automatically, they are used in physical examinations to test the condition of the nervous system.

BRAIN AREA	FUNCTION
Brain stem	
Medulla oblongata	Two-way conduction pathway between the spinal cord and higher brain centers; cardiac, respiratory, and vasomotor control center
Pons	Two-way conduction pathway between areas of the brain and other regions of the body; Influences respiration
Midbrain	Two-way conduction pathway; relay for visual and auditory Impulses
Diencephalon	
Hypothalamus	Regulation of body temperature, water balance, sleep cycle control appetite, and sexual arousal
Thalamus	Sensory relay station from various body areas to cerebral cortex; emotions and alerting or arousal mechanisms
Cerebellum	Muscle coordination; maintenance of equilibrium and posture
Cerebrum	Sensory perception, emotions willed movements, consciousness, and memory

Peripheral Nervous System

The nerves connecting the brain and the spinal cord to other parts pf the body constitutes the peripheral nervous system (PNS). This system includes cranial and spinal nerves that connect the brain and spinal cord, respectively, to peripheral structures such as the skin surface and the skeletal muscles.

In addition, other structures in the autonomic nervous system(ANS) are considered part of the PNS. These connect the brain and spinal cord to various glands in the body and to the cardiac and smooth muscle in the thorax and abdomen.



Nerves

A nerve is a bundle of nerve cell fibers located outside the CNS. Bundles of nerve cell fibers within the CNS are tracts. Tracts are located within the brain and also within the spinal cord to conduct impulses to and from the brain. A nerve or tract can be compared to an electric cable made up of many wires. As with muscles, the "wires," or nerve cell fibers in a nerve, are bound together with connective tissue.

A few of the cranial nerves have only sensory fibers for conducting impulses toward the brain. These are described as sensory, or afferent, nerves. A few of the cranial nerves contain only motor fibers for conducing impulses away from the brain and are classified as motor, or efferent, nerves.

However, the remainder of the cranial nerves and all of the spinal nerves contain both sensory and motor fibers and are referred to as mixed nerves.

❖ **Cranial Nerves Location of the Cranial Nerves**

Cranial nerves are nerves that are attached to the brain. There are 12 pairs of cranial nerves (henceforth, when a cranial nerve is identified, a pair is meant).

They are numbered according to their connection with the brain; beginning at the front and proceeding back .

The first 9 pairs and the 12th pair supply structures in the head. General Functions of the cranial nerves From the functional point of view, we may think of the kinds of messages the cranial nerves handle as belonging to one of four categories:

1. Special sensory impulses, such as for smell, taste, vision, and hearing

2. General sensory impulses, such as those for pain, touch, temperature, deep muscle sense, pressure, and vibrations

3. Somatic motor impulses resulting in voluntary control of skeletal muscles

4. Viscera motor impulses producing involuntary control of glands and involuntary muscles (cardiac and smooth muscle). These motor pathways are part pf the autonomic nervous system, parasympathetic division.

Names and Functions of the Cranial Nerves

The 12 cranial nerves are always numbered according to the traditional Roman style. A few of the cranial nerves−I, II, and VIll−contain only sensory fibers; some −III, IV, VI, XI and Xll−contain all or mostly motor fibers.

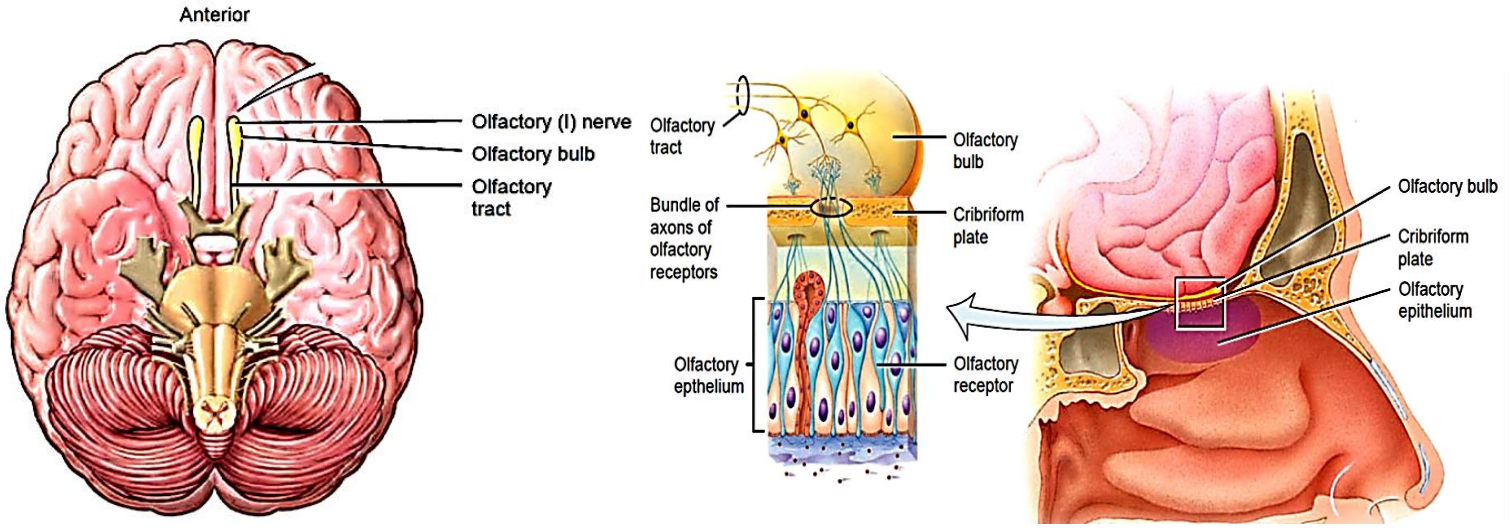
The remainder−V, VII, IX, and X− contain both sensory and motor fibers; they are known as mixed nerves.

All 12 nerves are listed below:

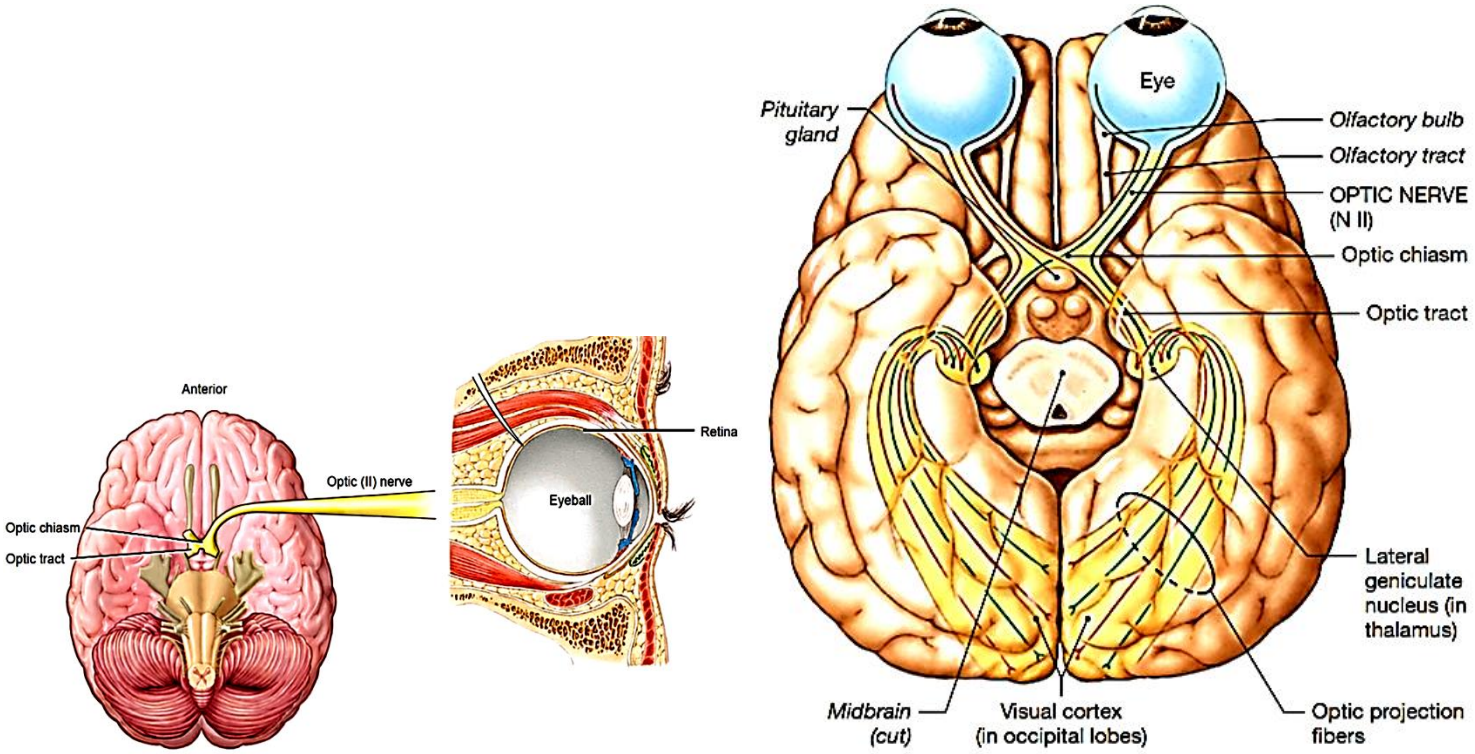
#	Cranial nerve name	Function	Specific function
I	**Olfactory**	S	Smell
II	**Optic**	S	Vision
III	**Oculomotor**	M, P	**Motor** to 4 extrinsic eye muscles and upper eyelids **Parasympathetic:** constrict pupil
IV	**Trochlear**	M	**Motor** to one extrinsic eye muscles
V	**Trigeminal**	M, S	**Sensory:** to face and teeth, **Motor:** to muscles of mastication
VI	**Abducens**	M	**Motor** to one extrinsic eye muscle
VII	**Facial**	M, S, P	**S: taste, M: muscles of facial expression,** **Parasympathetic:** salivary and tear glands
VIII	**Vestibulocochlear**	S	Hearing and balance
IX	**Glossopharyngeal**	M, S, P	**Sensory:** taste and touch to back of tongue **Motor**: pharyngeal muscles **Parasympathetic:** salivary gland
X	**Vagus**	S, M, P	**Sensory:** pharynx, larynx and viscera **Motor**: Palate, pharynx, and larynx **Parasympathetic**: viscera of thorax and abdomen
XI	**Spinal accessory**	M	Motor to neck and upper back
XII	**Hypoglossal)**	M	Motor to tongue muscles

S: Sensory, M: Motor, P: Parasympathetic

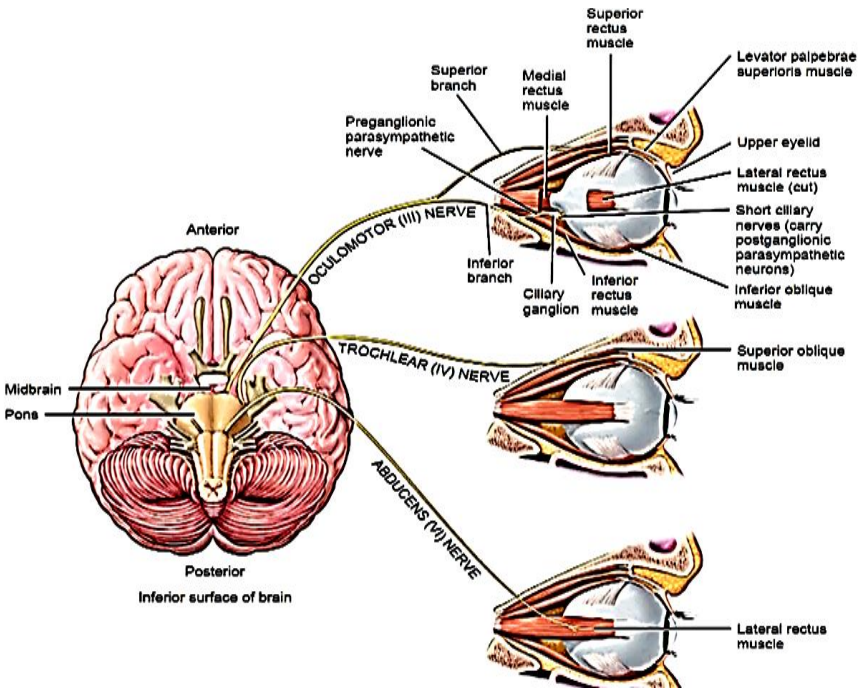
I. The olfactory nerve they supply the olfactory mucous membrane in the upper part of the nasal cavity. The nerve fibres originate I the bipolar olfactory cells of the mucosa and join to form 15-20 olfactory bundles which pass through the cribiform plate of the ethmoid bone to reach the olfactory bulb.



II. The optic nerve it is the sensory nerve of the retina. Its fibres originate in the ganglion layer and converge on the posterior part of the eye ball. The nerve passes backwards through the orbit and optic canal into the middle cranial fossa where it unites with the nerve of opposite side of the optic chiasma.

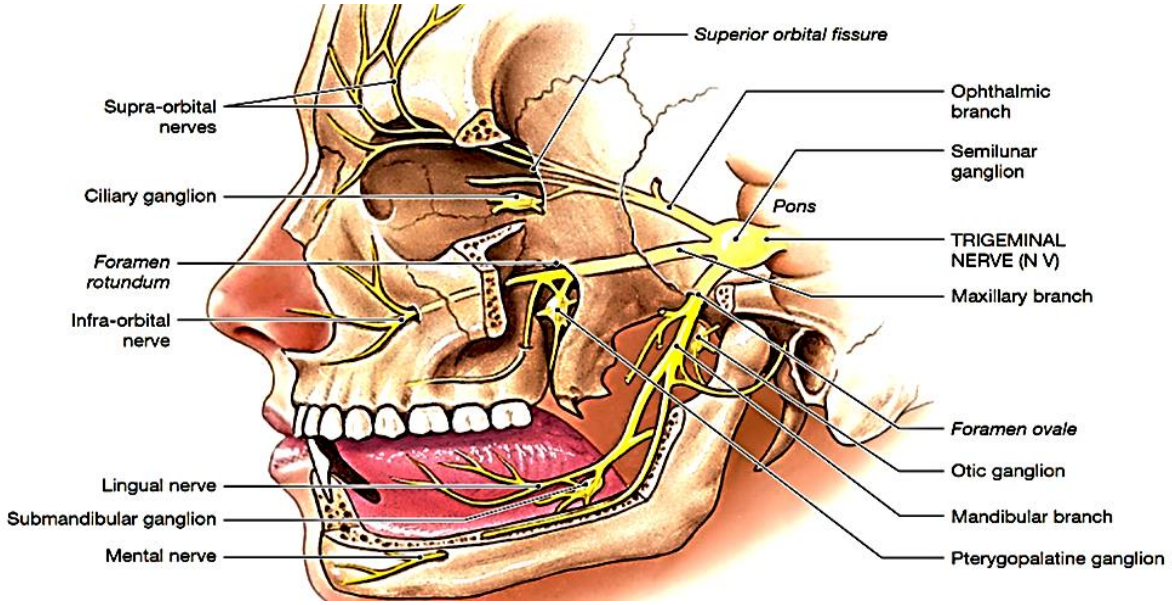


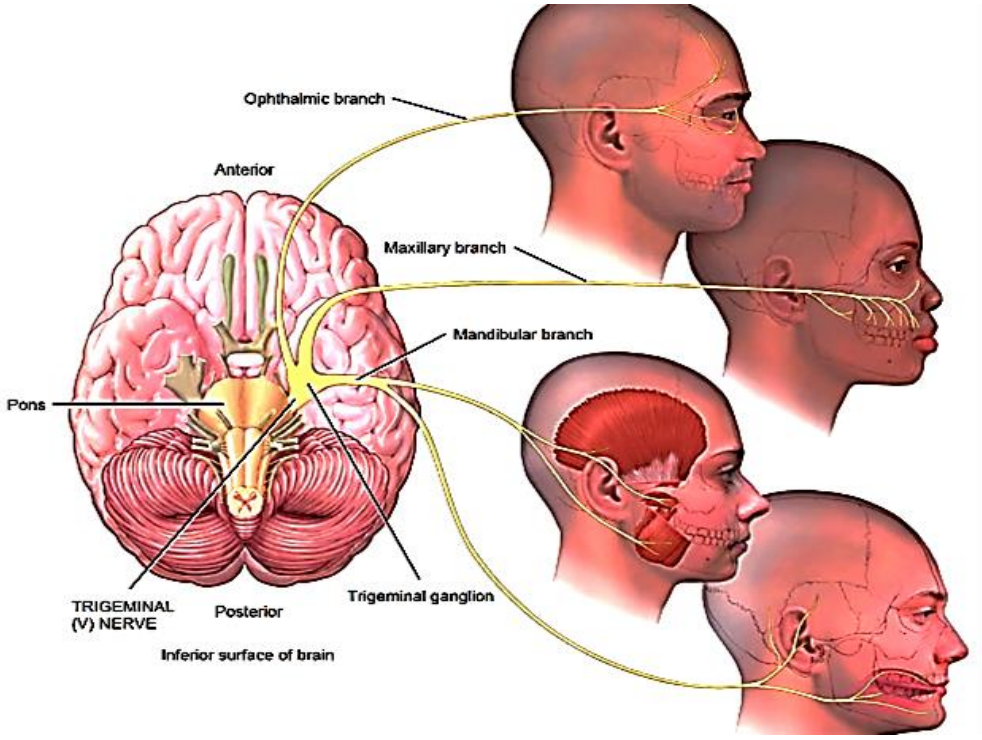
III. The oculomotor nerve this nerve has somatic motor and general visceral (parasympathetic) motor fibres. The somatic fibres supply the bulbar muscles, except superior oblique and lateral rectus. The parasympathetic fibres synapse in the ciliary ganglion and supply the sphincter pupillae and ciliary muscle. The nuclei of the nerve are situated in the upper midbrain in the peri-aqueductal grey matter. The nerve fibres pass forwards through the midbrain and leave it between the cerebral peduncles. The nerve pass through the posterior and middle cranial fossae and divides into superior and inferior divisions near the superior orbital fissure.



IV. The trochlear nerve this is the somatic motor nerve supply to the superior oblique. Its nucleus lies in the lower midbrain in the peri-aqueductal grey matter. The fibres pass posteriorly and undergo a dorsal decussation with the nerve of the opposite side caudal to the inferior colliculi the nerve then passes forwards through the posterior and middle cranial fossae,enters the orbit through the superior orbital fissure and supplies superior oblique.

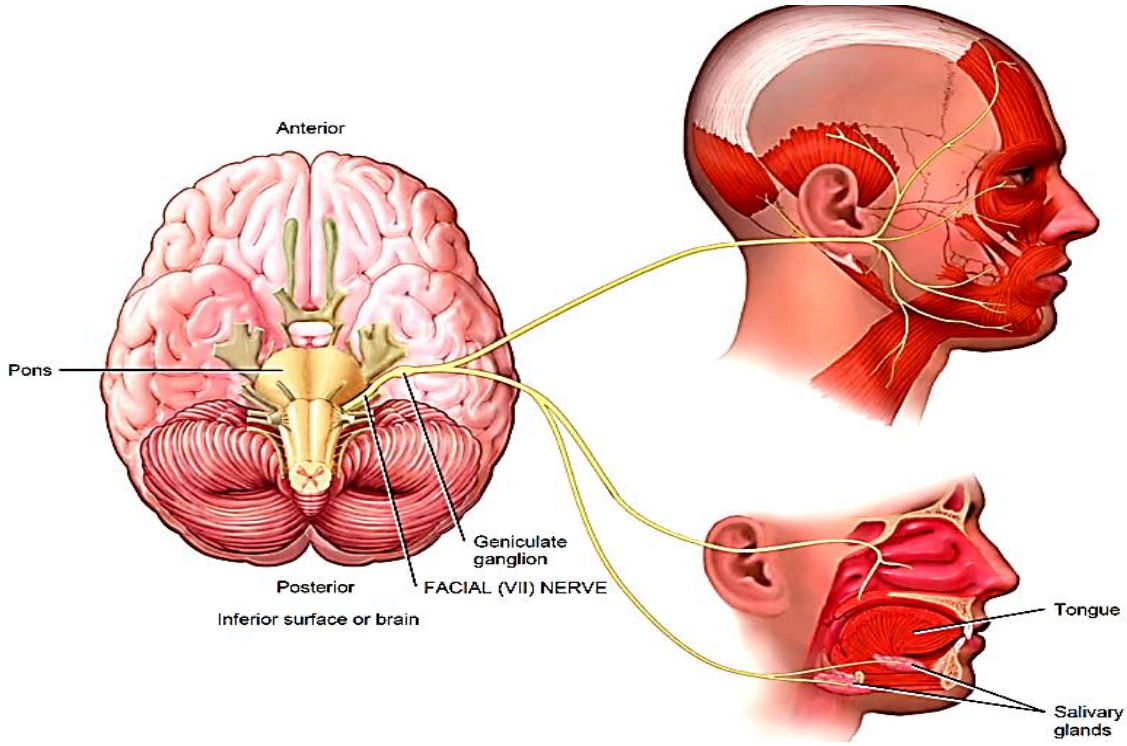
V. The trigeminal nerve is the great sensory nerve of the face and head. It has three branches that transport general sense impulses (e.g., pain, touch, temperature) from the eye, the upper jaw, and the lower jaw. The third branch is joined by motor fibers to the muscles of mastication (chewing).

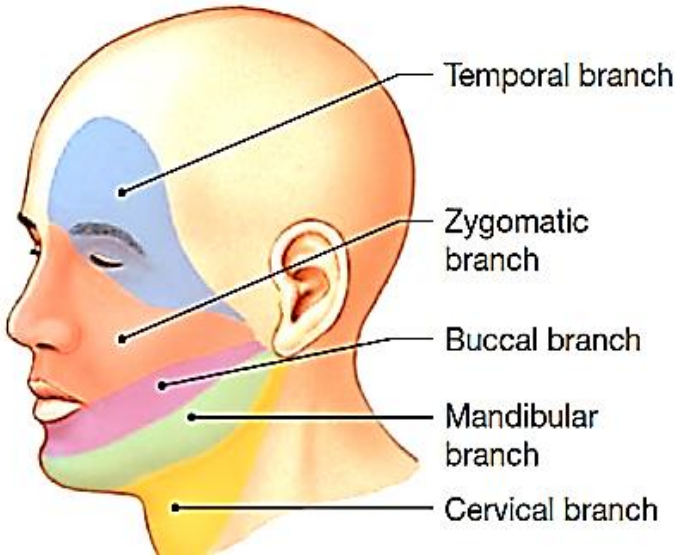




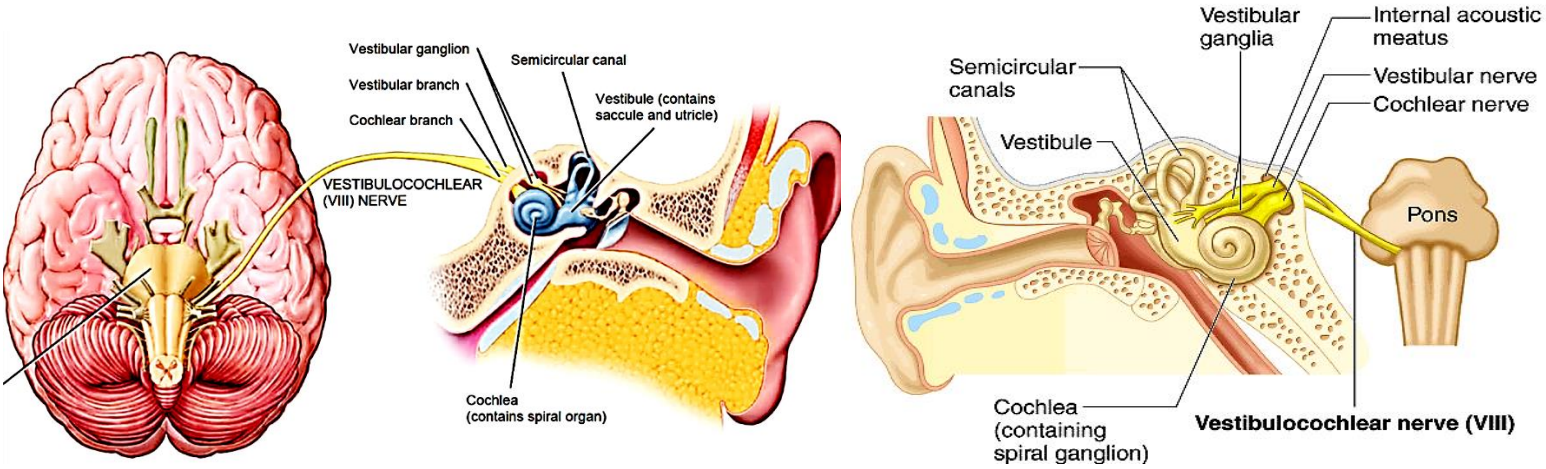
VI. The abducens nerve is a somatic motor nerve supplying lateral rectus. Its nucleus is situated in the lower pons. The nerve leaves the inferior border of the pons near the midline, passes forwards through the posterior and middle cranial fossae, the cavernous sinus and the orbit, and supplies lateral rectus.

VII. The facial nerve is largely motor. Branches from the facial nerve supply all the muscles of facial expression. This nerve also includes special sensory fibers for taste (anterior two thirds of the tongue), and it contains secretary fibers to the smaller salivary glands (the submandibular and sublingual) and to the lacrimal gland

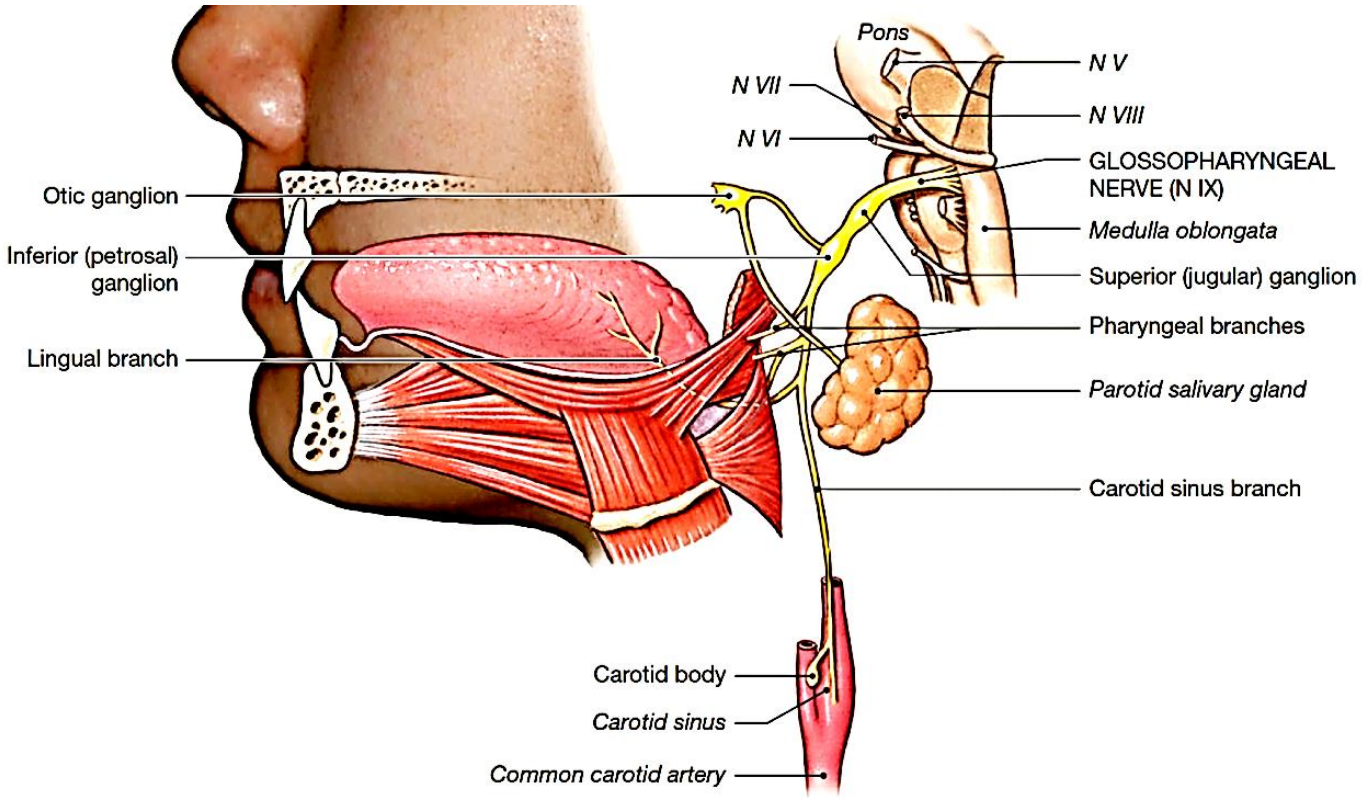




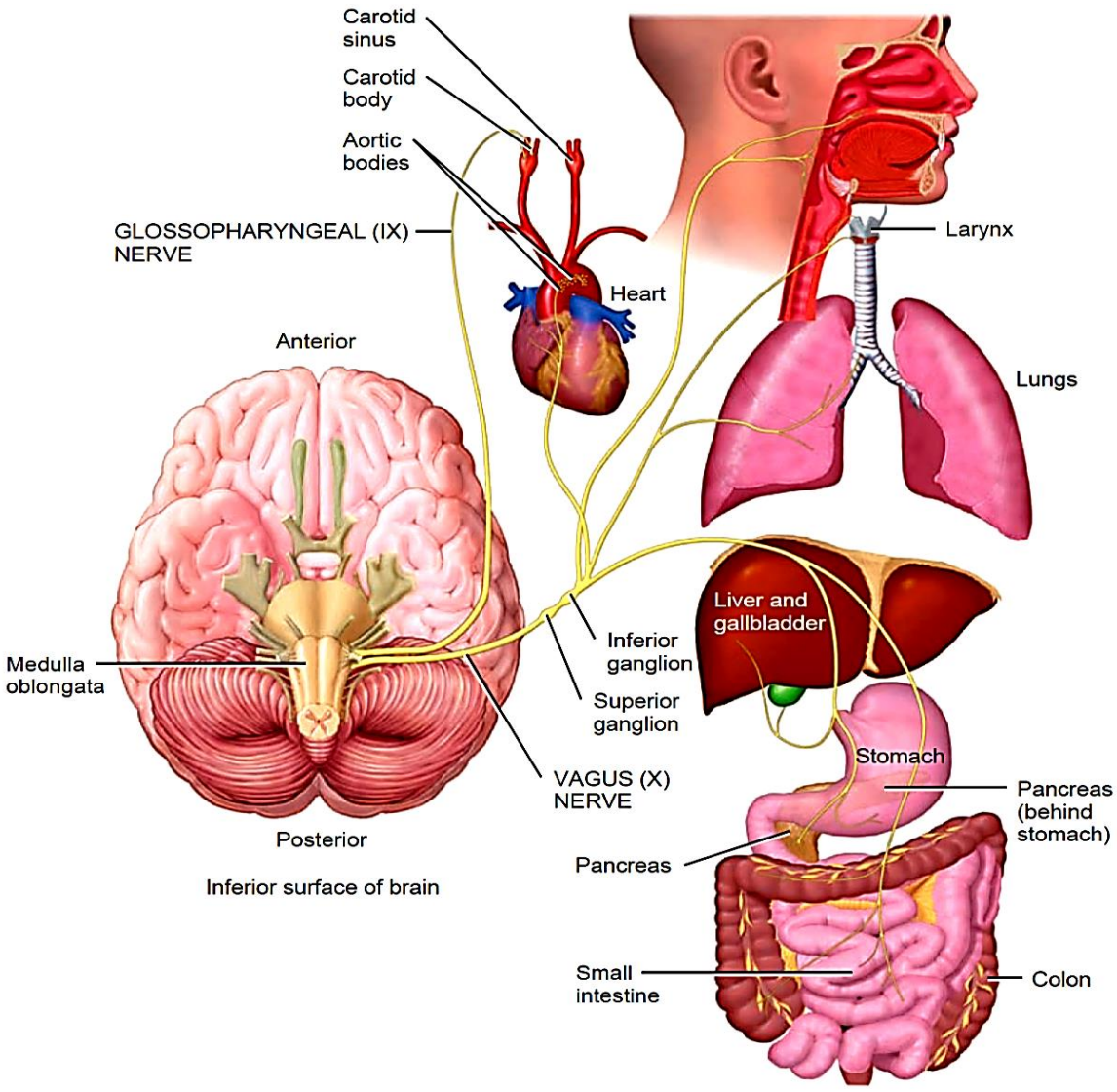
VIII. The vestibulocochlear nerve contains special sensory fibers for hearing as well as those for balance from the semicircular canals of the internal ear. This nerve is also called the auditory or acoustic nerve.



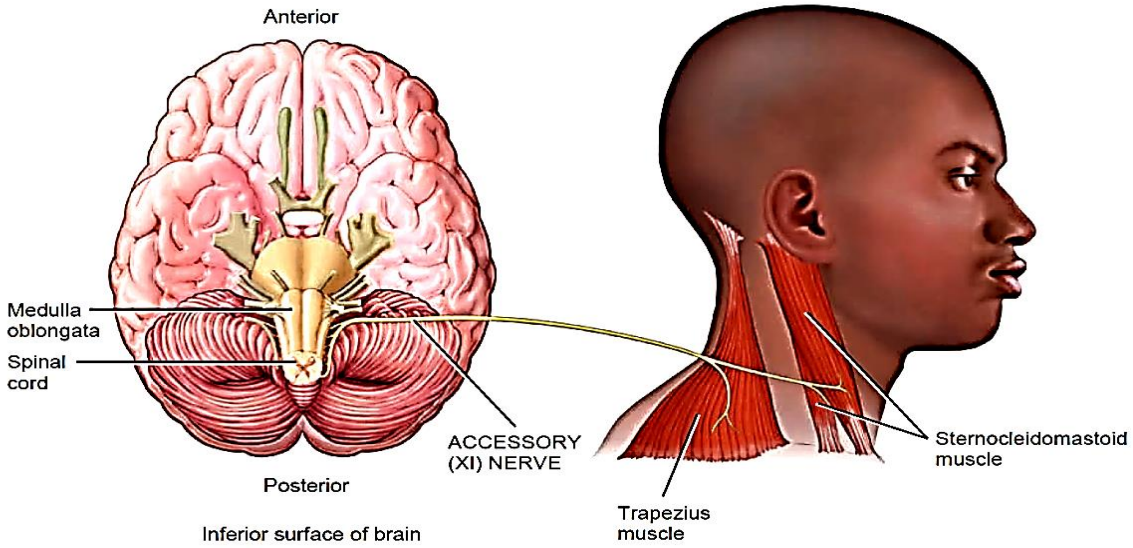
IX. The glossopharyngeal nerve contains general sensory fibers from the back of the tongue and the pharynx (throat). This nerve also contains sensory fibers for taste from the posterior third of the tongue, secretary fibers that supply the largest salivary gland (parotid), and motor nerve fibers to control the swallowing muscles in the pharynx.



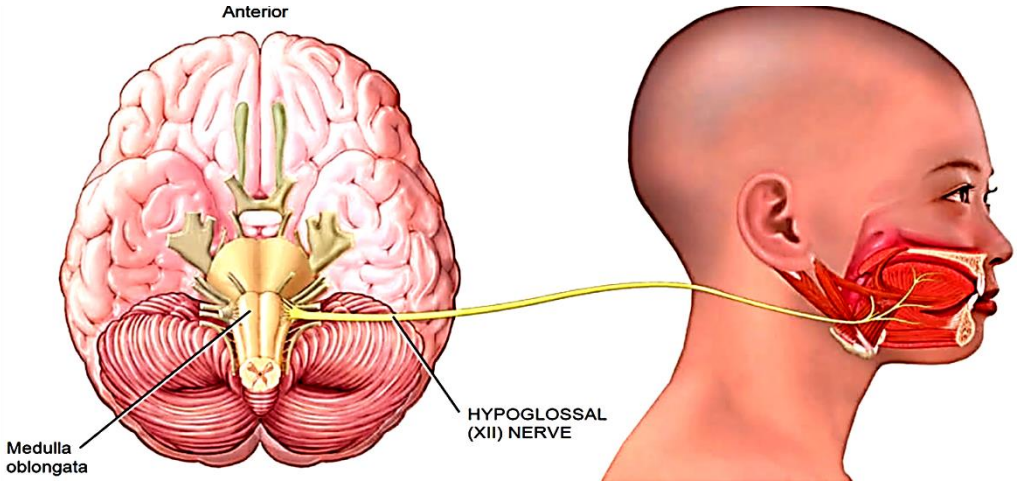
X. The vagus nerve is the longest cranial nerve. (Its name means "wanderer. ") It supplies most of the organs in the thoracic and abdominal cavities. This nerve also contains motor fibers to the larynx (voice box) and pharynx, and to glands that produce digestive juices and other secretions.



XI. The accessory nerve (formerly called the spinal accessory nerve) is a motor nerve with two branches. One branch control two muscles of the neck, the trapezius and sternocleidomastoid; the other supplies muscles of the larynx

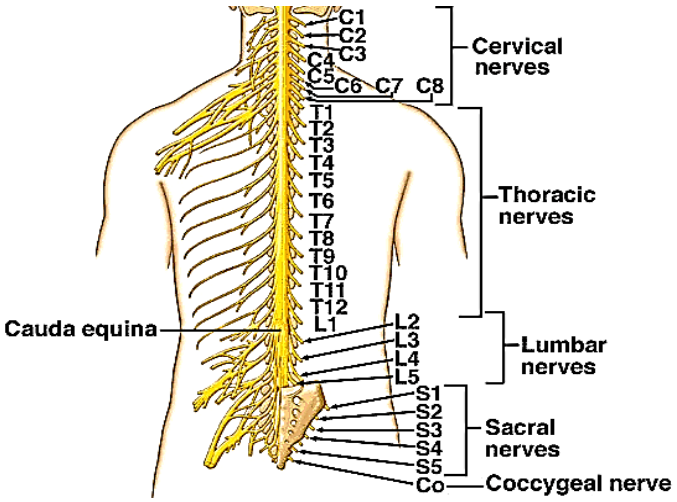


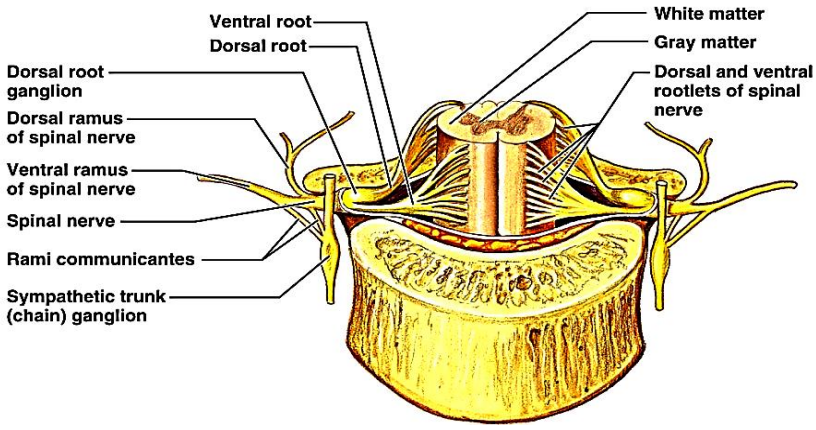
XII. The hypoglossal nerve, the last of the 12 cranial nerves, carries impulses controlling the muscles of the tongue.



Spinal Nerves

Location and Structure of Spinal Nerves Spinal nerves arise from spinal cord. There are 31 pairs of spinal nerves, each pair numbered according to the level of the spinal cord from which it arises .



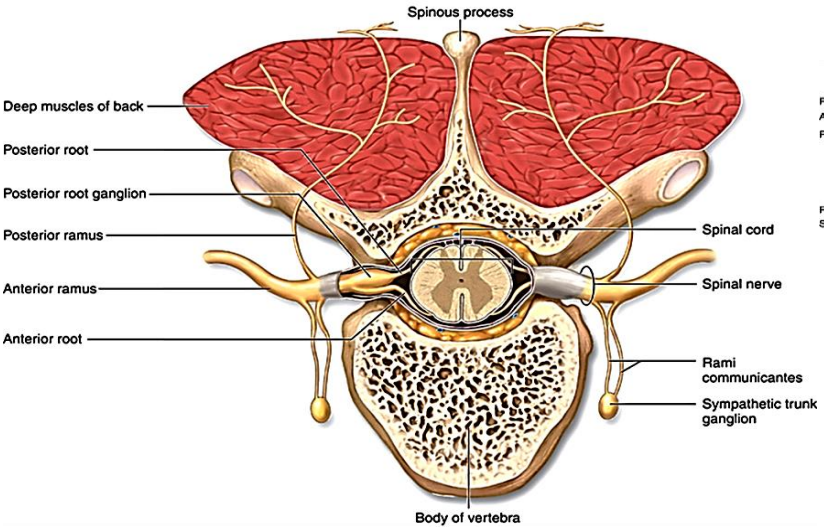


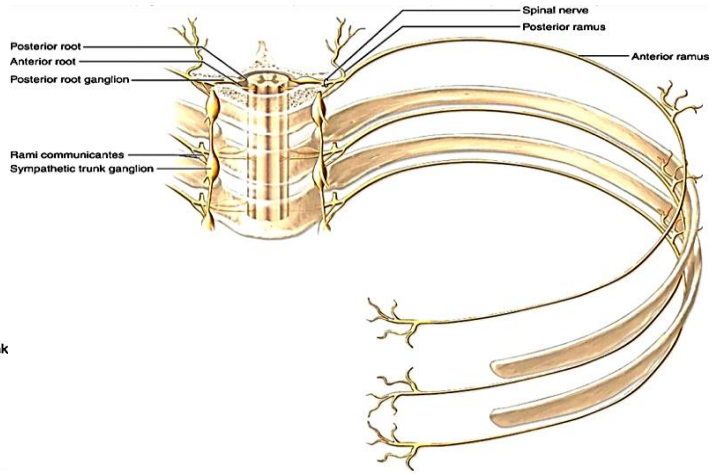
Each nerve is attached to the spinal cord by two roots; the dorsal root and the ventral root .

The roots are formed from a number of rootlets which emerge from the anterolateral and posterolateral sulci of the spinal cord. The ventral root carries efferent (motor) fibres from the cord and the dorsal root, afferent (sensory) fibres to the cord. The cell bodies of the sensory fibres are situated in a ganglion on the dorsal root.

The spinal nerves are therefore a mixture of motor and sensory fibres. Each nerve leaves the vertebral canal through an intervertebral foramen and soon divides into a large ventral and smaller dorsal ramus (branch). The adjacent ventral rami of most regions communicate to form plexuses (cervical, brachial and

lumbosacral) while those of the thoracic region become the intercostals and subcostal nerves. The dorsal rami pass backwards into the postvertebral muscles and divide into medial and lateral branches. These rami supply the muscles and skin over the posterior aspect of the body but give no branches to the limbs.





The ventral rami supply the anterior and lateral wall of the back and the lower limbs.

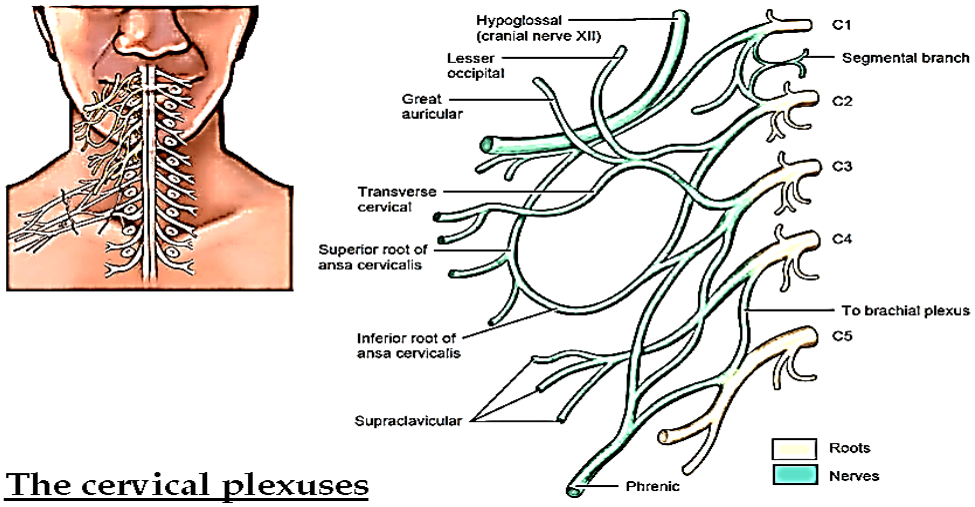
Branches of the Spinal Nerves

Each spinal nerve continues only a very short distance away from the spinal cord and then branches into small posterior divisions and rather large anterior divisions.

The larger anterior branches interlace to form networks called plexuses, which then distribute branches to the body parts.

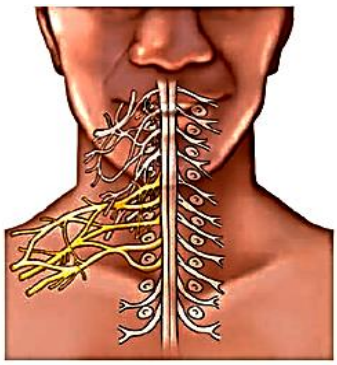
► **The three main plexuses are described as follows:**

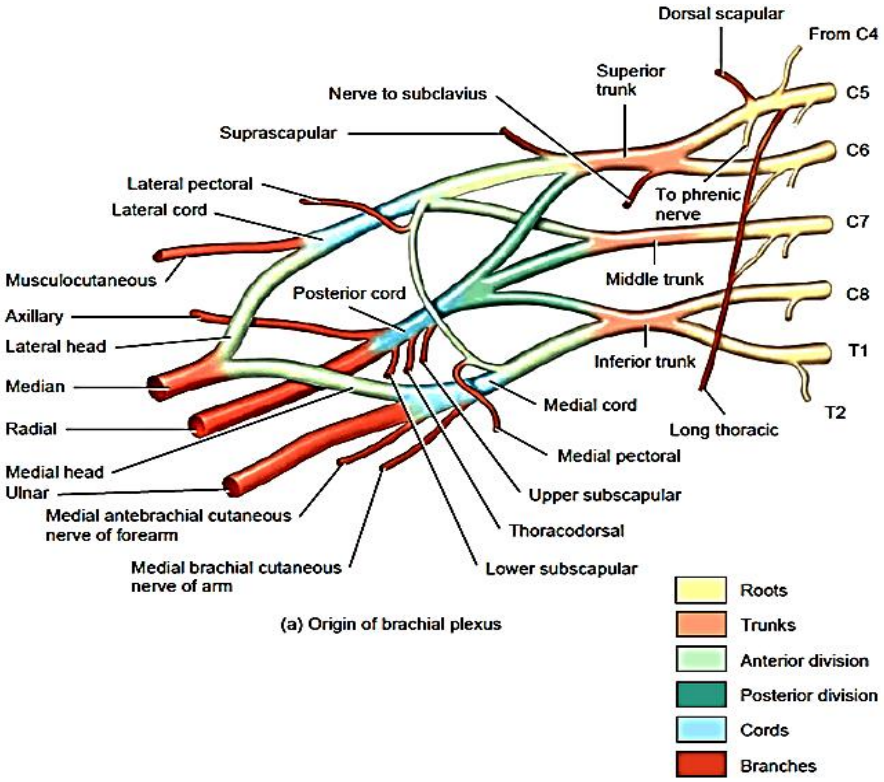
1. The cervical plexuses supplies motor impulses to the muscles of the neck and receive sensory impulses from the neck and the back of the head. The phrenic nerve, which activates the diaphragm, arises from this plexus.



The cervical plexuses

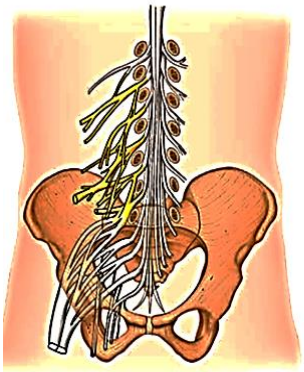
2. The brachial plexus sends numerous branches to the shoulder, arm, forearm, wrist, and hand. The radial nerve emerges from the brachial plexus.



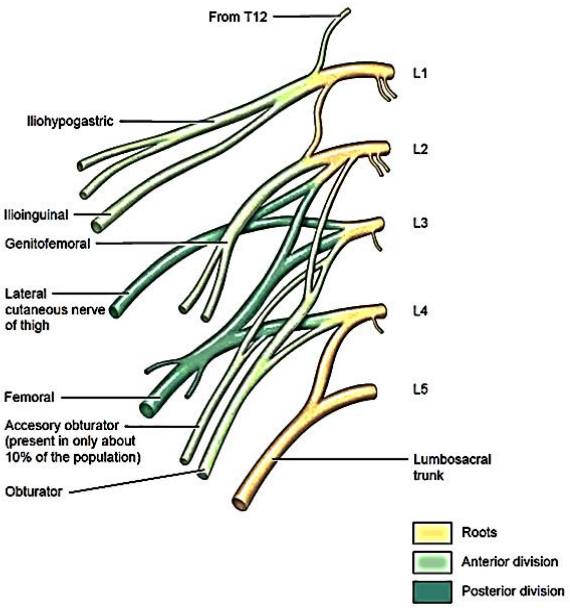


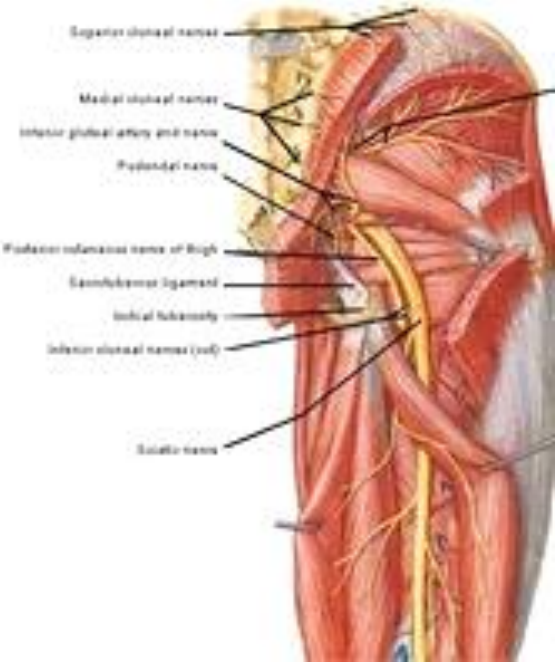
The brachial plexus

3. The lumbosacral plexus supplies nerves to the lower extremities. The largest of these branches is the sciatic nerve, which leaves the dorsal part of the pelvis, passes beneath the gluteus maximus muscle, and extends down the back of the thigh. At its beginning it is nearly 1 inch thick, but it soon branches to the thigh muscles; near the knee it forms two subdivisions that supply the leg and the foot.



The lumbosacral plexus





The Autonomic Nervous System

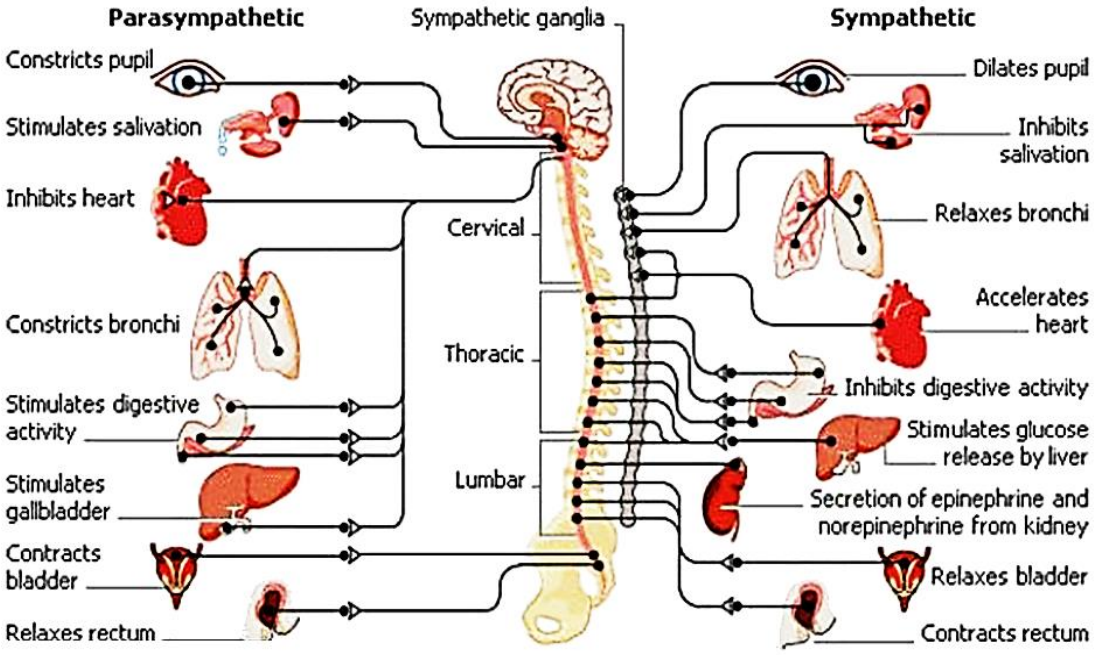
❖ **Parts of the Autonomic Nervous System** Although the internal organs such as the heart, lungs, and stomach contain nerve endings and nerve fibers for conducting sensory messages to the brain and cord, most of these impulses do not reach consciousness. These afferent impulses from the viscera are translated into reflex responses without reaching the higher center of the brain; the sensory neurons from the organs are grouped with those that come from the skin and voluntary muscles. In contrast, the efferent neurons, which supply the glands and the involuntary muscles, are arranged very differently from those that supply the voluntary muscles.

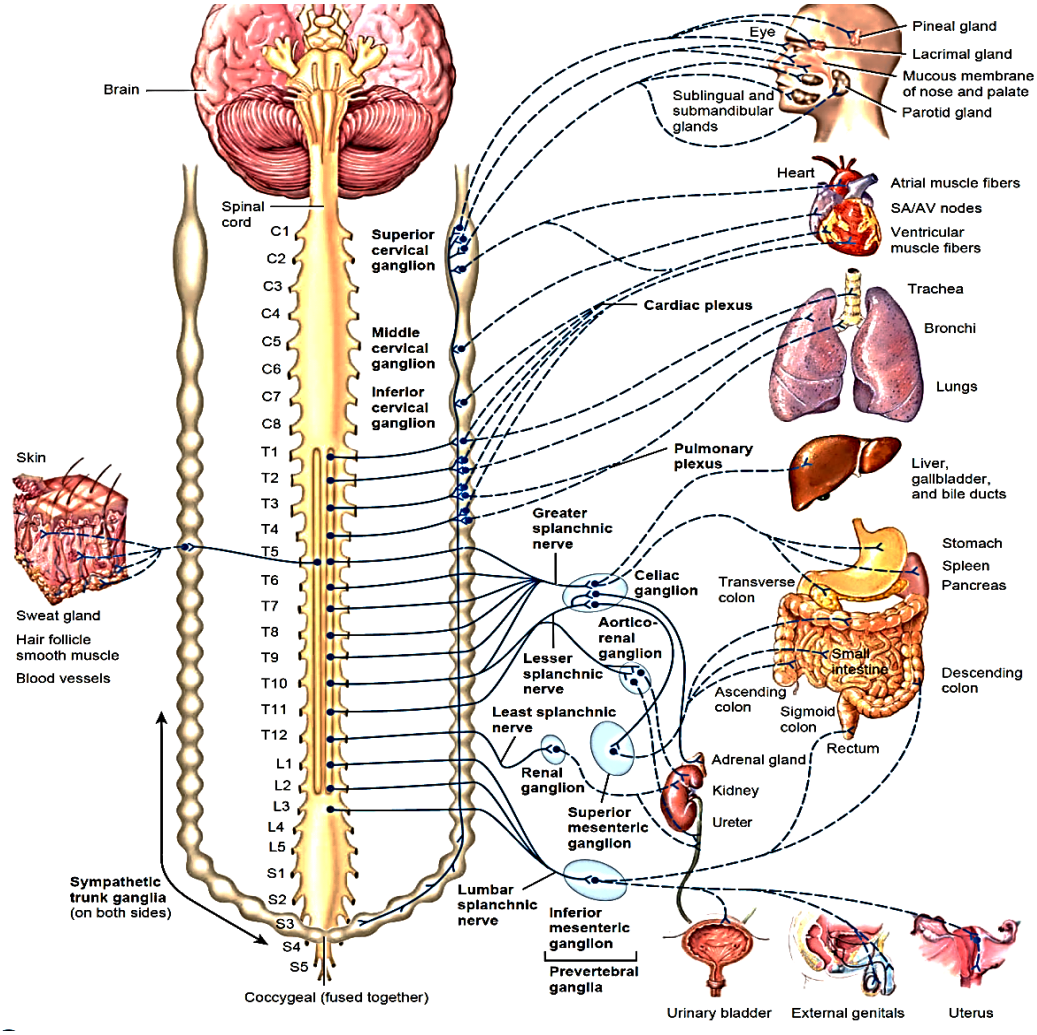
This variation in the location and arrangement of the visceral efferent neurons has led to their classification as part of a separate division called the autonomic nervous system (ANS) .

The ANS itself is comprised of sympathetic and parasympathetic divisions.

The autonomic nervous system has many ganglia that serve as relay stations. In these g~ each message is transferred at a synapse from the first neuron to a second one and from there to the muscle or gland cell. This differs from the voluntary (somatic nervous system, in which each motor nerve fiber extends all the way from the spinal cord to the skeletal muscle with no intervening synapse. Some of the autonomic fibers are within the spinal nerves; some are within the cranial nerves. The distribution of the two divisions of the ANS is as follows:

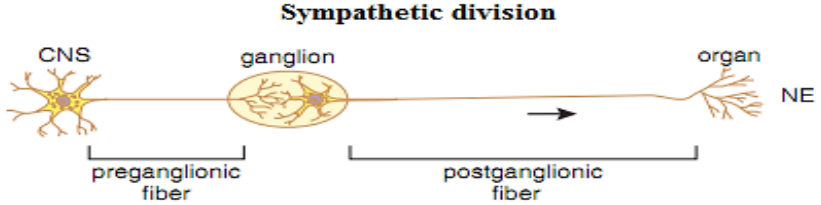
1. **The sympathetic pathways** begin in the spinal cord with cell bodies in the thoracic and lumbar regions, the thoracolumbar area.



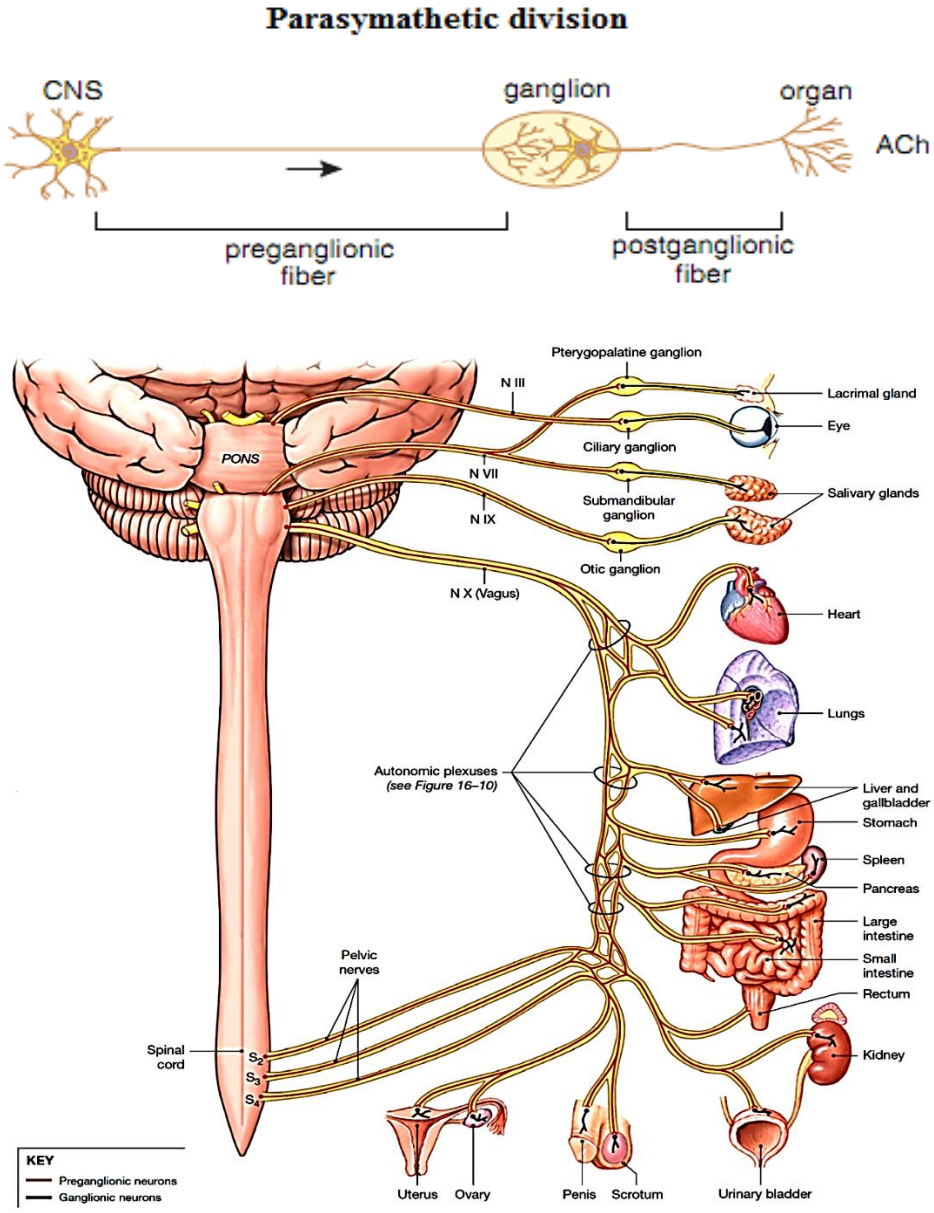


The sympathetic fibers arise from the spinal cord at the level of the first thoracic nerve down to the level of the second lumbar spinal nerve. From this part of the cord, nerve fibers extend to ganglia where they synapse with a second set of neurons, the fibers of which extend to the glands and involuntary muscle tissues.

Many of the sympathetic ganglia form the sympathetic chains, two cord like strands of ganglia that extend along either side of the spinal column from the lower neck to the upper abdominal region. The nerves that supply the organs of the abdominal and pelvic cavities synapse in three single ganglia farther from the spinal cord. The second neurons of the sympathetic nervous system act on the effectors by releasing the neurotransmitter epinephrine (adrenaline). This system is therefore described as adrenergic, which means "activated by adrenaline."



2. The parasympathetic pathways begin in the craniosacral areas, with fibers arising from cell bodies of the midbrain, medulla, and lower (sacral) part of the spinal cord. From these centers the first set of fibers extends to autonomic ganglia that are usually located near or within the walls of the effector organs. The pathways then continue along a second set of neurons that stimulate the involuntary tissues. These neurons release the neuro transmitter acetylcholine, leading to the description of this system as **cholinergic** (activated by acetylcholine).



Functions of the Autonomic Nervous System

The autonomic nervous system regulates the action of the glands, the smooth muscles of the hollow organs and vessels, and the heart muscle. These actions are all carried on automatically; whenever any changes occur that call for a regulatory adjustment, the adjustment is made without conscious awareness. The

sympathetic part of the autonomic nervous system tends to act as an accelerator for those organs needed to meet a stressful situation. It promotes what is called the fight-or-flight response. If you think of what happens to a person who is frightened or angry, you can easily remember the effects of impulses from the sympathetic nervous system:

1. Stimulation of the central portion of the adrenal gland.
2. This produces hormones, including epinephrine, that prepare the body to meet emergency situations in many ways. The sympathetic nerves and hormones from the adrenal gland reinforce each other.
3. Dilation of the pupil and decrease in focusing ability (for near objects)
4. Increase in the rate and force of heart contractions
5. Increase in blood pressure due partly to the more effective heartbeat and partly to constriction of small arteries in the skin and the internal organs
6. Dilation of blood vessels to skeletal muscles, bringing more blood to these tissues
7. Dilation of the bronchial tubes to allow more oxygen to enter
8. Increase in metabolism.

The sympathetic system also acts as a brake on those systems not directly involved in the response to stress, such as the urinary and digestive systems. If you try to eat while you are angry, you may note that your saliva is thick and so small in amount that you can swallow only with difficulty.

Under these circumstances, when food does reach the stomach, it seems to stay there longer than usual.

The parasympathetic part of the autonomic nervous system nonnal1y acts as a balance for the sympathetic system once a crisis has passed. The parasympathetic system brings about constriction of the pupils, slowing of the heart rate, and constriction of the bronchial tubes. It also stimulates the formation and release of urine and activity of the digestive tract. Saliva, for example, flows more easily and profusely and its quantity and fluidity increase.

Most organs of the body receive both sympathetic and parasympathetic stimulation, the effects of the two systems on a given organ generally being opposite. **Table below shows** some of the actions of these two systems.

Systems on Selected Organs

Effector	Sympathetic system	Parasympathetic System
Pupils of eye	Dilation	Constriction
Sweat glands	Stimulation	None
Digestive glands	Inhibition	Stimulation
Heart	Increased rate and strength of beat	Decreased rate and strength of beat
Bronchi of lungs	Dilation	Constriction
Muscles of digestive system	Decreased contraction (peristalsis)	Increased contraction
Kidneys	Decreased activity	None
Urinary bladder and emptying	Relaxation	Contraction
Liver	Increased release of glucose	None
Penis	Ejaculation	Erection
Adrenal medulla	Stimulation	None
Blood vessels to skeletal muscles	Dilation	Constriction
Skin	Constriction	None
Respiratory system	Dilation	Constriction
Digestive organs	Constriction	Dilation

Sense Organs

► Classification of sense organs

The sense organs are often classified as special sense organs and general sense organs. Special sense organs, such as the eye, are characterized by large and complex organs or by localized groupings of specialized receptors in areas such as the nasal mucosa or tongue. The general sense organs for detecting stimuli such as pain and touch are microscopic receptors widely distributed through out the body. Other general sense organs include receptors that indicate the tension on our muscles and tendons so that we can maintain balance and muscle tone and be aware of the positions of our body parts. Table classifies the special sense organs.

Converting stimulus into a sensation

All sense organs, regardless of size, type, or location, have in common some important functional characteristics. First, they must be able to sense or detect a stimulus in their environment. Of course, different sense organs detect and respond to different types of stimuli in different ways. Whether it is light, sound, temperature change, mechanical presence, or the presence of chemicals identified as taste or smell, the stimulus must be changed into an electrical signal or nerve impulse.

This signal is then transmitted over a nervous system "pathway" to the brain, where the sensation is perceived.

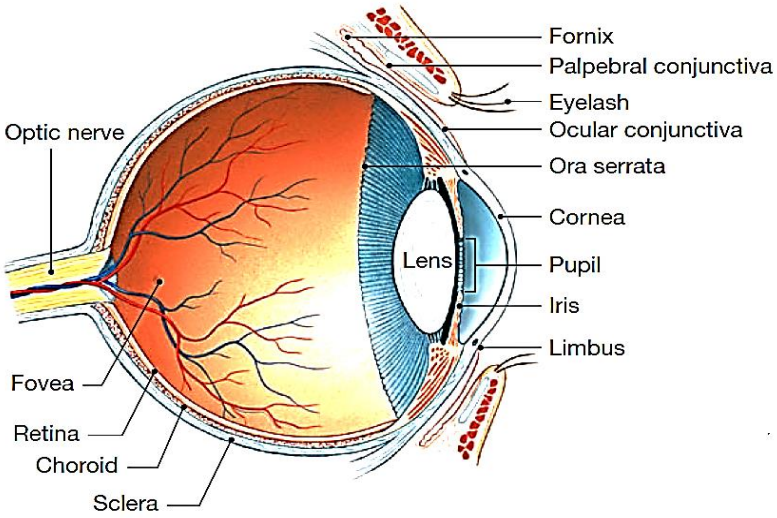
Special Sense Organs

SENSE ORGAN	SPECIFIC RECEPTOR	TYPE OF RECEPTOR	SENSE
Eye	Rods and Cons	Photoreceptor	Vision
Ear	Organ of Corti	Mechanoreceptor	Hearing
	Cristae ampularis	Mechanoreceptor	Balance
Nose	Olfactory cells	Chemoreceptor	Smell
Taste buds	Gustatory cells	Chemoreceptor	Taste

Special sense organs

The Eye

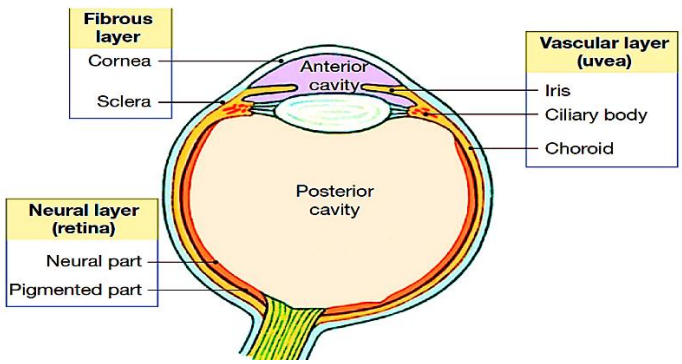
When you look at a person's eye you see only a small part of the whole eye. Three layers of tissue form the eye ball: the sclera, the choroids, and the retina .



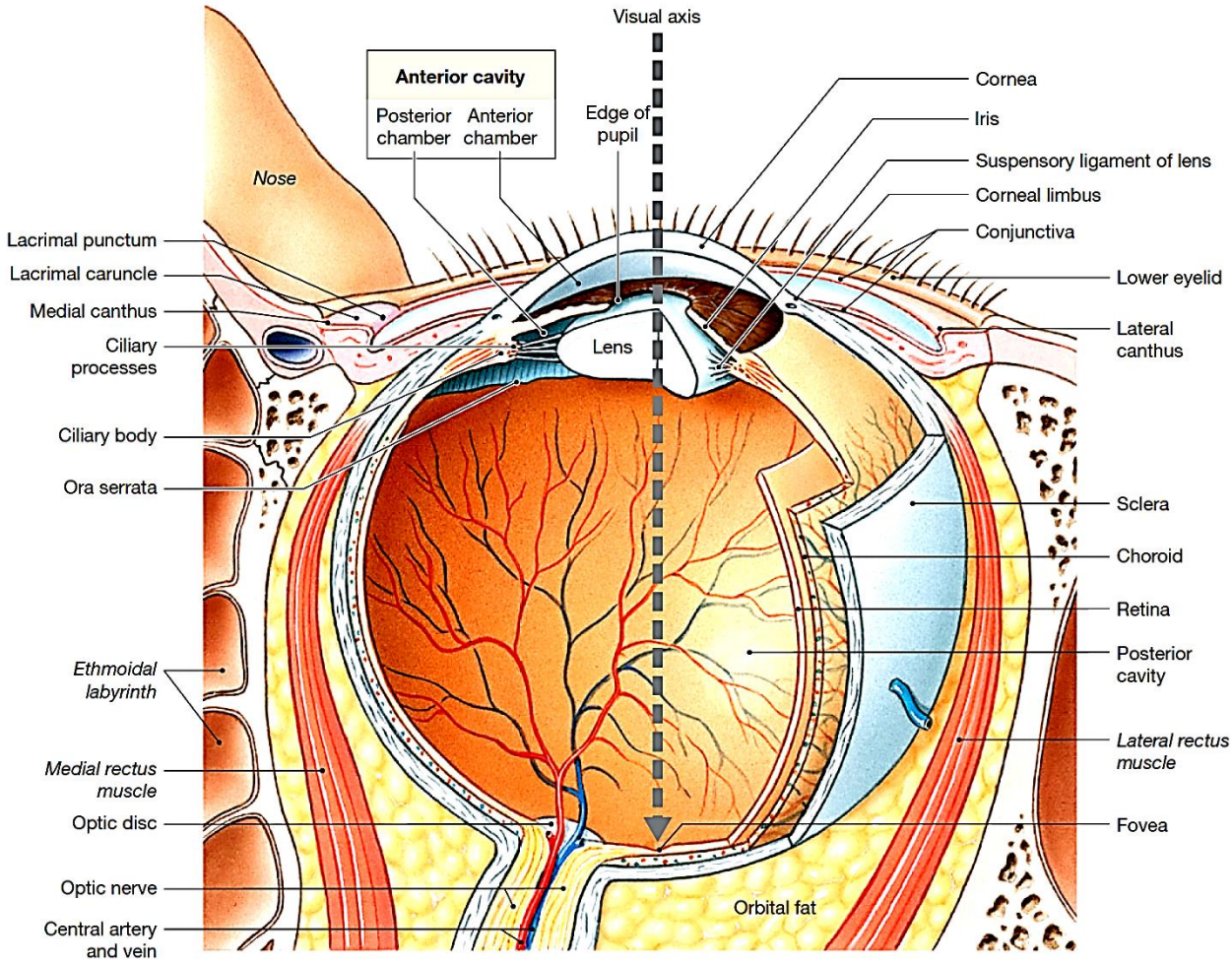
The outer layer of sclera consists of though fibrous tissue. The white of the eye is part of the front surface of the sclera. The other part of the front surface of the sclera is called the cornea and is sometimes spoken of as the window of the eye because of its transparency. At a casual glance, however, it does not look transparent but appears blue, brown, gray, or green because it lies over the iris, the colored part of the eye. A mucous membrane known as the conjunctiva lines the eyelids and covers the sclera in front. The conjunctiva is kept moist by tears formed in the lacrimal gland located in the upper lateral portion of the orbit.

The middle layer of the eyeball, the choroid, contains a dark pigment to prevent the scattering of incoming light rays. Two involuntary muscles make up the front part of the choroids.

One is the iris, the colored structure seen through **th**e cornea, and the othere is the ciliary muscle.

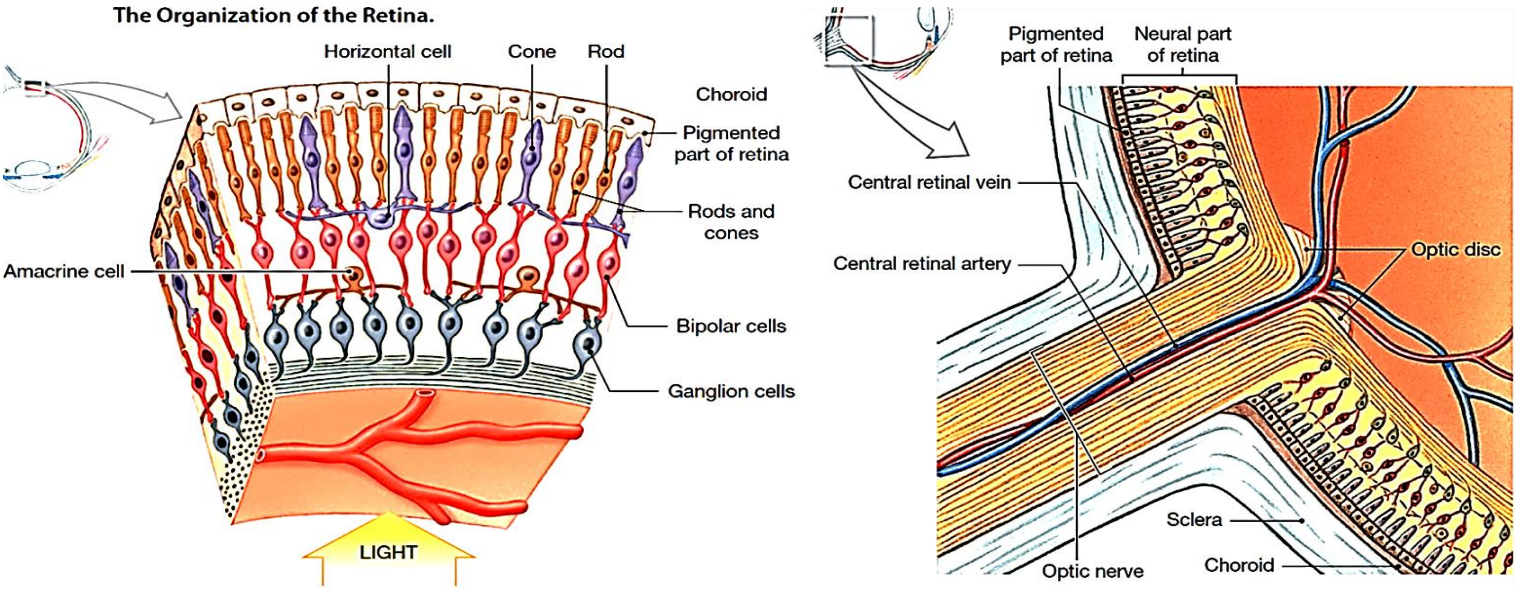


The black center of the iris is really a hole in this doughnut-shaped muscle; it is pupil of the eye. Some of the fibers of the iris are arranged like spokes in a wheel. When they contract the pupils dilate, letting in more light rays. Other fibers are circular. When they contract, the pupils constrict, letting fewer light rays. Normally, the pupils constrict in bright light and dilate in dim light. When we look at distant objects, the ciliary muscle is relaxed, and the lens has only a slightly curved shape.



To focus on near objects, however, the ciliary muscle contract. As it contracts, it pulls the choroids coat forward toward the lens, thus causing the lens to bulge and curve even more. Most of us become more farsighted as we grow older and lose the ability to focus on close objects because our lenses lose their elasticity and con no longer bulge enough to bring near objects into focus. Presbyopia or old sightedness is the name for this condition.

The retina or innermost layer of the eyeball contains microscopic receptor cells, called rods and cones because of their shapes. Dim light can stimulate the rods, but fairly bright light is necessary to stimulate the cones.



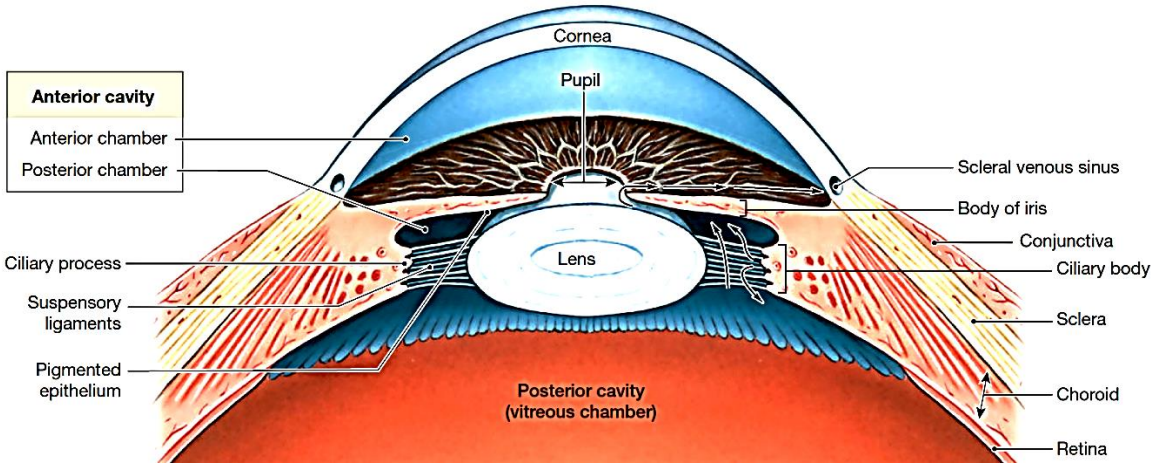
In other words, rods are the receptors for night vision and cones for daytime vision.

There are three kinds of cones; each is sensitive to a different colour: red, green, or blue. Scattered throughout the central portion of the retina, these three types of cones allow us to distinguish between different colors.

Fluids fill the hollow inside of the eyeball. They maintain the normal shape of the eyeball and help refract light rays; that is, the fluids bend light rays to bring them to focus on the retina.

Aqueous humor is the name of the watery fluid in front of the lens (in the anterior cavity of the eye), and vitreous humor is the name of the jellylike fluid behind the lens (in the posterior cavity).

The Circulation of Aqueous Humor. Aqueous humor, which is secreted at the ciliary body, circulates through the posterior and anterior chambers before it is reabsorbed through the scleral venous sinus.



Aqueous humor is constantly being formed, drained, and replaced in the anterior cavity. If drainage is blocked for any reason, the internal pressure within the eye will increase, and damage that could lead to blindness will occur. This condition is called glaucoma.

The lens of the eye lies directly behind the pupil. It is held in place by a ligament attached to the ciliary muscle. In most young people, the lens is transparent and somewhat elastic so that it is capable of changing shape. Exposure to ultraviolet (UV) radiation in sunlight may cause cataracts or milky spots on the lens. Large or numerous cataracts may cause blindness. Cataracts can be removed surgically and replaced with artificial lenses.

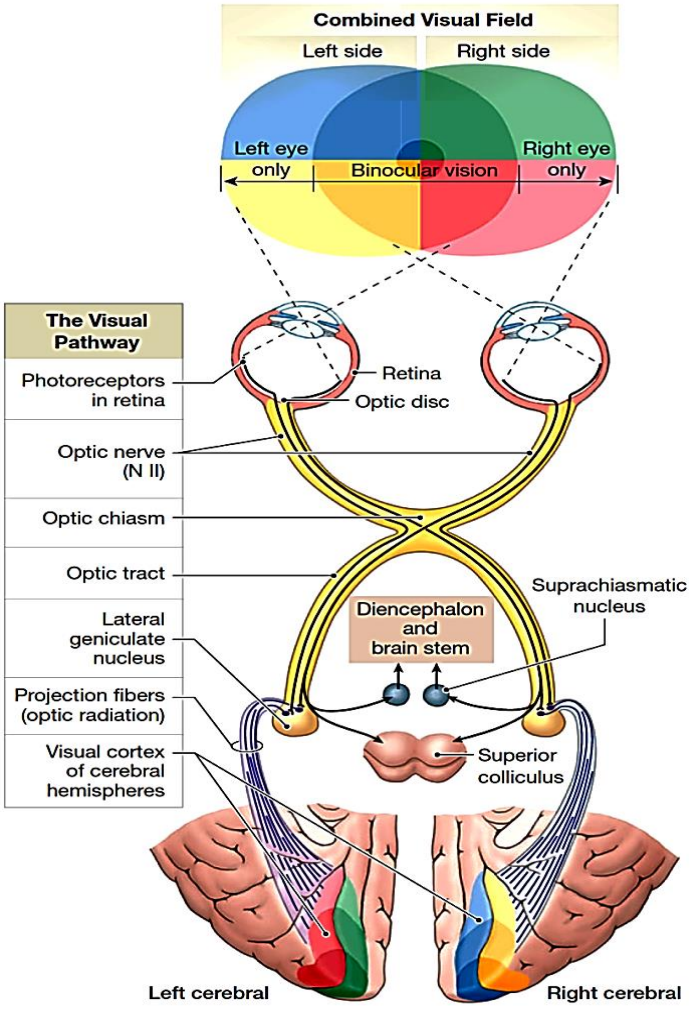
Visual Pathway

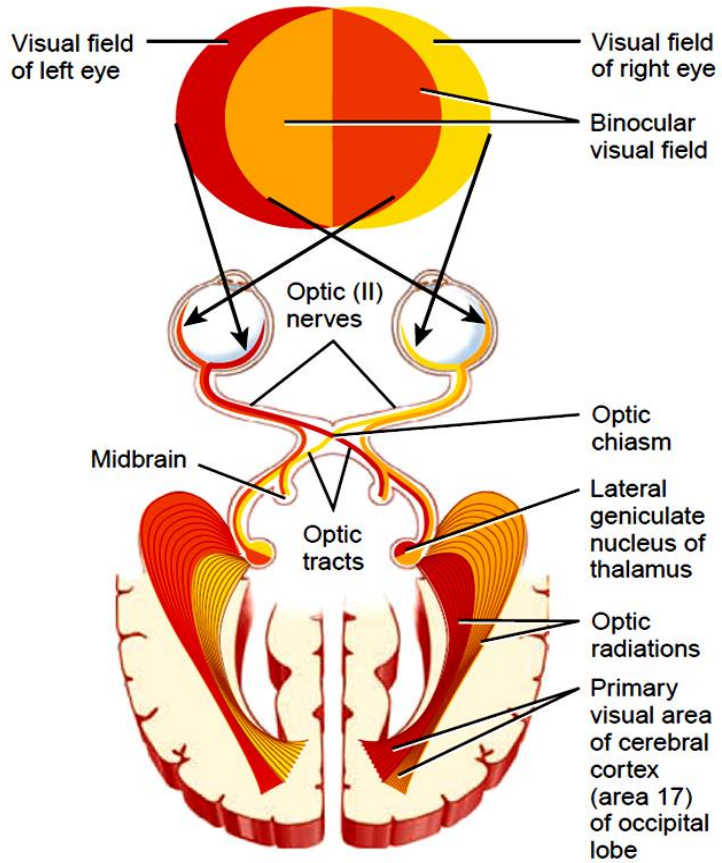
Light is the stimulus that results in vision (that is our ability to see objects as they exist in our environment). Light enters the eye through the pupil and is refracted or bent so that it is focused on the retina. Refraction occurs as light passes through the cornea, the aqueous humor, the lens, and the vitreous humor on its way to the retina.

The inner most layer of the retina contains the rods and cones, which are the photoreceptor cells of the eye .

They respond to a light stimulus by producing a nervous impulse. The rod and cone photoreceptor cells synapse with neurons in the bipolar and ganglionic layers of the retina.

Nervous signals eventually leave the retina and exit the eye through the optic nerve on the posterior surface of the eyeball. no rods or cones are present in the area of the retina where the optic nerve fibers exit. The result is a " blind spot" known as the optic disc .





Superior view of transverse section through eyeballs and brain

After leaving the eye, the optic nerves enter the brain and travel to the visual cortex of the occipital lobe. In this area of the brain, visual interpretation of the nervous impulses that were generated by light stimuli in the rods and cones of the retina result in "seeing".

The Ear

In addition to its role in hearing, the ear also functions as the sense organ of equilibrium and balance. As we shall later see, the stimulation or "trigger" that activates receptors involved with hearing and equilibrium is mechanical, and the receptors themselves are called mechanoreceptors.

Physical forces that involve sound vibrations and fluid movements are responsible

for initiating nervous impulses eventually perceived as sound and balance.

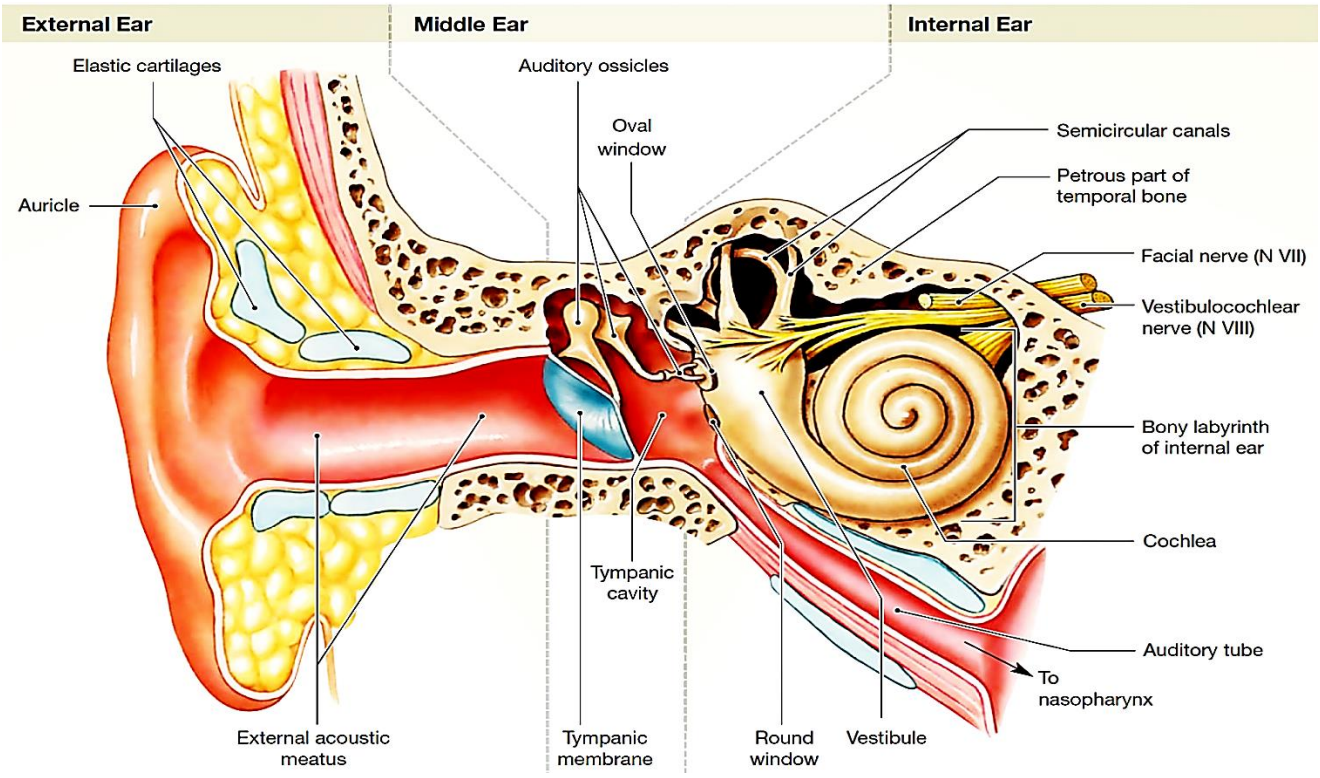
The ear is much more than a mere appendage on the side of the head. A large part of the ear, and by far its most important part, lies hidden from view deep inside the temporal bone.

It is divided into the following anatomical areas .

1. External ear

2. Middle ear

3. Inner (internal) ear



1.External ear

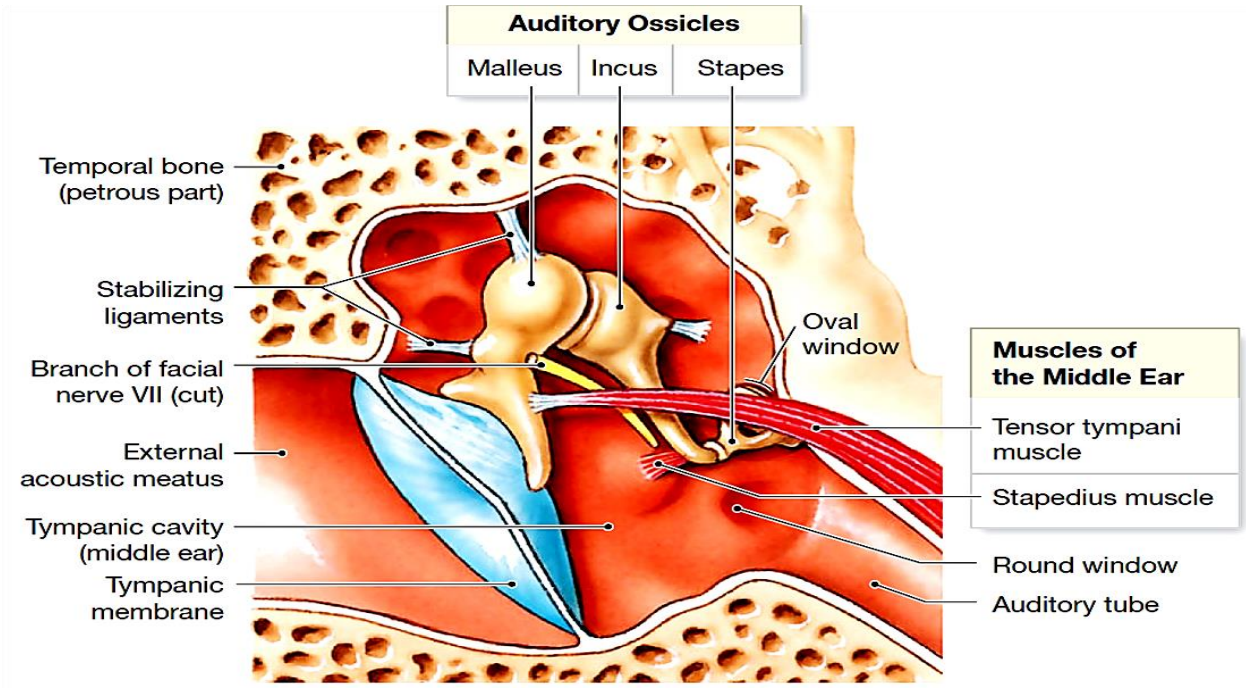
The external ear has two parts: the auricle or pinna and the external auditory canal. The auricle is the appendage on the side of the head surrounding the opening of the external auditory canal. The canal itself is a curve about 2.5 cm (1 inch) in length. It extends into the temporal bone and ends at the tympanic membrane or eardrum, which is a partition between the external and middle ear.

The skin of the auditory canal, especially in its outer one third, contains many short hairs and ceruminous glands that produce a waxy substance called cerumen that may collect in the canal and impair hearing by absorbing or blocking the passage of sound waves. Sound waves travelling through the external auditory canal strike the tympanic membrane and cause it to vibrate.

2.Middle Ear

The middle ear is a tiny and very thin epithelium lined cavity hollowed out of the temporal bone. It houses three very small bones. The names of these ear bones, called ossicles, describe their shapes − malleus (hammer), incus (anvil), and stapes (stirrup). The "handle" of the malleus attaches to the inside of the tympanic membrane, and the "head" attaches to the incus. The incus attaches to the stapes, and the stapes presses against a membrane that covers a small opening, the oval window. The oval window separates the middle ear from the inner ear.

When sound waves cause the eardrum to vibrate, that movement is transmitted and amplified by the ear ossicles as it passes through the middle ear.



Movement of the stapes against the oval window causes movement of fluid in the inner ear. A point worth mentioning, because it explains the frequent spread of infection from the throat to the ear, is the fact that a tube– the auditory or eustachian tube– connects the throat with the middle ear.

The epithelial lining of the middle ears, auditory tubes, and throat are extensions of one continuous membrane.

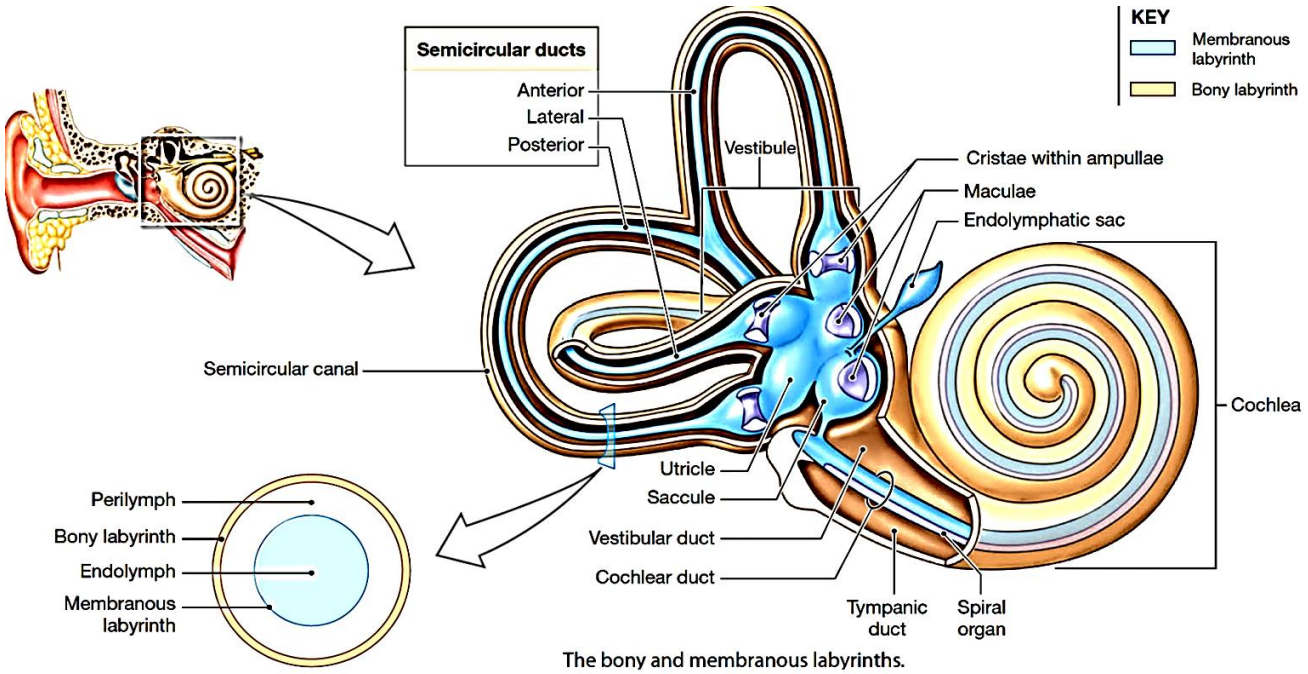
Consequently, a sore throat may spread to produce a middle ear infection called otitis media.

3.Inner Ear

The activation of specialized mechanoreceptors in the inner ear generates nervous impulses that result in hearing and equilibrium.

Anatomically, the inner ear consists of three spaces in the temporal bone, assembled in a complex maze called the bony labrynth. This odd shaped bony space is filled with a watery fluid called perilymph and is divided into the following parts: vestibule, semicircular canals, and cochlea.

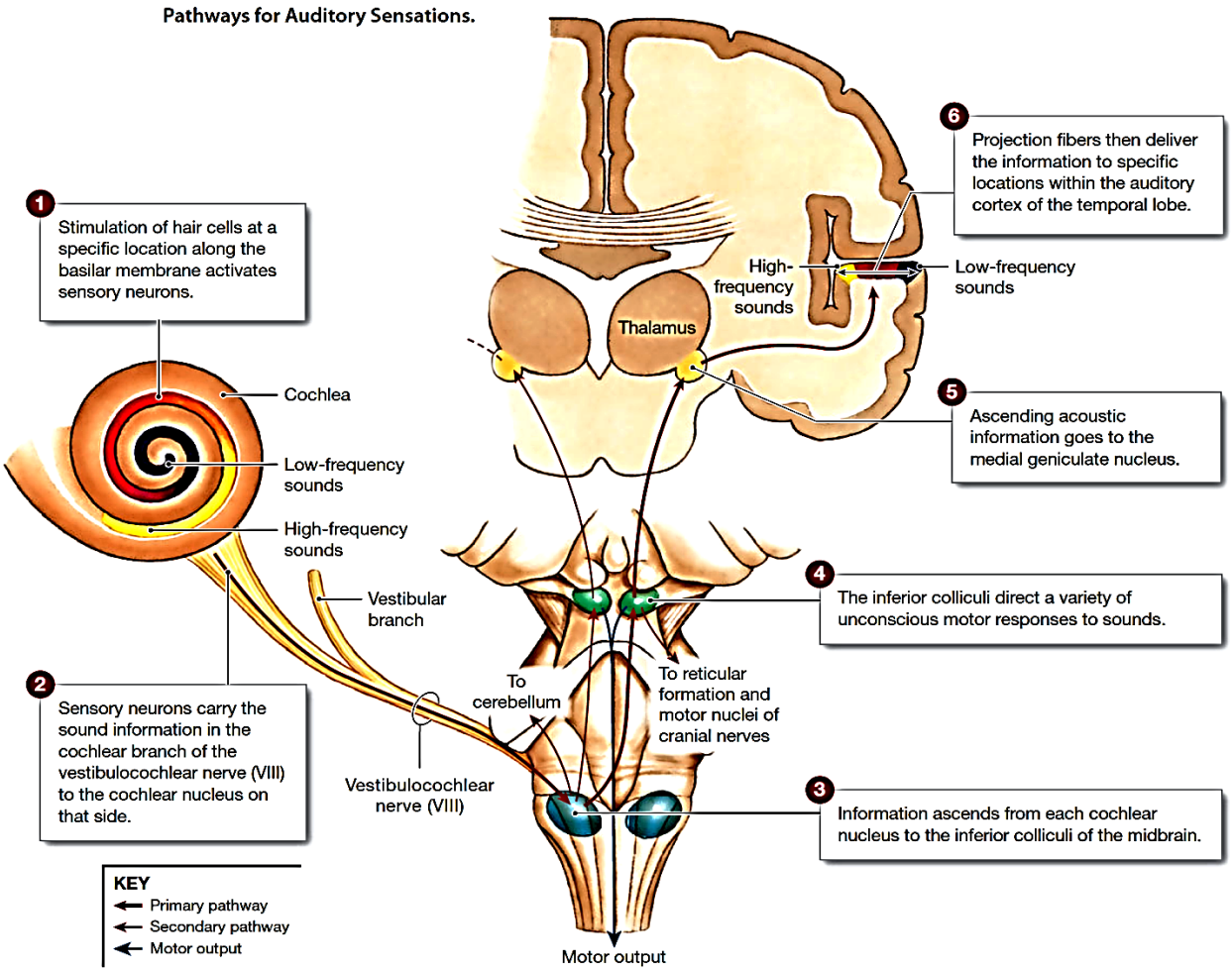
The vestibule is adjacent to the oval window between the semicircular canals and the cochlea. Ballonlike membranous sac is suspended in the perilymph and follows the shape of the bony labyrinth much like a "tube within a tube." This is a membranous labyrinth, and it is filled with a thicker fluid called endolymph.



The bony and membranous labyrinths.

The specialized membranoreceptors for balance and equilibrium are located in the three semicircular canals and the vestibule. The three half-circle semicircular canals are oriented at right angles to one another .

Within each canal is a specialized receptor called a crista ampullaris, which generates a nerve impulse when you move your head.



The sensory cells in the cristae ampullares have hair like extensions that are suspended in the endolymph. The sensory cells are stimulated when movement of the head causes the endolymph to move, thus causing the hairs to bend. Nerves from other receptors in the vestibule join those from the

semicircular canals to form the vestibular nerve, a division of the acoustic nerve or cranial nerve VIII.

Eventually, nervous impulses passing through this nerve reach the cerebellum and medulla. Other connections from these areas result in impulses reaching the cerebral cortex.

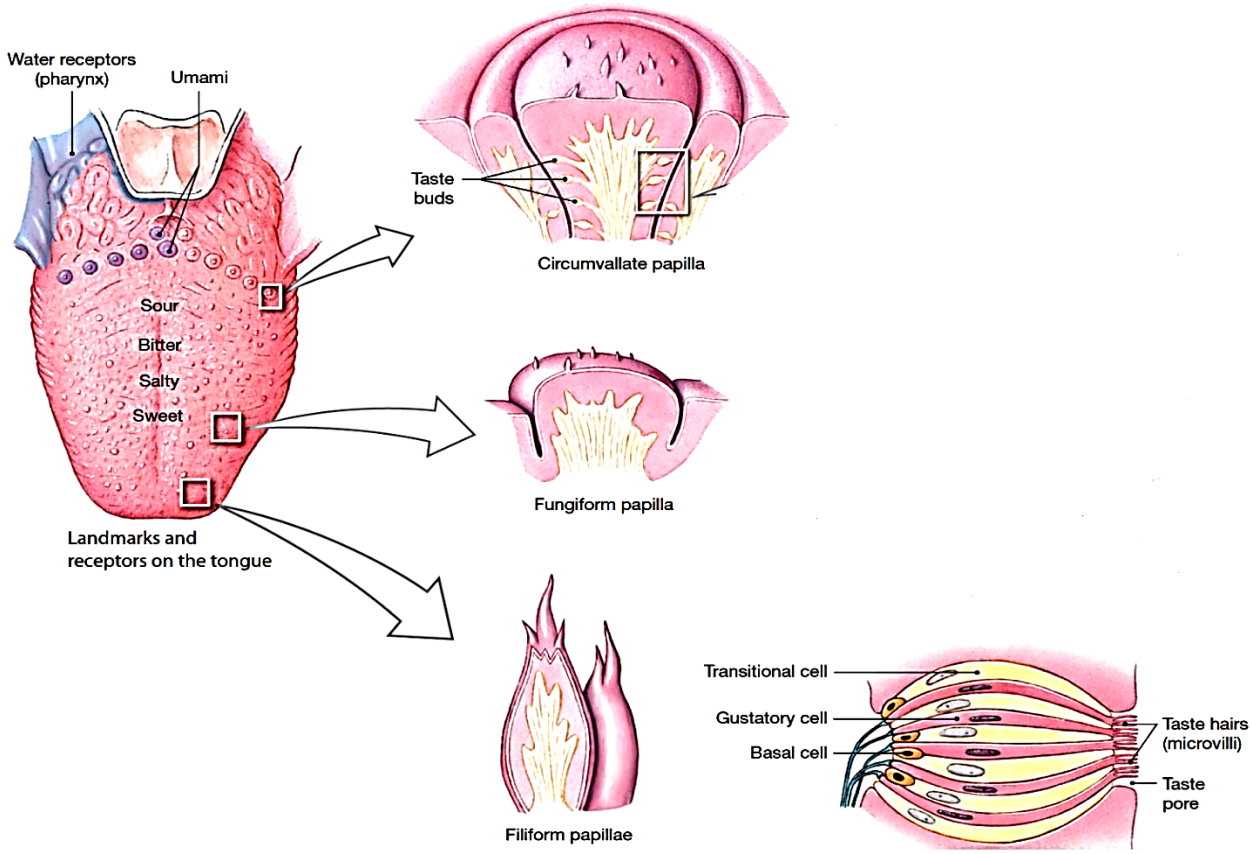
The organ of hearing, which lies in the snail shaped cochlea, is the organ of Corti. It is surrounded by endolymph filling the membranous cochlea or cochlear duct, which is the membranous tube within the bony cochlea. Specialized hair cells on the organ of Corti generate nerve impulses when they are bent by the movement or endolymph set in motion by sound waves.

The Taste Receptors

The chemical receptors that generate nervous impulses resulting in the sense of taste are called taste buds. About 10,000 of these microscopic receptors are found on the sides of much larger structure on the tongue called papillae and also as portions of other tissues in the mouth and throat.

Nervous impulses are generated by specialized cells in taste buds, called gustatory cells. They respond to dissolved chemicals in the saliva that bathe the tongue and mouth tissues .

Only four kinds of taste sensations−sweet, sour, bitter, and salty−result from stimulation of taste buds. All other flavors result from a combination of taste bud and olfacctory receptor stimulation. In other words, the myriads of tastes recognized are not tastes alone but tastes plus odors.



For this reason a cold that interferes with the stimulation of the olfactory receptors by odors from foods in the mouth markedly dulls taste sensations. Nervous impulses generated by stimulation of taste buds travel primarily through two cranial nerves (VII and IX) to end specialized taste area of the cerebral cortex.

The Smell Receptors

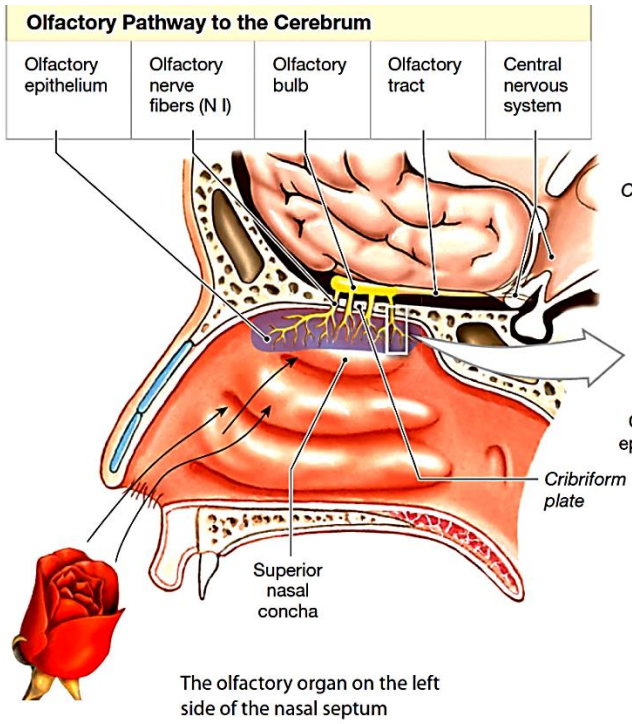
The chemical receptors responsible for the sense of smell are located in a small area of epithelial tissue in the upper part o the nasal cavity . The location of the olfactory receptors is somewhat hidden, and we are

often forced to forcefully sniff air to smell delicate odors. Each olfactory cell has a number of specialized cilia that sense different chemicals and cause the cell to respond by generating a nervous impulse. To be detected by olfactory receptors, chemicals must be dissolved in the watery mucus that lines the nasal cavity.

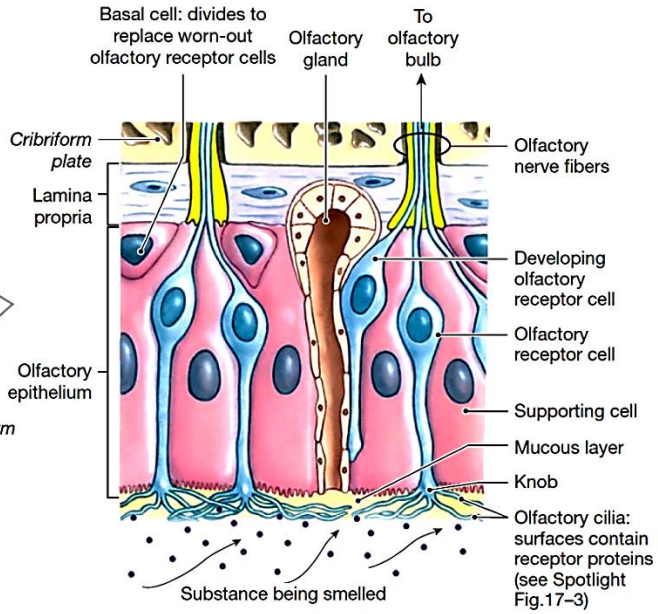
Although the olfactory receptors are extremely sensitive (that is, stimulated by even very slight odors), they are also easily fatigued —a fact that explains why odors that are at first very noticeable are not sensed at all after a short time.

After the olfactory cells are stimulated by odor-causing chemicals, the resulting nerve impulse travels through the olfactory nerves in the olfactory bulb and tract and then enters the thalamic and olfactory centers of the brain, where the nervous impulses are interpreted as specific odors.

The pathways taken by olfactory nerve impulses and the area where these impulses are interpreted are closely associated with areas of the brain important in memory and emotion. For this reason, we may retain vivid and long-lasting memories of particular smells and odors.



The olfactory organ on the left side of the nasal septum



An olfactory receptor is a modified neuron with multiple cilia extending from its free surface.

Temporary reduction of sensitivity to smells often results from colds and other nasal infections. Progressive reduction of the sense of smells often seen in smokers because of the damaging effects the pollutants in tobacco smoke.

In olfaction, as with all the special senses, advancing age often brings a structural degeneration that result in reduced function. It is no wonder that many older adults become isolated and depressed when their

contact with the outside world, the special senses, is gradually lost. Caring health professionals recognize these signs of aging and provide assistance needed by their aged patients to enjoy life.

General Sense Organs

Groups of highly specialized and localized receptors are typically associated with the special senses. In the sense organs, however, receptors are found in almost every part of the body. To demonstrate this fact, try touching any point of your skin with the tip of a toothpick. You can hardly miss stimulating at least one receptor and almost instantaneously experiencing a sensation of touch. Stimulation of some receptors leads to the sensation of heat; Stimulation of others gives the sensation of cold, and stimulation of still others gives the sensation of pain or pressure.

When special receptors in the muscles and joints are stimulated, you sense the position of the different parts of the body and know whether they are moving and in which direction they are moving without even looking at them. Perhaps you have never realized that you have this sense of position and movement a sense called proprioception or kinaesthesia.

Disruption of general sense organs can occur by a variety of mechanisms. For example, third degree burns can completely destroy general sense receptors throughout the affected area.

Temporary impairment of general sense receptors occurs when the blood flow to them is slowed. This commonly occurs when you put your legs in a position that presses your legs in a way that reduces blood flow. When you try to stand up, you cannot feel you legs because the general sense organs are temporarily impaired. You may not even be able to walk because you cannot tell where you legs are without looking at them. As blood flow returns, reactivation of the sense organs may produce a tingling sensation.